MATH 714: PROBLEMS, WITH SOLUTIONS

SANTIAGO ARANGO-PIÑEROS

Abstract. Math 714 is the graduate course on *The Arithmetic Of Elliptic Curves* taught at UMass Amherst. I taught this class in the Fall of 2026. The students read ahead, and we used the lecture time to work on the problems in this document. Each part below is one problem set (corresponding to one meeting), with its summaries, definitions, results and problems. Each problem is preceded by the definitions and results it uses, and each item is followed by its solution.

A note on how this was made. The problems were chosen and curated by Santiago Arango-Piñeros. The first draft typesetting and the solutions were written by Claude, and then reviewed by the author. Corrections and recommendations are welcome: send them to

santiago.arango.pineros@gmail.com.

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Part 1. Selmer elliptic curves

The problems below study a single family of plane cubics and use it to reconstruct, by hand, some of the main results of [Sil09, §§III.1–III.4]: Weierstrass equations, the group law, the Riemann–Roch construction of a Weierstrass model, and a first endomorphism. Section 4 is a set of worked `Magma` examples to read and re-run afterwards, with three short variations to try at the end.

Standing notation.

- A field k with $\operatorname{char} k \neq 3$, an algebraic closure $\overline{k}$ of k, and $d \in k^\times$.
- The cubic form $F := X^3 + Y^3 - dZ^3$, the plane cubic $C_d := \{F = 0\} \subset \mathbb{P}^2_k$, and its rational point $O := (1 : -1 : 0)$.
- A primitive cube root of unity $\zeta \in \overline{k}$.
- A Selmer elliptic curve: a pair (C_d, O) of this shape, after [Sel51]. Section 5 explains the name.
- Projective coordinates X, Y, Z, as usual, and lowercase letters for affine coordinates.
- The base change $\overline{C}_d := C_d \times_k \overline{k}$.

1. Warm-up: a plane cubic and a rational point on it

Definition 1.1 (Elliptic curves [Sil09, §III.3])**.** An elliptic curve over k is a pair (E, O) with E a smooth, projective, geometrically irreducible curve of genus one over k and $O \in E(k)$ a fixed rational point.

Problem 1 (It is an elliptic curve)**.**

(1) Check that $O \in C_d(k)$.

Solution. $1^3 + (-1)^3 - d \cdot 0 = 0$.

(2) Compute the three partial derivatives of F and deduce that C_d is smooth, using $\operatorname{char} k \neq 3$ and $d \neq 0$. Where is each hypothesis used?

Solution. **$\operatorname{char} k \neq 3$ to cancel the 3, and $d \neq 0$ to cancel d.**

$$\partial F/\partial X = 3X^2, \qquad \partial F/\partial Y = 3Y^2, \qquad \partial F/\partial Z = -3dZ^2.$$

If all three vanish at a point of $\mathbb{P}^2(\overline{k})$ then $3X^2 = 3Y^2 = 3dZ^2 = 0$. Since $3 \neq 0$ in k we get $X^2 = Y^2 = 0$, and since also $d \neq 0$ we get $Z^2 = 0$. So $X = Y = Z = 0$, which is not a point of $\mathbb{P}^2(\overline{k})$. Both hypotheses are used to divide: $\operatorname{char} k \neq 3$ to cancel the 3, and $d \neq 0$ to cancel d in the third equation.

Remark. If $d = 0$ the "curve" $X^3 + Y^3 = 0$ is three concurrent lines.

(3) Suppose instead that $\operatorname{char} k = 3$ (only for this problem). Show that over $\overline{k}$ the curve C_d is a *triple line*.

Solution. In characteristic three the Frobenius $a \mapsto a^3$ is a ring homomorphism, so for $\delta \in \overline{k}$ with $\delta^3 = d$,

$$(X + Y - \delta Z)^3 = X^3 + Y^3 + (-\delta Z)^3 = X^3 + Y^3 - dZ^3 = F.$$

Thus $\overline{C}_d$ is the triple line $\{X + Y - \delta Z = 0\}$: not reduced, hence certainly not smooth, and of arithmetic genus 1 for entirely uninteresting reasons.

Remark. Note that for $d \notin (k^\times)^3$ the polynomial F is still irreducible *over* k, so the failure is invisible until one passes to $\overline{k}$.

(4) Show that a smooth plane curve is geometrically irreducible.

Solution. If a plane curve C decomposes as $C_1 \cup C_2$ with C_i of degrees $e_i \geqslant 1$, then by Bézout $(C_1 \cap C_2)(\overline{k}) \neq \varnothing$, and any point of $(C_1 \cap C_2)(\overline{k})$ is a singular point of C. So a smooth plane curve is irreducible. Applied over $\overline{k}$ this gives geometric irreducibility.

(5) Conclude from the degree–genus formula that (C_d, O) is an elliptic curve over k.

Solution. A smooth plane curve of degree m has genus $(m-1)(m-2)/2$. For $m = 3$ this is 1. With (1), (2) and (4), the pair (C_d, O) is an elliptic curve over k.

The group law in Problem 2(4) is the chord–tangent construction of [Sil09, §III.2].

Proposition 1.2 (The group law and Pic^0 [Sil09, Proposition III.3.4])**.** *The map* $P \mapsto [P - O]$ *identifies the chord–tangent group law on* $C_d(\overline{k})$ *with addition in* $\mathrm{Pic}^0 \overline{C}_d$.

Problem 2 (The base point is an inflection point)**.**

(1) Show that the line $Z = 0$ meets C_d in three *distinct* points of $C_d(\overline{k})$, and write them down. For which k are all three in $C_d(k)$?

Solution. **All three lie in $C_d(k)$ exactly when $\zeta \in k$.** Setting $Z = 0$ gives $X^3 = -Y^3$, so $X/Y \in \{-1, -\zeta, -\zeta^2\}$ and the points of $C_d(\overline{k})$ on the line $Z = 0$ are

$$O = (1 : -1 : 0), \qquad Q_1 = (1 : -\zeta : 0), \qquad Q_2 = (1 : -\zeta^2 : 0).$$

They are distinct because $T^3 + 1$ is separable when $\operatorname{char} k \neq 3$.

(2) Compute the tangent line L to C_d at O, and show that $L \cap C_d = 3O$. (So O is an inflection point, or flex.)

Solution. **$L = \{X + Y = 0\}$.** From Problem 1(2), $\nabla F(O) = (3, 3, 0)$, so $L = \{X + Y = 0\}$. Substituting $X = -Y$ into F gives $(-Y)^3 + Y^3 - dZ^3 = -dZ^3$, which vanishes to order three at $Z = 0$. Hence $L \cap C_d = 3O$.

(3) In a Weierstrass model, which line plays the role of L? What is different here?

Solution. **The line at infinity $Z = 0$.** In a Weierstrass model it is the line at infinity $Z = 0$ that cuts $3O$. Here $Z = 0$ does something different: by (1) it meets C_d in three distinct points of $C_d(\overline{k})$. The flex tangent is instead $X + Y = 0$.

Remark. It is worth noticing that O being a flex is what makes the group law take the symmetric form in (4). It is not automatic for a plane cubic with a marked rational point.

(4) Let $\oplus$ denote the group law on C_d with identity O, defined by the chord–tangent construction. Using (2), show that for $P, Q, R \in C_d(\overline{k})$,

$$P \oplus Q \oplus R = O \iff P + Q + R \text{ is cut out on } C_d \text{ by a line.}$$

Solution. Write ℓ for a linear form cutting $P+Q+R$ and $\ell_O = X+Y$ for the flex tangent. Then $\operatorname{div}(\ell/\ell_O) = (P+Q+R) - 3O$ in $\operatorname{Div}\overline{C}_d$, so $P+Q+R \sim 3O$. Under the isomorphism $C_d(\overline{k}) \xrightarrow{\sim} \operatorname{Pic}^0 \overline{C}_d$, $P \mapsto [P-O]$, we get

$$P \oplus Q \oplus R \longmapsto [P+Q+R-3O] = 0,$$

so $P \oplus Q \oplus R = O$. Conversely, suppose $P \oplus Q \oplus R = O$. The line through P and Q meets C_d in a third point $R' \in C_d(\overline{k})$, and the direction just proved gives $P \oplus Q \oplus R' = O$. So $R' = R$, and $P+Q+R$ is cut out by that line.

2. The group law

Problem 3 (Negation, and 2-torsion)**.**

(1) Show that $\ominus(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$ for every point $(X_0 : Y_0 : Z_0) \in C_d(\overline{k})$. There are two things to check: that the right-hand side lies on C_d, and that it is the third intersection point of C_d with the line through O and $(X_0 : Y_0 : Z_0)$.

Solution. F is symmetric in X and Y, so $P' \coloneqq (Y_0 : X_0 : Z_0) \in C_d(\overline{k})$. Write $P \coloneqq (X_0 : Y_0 : Z_0)$. For collinearity,

$$\det \begin{pmatrix} 1 & -1 & 0 \\ X_0 & Y_0 & Z_0 \\ Y_0 & X_0 & Z_0 \end{pmatrix} = 1 \cdot (Y_0 Z_0 - X_0 Z_0) - (-1) \cdot (X_0 Z_0 - Y_0 Z_0) + 0 = 0.$$

If O, P, P' are distinct, the line through them therefore cuts out exactly $O+P+P'$. Otherwise $P' = P$ (note that $P' = O$ only when $P = O$): if $Z_0 \neq 0$ this means $X_0 = Y_0$, and if $Z_0 = 0$ it means $X_0 = \pm Y_0$, and of the three $\overline{k}$-points of Problem 2(1) only O qualifies. For $P = O$ the flex tangent cuts out $3O$ by Problem 2(2). For $X_0 = Y_0$ the two last rows coincide and the determinant says nothing, but the tangent line to C_d at P, $X_0^2 X + X_0^2 Y - d Z_0^2 Z = 0$, passes through O. So the line through O and P is tangent at P, and it cuts out $O+2P$. In every case $O+P+P'$ is cut out by a line, and since O is a flex, Problem 2(4) gives

$$O \oplus P \oplus P' = O, \quad \text{i.e.} \quad (Y_0 : X_0 : Z_0) = \ominus(X_0 : Y_0 : Z_0).$$

(2) Deduce that the nontrivial 2-torsion of C_d is cut out by the line $X = Y$, and that $\#C_d[2](\overline{k}) = 4$ when $\operatorname{char} k \neq 2$.

Solution. For $P = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$, $P = \ominus P$ means $(X_0 : Y_0 : Z_0) = (Y_0 : X_0 : Z_0)$ in $\mathbb{P}^2(\overline{k})$. If $Z_0 \neq 0$, normalize $Z_0 = 1$: then $X_0 = Y_0$ and $2X_0^3 = d$, which has three solutions in $\overline{k}$ since $\operatorname{char} k \neq 2$. If $Z_0 = 0$ then P is one of the three $\overline{k}$-points of Problem 2(1), and $(X_0 : Y_0 : 0) = (Y_0 : X_0 : 0)$ means $X_0^2 = Y_0^2$, i.e. $Y_0/X_0 = \pm 1$. Of $-1, -\zeta, -\zeta^2$ only -1 qualifies, so $P = O$. Hence the nontrivial 2-torsion consists of the three points of $C_d(\overline{k})$ on the line $X = Y$, and $\#C_d[2](\overline{k}) = 3+1 = 4$, as it must be for an elliptic curve in characteristic $\neq 2$.

(3) Take $k = \mathbb{Q}$. For which d does C_d have a nontrivial rational 2-torsion point? Write it down, and give the smallest positive integer d for which this happens.

Solution. **$d = 2c^3$ with $c \in \mathbb{Q}^\times$. Smallest: $d = 2$.** A nontrivial rational 2-torsion point exists if and only if $d/2 \in (\mathbb{Q}^\times)^3$, say $d = 2c^3$. It is then $(c : c : 1)$. The smallest positive integer is $d = 2$, with the rational point $(1 : 1 : 1)$, i.e. $1^3 + 1^3 = 2$.

Part (2) of Problem 4 is its heart. Parts (4) and (5) are then mechanical. If time is short, take (2) on faith and come back to its proof, but do not skip (3): it is what makes (5) painless. The

identity of (2) is the polarization of a cubic form. For a diagonal cubic the trilinear form can be written down by hand.

Problem 4 (The trilinear form, and the addition formulas)**.**
For vectors $P = (X_P, Y_P, Z_P)$, Q and R in $\overline{k}^3$, define

$$\langle P, Q, R\rangle := X_P X_Q X_R + Y_P Y_Q Y_R - d\, Z_P Z_Q Z_R. \tag{2.1}$$

(1) Check that (2.1) is symmetric in P, Q, R, linear in each argument separately, and satisfies $\langle P, P, P\rangle = F(P)$.

Solution. All three properties are visible: each monomial contains exactly one coordinate from each of P, Q, R, so the expression is linear in each argument; permuting P, Q, R permutes the factors within each monomial; and setting $P = Q = R$ gives $X_P^3 + Y_P^3 - dZ_P^3 = F(P)$.

(2) Prove the *expansion identity*: for scalars s, t,

$$F(sP + tQ) = s^3\langle P, P, P\rangle + 3s^2 t\,\langle P, P, Q\rangle + 3st^2\,\langle P, Q, Q\rangle + t^3\langle Q, Q, Q\rangle. \tag{2.2}$$

Where do the 3's come from?

Solution. **The 3's are binomial coefficients.** Expanding $\langle sP + tQ, sP + tQ, sP + tQ\rangle$ by linearity in each slot gives $2^3 = 8$ terms, indexed by which of sP, tQ goes into each slot. By the symmetry of (1), the value depends only on *how many* slots receive tQ: there are $\binom{3}{1} = 3$ terms with one tQ, each equal to $s^2 t\,\langle P, P, Q\rangle$, and $\binom{3}{2} = 3$ terms with two.

(3) Prove that $3\langle P, P, Q\rangle = \nabla F(P) \cdot Q$, and deduce that for $P \in C_d(\overline{k})$,

$$\langle P, P, Q\rangle = 0 \iff Q \text{ lies on the tangent line to } C_d \text{ at } P.$$

Solution. Both sides equal

$$3(X_P^2 X_Q + Y_P^2 Y_Q - dZ_P^2 Z_Q),$$

as $\nabla F(P) = (3X_P^2, 3Y_P^2, -3dZ_P^2)$. The tangent line to C_d at a smooth point P is $\{\nabla F(P) \cdot (X, Y, Z) = 0\}$, so $\langle P, P, Q\rangle = 0$ says exactly that Q lies on it.

(4) Let $P_1 = (X_1 : Y_1 : Z_1)$ and $P_2 = (X_2 : Y_2 : Z_2)$ be distinct points of $C_d(\overline{k})$, and parametrize the line through them as $sP_1 + tP_2$. Using (2), show that the third intersection point is at $(s : t) = (B : -A)$ where $A = \langle P_1, P_1, P_2\rangle$ and $B = \langle P_1, P_2, P_2\rangle$, and hence that

$$P_1 \oplus P_2 = (BY_1 - AY_2 : BX_1 - AX_2 : BZ_1 - AZ_2). \tag{2.3}$$

Why can A and B not both vanish?

Solution. Since $F(P_1) = F(P_2) = 0$, (2.2) reduces to

$$F(sP_1 + tP_2) = 3s^2 t\, A + 3st^2\, B = 3st(As + Bt).$$

This binary cubic in $(s : t)$ has roots $(1 : 0)$, $(0 : 1)$ and $(B : -A)$. The first two are P_1 and P_2, so the third intersection point is $BP_1 - AP_2$. Composing with $\ominus$ from Problem 3(1), which swaps the first two coordinates, gives (2.3). If $A = B = 0$ then F vanishes identically on the line, so the line would be a component of C_d, contradicting irreducibility (Problem 1(4)).

(5) Let $P = (X_0 : Y_0 : Z_0) \in C_d(\overline{k})$ and put $Q := (Y_0^2 : -X_0^2 : 0)$. Verify that Q lies on the tangent to C_d at P, and use (2) and (3) to derive

$$[2]P = \big(-Y_0(2X_0^3 + Y_0^3) : X_0(X_0^3 + 2Y_0^3) : Z_0(X_0^3 - Y_0^3)\big). \tag{2.4}$$

Solution. $\langle P,P,Q\rangle = X_0^2Y_0^2 + Y_0^2(-X_0^2) - dZ_0^2\cdot 0 = 0$, so Q is on the tangent by (3). Now (2.2) loses two terms:

$$F(sP+tQ) = 3st^2\,\langle P,Q,Q\rangle + t^3F(Q).$$

The double root at $t=0$ is the tangency at P. The third root gives the third intersection point of the tangent with C_d, call it $P*P$:

$$P*P = -F(Q)\,P + 3\langle P,Q,Q\rangle\,Q.$$

Here

$$F(Q) = Y_0^6 - X_0^6, \qquad \langle P,Q,Q\rangle = X_0Y_0^4 + Y_0X_0^4 = X_0Y_0(X_0^3+Y_0^3) = dX_0Y_0Z_0^3.$$

Substituting and using $dZ_0^3 = X_0^3+Y_0^3$ to simplify, each coordinate acquires the factor $X_0^3+Y_0^3$. Cancelling it and applying $\ominus$ gives (2.4). Explicitly, the first coordinate before cancelling is

$$X_0(X_0^6 - Y_0^6) + 3dX_0Y_0^3Z_0^3 = X_0(X_0^3+Y_0^3)(X_0^3+2Y_0^3).$$

The cancellation needs $X_0^3+Y_0^3 = dZ_0^3 \neq 0$, that is, $Z_0\neq 0$. On the line $Z=0$ the derivation says nothing: there $Y_0^3 = -X_0^3$, so $Q = (Y_0^2 : -X_0^2 : 0) = (Y_0^3 : -X_0^2Y_0 : 0) = (X_0 : Y_0 : 0)$ is P itself and $F(sP+tQ)$ vanishes identically. But (2.4) holds there too. The tangent line at $P=(X_0:Y_0:0)$ is $X_0^2X + Y_0^2Y = 0$. On it $X^3 = -Y_0^6Y^3/X_0^6 = -Y^3$, so F restricts to $-dZ^3$ and the tangent cuts out $3P$, exactly as at O in Problem 2(2). Hence $P*P = P$ and $[2]P = \ominus P = (Y_0 : X_0 : 0)$, while the right-hand side of (2.4) is $(-Y_0X_0^3 : -X_0^4 : 0) = (Y_0 : X_0 : 0)$.

(6) Check (2.4) against Problem 3: what does it give on the line $X=Y$? (The line $X=0$ is taken up in Problem 6.)

Solution. **$[2]P = O$.** If $X_0 = Y_0$, then $[2]P = (-3X_0^4 : 3X_0^4 : 0) = (1 : -1 : 0) = O$, consistent with Problem 3(2).

(7) Where does the whole method break down if $\operatorname{char} k = 3$? Compare your answer with Problem 1(3).

Solution. **The middle coefficients of (2.2) are literally 3.** In characteristic three the identity collapses to $F(sP+tQ) = s^3F(P) + t^3F(Q)$, which vanishes identically once P and Q represent points of $C_d(\bar{k})$: every line "meets" the cubic everywhere and the chord–tangent construction carries no information at all. This is the same degeneration as in Problem 1(3), seen from the side of the group law.

Problem 5 (Arithmetic on C_7)**.**
Let $d=7$, so that $P := (2:-1:1) \in C_7(\mathbb{Q})$ because $2^3 + (-1)^3 = 7$.

(1) Compute $[2]P$ using (2.4), and simplify. Verify your answer by exhibiting an identity between cubes of integers.

Solution. **$[2]P = (5:4:3)$.** With $X_0=2$, $Y_0=-1$, $Z_0=1$ we have $X_0^3 = 8$ and $Y_0^3 = -1$, so

$$[2]P = \big(-(-1)(16-1) : 2(8-2) : 1\cdot(8+1)\big) = (15:12:9) = (5:4:3),$$

and indeed $5^3+4^3 = 125+64 = 189 = 7\cdot 27 = 7\cdot 3^3$.

(2) Compute $[3]P$ using (2.3) applied to P and $[2]P$, and verify it the same way.

Solution. **$[3]P = (-17:73:38)$.** With $P_1 = (2,-1,1)$ and $P_2 = (5,4,3)$,

$$A = 4\cdot 5 + 1\cdot 4 - 7\cdot 1\cdot 3 = 20+4-21 = 3, \qquad B = 2\cdot 25 + (-1)\cdot 16 - 7\cdot 1\cdot 9 = 50-16-63 = -29.$$

Then (2.3) gives

$$[3]P = (-29(-1) - 3\cdot 4 \;:\; -29\cdot 2 - 3\cdot 5 \;:\; -29 - 9) = (17 : -73 : -38) = (-17 : 73 : 38),$$

and $73^3 - 17^3 = 389017 - 4913 = 384104 = 7 \cdot 54872 = 7 \cdot 38^3$.

(3) Compute $[4]P$.

SOLUTION. **$[4]P = (-1256 : 1265 : 183)$.** Doubling $(5 : 4 : 3)$: $X_0^3 = 125$, $Y_0^3 = 64$, so

$$[4]P = \big(-4(250 + 64) : 5(125 + 128) : 3(125 - 64)\big) = (-1256 : 1265 : 183),$$

and $1265^3 - 1256^3 = 7 \cdot 183^3$.

(4) The three rational points you have found have numerators and denominators of rapidly increasing size. What does this suggest about the order of P in $C_7(\mathbb{Q})$? (You are not asked to prove it.)

SOLUTION. **That P has infinite order.** If P were a torsion point, its multiples would run through a finite set, so their heights would be bounded. They visibly are not. So P has infinite order, and $C_7(\mathbb{Q})$ has positive rank.

Remark. In fact the rank is one and P generates modulo torsion. For the heights behind this argument, see [Sil09, §VIII.9] or [HS00, Part B].

In Problem 6, parts (1)–(3) are the general theory and (5)–(7) the payoff for our family. If time is short, take (3) on faith and go to (5)–(7). Part (4) is the most satisfying: two completely different counts turn out to agree. Part (1) uses Bézout's theorem [Har77, §I.7].

Proposition 2.1 (The Hessian criterion)**.** *Let $C = V(F) \subset \mathbb{P}^2_{\bar{k}}$ be a smooth plane curve of degree m over a field of characteristic 0 or > 3, and let*

$$H(F) = \det\left(\frac{\partial^2 F}{\partial X_i\, \partial X_j}\right)_{0\leqslant i,j\leqslant 2}.$$

Then a point $P \in C(\bar{k})$ is an inflection point if and only if $H(F)(P) = 0$. This is classical. Characteristic three is taken up in Problem 6(8) below.

Problem 6 (Inflection points and 3-torsion)**.**

(1) Let $C = V(F)$ be as in Proposition 2.1. Show that $H(F)$ is a form of degree $3(m-2)$, so of degree 3 when $m = 3$. Assuming C is a smooth plane cubic, explain why $C \not\subseteq V(H(F))$, and conclude from Bézout's theorem that C has exactly 9 inflection points *counted with multiplicity*.

SOLUTION. Each entry $\partial^2 F/\partial X_i \partial X_j$ is a form of degree $m-2$, and the determinant of a 3×3 matrix is a sum of products of three entries, so $H(F)$ has degree $3(m-2)$. For $m = 3$ this is 3. If $C \subseteq V(H(F))$ then by Proposition 2.1 every point of $C(\bar{k})$ would be a flex, which is false for a smooth cubic: a curve all of whose $\bar{k}$-points are flexes is a line (and (2)–(4) below show a smooth cubic has only nine). Hence C and $V(H(F))$ share no component and Bézout applies, giving $\deg\big(C \cap V(H(F))\big) = 3\cdot 3 = 9$.

(2) Now forget the Hessian. Let C be a smooth plane cubic over $\bar{k}$ and suppose $O \in C(\bar{k})$ is a flex, so that (C, O) is an elliptic curve with the group law of Problem 2(4). Show that

$$P \text{ is a flex} \iff [3]P = O,$$

so that the set of flexes is exactly $C[3](\bar{k})$.

Solution. By definition P is a flex when the tangent line at P meets C at P with multiplicity three, i.e. when the tangent cuts out the divisor $3P$. By Problem 2(4) that says $P \oplus P \oplus P = O$, i.e. $[3]P = O$.

(3) Show that a smooth plane cubic over $\overline{k}$ always *has* a flex, without using the Hessian. Deduce that, with no flex singled out, the set of flexes is a torsor under $\mathrm{Pic}^0(C)[3]$. In particular the number of flexes is $\#C[3](\overline{k})$.

Solution. Fix any $P_0 \in C(\overline{k})$ and consider the degree-zero class $\mathcal{O}_C(1) - 3[P_0] \in \mathrm{Pic}^0 C$ (here C is already over $\overline{k}$, so no base change is involved). Multiplication by 3 is a nonzero isogeny of the elliptic curve $\mathrm{Pic}^0 C$, hence surjective on $\overline{k}$-points, so there is a class δ with $3\delta = \mathcal{O}_C(1) - 3[P_0]$. On a genus-one curve with a rational point every degree-zero class has the form $[Q] - [P_0]$, so $\delta = [Q] - [P_0]$ for some $Q \in C(\overline{k})$, and then $3[Q] \sim \mathcal{O}_C(1)$: the tangent at Q cuts $3Q$, so Q is a flex. Given one flex O, part (2) identifies the flexes with $C[3]$. Changing the chosen flex translates this identification, so without a choice the flexes form a torsor under $\mathrm{Pic}^0(C)[3]$ and their number is $\#C[3](\overline{k})$.

(4) Combine (2) and (3) with the structure of $C[3]$ in characteristic $\neq 3$ to conclude that a smooth plane cubic has exactly 9 flexes. What does comparing this with (1) tell you about the intersection $C \cap V(H(F))$?

Solution. **Exactly 9 flexes. Each multiplicity in $C \cap V(H(F))$ is 1.** For $\mathrm{char}\, k \neq 3$ the multiplication-by-3 map is separable of degree 9, so $C[3](\overline{k}) \simeq (\mathbb{Z}/3)^2$ has exactly 9 elements. By (3) there are exactly 9 flexes. Comparing with (1): nine flexes appear in an intersection of total degree nine, so every intersection multiplicity is 1.

Remark. That is, C meets its Hessian *transversally*, something the Bézout count alone could not tell us.

(5) Specialize to C_d. Compute $H(F)$ for $F = X^3 + Y^3 - dZ^3$ and deduce that, as subsets of $C_d(\overline{k})$,

$$C_d[3](\overline{k}) = (C_d \cap \{XYZ = 0\})\,(\overline{k}).$$

Write down all nine flexes.

Solution. $\boldsymbol{H(F) = -216\,d\,XYZ.}$ The Hessian matrix of $F = X^3 + Y^3 - dZ^3$ is $\mathrm{diag}(6X,\, 6Y,\, -6dZ)$, so

$$H(F) = -216\,d\,XYZ.$$

Since $d \neq 0$ and $216 = 2^3 3^3$ is invertible for $\mathrm{char}\, k \neq 2, 3$, we get $V(H(F)) = \{XYZ = 0\}$, and by (2), (4) the nine flexes are $C_d \cap \{XYZ = 0\}$. (In characteristic 2 the Hessian vanishes identically. There, (6) below and the analogue of the tangent computation of Problem 2(2) at Q_1, Q_2 give the same nine flexes.) Explicitly, with $\delta \in \overline{k}$ a cube root of d:

$$\begin{aligned}
Z = 0: &\quad (1:-1:0), \quad (1:-\zeta:0), \quad (1:-\zeta^2:0),\\
X = 0: &\quad (0:\delta:1), \quad (0:\zeta\delta:1), \quad (0:\zeta^2\delta:1),\\
Y = 0: &\quad (\delta:0:1), \quad (\zeta\delta:0:1), \quad (\zeta^2\delta:0:1).
\end{aligned}$$

Three on each line, nine in all.

(6) Check (5) against the group law directly: use (2.4) to show that every point $P = (0 : Y_0 : Z_0) \in C_d(\overline{k})$ satisfies $[2]P = \ominus P$.

Solution. Put $X_0 = 0$ in (2.4). Then $Y_0 \neq 0$, since $(0:0:1) \notin C_d$, and the coordinates become

$$\left(-Y_0 \cdot Y_0^3 : 0 : Z_0(0 - Y_0^3)\right) = (-Y_0^4 : 0 : -Y_0^3 Z_0) = (Y_0 : 0 : Z_0),$$

which is $\ominus(0 : Y_0 : Z_0)$ by Problem 3(1). So $[2]P = \ominus P$, hence $[3]P = O$, confirming that the points of $C_d(\overline{k})$ on the line $X = 0$ are 3-torsion. And by the symmetry $X \leftrightarrow Y$, likewise those on $Y = 0$.

(7) Which of the nine flexes lie in $C_d(k)$? Treat the three lines separately, and say what happens for $k = \mathbb{Q}$.

Solution. **Over $\mathbb{Q}$: only O, unless $d = \delta^3$, when also $(0 : \delta : 1)$ and $(\delta : 0 : 1)$.** On $Z = 0$: O is always rational, and Q_1, Q_2 are rational exactly when $\zeta \in k$, by Problem 2(1). On $X = 0$ and $Y = 0$: the flex $(0 : \delta : 1)$ is rational if and only if $\delta \in k$, i.e. if and only if $d \in (k^\times)^3$. The remaining four need ζ as well. So over $k = \mathbb{Q}$, where $\zeta \notin \mathbb{Q}$, there are only two possibilities: if d is not a rational cube the only rational flex is O, and if $d = \delta^3$ then $C_d(\mathbb{Q})$ contains exactly three flexes,

$$O, \qquad (0 : \delta : 1), \qquad (\delta : 0 : 1),$$

which by (6) form a subgroup isomorphic to $\mathbb{Z}/3\mathbb{Z}$ (the last two are inverse to each other). One can never see all nine over $\mathbb{Q}$.

(8) ($\star$) In characteristic three, $\#C[3](\overline{k})$ is 3 or 1 according as C is ordinary or supersingular, so a smooth plane cubic has 3 or 1 flexes. How is that compatible with (1)?

Solution. **No contradiction: the nine are counted with multiplicity, here 3, 3, 3 or 9.** Part (1) rests on Proposition 2.1, which we stated only in characteristic 0 or > 3, so as it stands it is not available. The argument of (2) and (3), by contrast, never mentions the Hessian and is valid in every characteristic. It gives $\#C[3](\overline{k})$ flexes, which is 3 or 1 when $\operatorname{char} k = 3$ because [3] is then inseparable. In fact, for cubics the Hessian still finds the flexes in characteristic three. What breaks is the transversality of (4). Take C in Weierstrass form $y^2 = x^3 + a_2x^2 + a_4x + a_6$. (By (3) C has a flex, and a linear change of coordinates as in Problem 8 reaches this form. Then $H(F)$ changes by the square of the determinant, so where it vanishes does not depend on the coordinates.) In characteristic three,

$$H(F) = Z\,(a_2^2X^2 + a_2a_4XZ + a_4^2Z^2 - a_2Y^2),$$

and on C the second factor equals $a_4^2 - a_2a_6 - a_2x^3$, which is minus the 3-division polynomial $\psi_3 = b_2x^3 + b_8 = a_2x^3 + a_2a_6 - a_4^2$ of C in characteristic three. So the second factor vanishes exactly at the points of order 3 [Sil09, Exercise 3.7]. The intersection $C \cap V(H(F))$ thus consists of the flexes only, but with multiplicities: the line $Z = 0$ is the flex tangent at O and meets C there with multiplicity 3; if $a_2 \neq 0$ (the ordinary case) the conic meets C in the two points of order 3, which are swapped by $y \mapsto -y$, so with multiplicity 3 each; if $a_2 = 0$ (the supersingular case) $H(F) = a_4^2Z^3$ with $a_4 \neq 0$, since C is smooth, and all nine lie at O. The Hessian criterion for cubics really does fail in characteristic 2, where $H(F)$ vanishes identically (see (5)).

In Problem 7 you may use the following two results. Part (4) asks what the classification of automorphism groups in [Sil09, §III.10] predicts.

Theorem 2.2 (Morphisms fixing O [Sil09, Theorem III.4.8])**.** *A morphism of elliptic curves taking O to O is a group homomorphism.*

Proposition 2.3 (Endomorphism rings [Sil09, §III.4])**.** *The endomorphism ring of an elliptic curve has no zero divisors.*

Problem 7 (Complex multiplication, $\star$)**.**
Let $\alpha(X : Y : Z) = (X : Y : \zeta Z)$, a map defined over $k(\zeta)$.

(1) Show that α maps C_d to itself and fixes O.

Solution. $F(X, Y, \zeta Z) = X^3 + Y^3 - d\zeta^3 Z^3 = F(X, Y, Z)$ since $\zeta^3 = 1$, so α carries C_d into C_d. And $\alpha(1 : -1 : 0) = (1 : -1 : 0)$.

(2) Conclude that α is a group homomorphism, hence an automorphism of the elliptic curve (C_d, O) over $k(\zeta)$.

Solution. α is a morphism of curves sending O to O. By [Sil09, Theorem III.4.8] any such morphism is a group homomorphism. It is invertible (with inverse given by ζ^2), so it is an automorphism of (C_d, O).

(3) Show that $\alpha^2 + \alpha + 1 = 0$ in $\mathrm{End}(C_d \times_k k(\zeta))$, the ring of endomorphisms of C_d defined over $k(\zeta)$.

Solution. $\alpha^3 = \mathrm{id}$ because $\zeta^3 = 1$, so $(\alpha - 1)(\alpha^2 + \alpha + 1) = \alpha^3 - 1 = 0$ in $\mathrm{End}(C_d \times_k k(\zeta))$. The endomorphism ring of an elliptic curve is a ring without zero divisors (degrees multiply), and $\alpha \neq 1$ since $\zeta \neq 1$. Hence $\alpha^2 + \alpha + 1 = 0$.

(4) Deduce that $\mathbb{Z}[\zeta] \subseteq \mathrm{End}(C_d \times_k k(\zeta))$, so every Selmer curve has complex multiplication. What does the classification of automorphism groups then predict for $j(C_d)$ and for $\#\operatorname{Aut}(C_d)$ and $\#\operatorname{Aut}(C_d \times_k k(\zeta))$?

Solution. **$j(C_d) = 0$. For $k = \mathbb{Q}$, $\#\operatorname{Aut}(C_d) = 2$ and $\#\operatorname{Aut}(C_d \times_{\mathbb{Q}} \mathbb{Q}(\zeta)) = 6$.** By (3) the subring of $\mathrm{End}(C_d \times_k k(\zeta))$ generated by α is a quotient of $\mathbb{Z}[T]/(T^2 + T + 1) = \mathbb{Z}[\zeta]$. It is torsion-free, because the endomorphism ring is [Sil09, §III.4], and every proper quotient of $\mathbb{Z}[\zeta]$ is finite. So $\mathbb{Z}[\zeta] \subseteq \mathrm{End}(C_d \times_k k(\zeta)) \subseteq \mathrm{End}(\overline{C}_d)$. In particular the endomorphism ring is strictly larger than $\mathbb{Z}$, which is complex multiplication. A curve admitting an automorphism of order three has $j = 0$. And by [Sil09, Theorem III.10.1], in characteristic $\neq 2, 3$ a curve with $j = 0$ has automorphism group of order 6 over a field containing ζ, and order 2 otherwise. For $k = \mathbb{Q}$ we expect $\#\operatorname{Aut}(C_d) = 2$ and $\#\operatorname{Aut}(C_d \times_{\mathbb{Q}} \mathbb{Q}(\zeta)) = 6$.

Remark. Example 4.2 confirms this in `Magma`.

3. From Riemann–Roch to a Weierstrass equation

This section reproduces, for our family, the proof of [Sil09, Proposition III.3.1]: every elliptic curve is isomorphic to a curve in Weierstrass form.

Problem 8 is the longest problem here. Parts (1)–(3) go quickly once you see that Proposition 3.2 converts "pole order at O" into "multiplicity of O in $h \cap C_d$". Part (5) is a two-line computation with a surprising answer, and (8) is worth coming back to. Weierstrass equations and their coefficients $a_1, \ldots, a_6$ are as in [Sil09, §III.1].

Definition 3.1 (Riemann–Roch spaces)**.** For $D \in \operatorname{Div} C_d$,

$$\mathcal{L}(D) = \left\{ f \in k(C_d)^\times : \operatorname{div}(f) \geqslant -D \right\} \cup \{0\}, \qquad \operatorname{div}(f) \in \operatorname{Div} C_d.$$

This is a space of functions defined *over* k.

You may use the following two results without proof.

Proposition 3.2 (Bézout)**.** *For a nonzero linear form h, the scheme-theoretic intersection $h \cap C_d$ is an effective divisor of degree 3 in $\operatorname{Div} C_d$. Over $\overline{k}$ it becomes a sum of three points of $C_d(\overline{k})$ counted*

with multiplicity. Consequently, since $X+Y$ cuts out $3O$ by Problem 2(2),

$$\operatorname{div}\left(\frac{h}{X+Y}\right) = (h \cap C_d) - 3O.$$

Proposition 3.3 (Riemann–Roch for genus one [Sil09, §II.5 and proof of Proposition III.3.1])**.** *For every $n \geqslant 1$ we have $\dim_k \mathcal{L}(nO) = n$. Since C_d is geometrically integral, $\mathcal{L}(D) \otimes_k \overline{k} = \mathcal{L}(D_{\overline{k}})$, so the dimension is the same whether computed over k or over $\overline{k}$.*

Problem 8 (The Weierstrass model of C_d)**.**
Consider the rational functions in $k(C_d)$

$$w \coloneqq \frac{Z}{X+Y}, \quad v \coloneqq \frac{X-Y}{X+Y}.$$

For parts (2)–(6) and (8) assume $\operatorname{char} k \neq 2$. Part (7) replaces v by $X/(X+Y) = (1+v)/2$, which also makes sense in characteristic 2, and reaches a model valid whenever $\operatorname{char} k \neq 3$.

(1) Using Problem 2(1) and Proposition 3.2, compute $\operatorname{div}(w)$. Deduce that w has a pole of order exactly 2 at O and no other pole, so $w \in \mathcal{L}(2O)$. Where are its zeros, and what are they in the language of Problem 6?

Solution. **$\operatorname{div}(w) = Q_1 + Q_2 - 2O$. Its zeros are the flexes Q_1, Q_2.** By Problem 2(1), $Z \cap C_d = O + Q_1 + Q_2$ with $Q_1, Q_2 \neq O$. So

$$\operatorname{div}(w) = O + Q_1 + Q_2 - 3O = Q_1 + Q_2 - 2O.$$

Hence the only pole is at O, of order exactly 2. This last computation is an identity in $\operatorname{Div} \overline{C}_d$. In $\operatorname{Div} C_d$, when $\zeta \notin k$, the pair $\{Q_1, Q_2\}$ is a *single* closed point $\mathfrak{q}$ of degree 2 (see Warning 4.1), and the same statement reads $\operatorname{div}(w) = \mathfrak{q} - 2O$. Either way the degree is 0 and the pole order at O is 2. Its zeros Q_1, Q_2 are, by Problem 6(5) and Problem 2(1), the nontrivial 3-torsion points of $C_d(\overline{k})$ on the line $Z = 0$.

(2) Show that the line $X - Y = 0$ does not pass through O. Deduce that v has a pole of order exactly 3 at O, so $v \in \mathcal{L}(3O)$. Where are its zeros?

Solution. **At the three points of order two, on the line $X = Y$.** The form $X - Y$ takes the value $1 - (-1) = 2$ at O, which is nonzero since $\operatorname{char} k \neq 2$. So O does not appear in $(X-Y) \cap C_d$ and $\operatorname{div}(v) = ((X-Y) \cap C_d) - 3O$ has a pole of order exactly 3 at O. Its zeros are the three points of $C_d(\overline{k})$ on the line $X = Y$, i.e. the nontrivial 2-torsion of Problem 3(2). In $\operatorname{Div} C_d$ these assemble into closed points of total degree 3.

Remark. In characteristic 2, $X - Y = X + Y$ vanishes at O. That is why (7) avoids v.

(3) Using Proposition 3.3, explain why $\{1, w\}$ is a basis of $\mathcal{L}(2O)$ and $\{1, w, v\}$ is a basis of $\mathcal{L}(3O)$.

Solution. $\mathcal{L}(O)$ has dimension 1 by Proposition 3.3 and contains the constants, so $\mathcal{L}(O) = k$. Now $\dim \mathcal{L}(2O) = 2$ and $1, w \in \mathcal{L}(2O)$ are independent because $w \notin k$ (it has a pole). So $\{1, w\}$ is a basis. Similarly $\dim \mathcal{L}(3O) = 3$ and $v \in \mathcal{L}(3O) \setminus \mathcal{L}(2O)$ since its pole order is 3, so $\{1, w, v\}$ is a basis.

(4) The seven functions $1, w, v, w^2, wv, w^3, v^2$ all lie in $\mathcal{L}(6O)$. Why must they satisfy a nontrivial linear relation over k? List their pole orders at O. Which two collide, and why does that force the relation to involve both?

SOLUTION. **Pole orders $0, 2, 3, 4, 5, 6, 6$. Here w^3 and v^2 collide.** The seven functions lie in $\mathcal{L}(6O)$, which has dimension 6 by Proposition 3.3. Seven vectors in a 6-dimensional space are dependent. Their pole orders at O are

$$1,\ w,\ v,\ w^2,\ wv,\ w^3,\ v^2 \quad\longmapsto\quad 0,\ 2,\ 3,\ 4,\ 5,\ 6,\ 6.$$

The first five have *distinct* pole orders, hence are linearly independent (a nontrivial combination has a pole of order equal to the largest one occurring). So any relation must involve w^3 or v^2. And since these are the only two with pole order exactly 6, a relation involving one must involve the other, with the leading terms cancelling. It also follows that the space of relations is exactly one-dimensional.

(5) Find the relation.

SOLUTION. **$3v^2 + 1 = 4d\,w^3$.** On the chart $X + Y = 1$ we have $w = Z$ and $v = X - Y$, so $X = (1 + v)/2$, $Y = (1 - v)/2$, and

$$X^3 + Y^3 = \frac{(1+v)^3 + (1-v)^3}{8} = \frac{2 + 6v^2}{8} = \frac{3v^2 + 1}{4}.$$

The defining equation $X^3 + Y^3 = dZ^3$ therefore reads

$$3v^2 + 1 = 4d\,w^3. \tag{3.1}$$

(6) Let $E_1 \colon y_1^2 = x_1^3 - 432d^2$. Show that the relation of (5) says that there is a morphism $\varphi_1 \colon C_d \to E_1$ with $\varphi_1^* x_1 = 12dw$ and $\varphi_1^* y_1 = 36dv$. Write down φ_1 and its inverse explicitly.

SOLUTION. **$\varphi_1^{-1}(x_1, y_1) = (36d + y_1 : 36d - y_1 : 6x_1)$.** Multiplying (3.1) by $432d^2$ gives

$$(36dv)^2 + 432d^2 = (12dw)^3,$$

so the functions $12dw$ and $36dv$ satisfy the equation of E_1, and this defines φ_1. Explicitly,

$$\varphi_1(X : Y : Z) = \left(\frac{12dZ}{X+Y},\ \frac{36d(X-Y)}{X+Y}\right), \qquad \varphi_1^{-1}(x_1, y_1) = (36d + y_1 : 36d - y_1 : 6x_1).$$

That φ_1^{-1} lands on C_d is a one-line check:

$$(36d + y_1)^3 + (36d - y_1)^3 = 2(36d)^3 + 216d\,y_1^2, \qquad d(6x_1)^3 = 216d(y_1^2 + 432d^2),$$

and $2 \cdot 36^3 = 216 \cdot 432 = 93312$.

(7) Where exactly did you use $\operatorname{char} k \neq 2$? Show that there is instead a morphism φ_2 with $\varphi_2^* x_2 = 3dw$ and $\varphi_2^* y_2 = 9dX/(X + Y)$ to

$$E_2 \colon y_2^2 - 9d\,y_2 = x_2^3 - 27d^2, \tag{3.2}$$

which is a Weierstrass equation valid whenever $\operatorname{char} k \neq 3$. Which of $a_1, \dots, a_6$ are nonzero?

SOLUTION. **Only $a_3 = -9d$ and $a_6 = -27d^2$.** The hypothesis $\operatorname{char} k \neq 2$ was used three times: in (2), where $X - Y$ takes the value 2 at O (in characteristic 2, $X - Y = X + Y$ and $v = 1$ has no pole at all); to divide by 2 when solving for X and Y in terms of v; and, most seriously, to complete the square, which is what reaches the form $y^2 = x^3 + a_6$. For φ_2, work on the chart $X + Y = 1$, where $\varphi_2^* y_2 = 9dX$ and $X - 1 = -Y$, so

$$(\varphi_2^* y_2)^2 - 9d\,\varphi_2^* y_2 = 81d^2 X(X - 1) = -81d^2 XY.$$

Meanwhile $X^3 + Y^3 = (X + Y)^3 - 3XY(X + Y) = 1 - 3XY$, and this equals dw^3, so $XY = (1 - dw^3)/3$ and

$$(\varphi_2^* y_2)^2 - 9d\,\varphi_2^* y_2 = -27d^2(1 - dw^3) = 27d^3 w^3 - 27d^2 = (\varphi_2^* x_2)^3 - 27d^2.$$

No division by 2 occurred, and the identity holds whenever $\operatorname{char} k \neq 3$. In the notation of [Sil09, §III.1] this is $a_1 = a_2 = a_4 = 0$, $a_3 = -9d$, $a_6 = -27d^2$.

Remark. This is exactly why the general Weierstrass equation carries the coefficients a_1 and a_3 that one is tempted to discard.

(8) ($\star$) The relation in (5) has no w^2, wv, w or v term, although a general Weierstrass equation allows all four. Explain why, using the automorphisms $\sigma = \ominus$ of Problem 3 and α of Problem 7.

Solution. From the definitions, $\sigma(X : Y : Z) = (Y : X : Z)$ and $\alpha(X : Y : Z) = (X : Y : \zeta Z)$ give

$$\sigma^* w = w, \quad \sigma^* v = -v, \qquad \alpha^* w = \zeta w, \quad \alpha^* v = v.$$

By (4) the space of relations among the seven monomials is one-dimensional. Both σ^* and α^* multiply each monomial by a scalar, so they preserve that space and act on it by a scalar: the relation is a simultaneous eigenvector. It involves w^3 and v^2, and

$$\sigma^* w^3 = w^3, \quad \sigma^* v^2 = v^2, \qquad \alpha^* w^3 = w^3, \quad \alpha^* v^2 = v^2.$$

Both are fixed. So both eigenvalues are 1, and every monomial that is *not* fixed by both must have coefficient zero. Those are v and wv (negated by σ^*) and w, w^2, wv (scaled by α^*): precisely the four missing terms.

Remark. The shape of the Weierstrass equation is dictated by the automorphism group.

4. Worked examples in Magma

Most of you have never opened `Magma`, so this section is written as worked examples rather than exercises: read them, then re-run them yourself and change things. Every line of output below is reproduced from `Magma` V2.28-8 [BCP97], with whitespace tidied and one long type name elided. Three short variations to try on your own are collected at the end.

Warning 4.1 (Closed points in `Magma`)**.** A closed point P of C_d is a point of the scheme C_d. Its residue field $k(P)$ is a finite extension of k, and $\deg P := [k(P) : k]$. The closed points of degree one are the rational points, and a closed point P with $k(P)/k$ separable corresponds to an orbit of $\deg P$ points of $C_d(\overline{k})$ under $\mathrm{Gal}(k^{\mathrm{sep}}/k)$. The divisor group $\operatorname{Div} C_d$ is the free abelian group on the closed points [Poo06, §1.9], whereas [Sil09, §II.3] works with points of $C_d(\overline{k})$ and Galois-invariant divisors on $\overline{C}_d$. `Magma` follows the first convention: `Places` returns closed points, `Divisor` builds elements of $\operatorname{Div} C_d$, and `RiemannRochSpace` returns $\mathcal{L}(D)$ as a k-vector space.

Each example continues the previous one. Begin with

```
QQ := Rationals();
P2<X,Y,Z> := ProjectiveSpace(QQ,2);
d := 7;
C := Curve(P2, X^3 + Y^3 - d*Z^3);
O := C![1,-1,0];
```

The notation `P2<X,Y,Z>` names the coordinates so that we can write equations in them, and `C!` coerces a list of three numbers into a rational point of `C`. Typing `O := (1:-1:0);` does not work.

Example 4.2 (Setting up, and a trap)**.** First, Problem 1:

```
IsNonSingular(C);  // true
Genus(C);          // 1
O;                 // (-1 : 1 : 0)
```

The last line is worth a pause. We asked for $(1:-1:0)$ and got $(-1:1:0)$: a point of $\mathbb{P}^2(\mathbb{Q})$ is defined only up to scaling, and `Magma` normalizes the representative it stores, here by -1.

Now ask for the Weierstrass model. The second return value `mp` is the isomorphism from `C` to `E`.

```
E, mp := EllipticCurve(C, 0);
E;
// Elliptic Curve defined by y^2 - 9/49*y = x^3 - 27/2401 over Rational Field
jInvariant(E);                          // 0
E eq EllipticCurve([0,0,0,0,-432*d^2]);   // false
```

That `false` looks alarming, since Problem 8(6) says the answer should be $E_1\colon y_1^2 = x_1^3 - 432d^2$. It is not a contradiction: `eq` compares Weierstrass *coefficients*, and by [Sil09, Proposition III.3.1(b)] a Weierstrass model is unique only up to the changes of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$. The right question is whether the curves are isomorphic, and against which model:

```
Eours := EllipticCurve([0,0,-9*d,0,-27*d^2]);
Eours;                          // y^2 - 63*y = x^3 - 1323
bool, iso := IsIsomorphic(Eours, E);
bool;                           // true
IsomorphismData(iso);           // [ 0, 0, 0, 1/7 ]
```

So $r = s = t = 0$ and $u = 1/7 = 1/d$: `Magma` found exactly the model E_2 of Problem 8(7), with x_2 and y_2 divided by d^2 and d^3. The two maps agree as well,

```
DefiningEquations(mp);   // [ 3/7*Z, 9/49*X, X + Y ]
```

against our $(3dZ : 9dX : X + Y)$, the ratios being d^2 and d^3 exactly as $u = 1/d$ demands. The moral: for elliptic curves, `eq` is almost never the question you mean to ask.

We can even watch this happen for all d at once, by working over the function field $\mathbb{Q}(d)$ instead of over $\mathbb{Q}$:

```
Fd<D> := FunctionField(Rationals());
P2d<Xd,Yd,Zd> := ProjectiveSpace(Fd,2);
Cd := Curve(P2d, Xd^3 + Yd^3 - D*Zd^3);
Ed, mpd := EllipticCurve(Cd, Cd![1,-1,0]);
Ed;
// Elliptic Curve defined by y^2 - 9/D^2*y = x^3 - 27/D^4
DefiningEquations(mpd);
// [ 3/D*Zd, 9/D^2*Xd, Xd + Yd ]
```

which is (3.2) rescaled by $u = d$, as an identity in d.

Finally, the predictions of Problem 7(4):

```
#AutomorphismGroup(E);                  // 2
K<zeta> := CyclotomicField(3);
#AutomorphismGroup(BaseChange(E,K));    // 6
```

Example 4.3 (The addition formulas, checked)**.** A `Magma` `function` takes arguments and returns a value. Here both are sequences of three coordinates, so that (2.3) and (2.4) can be typed almost verbatim.

```
dbl := function(P)
  x := P[1]; y := P[2]; z := P[3];
  return [-y*(2*x^3+y^3), x*(x^3+2*y^3), z*(x^3-y^3)];
end function;

add := function(P1,P2)
  A := P1[1]^2*P2[1] + P1[2]^2*P2[2] - d*P1[3]^2*P2[3];
  B := P1[1]*P2[1]^2 + P1[2]*P2[2]^2 - d*P1[3]*P2[3]^2;
  return [B*P1[2]-A*P2[2], B*P1[1]-A*P2[1], B*P1[3]-A*P2[3]];
end function;

P := [2,-1,1];
Q := dbl(P);      C!Q;   //  (5/3 : 4/3 : 1)
R := dbl(Q);      C!R;   //  (-1256/183 : 1265/183 : 1)
S := add(P,Q);    C!S;   //  (-17/38 : 73/38 : 1)
```

These are the rational points of Problem 5. Coercing through `C!` normalizes them, which is why $(15 : 12 : 9)$ prints as $(5/3 : 4/3 : 1)$. Now transport everything to E through `mp` and compare with `Magma`'s own group law:

```
EP := mp(C!P);
mp(C!Q) eq 2*EP;          // true
mp(C!S) eq 3*EP;          // true
mp(C!R) eq 4*EP;          // true
```

Three `true`s: formulas derived from nothing but (2.1) and a 3×3 determinant agree with a general-purpose implementation.

Example 4.4 (Riemann–Roch, mechanically)**.** `Magma` can carry out Problem 8 without being told the answer. The generators `x` and `y` of `FunctionField(C)` are the affine coordinates X/Z and Y/Z of C_d (not Weierstrass coordinates), so our two functions are built from those.

```
FF<x,y> := FunctionField(C);    // x = X/Z,  y = Y/Z
w := 1/(x+y);                   // Z/(X+Y)
v := (x-y)/(x+y);               // (X-Y)/(X+Y)
pl0 := Place(0);
Valuation(w, pl0);              // -2
Valuation(v, pl0);              // -3
```

Those are the pole orders of Problem 8(1), (2). A valuation of -2 is a pole of order 2. As in Warning 4.1, `Place(0)` is the closed point O and `Divisor` builds an element of $\operatorname{Div} C_d$, so everything here happens over $\mathbb{Q}$. Next, Proposition 3.3 itself:

```
D := Divisor(pl0);
for n in [1..6] do print n, Dimension(RiemannRochSpace(n*D)); end for;
// 1 1
// 2 2
// 3 3
// 4 4
```

```
// 5 5
// 6 6
```

Now ask for a basis of $\mathcal{L}(2O)$ without suggesting one:

```
V2, m2 := RiemannRochSpace(2*D);
[m2(b) : b in Basis(V2)];      // [ x^2 - y*x + y^2, 1 ]
```

which looks nothing like our w. But $X^3+Y^3=(X+Y)(X^2-XY+Y^2)$, and on C_d the left side is dZ^3, so

$$\frac{X^2-XY+Y^2}{Z^2}=\frac{dZ^3}{Z^2(X+Y)}=d\,\frac{Z}{X+Y}=d\,w,$$

and indeed

```
(x^2 - x*y + y^2) eq d*w;      // true
```

Finally the relation. We list the seven functions of Problem 8(4), express each in a basis of $\mathcal{L}(6O)$ (that is what `@@` does, applying the inverse of the map `mV`), and take the kernel of the resulting matrix.

```
V, mV := RiemannRochSpace(6*D);
fns := [FF!1, w, v, w^2, w*v, w^3, v^2];
[-Valuation(f, pl0) : f in fns];   // [ 0, 2, 3, 4, 5, 6, 6 ]
M := Matrix([Eltseq(f @@ mV) : f in fns]);
K := Kernel(M);
Dimension(K);                      // 1
Basis(K);                          // [ ( 1  0  0  0  0  -28  3) ]
```

The pole orders are $0,2,3,4,5,6,6$: seven functions, six available orders, the collision at 6 forcing the relation exactly as in Problem 8(4). And the kernel is one-dimensional, spanned by the vector reading

$$1-28w^3+3v^2=0,$$

which is $3v^2+1=4dw^3$ at $d=7$. We got it by hand from a factorization. `Magma` got it from linear algebra in a Riemann–Roch space.

Now try these. Each one is a small edit to Example 4.2 or Example 4.3, and in each case the worksheet already tells you what the answer must be.

(1) Set `d := 2`. Problem 3(3) predicts a rational point of order two. Find it, verify with the group law that it has order two, and compute its image under φ_1 by hand.

Solution. **The rational point $(1:1:1)$, with $\varphi_1(1:1:1)=(12,0)$.** $d=2=2\cdot 1^3$, so by Problem 3(3) the rational 2-torsion point is $(1:1:1)$, and indeed $1^3+1^3=2$. Then

$$\varphi_1(1:1:1)=\left(\frac{12\cdot 2\cdot 1}{1+1},\ \frac{36\cdot 2\cdot 0}{1+1}\right)=(12,\,0),$$

which lies on $E_1\colon y_1^2=x_1^3-432\cdot 4=x_1^3-1728$ because $12^3=1728$. A point of $E_1(\bar{k})$ with $y_1=0$ is exactly a point of order two. Equivalently `dbl([1,1,1])` returns $(-3:3:0)=O$.

(2) Set `d := 8`. Problem 6(7) predicts rational 3-torsion. Find the two rational 3-torsion points other than O, use your `dbl` function to check that $[2]P=\ominus P$, and compute $\varphi_1(P)$ by hand.

Solution. **(2 : 0 : 1) and (0 : 2 : 1), with $\varphi_1(2:0:1) = (48, 288)$.** $d = 8 = 2^3$, so $\delta = 2$ and the rational flexes are $(2:0:1)$ and $(0:2:1)$ besides O. Doubling: `dbl([2,0,1])` gives

$$\big(-0 : 2(8+0) : 1(8-0)\big) = (0 : 16 : 8) = (0 : 2 : 1),$$

which is $\ominus(2:0:1)$, so $[3](2:0:1) = O$. By hand,

$$\varphi_1(2:0:1) = \left(\frac{12\cdot 8\cdot 1}{2},\ \frac{36\cdot 8\cdot 2}{2}\right) = (48,\ 288),$$

and $48^3 - 432\cdot 64 = 110592 - 27648 = 82944 = 288^2$.

(3) Use `IsIsomorphic` to decide whether $C_1 \simeq C_8$ and whether $C_1 \simeq C_2$ over $\mathbb{Q}$. Pass it the *Weierstrass* models: on the plane curves themselves Magma answers `For genus 0 or 1 curves the basefield must be finite`. Reconcile the answers with the cubic-twist criterion of Section 5.

Solution. **$C_1 \simeq C_8$ over $\mathbb{Q}$, but $C_1 \not\simeq C_2$.** Over $\mathbb{Q}$ the curves $y_1^2 = x_1^3 + A_1$ and $y_2^2 = x_2^3 + A_2$ are isomorphic exactly when A_1/A_2 is a sixth power. For C_1 and C_8 we get $-432/(-432\cdot 64) = 1/64 = (1/2)^6$, so `IsIsomorphic(EllipticCurve(C1,O1), EllipticCurve(C8,O8))` returns `true`. Consistently, $1/8 = (1/2)^3$ is a cube. For C_1 and C_2 we get $1/4$, which is not a sixth power in $\mathbb{Q}^\times$ (that would need $2^{-1/3} \in \mathbb{Q}$), so the answer is `false`. Consistently, 2 is not a rational cube. This is the criterion of Section 5: $C_d \simeq C_{d'}$ over k if and only if $d/d' \in (k^\times)^3$.

5. Where the name comes from

Selmer [Sel51] studied the general diagonal cubic $aX^3 + bY^3 + cZ^3 = 0$, and produced the standard counterexample to the Hasse principle: the curve $3X^3 + 4Y^3 + 5Z^3 = 0$ has real points and $\mathbb{Q}_p$-points for every prime p, but no rational point. Such a curve is a torsor under its Jacobian [Poo17, §5.12], and that Jacobian is a Selmer curve in our sense:

$$\mathrm{Jac}(aX^3 + bY^3 + cZ^3) \simeq \big(y^2 = x^3 - 432(abc)^2\big).$$

Two ways to test this in Magma: the curve $X^3 + 2Y^3 + 3Z^3 = 0$ *does* have the rational point $(1:1:-1)$, so it is its own Jacobian and should be isomorphic to C_6, that is to $y^2 = x^3 - 432\cdot 6^2$; and for Selmer's own curve one can call `Jacobian(GenusOneModel(3*X^3 + 4*Y^3 + 5*Z^3))` and compare with $y^2 = x^3 - 432\cdot 60^2$. Both come out `true`.

Finally, by Problem 8(6) together with the classification of twists of $y^2 = x^3 + A$ [Sil09, §X.5], for $\mathrm{char}\, k \neq 2, 3$ the curves C_d and $C_{d'}$ are isomorphic over k exactly when $d/d' \in (k^\times)^3$, so the family is parametrized by $k^\times/(k^\times)^3$. These are the cubic twists of $X^3 + Y^3 = Z^3$, and they are the natural first examples for the descent machinery of [Sil09, §X] and [Poo17, Ch. 8].

Part 2. The Weil pairing, endomorphisms and automorphisms

Section 6 shows that the determinant of the mod N Galois representation is the cyclotomic character. Section 7 exhibits an endomorphism ring that is an order in a definite quaternion algebra, the third case of the classification [Sil09, Theorem III.9.3], on a single curve over $\mathbb{F}_9$. Section 8 computes the automorphism groups of order 12 and 24 in characteristic 3 and 2, the two cases of [Sil09, Theorem III.10.1] in which the group is not cyclic.

Reading. [Sil09, §§III.8–III.10], with [Sil09, Appendix A.1] for the automorphism groups in characteristic 2 and 3. For the quaternion background: [Voi21, Definition 2.2.1] for the algebra $(a, b \mid \mathbb{Q})$, [Voi21, §§3.2–3.3] for the standard involution, the reduced trace and the reduced norm, [Voi21, Main Theorem 5.4.4 and §14.1] for split and definite algebras, and [Voi21, Definitions 10.2.1 and 10.4.1] for orders and maximal orders. Proposition 7.6, Theorem 7.7 and Proposition 7.8 are the results we actually use.

`Magma`. If you have no local installation, every block below runs in the free online calculator, which caps how long a single job may take.

Standing notation.

- A field k, with an algebraic closure $\overline{k}$ and the separable closure $k^{\mathrm{sep}} \subseteq \overline{k}$.
- Multiplication by m, written $[m]$, and the dual $\hat{\varphi}$ of an isogeny φ.
- The q-power Frobenius endomorphism $\mathsf{F} \in \mathrm{End}(E)$ of a curve over $\mathbb{F}_q$. Beware that φ_ℓ already means the map induced by φ on $T_\ell(E)$, so the Frobenius *endomorphism* is F and never φ_q.

6. The Weil pairing and the cyclotomic character

Fix $N \geqslant 2$ with $\mathrm{char}\, k \nmid N$, so that $E[N](k^{\mathrm{sep}}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.

Definition 6.1 (The Weil pairing)**.** We write

$$e_N \colon E[N](k^{\mathrm{sep}}) \times E[N](k^{\mathrm{sep}}) \to \mu_N(k^{\mathrm{sep}})$$

for the Weil pairing.

Definition 6.2 (Galois representations)**.** We write

$$\bar{\rho}_{E,N} \colon \mathrm{Gal}_k \to \mathrm{Aut}\big(E[N](k^{\mathrm{sep}})\big) \cong \mathrm{GL}_2(\mathbb{Z}/N\mathbb{Z})$$

for the mod N Galois representation, the isomorphism depending on a choice of basis, and $\rho_{E,\ell}$ for the ℓ-adic representation on $T_\ell(E)$.

Proposition 6.3 (Properties of the Weil pairing [Sil09, Propositions III.8.1 and III.8.2])**.** *The Weil pairing e_N is*

(1) bilinear;
(2) alternating: $e_N(T,T) = 1$, so $e_N(S,T) = e_N(T,S)^{-1}$;
(3) non-degenerate: if $e_N(S,T) = 1$ for all $S \in E[N](k^{\mathrm{sep}})$, then $T = O$;
(4) Galois equivariant: $\sigma\big(e_N(S,T)\big) = e_N(\sigma(S),\sigma(T))$ for $\sigma \in \mathrm{Gal}_k$;
(5) compatible: $e_{NN'}(S,T) = e_N([N']S,T)$;
(6) adjoint for isogenies: $e_N(\varphi S,T) = e_N(S,\hat{\varphi}T)$.

These properties are.

Proposition 6.4 (The determinant of φ_ℓ [Sil09, Proposition III.8.6])**.** *Let $\varphi \in \mathrm{End}(E)$ and let $\ell \neq \mathrm{char}\, k$ be a prime. Then $\det \varphi_\ell = \deg \varphi$.*

Problem 9 (The determinant is the cyclotomic character)**.**

(1) Let $\zeta \in \mu_N(k^{\mathrm{sep}})$ be primitive. Show that for each $\sigma \in \mathrm{Gal}_k$ there is a unique $\chi_N(\sigma) \in \mathbb{Z}/N\mathbb{Z}$ with $\sigma(\zeta) = \zeta^{\chi_N(\sigma)}$, that $\chi_N(\sigma)$ is a unit, and that the resulting map $\chi_N \colon \mathrm{Gal}_k \to (\mathbb{Z}/N\mathbb{Z})^\times$ is a homomorphism independent of the choice of ζ. This is the mod N cyclotomic character.

Solution. Since σ fixes 1 and is a ring homomorphism, it permutes the roots of $X^N - 1$, so $\sigma(\zeta) \in \mu_N(k^{\mathrm{sep}})$. As ζ is primitive, $\mu_N(k^{\mathrm{sep}}) = \{\zeta^j : 0 \leqslant j < N\}$ and these are distinct, so $\sigma(\zeta) = \zeta^j$ for a unique $j =: \chi_N(\sigma) \bmod N$. Applying σ^{-1} shows $\zeta \mapsto \zeta^{\chi_N(\sigma)}$ is bijective on $\mu_N(k^{\mathrm{sep}})$, so $\gcd(\chi_N(\sigma), N) = 1$. For the homomorphism property, $\sigma\tau(\zeta) = \sigma(\zeta^{\chi_N(\tau)}) = \zeta^{\chi_N(\sigma)\chi_N(\tau)}$. Independence of ζ: any other primitive root is ζ^u with u a unit, and $\sigma(\zeta^u) = (\zeta^u)^{\chi_N(\sigma)}$.

(2) Let S, T be a basis of $E[N](k^{\mathrm{sep}})$ and put $\zeta \coloneqq e_N(S,T)$. Show that ζ is a *primitive* N-th root of unity, using only parts (1)–(3) of Proposition 6.3.

Solution. Let m be the order of $\zeta = e_N(S,T)$, so $m \mid N$. By bilinearity, $e_N(S,[m]T) = e_N(S,T)^m = 1$, and by (2) together with bilinearity, $e_N(T,[m]T) = e_N(T,T)^m = 1$. Since

S, T generate $E[N](k^{\mathrm{sep}})$, bilinearity gives $e_N(X, [m]T) = 1$ for every $X \in E[N](k^{\mathrm{sep}})$. Non-degeneracy ((3), in the first variable, via (2)) forces $[m]T = O$. But T has order N, so $N \mid m$, and hence $m = N$.

(3) Write $\sigma(S) = aS + cT$ and $\sigma(T) = bS + dT$, so that $\bar\rho_{E,N}(\sigma) = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ in the basis S, T. Expand $e_N(\sigma(S), \sigma(T))$ and deduce

$$\det \bar\rho_{E,N} = \chi_N.$$

Solution. Using bilinearity and then $e_N(S,S) = e_N(T,T) = 1$ and $e_N(T,S) = e_N(S,T)^{-1}$,

$e_N(\sigma(S), \sigma(T)) = e_N(aS + cT,\, bS + dT) = e_N(S,S)^{ab} e_N(S,T)^{ad} e_N(T,S)^{cb} e_N(T,T)^{cd} = \zeta^{ad-bc}$.

On the other hand Galois equivariance gives $e_N(\sigma(S), \sigma(T)) = \sigma(\zeta) = \zeta^{\chi_N(\sigma)}$. So $\zeta^{ad-bc} = \zeta^{\chi_N(\sigma)}$, and since ζ has order exactly N by (2), we get $ad - bc \equiv \chi_N(\sigma) \pmod N$, i.e. $\det \bar\rho_{E,N}(\sigma) = \chi_N(\sigma)$.

(4) Take $k = \mathbb{Q}$. Show that $\det \bar\rho_{E,N}$ is surjective for *every* elliptic curve E over $\mathbb{Q}$, that $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}\big(E[N]\big)$, and that $\det \bar\rho_{E,N}(c) = -1$ for c a complex conjugation: the representation is odd.

Solution. Over $\mathbb{Q}$, restriction gives $\mathrm{Gal}_{\mathbb{Q}} \twoheadrightarrow \mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times$, and this isomorphism *is* χ_N. So χ_N, hence $\det \bar\rho_{E,N}$, is surjective, for every E. Since $\chi_N = \det \bar\rho_{E,N}$ factors through $\mathrm{Gal}\big(\mathbb{Q}(E[N])/\mathbb{Q}\big)$, its kernel contains $\mathrm{Gal}\big(\overline{\mathbb{Q}}/\mathbb{Q}(E[N])\big)$, so $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}(E[N])$. Finally $c(\zeta_N) = \zeta_N^{-1}$, so $\chi_N(c) = -1$. Taking $N = \ell^n$ and passing to the limit, $\det \rho_{E,\ell}(c) = -1$.

Remark. The determinant never degenerates, however small the image of $\bar\rho_{E,N}$ itself may be.

(5) Take $k = \mathbb{F}_q$ with q prime to N. Show that $\det \bar\rho_{E,N}(\mathsf{F}) = q$, and check this against the identity $\det \varphi_\ell = \deg \varphi$.

Solution. $\mathrm{Gal}_{\mathbb{F}_q}$ is topologically generated by the q-power Frobenius, which acts on $\mu_N(\overline{\mathbb{F}}_q)$ by $\zeta \mapsto \zeta^q$, so $\chi_N(\mathsf{F}) = q$ and (3) gives $\det \bar\rho_{E,N}(\mathsf{F}) = q$. The second computation is different: $\det \mathsf{F}_\ell = \deg \mathsf{F} = q$ because the pairing on $T_\ell(E)$ is alternating, so an endomorphism scales it by its determinant, while adjointness scales it by $\widehat{\mathsf{F}}\,\mathsf{F} = [\deg \mathsf{F}]$.

Remark. So the first computation uses Galois equivariance (4) and the second uses adjointness (6). That the two agree is the statement that $\det \rho_{E,\ell}$ is the cyclotomic character.

(6) Check (5) numerically in `Magma`, taking $N = 3$, $q = 5$ and $E\colon y^2 = x^3 + 1$ over $\mathbb{F}_5$.

```
q := 5;  N := 3;
E := EllipticCurve([GF(q) | 0, 1]);      // y^2 = x^3 + 1
#E;                                      // 6
Ek := ChangeRing(E, GF(q^2));
#Ek, AbelianGroup(Ek);                   // 36, Z/6 + Z/6
T3 := [P : P in Points(Ek) | N*P eq Ek!0];
#T3;                                     // 9
S := [P : P in T3 | P ne Ek!0][1];
T := [P : P in T3 | P ne Ek!0 and P ne S and P ne -S][1];
zeta := WeilPairing(S, T, N);  Order(zeta);    // 3: a primitive cube root
Frob := func<P | P eq Ek!0 select Ek!0 else Ek![P[1]^q, P[2]^q]>;
WeilPairing(Frob(S), Frob(T), N) eq zeta^q;    // true
```

Two lines carry the mathematics. That `Order(zeta)` is 3 is (2). That the last line is `true` is $\det \bar\rho_{E,3}(\mathsf{F}) = q$, which is (5) on this curve. Note that `Frob` raises both coordinates to the q-th

power: that is the q-power Frobenius, an endomorphism because E is defined over $\mathbb{F}_q$. Now write the matrix of F in the basis S, T and read its determinant off:

```
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(S) ];   // [ <1, 0> ]
[ <a, b> : a, b in [0..2] | a*S + b*T eq Frob(T) ];   // [ <0, 2> ]
GF(3)!q;                                              // 2
```

Check that a different choice of basis moves the matrix but not the determinant.

SOLUTION. $\left[\begin{smallmatrix}1&0\\0&2\end{smallmatrix}\right]$**, with determinant** $2 = q$ **mod 3.** The pairing $e_3(S,T)$ has order exactly 3, as (2) predicts, and `WeilPairing(Frob(S), Frob(T), N) eq zeta^q` is exactly $\det \bar{\rho}_{E,3}(\mathsf{F}) = q$. The last block gives $\mathsf{F}S = S$ and $\mathsf{F}T = [2]T$, so the matrix of F in the basis S, T is $\left[\begin{smallmatrix}1&0\\0&2\end{smallmatrix}\right]$, with determinant $2 = q \bmod 3$. Another basis gives another matrix, never another determinant.

Remark 6.5. Part (3) of Problem 9 gives back [Sil09, Corollary III.8.1.1]: if $E[N](k^{\mathrm{sep}}) \subseteq E(k)$, then $\bar{\rho}_{E,N}$ is trivial, so χ_N is trivial, so every $\sigma \in \mathrm{Gal}_k$ fixes a primitive N-th root of unity, and $\mu_N(k^{\mathrm{sep}}) \subseteq k$.

A word, for the curious, on the example in Problem 9(6): choosing it was half of that part. We want $q \not\equiv 1 \pmod N$, so that χ_N is *not* trivial and the check has content. By Remark 6.5 that already forces $E[N](k^{\mathrm{sep}}) \not\subseteq E(\mathbb{F}_q)$, so the N-torsion becomes rational only over an extension. The curve $E\colon y^2 = x^3 + 1$ over $\mathbb{F}_5$ has $\#E(\mathbb{F}_5) = 6$ and $E(\mathbb{F}_{25}) \cong (\mathbb{Z}/6\mathbb{Z})^2$, so all of $E[3](\overline{\mathbb{F}}_5)$ is already rational over the quadratic extension: small enough to list, large enough to have something to say.

7. A QUATERNIONIC ENDOMORPHISM RING

The classification of [Sil09, Theorem III.9.3] puts the endomorphism ring of an elliptic curve in one of three cases: $\mathbb{Z}$, an order in an imaginary quadratic field, or an order in a definite quaternion algebra over $\mathbb{Q}$. The first two are easy to exhibit: a generic curve has endomorphism ring $\mathbb{Z}$, and $y^2 = x^3 + x$ has $\mathbb{Z}[i]$ inside its endomorphism ring. Here is an example of the third, on the smallest curve that has one.

Definition 7.1 (The quaternion algebra D [Voi21, Definition 2.2.1])**.** Let $D := (-1, -3 \mid \mathbb{Q})$, the quaternion algebra with $i^2 = -1$, $j^2 = -3$ and $ij = -ji$. Its reduced discriminant is the product of the primes p at which $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is not split.

Definition 7.2 (Definite quaternion algebras [Voi21, Main Theorem 5.4.4 and §14.1])**.** A quaternion algebra B over $\mathbb{Q}$ is definite when $B \otimes_{\mathbb{Q}} \mathbb{R}$ is Hamilton's $\mathbb{H}$ rather than $M_2(\mathbb{R})$, equivalently when its reduced norm is a positive definite quadratic form in the four coordinates. Otherwise B is indefinite.

Definition 7.3 (Orders and the reduced discriminant [Voi21, §15.2])**.** Orders are as in [Voi21, Definition 10.2.1]. The reduced discriminant $\mathrm{discrd}(\mathcal{O})$ of an order $\mathcal{O} \subseteq D$ is the positive integer with $\mathrm{discrd}(\mathcal{O})^2 = \left|\det\big(\mathrm{trd}(\alpha_k \alpha_l)\big)_{k,l}\right|$ for a $\mathbb{Z}$-basis $\alpha_1, \ldots, \alpha_4$ of $\mathcal{O}$.

Proposition 7.4 (The characteristic polynomial of Frobenius [Sil09, §V.2])**.** *Let E be an elliptic curve over $\mathbb{F}_q$ with F its q-power Frobenius, $a := q + 1 - \#E(\mathbb{F}_q)$, and $\ell \nmid q$ a prime. Then F satisfies $\mathsf{F}^2 - [a]\mathsf{F} + [q] = 0$ in $\mathrm{End}(E)$: by [Sil09, Proposition III.8.6] the characteristic polynomial of F_ℓ is $T^2 - aT + q$, and every $\varphi \in \mathrm{End}(E)$ satisfies $\varphi^2 - [\mathrm{tr}\,\varphi_\ell]\varphi + [\deg \varphi] = 0$ in $\mathrm{End}(E)$, not merely on $T_\ell(E)$.*

Corollary 7.5 (Factoring through a separable isogeny [Sil09, Corollary III.4.11])**.** *If $\varphi\colon E_1 \to E_2$ is separable and $\psi\colon E_1 \to E_3$ is an isogeny with $\ker\varphi \subseteq \ker\psi$, then $\psi = \lambda \circ \varphi$ for a unique isogeny $\lambda\colon E_2 \to E_3$.*

Proposition 7.6 (Where D is split [Voi21, Exercise 11.11 and §14.1]). *The algebra $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is split for every prime $p \neq 3$, and $D \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is not. So D is a division algebra, and its reduced discriminant is 3.*

Theorem 7.7 (Maximal orders [Voi21, Theorem 15.5.5]). *An order $\mathcal{O} \subseteq D$ is maximal if and only if* $\operatorname{discrd}(\mathcal{O}) = 3$.

Proposition 7.8 (The index and the discriminant [Voi21, §15.2]). *If $\mathcal{O} \subseteq \mathcal{O}'$ are orders of D, then* $\operatorname{discrd}(\mathcal{O}) = [\mathcal{O}' : \mathcal{O}] \operatorname{discrd}(\mathcal{O}')$.

For the rest of this section

$$E\colon y^2 = x^3 - x \quad \text{over } \mathbb{F}_3,$$

so that the generic E of Proposition 7.4 is this curve, with $q = 3$.

Problem 10 (End($E_{\mathbb{F}_9}$) is a maximal order in a quaternion algebra).
Every part below has a hand computation and a check to run in `Magma`. The curve is small enough that $[i]$ and F can be verified on all sixteen points of $E(\mathbb{F}_9)$.

(1) Check that E is an elliptic curve over $\mathbb{F}_3$, list $E(\mathbb{F}_3)$, and deduce that $a = 0$ and that

$$\mathsf{F}^2 = [-3].$$

Record also that every point of $E[2](\overline{\mathbb{F}}_3)$ lies in $E(\mathbb{F}_3)$: this is what makes part (5) work.

```
E3 := EllipticCurve([GF(3) | -1, 0]);     // prints as y^2 = x^3 + 2*x
Points(E3);
#E3;                                      // 4
3 + 1 - #E3;                              // 0        (so a = 0)
jInvariant(E3);                           // 0
```

`Magma` hands you the point count and the j-invariant. The identity $\mathsf{F}^2 = [-3]$ is yours to deduce, from Proposition 7.4. Read $E[2](\overline{\mathbb{F}}_3) \subseteq E(\mathbb{F}_3)$ off the list of rational points — which points of $y^2 = f(x)$ have order 2?

Solution. ***$E(\mathbb{F}_3) = \{O, (0,0), (1,0), (2,0)\}$, so $a = 0$.*** Here $A = -1$ and $B = 0$, so $\Delta = -16(4A^3 + 27B^2) = 64 \equiv 1 \pmod 3$, which is non-zero: E is an elliptic curve. Every $x \in \mathbb{F}_3$ satisfies $x^3 = x$, so $x^3 - x = 0$ for all three values, giving

$$E(\mathbb{F}_3) = \{O, (0,0), (1,0), (2,0)\}, \qquad \#E(\mathbb{F}_3) = 4.$$

Hence $a = 3 + 1 - 4 = 0$, and Proposition 7.4 gives $\mathsf{F}^2 + [3] = 0$, i.e. $\mathsf{F}^2 = [-3]$. Finally the three affine rational points listed above are precisely the points of order 2, since a point (x, y) of $E(\overline{\mathbb{F}}_3)$ has order 2 if and only if $y = 0$. All three are in $E(\mathbb{F}_3)$, so $E[2](\overline{\mathbb{F}}_3) \subseteq E(\mathbb{F}_3)$, which is what `Magma`'s list of rational points shows.

(2) Show that $x^2 + 1$ is irreducible over $\mathbb{F}_3$, so that $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$. Show that

$$[i](x, y) \coloneqq (-x, iy)$$

defines an endomorphism of $E_{\mathbb{F}_9}$, and that $[i]^2 = [-1]$. Why is a morphism $E \to E$ fixing O automatically a group homomorphism? Mind Warning 7.9 before you start typing.

```
F9 := GF(9);  i := Sqrt(F9!(-1));
i^2, MinimalPolynomial(i);                // 2 ($.1^2 + 1), and 2 = -1
E9 := ChangeRing(E3, F9);  #E9;           // 16
imap := func<P | P eq E9!0 select E9!0 else E9![-P[1], i*P[2]]>;
```

```
forall{P : P in Points(E9) | imap(P) in Points(E9)};        // [i] is well defined
forall{<P,Q> : P, Q in Points(E9) | imap(P+Q) eq imap(P)+imap(Q)};  // additive
forall{P : P in Points(E9) | imap(imap(P)) eq -P};          // [i]^2 = [-1]
// all true
```

The second `forall` is additivity, checked on all 16^2 pairs. That is a proof only because the curve is finite, and it is no answer to the question above: additivity is not a coincidence to be confirmed point by point but a consequence of fixing O.

Solution. x^2+1 has no root in $\mathbb{F}_3$: the values of x^2+1 are $1, 2, 2$. So it is irreducible and $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$. For the map, substitute:

$$(iy)^2 = i^2y^2 = -(x^3 - x) = -x^3 + x = (-x)^3 - (-x).$$

So $[i]$ carries points of E to points of E, and it is given by polynomials in x, y with coefficients in $\mathbb{F}_9$, hence is a morphism $E_{\mathbb{F}_9} \to E_{\mathbb{F}_9}$. It fixes $O = (0:1:0)$. A morphism $\varphi\colon E_{\mathbb{F}_9} \to E_{\mathbb{F}_9}$ with $\varphi(O) = O$ is a group homomorphism because the group law of E is the one of $\operatorname{Pic}^0 E$, transported along the isomorphism of groups $\kappa\colon E(\overline{\mathbb{F}}_3) \to \operatorname{Pic}^0 E_{\overline{\mathbb{F}}_3}$, $P \mapsto [(P) - (O)]$. If φ is constant it is the zero map. Otherwise it is finite, so the push-forward of divisors φ_* is a homomorphism $\operatorname{Pic}^0 E_{\overline{\mathbb{F}}_3} \to \operatorname{Pic}^0 E_{\overline{\mathbb{F}}_3}$, and since $\varphi(O) = O$ it sends $[(P) - (O)]$ to $[(\varphi(P)) - (O)]$ for every point P of $E(\overline{\mathbb{F}}_3)$. Hence $\varphi = \kappa^{-1} \circ \varphi_* \circ \kappa$ is a composition of homomorphisms [Sil09, Theorem III.4.8]. So $[i] \in \operatorname{End}(E_{\mathbb{F}_9})$. Composing,

$$[i]^2(x, y) = [i](-x, iy) = (x, i^2y) = (x, -y) = [-1](x, y),$$

so $[i]^2 = [-1]$. In particular $[i]$ is an automorphism, of order 4.

(3) Show that $\mathsf{F}[i] = -[i]\mathsf{F}$. Conclude that $\operatorname{End}(E_{\mathbb{F}_9})$ is not commutative, so that $E_{\mathbb{F}_9}$ falls into the third case of the classification.

```
Frob := func<P | P eq E9!0 select E9!0 else E9![P[1]^3, P[2]^3]>;
forall{P : P in Points(E9) | Frob(Frob(P)) eq -3*P};            // Frob^2 = [-3]
forall{P : P in Points(E9) | Frob(imap(P)) eq -imap(Frob(P))};  // Frob[i] = -[i]Frob
// both true
```

The first line verifies the $\mathsf{F}^2 = [-3]$ you deduced in (1) from the characteristic polynomial, which is worth having independently. For the second, do the computation by hand as well: it comes down to a single fact about i^3.

Solution. $\mathsf{F}(x, y) = (x^3, y^3)$, so

$$[i]\mathsf{F}(x, y) = [i](x^3, y^3) = (-x^3, iy^3),$$
$$\mathsf{F}[i](x, y) = \mathsf{F}(-x, iy) = \big((-x)^3, (iy)^3\big) = (-x^3, i^3y^3).$$

Now $i^3 = i^2 \cdot i = -i$, so $\mathsf{F}[i](x, y) = (-x^3, -iy^3) = -[i]\mathsf{F}(x, y)$. This shows that

$$\mathsf{F}[i] = -[i]\mathsf{F}.$$

Since $[2][i]\mathsf{F} \neq 0$ (it has degree 12), $-[i]\mathsf{F} \neq [i]\mathsf{F}$: the two endomorphisms $[i]$ and F do not commute, so $\operatorname{End}(E_{\mathbb{F}_9})$ is non-commutative. By the classification it is therefore an order in a definite quaternion algebra: cases $\mathbb{Z}$ and an imaginary quadratic order are both commutative.

(4) Deduce that

$$\operatorname{End}(E_{\mathbb{F}_9}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong D = (-1, -3 \mid \mathbb{Q}).$$

Write down the reduced norm of D and check that D is definite. How does the reduced norm compare with the degree of an endomorphism?

```
D<ii,jj> := QuaternionAlgebra<Rationals() | -1, -3>;
RamifiedPrimes(D), Discriminant(D);      // [ 3 ] 3
IsDefinite(D);                           // true
Trace(jj), Norm(jj);                     // 0 3
```

The two arguments are the squares of the generators, so the algebra `Magma` builds is the D of Definition 7.1. Here `Trace` and `Norm` on a quaternion algebra are the reduced trace and the reduced norm, and the pair they return for `jj` should be recognizable as $(\operatorname{tr}\mathsf{F}, \deg\mathsf{F})$. All of this is about D alone. That D is the *whole* of $\mathrm{End}(E_{\mathbb{F}_9})\otimes\mathbb{Q}$ is the one claim here `Magma` cannot be asked, and it needs a rank count.

Solution. **$\mathbf{nrd}(t+xi+yj+zij) = t^2+x^2+3y^2+3z^2$. So nrd = deg.** By (2), (1) and (3) the elements $[i]$ and F of $\mathrm{End}(E_{\mathbb{F}_9})\otimes\mathbb{Q}$ satisfy $i^2=-1$, $\mathsf{F}^2=-3$ and $i\mathsf{F}=-\mathsf{F}i$, which are the defining relations of $D=(-1,-3\mid\mathbb{Q})$. So $i\mapsto[i]$, $j\mapsto\mathsf{F}$ defines a ring homomorphism $D\to\mathrm{End}(E_{\mathbb{F}_9})\otimes\mathbb{Q}$. It is injective because D is a division algebra (Proposition 7.6), so its only two-sided ideals are 0 and D. And it is surjective because $\mathrm{End}(E_{\mathbb{F}_9})$ has rank at most 4 over $\mathbb{Z}$ ([Sil09, Corollary III.7.5]) while D has dimension exactly 4. The reduced norm of D is

$$\mathrm{nrd}(t+xi+yj+zij) = t^2+x^2+3y^2+3z^2,$$

which is positive definite: $(a,b\mid\mathbb{Q})$ is definite exactly when $a<0$ and $b<0$ [Voi21, Main Theorem 5.4.4 and §14.1], and here $a=-1$, $b=-3$. Under the isomorphism, nrd is the degree: the dual satisfies

$$\hat\varphi\varphi = [\deg\varphi], \qquad \varphi+\hat\varphi = [1+\deg\varphi-\deg(1-\varphi)]\in\mathbb{Z}$$

[Sil09, Theorem III.6.2 and Corollary III.6.3], so $\varphi\mapsto\hat\varphi$ is the standard involution of D and $\mathrm{nrd}(\varphi)=\hat\varphi\varphi=\deg\varphi$. For instance $\deg\mathsf{F}=3=\mathrm{nrd}(j)$ and $\deg[i]=1=\mathrm{nrd}(i)$.
In the code, `Trace(jj), Norm(jj)` returns 0 and 3: that is $\mathrm{trd}(\mathsf{F})=0=a$ and $\mathrm{nrd}(\mathsf{F})=3=\deg\mathsf{F}$, as the identification of nrd with the degree requires.

Remark. Note that the prime at which D is not split, by Proposition 7.6, is the characteristic: in general, when the endomorphism algebra of an elliptic curve in characteristic p is a quaternion algebra, it is the one over $\mathbb{Q}$ ramified exactly at p and ∞ [Sil09, Theorem V.3.1], [Voi21, Theorem 42.1.9].

(5) Show that $\frac{1+\mathsf{F}}{2}$ and $\frac{i+i\mathsf{F}}{2}$ lie in $\mathrm{End}(E_{\mathbb{F}_9})$. Then show that

$$\mathcal{O} \coloneqq \mathbb{Z}+\mathbb{Z}i+\mathbb{Z}\frac{1+\mathsf{F}}{2}+\mathbb{Z}\frac{i+i\mathsf{F}}{2}$$

is an order of D, compute $\mathrm{discrd}(\mathcal{O})$, and conclude with Theorem 7.7 that $\mathrm{End}(E_{\mathbb{F}_9})=\mathcal{O}$ is a maximal order.

```
OO := QuaternionOrder([D!1, ii, jj, ii*jj]);
Discriminant(OO);                        // 12
O  := QuaternionOrder([D!1, ii, (1+jj)/2, (ii+ii*jj)/2]);
Discriminant(O), IsMaximal(O);           // 3 true
```

`QuaternionOrder` takes a $\mathbb{Z}$-basis and `Discriminant` of an order is its reduced discriminant, so the two lists above are the $\mathbb{Z}$-bases of $\mathcal{O}_0$ and $\mathcal{O}$. Do not reach for `Index` to compare the two orders:

`Magma` has no index for quaternion orders, and answers `Runtime error in 'Index': Bad argument types`. Recover $[\mathcal{O} : \mathcal{O}_0]$ from the two discriminants instead, by Proposition 7.8, and check it against the determinant of the change of basis.

The first sentence of this part is where the real work is, and none of the code touches it: $\frac{1+\mathsf{F}}{2}$ is a quotient in a ring with no division, and only the kernel argument puts it in $\mathrm{End}(E_{\mathbb{F}_9})$.

SOLUTION. **discrd($\mathcal{O}$) = 3, so End($E_{\mathbb{F}_9}$) = $\mathcal{O}$.** By (1) every $P \in E[2](\overline{\mathbb{F}}_3)$ lies in $E(\mathbb{F}_3)$, so F fixes it, and therefore

$$(1+\mathsf{F})P = P + \mathsf{F}P = P + P = [2]P = O.$$

Hence $\ker[2] = E[2] \subseteq \ker(1+\mathsf{F})$. Since the characteristic 3 does not divide 2, the isogeny $[2]$ is separable, so Corollary 7.5 gives an isogeny λ with $1 + \mathsf{F} = \lambda \circ [2]$. That is, $\frac{1+\mathsf{F}}{2} = \lambda$ lies in $\mathrm{End}(E_{\mathbb{F}_9})$. Multiplying by $[i] \in \mathrm{End}(E_{\mathbb{F}_9})$ shows $\frac{i+i\mathsf{F}}{2} = [i] \circ \lambda$ lies there too.
So $\mathcal{O} \subseteq \mathrm{End}(E_{\mathbb{F}_9})$, and $\mathcal{O}$ is an order: it is a free $\mathbb{Z}$-module of rank 4 spanning D over $\mathbb{Q}$, and it is closed under multiplication, since the pairwise products of 1, i, $\frac{1+\mathsf{F}}{2}$ and $\frac{i+i\mathsf{F}}{2} = i\,\frac{1+\mathsf{F}}{2}$ lie in their $\mathbb{Z}$-span: it suffices to check the products of i and $\frac{1+\mathsf{F}}{2}$, and

$$i^2 = -1, \qquad \left(\frac{1+\mathsf{F}}{2}\right)^2 = \frac{1+2\mathsf{F}+\mathsf{F}^2}{4} = \frac{-2+2\mathsf{F}}{4} = \frac{1+\mathsf{F}}{2} - 1,$$
$$i\,\frac{1+\mathsf{F}}{2} = \frac{i+i\mathsf{F}}{2}, \qquad \frac{1+\mathsf{F}}{2}\,i = \frac{i-i\mathsf{F}}{2} = i - \frac{i+i\mathsf{F}}{2}.$$

To see that $\mathcal{O}$ is maximal, compute reduced discriminants, starting with the obvious suborder $\mathcal{O}_0 \coloneqq \mathbb{Z} + \mathbb{Z}i + \mathbb{Z}\mathsf{F} + \mathbb{Z}i\mathsf{F}$. Its reduced trace form $\big(\mathrm{trd}(\alpha_k\alpha_l)\big)$ in the basis $1, i, \mathsf{F}, i\mathsf{F}$ is diagonal, because distinct basis products have reduced trace 0:

$$\mathrm{trd}(1) = 2, \quad \mathrm{trd}(i^2) = -2, \quad \mathrm{trd}(\mathsf{F}^2) = -6, \quad \mathrm{trd}\big((i\mathsf{F})^2\big) = -6,$$

so the Gram determinant is $2 \cdot (-2) \cdot (-6) \cdot (-6) = -144$ and $\mathrm{discrd}(\mathcal{O}_0) = 12$. Expressing the basis of $\mathcal{O}$ in terms of that of $\mathcal{O}_0$ gives the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix}, \qquad \det = \tfrac{1}{4},$$

so $[\mathcal{O} : \mathcal{O}_0] = 4$, and by Proposition 7.8

$$\mathrm{discrd}(\mathcal{O}) = \frac{\mathrm{discrd}(\mathcal{O}_0)}{[\mathcal{O} : \mathcal{O}_0]} = \frac{12}{4} = 3.$$

By Theorem 7.7, $\mathcal{O}$ is therefore a maximal order. A maximal order is not properly contained in any order, and $\mathcal{O} \subseteq \mathrm{End}(E_{\mathbb{F}_9})$ is an order, so $\mathcal{O} = \mathrm{End}(E_{\mathbb{F}_9})$.
`Magma` returns 12 and 3 for the two reduced discriminants, and the ratio $12/3 = 4$ is the index $[\mathcal{O} : \mathcal{O}_0]$ computed above from the change of basis, so the two routes agree.

Remark. Note that `IsMaximal` is not telling us anything the discriminant did not: by Theorem 7.7 maximality *is* the statement discrd = 3.
A closing word on scope: when the endomorphism ring of a curve over $\overline{\mathbb{F}}_p$ is non-commutative, all of its endomorphisms are already defined over $\mathbb{F}_{p^2}$, so this also computes $\mathrm{End}(E_{\overline{\mathbb{F}}_3}) = \mathcal{O}$, in agreement with Deuring's theorem that such a ring is a maximal order in the quaternion algebra ramified exactly at p and ∞ [Voi21, Theorem 42.1.9].

Warning 7.9 ($\mathbb{F}_9$ does not come with i)**.** `GF(9)` does *not* come with i. `Magma`'s default generator of $\mathbb{F}_9$ is a root of the Conway polynomial $x^2 + 2x + 2$, so writing `F9<i> := GF(9)` gives an i with

$i^2 = i + 1$, not $i^2 = -1$, and every check in Problem 10 would silently fail. Construct the square root of -1 explicitly, and confirm it: `MinimalPolynomial` of the element you build must be $x^2 + 1$, and `i^2` must print as 2, which is -1 in $\mathbb{F}_9$.

8. Automorphisms in characteristic 2 and 3

For $\operatorname{char} k = 2, 3$ the geometric automorphism group of an elliptic curve may be non-abelian, which is the subject of Problem 11. In the following statements, E is an elliptic curve over k, given by a Weierstrass equation with coefficients a_1, a_2, a_3, a_4, a_6.

Definition 8.1 (Changes of variables)**.** A change of variables preserving a Weierstrass equation is a substitution

$$x = u^2 x' + r, \qquad y = u^3 y' + u^2 s x' + t, \qquad u \in \overline{k}^\times,\ r, s, t \in \overline{k}. \tag{8.1}$$

For such u, r, s, t we write

$$\Psi_{u,r,s,t}(x, y) \coloneqq (u^2 x + r,\ u^3 y + u^2 s x + t).$$

Theorem 8.2 (The automorphism group [Sil09, Theorem III.10.1])**.** *The group* $\operatorname{Aut}(E_{\overline{k}})$ *is finite, of order*

$$\#\operatorname{Aut}(E_{\overline{k}}) = \begin{cases} 2 & \text{if } j(E) \neq 0, 1728, \\ 4 & \text{if } j(E) = 1728 \text{ and } \operatorname{char} k \neq 2, 3, \\ 6 & \text{if } j(E) = 0 \text{ and } \operatorname{char} k \neq 2, 3, \\ 12 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 3, \\ 24 & \text{if } j(E) = 0 = 1728 \text{ and } \operatorname{char} k = 2. \end{cases}$$

Corollary 8.3 (The cyclic cases [Sil09, Corollary III.10.2])**.** *If* $\operatorname{char} k \neq 2, 3$*, then* $\operatorname{Aut}(E_{k^{\mathrm{sep}}}) \cong \mu_n(k^{\mathrm{sep}})$ *with* $n = \#\operatorname{Aut}(E_{\overline{k}})$*, via* $[\zeta](x, y) \coloneqq (\zeta^2 x, \zeta^3 y)$*. In particular, the geometric automorphism group is cyclic in that case.*

Proposition 8.4 (The action on the coefficients [Sil09, Proposition III.3.1(b) and Table 3.1])**.** *Under a change of variables* (8.1) *the coefficients transform by*

$$\begin{aligned} u a_1' &= a_1 + 2s, \\ u^2 a_2' &= a_2 - s a_1 + 3r - s^2, \\ u^3 a_3' &= a_3 + r a_1 + 2t, \\ u^4 a_4' &= a_4 - s a_3 + 2r a_2 - (t + rs) a_1 + 3r^2 - 2st, \\ u^6 a_6' &= a_6 + r a_4 + r^2 a_2 + r^3 - t a_3 - t^2 - rt a_1. \end{aligned} \tag{8.2}$$

A substitution (8.1) *carries E to the curve with coefficients a_i', so it is a change of coordinates from E to itself exactly when $a_i' = a_i$ for every i, and then $\Psi_{u,r,s,t}$ is the automorphism of E that it defines. Setting $a_i' = a_i$ in* (8.2) *therefore turns those five lines into five equations in the four unknowns u, r, s, t, whose solutions are exactly the elements of* $\operatorname{Aut}(E_{\overline{k}})$*.*

The whole point of this section is visible in (8.2): observe the 2's and 3's in the coefficients. The characteristic enters through the terms $2s$, $2t$ and $3r$. Away from 2 and 3 they are invertible multiples of s, t and r, which determines those three and leaves u as the only free parameter: that is why the group is cyclic there. In characteristic 3 the term $3r$ vanishes, in characteristic 2 the terms $2s$ and $2t$ do, and the parameters they no longer pin down are what make the two groups of Problem 11 large, and non-cyclic.

Problem 11 (The two non-cyclic cases)**.**
As in Problem 10, each part pairs a hand computation with a check in `Magma`. Here the machine is best at naming the group, which is the trickiest thing to do by hand.

(1) Let $\operatorname{char} k = 3$. Show that every short Weierstrass equation $y^2 = x^3 + Ax + B$ has $j = 0$, and explain why a curve with $j \neq 0$ in characteristic 3 needs an a_2x^2 term. So the case "$j = 0 = 1728$, $\operatorname{char} k = 3$" of Theorem 8.2 is exactly the short-Weierstrass one. The first claim is a one-line sweep over the whole family:

```
{ jInvariant(EllipticCurve([GF(3) | A, B])) : A, B in GF(3) | A ne 0 };  // { 0 }
jInvariant(EllipticCurve([GF(3) | 0, 1, 0, 0, -1]));   // 1
```

The condition `A ne 0` is what keeps the equation non-singular: work out what $\Delta \neq 0$ becomes for $y^2 = x^3 + Ax + B$ once $27 = 0$, and you will find that B drops out. The set that comes back has one element, which is the first claim. The second line is a curve with an a_2x^2 term, and its j-invariant is not in that set.

Solution. For $y^2 = x^3 + Ax + B$ we have $c_4 = -48A$ and $c_6 = -864B$. In characteristic 3, $48 = 3 \cdot 16 \equiv 0$, so $c_4 = 0$, and therefore $j = c_4^3/\Delta = 0$. (Also $1728 = 3^3 \cdot 64 \equiv 0$, so $j = 0 = 1728$: the two exceptional values of Theorem 8.2 collide.) A curve with $j \neq 0$ in characteristic 3 therefore cannot be put in short Weierstrass form. Indeed, completing the cube is what removes a_2, and it requires dividing by 3. In characteristic 3 one can only complete the square, reaching $y^2 = x^3 + a_2x^2 + a_4x + a_6$, and $j \neq 0$ forces $a_2 \neq 0$. So the case at hand is the short-Weierstrass one, which is where $E\colon y^2 = x^3 - x$ of Problem 10 lives: Problem 10(1) found $j = 0$ for it.
As for the sweep: in characteristic 3 we have $27 = 0$, so

$$\Delta = -16(4A^3 + 27B^2) = -64A^3 = 2A^3,$$

and non-singularity is just $A \neq 0$ — B is unconstrained, which is why it is not mentioned in the condition. The one-element set that comes back says that every short Weierstrass curve over $\mathbb{F}_3$ has $j = 0$, while $y^2 = x^3 + x^2 - 1$, which has an a_2 term, has $j = 1 \neq 0$.

(2) Let $\operatorname{char} k = 3$ and $E\colon y^2 = x^3 - x$. Using (8.2), show first that $s = t = 0$, and then that (8.1) is an automorphism of E if and only if

$$u^4 = 1 \qquad \text{and} \qquad r^3 - r = 0,$$

i.e. $u \in \mu_4(\overline{\mathbb{F}}_3) \subseteq \mathbb{F}_9^\times$ and $r \in \mathbb{F}_3$. Conclude $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) = 12$, and that all 12 are already defined over $\mathbb{F}_9$. Where did r become free, and what would have forced $r = 0$ if $\operatorname{char} k \neq 3$? Two counts settle both claims at once:

```
E3 := EllipticCurve([GF(3) | -1, 0]);
#AutomorphismGroup(E3), #AutomorphismGroup(ChangeRing(E3, GF(9)));   // 6 12
```

Predict both numbers from your conditions on u and r before running it. Over $\mathbb{F}_3$ the fourth roots of unity are only ± 1, while r already ranges over all of $\mathbb{F}_3$. Over $\mathbb{F}_9$ all four roots of unity appear. The second number being 12, and being reached over $\mathbb{F}_9$ rather than over some larger field, is exactly the claim that all twelve automorphisms are defined there.

Solution. **$\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) = 12$, all defined over $\mathbb{F}_9$. Over $\mathbb{F}_3$ there are 6.** Here $a_1 = a_2 = a_3 = 0$, $a_4 = A = -1$, $a_6 = B = 0$. From (8.2), $ua_1' = 2s$ and $u^3a_3' = 2t$. Since 2 is invertible in

characteristic 3, requiring $a_1' = a_3' = 0$ gives $s = t = 0$. With $s = t = 0$ the remaining equations read

$$u^2 a_2' = 3r = 0, \qquad u^4 a_4' = A + 3r^2 = A, \qquad u^6 a_6' = B + rA + r^3.$$

The first is automatic—this is where r becomes free, because $3r = 0$ in characteristic 3. If $\operatorname{char} k \neq 2, 3$, the same equation would read $3r = 0$ with 3 invertible, forcing $r = 0$, and only u would survive: that is exactly why $\operatorname{Aut}(E_{\overline{k}}) \cong \mu_n(\overline{k})$ away from 2 and 3. Requiring $a_4' = A$ gives $A/u^4 = A$, so $u^4 = 1$ as $A \neq 0$. Requiring $a_6' = B$ gives $r^3 + Ar + B = u^6 B$, which for $A = -1$, $B = 0$ is $r^3 - r = 0$, i.e. $r \in \mathbb{F}_3$. So there are 4 choices of u (namely $\mu_4(\overline{\mathbb{F}}_3) = \{\pm 1, \pm i\} \subseteq \mathbb{F}_9^\times$, using i from Problem 10(2)) and 3 of r: twelve in all, and since $\mu_4(\overline{\mathbb{F}}_3) \cup \mathbb{F}_3 \subseteq \mathbb{F}_9$, all twelve are defined over $\mathbb{F}_9$.

The prediction for the two counts: over $\mathbb{F}_3$ only $u = \pm 1$ satisfy $u^4 = 1$, while r still ranges over $\mathbb{F}_3$, giving $2 \cdot 3 = 6$; over $\mathbb{F}_9$ all of $\mu_4(\overline{\mathbb{F}}_3)$ is available, giving $4 \cdot 3 = 12$. `Magma` returns `6 12`. The 12 is $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_3})$ by Theorem 8.2, so passing to any field larger than $\mathbb{F}_9$ cannot produce more: that is the precise sense in which all twelve are already defined over $\mathbb{F}_9$.

(3) Write $\varphi_{u,r} \coloneqq \Psi_{u,r,0,0}$, so that $\varphi_{u,r}(x, y) = (u^2 x + r, u^3 y)$. Compute $\varphi_{u',r'} \circ \varphi_{u,r}$ and deduce

$$\operatorname{Aut}(E_{\overline{\mathbb{F}}_3}) \cong \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z},$$

where $\mathbb{Z}/3\mathbb{Z} = \mathbb{F}_3$ is the group of translations r, where $\mathbb{Z}/4\mathbb{Z} = \mu_4(\overline{\mathbb{F}}_3)$ is the group of the u, and where u acts on r by $r \mapsto u^2 r$. Since $u^4 = 1$ forces $u^2 = \pm 1$, that action is not faithful: it factors through the squaring map $\mu_4(\overline{\mathbb{F}}_3) \twoheadrightarrow \{\pm 1\} = \operatorname{Aut}(\mathbb{Z}/3\mathbb{Z})$, so ± 1 act trivially and $\pm i$ act by $r \mapsto -r$. Show this group is non-abelian with a *unique* element of order 2, and deduce that it is neither cyclic, nor dihedral, nor A_4: it is the dicyclic group of order 12. Which automorphism is the unique involution? `AutomorphismGroup` returns a polycyclic group and `IdentifyGroup` gives its label in the small-groups database.

```
G3 := AutomorphismGroup(ChangeRing(E3, GF(9)));
IdentifyGroup(G3), GroupName(G3);        // <12, 1>  C3:C4
IsAbelian(G3), IsCyclic(G3);             // false false
{* Order(g) : g in G3 *};                // {* 1, 2, 3^^2, 4^^6, 6^^2 *}
```

The last line is the multiset of element orders. Say how it rules out the dihedral group and A_4 — how many involutions does each of those have? — and match its single element of order 2 against the automorphism you named.

Solution. **The dicyclic group $\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. Its unique involution is $[-1]$.** Directly,

$$\varphi_{u',r'}\big(\varphi_{u,r}(x, y)\big) = \varphi_{u',r'}(u^2 x + r,\, u^3 y) = \big(u'^2 u^2 x + u'^2 r + r',\, u'^3 u^3 y\big) = \varphi_{u'u,\, u'^2 r + r'}(x, y),$$

so the composition law is $(u', r') \cdot (u, r) = (u'u,\, u'^2 r + r')$. This is the semidirect product $\mathbb{F}_3 \rtimes \mu_4(\overline{\mathbb{F}}_3) \cong \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$, where $\mu_4(\overline{\mathbb{F}}_3)$ acts on $r \in \mathbb{F}_3$ by multiplication by u^2. Since $u^4 = 1$ we have $u^2 = \pm 1$, so the action is through the quotient $\mu_4(\overline{\mathbb{F}}_3) \to \{\pm 1\} = \operatorname{Aut}(\mathbb{Z}/3\mathbb{Z})$. For the involutions: $(u, r)^2 = (u^2,\, u^2 r + r)$, which is the identity if and only if $u^2 = 1$ and $(u^2 + 1)r = 2r = 0$, hence $r = 0$ because 2 is invertible. So the elements of order dividing 2 are $(\pm 1, 0)$, and the unique involution is $\varphi_{-1,0}(x, y) = (x, -y) = [-1]$. The group is not cyclic, being non-abelian ($\mu_4(\overline{\mathbb{F}}_3)$ acts non-trivially on $\mathbb{F}_3$). Its unique involution rules out D_6, which has seven, and A_4, which has three. The remaining group of order 12 is the dicyclic group $\mathrm{Dic}_3 = \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. `Magma`'s label `<12, 1>` is that group, and its multiset of element orders shows a single element of order 2, against seven in D_6 and three in A_4: the counting argument above, in one line.

(4) Let $\operatorname{char} k = 2$ and $E\colon y^2 + y = x^3$. Using (8.2), show that $a_1' = a_1$ automatically, that $a_2' = 0$ forces $r = s^2$, and then that (8.1) is an automorphism if and only if

$$u^3 = 1, \qquad s^4 = s, \qquad t^2 + t = s^6.$$

Deduce that $s \in \mathbb{F}_4$, count the solutions, and conclude $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$, all defined over $\mathbb{F}_4$. The same two counts as in (2):

```
E2 := EllipticCurve([GF(2) | 0, 0, 1, 0, 0]);          // y^2 + y = x^3
#AutomorphismGroup(E2), #AutomorphismGroup(ChangeRing(E2, GF(4)));   // 2 24
```

Predict both first. Over $\mathbb{F}_2$ the three conditions bite hard: $u^3 = 1$ leaves only $u = 1$, and of the values $s \in \mathbb{F}_2$ one of them makes $t^2 + t = s^6$ unsolvable in $\mathbb{F}_2$. The gap between the two numbers here is much wider than in characteristic 3, and that gap is the content of "all defined over $\mathbb{F}_4$".

Solution. **$\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$, all defined over $\mathbb{F}_4$. Over $\mathbb{F}_2$ there are 2.** Here $a_1 = a_2 = 0$, $a_3 = 1$, $a_4 = a_6 = 0$. From (8.2), $ua_1' = a_1 + 2s = 0$ automatically, because $2 = 0$: this is where s becomes free, the characteristic-2 counterpart of the free r in part (2). Next $u^2 a_2' = 3r - s^2 = r - s^2$ using $3 = 1$, so $a_2' = 0$ forces $r = s^2$ — which is why the substitution is usually written with s^2 in place of r. The remaining three equations become, with $r = s^2$,

$$u^3 a_3' = 1, \qquad u^4 a_4' = -s + 3r^2 = s + s^4, \qquad u^6 a_6' = r^3 - t - t^2 = s^6 + t + t^2.$$

So $a_3' = 1$ gives $u^3 = 1$; $a_4' = 0$ gives $s^4 = s$, i.e. $s \in \mathbb{F}_4$; and $a_6' = 0$ gives $t^2 + t = s^6$. Counting: if $s = 0$ then $t^2 + t = 0$, so $t \in \mathbb{F}_2$, two values. If $s \neq 0$ then $s \in \mathbb{F}_4^\times$ has $s^3 = 1$, so $s^6 = 1$ and $t^2 + t = 1$, whose two roots generate $\mathbb{F}_4$ over $\mathbb{F}_2$. Again two values. So there are $4 \cdot 2 = 8$ pairs (s, t) and 3 cube roots of unity u, giving $\#\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) = 24$. All the parameters lie in $\mathbb{F}_4$, so all 24 are defined over $\mathbb{F}_4$.

Over $\mathbb{F}_2$ the count collapses: $u^3 = 1$ forces $u = 1$, and of $s \in \mathbb{F}_2$ only $s = 0$ survives, since $s = 1$ gives $s^6 = 1$ and $t^2 + t = 1$ has no root in $\mathbb{F}_2$. That leaves $s = 0$ with $t \in \mathbb{F}_2$, so 2 automorphisms — just ± 1, as for any curve.

Remark. From 2 to 24 on a quadratic extension: in characteristic 2 essentially none of the automorphism group is visible over the prime field.

(5) Write $\psi_{s,t} \coloneqq \Psi_{1,s^2,s,t}$ for the automorphisms with $u = 1$, so that $\psi_{s,t}(x, y) = (x + s^2, y + sx + t)$. Show that

$$\psi_{s',t'} \circ \psi_{s,t} = \psi_{s+s',\, t+t'+s's^2}, \qquad \psi_{s,t}^2 = \psi_{0,s^3},$$

and deduce that these 8 automorphisms form a copy of the quaternion group Q_8. For the remaining factor, let $\omega \in \mathbb{F}_4$ be a primitive cube root of unity and $\varphi_\omega \coloneqq \Psi_{\omega,0,0,0}$, which has order 3. Check that conjugation by it is

$$\varphi_\omega \circ \psi_{s,t} \circ \varphi_\omega^{-1} = \psi_{\omega s,\, t},$$

so φ_ω normalizes Q_8, fixes its center $\langle \psi_{0,1} \rangle$, and cycles the three pairs $\pm i, \pm j, \pm k$ — the order-3 automorphism of Q_8. Since $\langle \varphi_\omega \rangle \cap Q_8 = 1$, conclude that

$$\operatorname{Aut}(E_{\overline{\mathbb{F}}_2}) \cong Q_8 \rtimes \mathbb{Z}/3\mathbb{Z} \cong \mathrm{SL}_2(\mathbb{F}_3).$$

As before:

```
G2 := AutomorphismGroup(ChangeRing(E2, GF(4)));
IdentifyGroup(G2), GroupName(G2);        // <24, 3>  SL(2,3)
```

```
{* Order(g) : g in G2 *};                    // {* 1, 2, 3^^8, 4^^6, 6^^8 *}
```

Exactly one element of order 2 again. Explain why that alone pins down Q_8 among the groups of order 8, and why it does *not* by itself pin down $\mathrm{SL}_2(\mathbb{F}_3)$ among the groups of order 24 — what extra fact did you use?

SOLUTION. **$\mathrm{SL}_2(\mathbb{F}_3)$. The extra fact is a normal Q_8 on which $\mathbb{Z}/3\mathbb{Z}$ acts faithfully.** With $u = 1$ and $r = s^2$ the substitution (8.1) reads $\psi_{s,t}(x, y) = (x + s^2,\ y + sx + t)$. Composing,

$$\psi_{s',t'}\big(\psi_{s,t}(x,y)\big) = \big(x+s^2+s'^2,\ y+sx+t+s'(x+s^2)+t'\big) = \big(x+(s+s')^2,\ y+(s+s')x+t+t'+s's^2\big),$$

using $s^2 + s'^2 = (s + s')^2$ in characteristic 2. That is $\psi_{s',t'} \circ \psi_{s,t} = \psi_{s+s',\, t+t'+s's^2}$. Squaring, $\psi_{s,t}^2 = \psi_{0,\, 2t+s^3} = \psi_{0,s^3}$. Now $\psi_{0,t}$ is $(x, y) \mapsto (x, y + t)$ with $t \in \mathbb{F}_2$, so $\psi_{0,0} = \mathrm{id}$ and $\psi_{0,1}$ is an involution. For $s \neq 0$ we have $s^3 = 1$, so $\psi_{s,t}^2 = \psi_{0,1} \neq \mathrm{id}$ and $\psi_{s,t}$ has order 4. So the group of order 8 has exactly one element of order 2 and six of order 4, which characterizes Q_8 among groups of order 8. For the conjugation, φ_ω^{-1} is $(x, y) \mapsto (\omega x, y)$ because $\omega^{-2} = \omega$, so

$$\varphi_\omega\big(\psi_{s,t}(\omega x, y)\big) = \varphi_\omega(\omega x + s^2,\ y + s\omega x + t) = \big(x + (\omega s)^2,\ y + (\omega s)x + t\big) = \psi_{\omega s,\, t}(x, y),$$

using $\omega^3 = 1$ and $\omega^2 s^2 = (\omega s)^2$. So conjugation by φ_ω multiplies s by ω: it fixes $\psi_{0,0}$ and $\psi_{0,1}$, the center of Q_8, and cycles the three pairs $\pm i, \pm j, \pm k$, which is the order-3 automorphism of Q_8. As $\langle \varphi_\omega \rangle \cap Q_8 = 1$, we get $\mathrm{Aut}(E_{\overline{\mathbb{F}}_2}) = Q_8 \rtimes \mathbb{Z}/3\mathbb{Z}$, a group of order 24 with a normal Q_8 on which $\mathbb{Z}/3\mathbb{Z}$ acts faithfully. That is $\mathrm{SL}_2(\mathbb{F}_3)$, the binary tetrahedral group. Magma's label `<24, 3>` is $\mathrm{SL}_2(\mathbb{F}_3)$, as computed by hand. As to the question in the part: among groups of order 8 a unique involution characterizes Q_8, but among groups of order 24 it does not — the dicyclic group of order 24 also has one — and what settled $\mathrm{SL}_2(\mathbb{F}_3)$ was the normal Q_8 with $\mathbb{Z}/3\mathbb{Z}$ acting faithfully on it.

Remark 8.5 (Both groups are unit groups)**.** Automorphisms are the units of the endomorphism ring, $\mathrm{Aut}(E_{\overline{k}}) = \mathrm{End}(E_{\overline{k}})^\times$. In characteristic 2 and 3 the curve with $j = 0$ has $\mathrm{End}(E_{\overline{k}})$ a maximal order in the quaternion algebra over $\mathbb{Q}$ that is not split exactly at p and ∞ [Sil09, Theorem V.3.1], [Voi21, Theorem 42.1.9]. For $p = 3$ that order is the one computed in Problem 10(5), and its unit group is the dicyclic group of order 12 of Problem 11(3). For $p = 2$ it is the Hurwitz order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}\frac{1+i+j+ij}{2}$ in $(-1, -1 \mid \mathbb{Q})$ [Voi21, §11.1], whose unit group is the binary tetrahedral group of order 24 of Problem 11(5). This explains the shape of Theorem 8.2: when $\mathrm{End}(E_{\overline{k}})$ is commutative it is $\mathbb{Z}$ or an imaginary quadratic order, whose unit groups are the cyclic groups of orders 2, 4 and 6, giving its first three lines, while a non-commutative $\mathrm{End}(E_{\overline{k}})$ is a quaternion order, and its unit group need not be, and here is not, cyclic.

Both counts can be recovered from the orders alone, with no elliptic curve anywhere in the computation:

```
for d in [-1, -3] do
  Dd := QuaternionAlgebra<Rationals() | -1, d>;
  Od := MaximalOrder(Dd);
  print RamifiedPrimes(Dd), #UnitGroup(Od);
end for;
// [ 2 ] 24      <- Hurwitz order, characteristic 2
// [ 3 ] 12      <- the endomorphism ring of Problem 10, characteristic 3
```

So the 24 and the 12 of Theorem 8.2 are the unit counts of the two maximal quaternion orders.

Looking ahead. Curves whose Frobenius has $a = 0$, like the one of Section 7, have a name and a theory of their own. They come later in the course, with Hasse's bound and the structure of End over $\mathbb{F}_q$. In characteristic 0 the third case of the classification never occurs, and $\mathrm{End}(E_{\overline{k}}) \neq \mathbb{Z}$ becomes complex multiplication. The oddness of $\rho_{E,\ell}$ from Problem 9(4) is the first hint of modularity.

Part 3. Differentials, dual isogenies, and the Tate module

Here are three computations by hand, on the invariant differential, the dual isogeny and the Tate module. Each compares the multiplicative group $\mathbb{G}_{\mathrm{m}}$ with the elliptic curve

$$E\colon y^2 = x^3 - x, \qquad \omega_E := \frac{\mathrm{d}x}{2y}.$$

Throughout, k is a field with $\operatorname{char} k \neq 2$, $\overline{k}$ an algebraic closure and $k^{\mathrm{sep}} \subseteq \overline{k}$ the separable closure. Here $\mathbb{G}_{\mathrm{m}} = \mathbb{P}^1_k \smallsetminus \{0, \infty\}$ has coordinate t and invariant differential $\omega_{\mathbb{G}_{\mathrm{m}}} := \mathrm{d}t/t$.

Reading. [Sil09, §§III.4–III.7], and [Sil09, §III.8] for the Weil pairing of Proposition 11.1.

A formula. We use the following duplication formula on E, derived in Problem 12(5): if $x \in k(E)$ is the x-coordinate and $[2]\colon E \to E$ is multiplication by 2, then

$$[2]^* x := x \circ [2] = \frac{(x^2+1)^2}{4(x^3 - x)}. \tag{8.3}$$

9. Invariant differentials

Problem 12 (Poles and pullbacks)**.**

(1) Rewrite $\omega_{\mathbb{G}_{\mathrm{m}}}$ in the coordinate $s := 1/t$ near ∞, and compute $\mathrm{div}(\omega_{\mathbb{G}_{\mathrm{m}}})$ on $\mathbb{P}^1_k$. Compute $[m]^*\omega_{\mathbb{G}_{\mathrm{m}}}$ for $[m](t) = t^m$, $m \in \mathbb{Z}$.

Solution. $\boldsymbol{\omega_{\mathbb{G}_{\mathrm{m}}} = -\,\mathrm{d}s/s}$**, div$\boldsymbol{(\omega_{\mathbb{G}_{\mathrm{m}}}) = -(0) - (\infty)}$ and $\boldsymbol{[m]^*\omega_{\mathbb{G}_{\mathrm{m}}} = m\,\omega_{\mathbb{G}_{\mathrm{m}}}}$.** $t = 1/s$ gives $\mathrm{d}t = -\,\mathrm{d}s/s^2$, so $\omega_{\mathbb{G}_{\mathrm{m}}} = -\,\mathrm{d}s/s$. Hence $\omega_{\mathbb{G}_{\mathrm{m}}}$ has simple poles at 0 and ∞, with residues 1 and -1, and no other zeros or poles: $\mathrm{div}(\omega_{\mathbb{G}_{\mathrm{m}}}) = -(0) - (\infty)$. Also $[m]^*\omega_{\mathbb{G}_{\mathrm{m}}} = \mathrm{d}(t^m)/t^m = m\,\omega_{\mathbb{G}_{\mathrm{m}}}$.

(2) Show that $\omega_E = \mathrm{d}y/(3x^2 - 1)$, and that y and $3x^2 - 1$ have no common zero on E. Deduce that ω_E has no zeros or poles on $E \smallsetminus \{O\}$.

Solution. Differentiating $y^2 = x^3 - x$ gives $2y\,\mathrm{d}y = (3x^2 - 1)\,\mathrm{d}x$. If $y = 0$, then $x \in \{0, \pm 1\}$ and $3x^2 - 1 \in \{-1, 2\}$, which is non-zero. On the open set $y \neq 0$ the differential $\mathrm{d}x$ generates $\Omega_{E/k}$, and $\omega_E = \mathrm{d}x/(2y)$ is a unit times it. On $3x^2 - 1 \neq 0$, likewise with $\mathrm{d}y$ and $\mathrm{d}y/(3x^2 - 1)$. These open sets cover $E \smallsetminus \{O\}$.

(3) Using $\mathrm{ord}_O\, x = -2$ and $\mathrm{ord}_O\, y = -3$, show that $\mathrm{ord}_O\, \omega_E = 0$.

Solution. With a uniformizer w at O, write $x = w^{-2}u$ with $u(O) \neq 0$. Then

$$\mathrm{d}x = (-2w^{-3}u + w^{-2}u')\,\mathrm{d}w$$

has order -3, because $2 \neq 0$ in k. So $\mathrm{ord}_O\, \omega_E = -3 - (-3) = 0$.

(4) Compute $[-1]^*\omega_E$ from $[-1](x, y) = (x, -y)$, and compare with $[-1]^*\omega_{\mathbb{G}_{\mathrm{m}}}$.

Solution. $\boldsymbol{[-1]^*\omega_E = -\omega_E}$**, and $\boldsymbol{[-1]^*\omega_{\mathbb{G}_{\mathrm{m}}} = -\omega_{\mathbb{G}_{\mathrm{m}}}}$.** $[-1]^*\omega_E = \mathrm{d}x/(-2y) = -\omega_E$. On $\mathbb{G}_{\mathrm{m}}$, $[-1](t) = 1/t$ and $[-1]^*\omega_{\mathbb{G}_{\mathrm{m}}} = -\omega_{\mathbb{G}_{\mathrm{m}}}$ as well.

(5) We derive (8.3) by hand. Here $x, y \in k(E)$ are functions. For a point $P \in E(\overline{k})$ write $x_P := x(P)$ and $y_P := y(P)$ for its coordinates.

(a) Let $P \in E(\overline{k})$ with $y_P \neq 0$. Show that the tangent line to E at P has slope $\lambda_P = (3x_P^2 - 1)/(2y_P)$, and use that its third point of intersection with E is $-[2]P$ to show that $x([2]P) = \lambda_P^2 - 2x_P$.

SOLUTION. In the affine plane with coordinates x, y, differentiating $y^2 = x^3 - x$ at P gives $2y_P\,\mathrm{d}y = (3x_P^2 - 1)\,\mathrm{d}x$, so the tangent line at P is

$$y = \lambda_P x + \nu_P, \qquad \lambda_P = \frac{3x_P^2 - 1}{2y_P}, \qquad \nu_P = y_P - \lambda_P x_P.$$

Substituting into $y^2 = x^3 - x$ gives

$$x^3 - \lambda_P^2 x^2 - (1 + 2\lambda_P\nu_P)x - \nu_P^2 = 0,$$

whose roots are x_P, twice because the line is tangent, and $x(-[2]P)$. Comparing coefficients of x^2 gives $x(-[2]P) = \lambda_P^2 - 2x_P$, and $x(-[2]P) = x([2]P)$ since $[-1](x, y) = (x, -y)$.

(b) Let $\lambda := (3x^2 - 1)/(2y) \in k(E)$, so that $\lambda(P) = \lambda_P$. Explain why (5)(a) gives the identity $[2]^*x = \lambda^2 - 2x$ of functions in $k(E)$, and simplify it using $y^2 = x^3 - x$.

SOLUTION. $\boldsymbol{[2]^*x = (x^2+1)^2/(4(x^3-x))}$. By definition of the pullback, $([2]^*x)(P) = x([2]P)$, and $(\lambda^2 - 2x)(P) = \lambda_P^2 - 2x_P$. So by (5)(a) the function $[2]^*x - (\lambda^2 - 2x) \in k(E)$ vanishes at every $P \in E(\overline{k})$ with $y_P \neq 0$, that is, at every point of $E(\overline{k})$ outside the finite set $E[2](\overline{k})$. A non-zero rational function has only finitely many zeros, so $[2]^*x = \lambda^2 - 2x$ in $k(E)$. Finally, in $k(E)$ we have $4y^2 = 4(x^3 - x)$, so

$$[2]^*x = \lambda^2 - 2x = \frac{(3x^2-1)^2 - 8x(x^3 - x)}{4(x^3 - x)} = \frac{x^4 + 2x^2 + 1}{4(x^3 - x)} = \frac{(x^2+1)^2}{4(x^3 - x)}.$$

Remark. This is the precise meaning of the shorthand "$x([2]P) = \lambda^2 - 2x$ for $P = (x, y)$".

(6) Solve this part in `Magma`, working in the function field $\mathbb{Q}(E) = \mathbb{Q}(x)[y]/(y^2 - x^3 + x)$. Compute $[2]^*x$ and $[2]^*y$ from the function λ of (5), check (8.3), and show that

$$[2]^*x - e = \frac{g_e(x)^2}{4(x^3 - x)}, \qquad g_0 = x^2 + 1, \quad g_1 = x^2 - 2x - 1, \quad g_{-1} = x^2 + 2x - 1,$$

for $e \in \{0, 1, -1\}$. Show that $[2]^*y = \pm(x^2+1)(x^4 - 6x^2 + 1)/(8y^3)$ and $[2]^*\omega_E = \pm 2\omega_E$. Which sign is the correct one?

SOLUTION. **Both signs are +.** The identities have coefficients in $\mathbb{Z}[1/2]$, so it suffices to check them over $\mathbb{Q}$. By (5), $[2]^*x = \lambda^2 - 2x$ with $\lambda = (3x^2 - 1)/(2y)$, and the same argument applied to the y-coordinate of the third point of intersection gives $[2]^*y = -\big(y + \lambda([2]^*x - x)\big)$. Since $\omega_E = \mathrm{d}x/(2y)$, we have

$$\frac{[2]^*\omega_E}{\omega_E} = \frac{\mathrm{d}([2]^*x)/\,\mathrm{d}x}{2\,[2]^*y} \cdot 2y.$$

```
Kx<x> := FunctionField(Rationals());
R<t>  := PolynomialRing(Kx);
F<y>  := ext<Kx | t^2 - (x^3 - x)>;                    // Q(E)
lam := (3*x^2 - 1)/(2*y);                              // tangent slope
x2 := lam^2 - 2*x;  y2 := -(y + lam*(x2 - x));         // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x));                       // true
[ Factorization(Numerator(Kx!(x2 - e))) : e in [0, 1, -1] ];
// [ [ <x^2 + 1, 2> ], [ <x^2 - 2*x - 1, 2> ], [ <x^2 + 2*x - 1, 2> ] ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3);             // true
```

```
(F!Derivative(Kx!x2))/(2*y2) * 2*y;                    // 2
```

The denominator of $[2]^*x - e$ is that of $[2]^*x$, namely $4(x^3 - x)$, and the factorizations show that its numerator is $g_e(x)^2$. So $[2]^*y = (x^2+1)(x^4 - 6x^2 + 1)/(8y^3)$ and $[2]^*\omega_E = 2\omega_E$, as [Sil09, Corollary III.5.3] predicts.

Remark. Alternatively, `Magma`'s elliptic curve functionality provides [2] as an isogeny, and pulls back both functions and differentials along it, so no group law is typed in.

```
E := EllipticCurve([-1,0]);                            // y^2 = x^3 - x
m2 := MultiplicationByMMap(E, 2);                      // [2] : E -> E
KE<x,y> := FunctionField(E);                           // Q(E)
x2 := Pullback(m2, x);  y2 := Pullback(m2, y);         // [2]^*x, [2]^*y
x2 eq (x^2 + 1)^2/(4*(x^3 - x));                       // true
g := [x^2 + 1, x^2 - 2*x - 1, x^2 + 2*x - 1];          // g_0, g_1, g_{-1}
[ x2 - [0, 1, -1][i] eq g[i]^2/(4*(x^3 - x)) : i in [1..3] ];
// [ true, true, true ]
y2 eq (x^2 + 1)*(x^4 - 6*x^2 + 1)/(8*y^3);             // true
omega := Differential(x)/(2*y);                        // omega_E
Pullback(m2, omega)/omega;                             // 2
```

10. Dual isogenies

Consider the elliptic curves

$$E_1\colon y_1^2 = x_1^3 - x_1, \qquad E_2\colon y_2^2 = x_2^3 + 4x_2, \qquad E_3\colon y_3^2 = x_3^3 - 16x_3,$$

so that $E_1 = E$, and write $\omega_i := \mathrm{d}x_i/(2y_i)$ for the invariant differential of E_i. Consider the maps

$$\begin{aligned}
\varphi\colon E_1 \to E_2, &\quad (x_1, y_1) \mapsto \left(\frac{x_1^2 - 1}{x_1}, \frac{y_1(x_1^2+1)}{x_1^2}\right),\\
\psi\colon E_2 \to E_3, &\quad (x_2, y_2) \mapsto \left(\frac{y_2^2}{x_2^2}, \frac{y_2(x_2^2 - 4)}{x_2^2}\right),\\
\iota\colon E_3 \to E_1, &\quad (x_3, y_3) \mapsto \left(\frac{x_3}{4}, \frac{y_3}{8}\right).
\end{aligned}$$

The maps φ and ψ are special cases of [Sil09, Example III.4.5], which treats $y^2 = x^3 + ax^2 + bx$ in general, so they are isogenies. Also ι is an isomorphism. That example computes the kernels and the dual, but not the constants c_α of Definition 10.1: that is part (2) of Problem 13.

Definition 10.1 (The constant c_α)**.** For an isogeny $\alpha\colon E_i \to E_j$, define c_α by $\alpha^*\omega_j = c_\alpha\,\omega_i$.

Problem 13 (A 2-isogeny and its dual)**.**
(1) Check that φ lands on E_2. Show that $\ker\varphi = \{O_1, (0,0)\}$, that $\varphi(\pm 1, 0) = (0,0)$, and that $\deg\varphi = 2$.

Solution. Since $y_1^2 = x_1(x_1^2 - 1)$, the functions $\varphi^*x_2 = (x_1^2 - 1)/x_1$ and $\varphi^*y_2 = y_1(x_1^2+1)/x_1^2$ satisfy

$$(\varphi^*y_2)^2 = \frac{x_1(x_1^2-1)(x_1^2+1)^2}{x_1^4}, \qquad (\varphi^*x_2)^3 + 4\varphi^*x_2 = \frac{x_1^2-1}{x_1}\cdot\frac{(x_1^2+1)^2}{x_1^2},$$

which agree. The function $\varphi^* x_2 = x_1 - 1/x_1$ has poles only at $(0,0)$ and at O_1, so $\ker\varphi = \{O_1, (0,0)\}$. And at $(\pm 1, 0)$ both coordinates of φ vanish. Next,

$$y_1 = \frac{x_1^2\,\varphi^* y_2}{x_1^2+1}, \qquad x_1^2 - (\varphi^* x_2)x_1 - 1 = 0,$$

so $k(E_1) = \varphi^* k(E_2)(x_1)$ has degree at most 2 over $\varphi^* k(E_2)$. Since $\#\ker\varphi = \deg_s\varphi \leqslant \deg\varphi$, we get $\deg\varphi = 2$.

(2) Compute c_φ.

Solution. **$c_\varphi = 1$.**

$$\varphi^*\,\mathrm{d}x_2 = \left(1 + \frac{1}{x_1^2}\right)\mathrm{d}x_1 = \frac{(x_1^2+1)\,\mathrm{d}x_1}{x_1^2}, \qquad 2\varphi^* y_2 = \frac{2y_1(x_1^2+1)}{x_1^2},$$

so $\varphi^*\omega_2 = \omega_1$ and $c_\varphi = 1$.

Remark. This is why the second coordinate of φ has the sign it does: the opposite sign would give $c_\varphi = -1$.

(3) Check that ψ lands on E_3 and that ι lands on E_1, and compute c_ψ and c_ι.

Solution. **$c_\psi = 1$ and $c_\iota = 2$.** φ and ψ are the same construction for $y^2 = x^3 + bx$, with $b = -1$ and $b = 4$ respectively, and target $y^2 = x^3 - 4bx$. The computation of (1) applies verbatim, and that of (2) gives $c_\psi = 1$. Also $(y_3/8)^2 = (x_3^3 - 16x_3)/64 = (x_3/4)^3 - x_3/4$, so ι lands on E_1, and

$$\iota^*\omega_1 = \frac{\mathrm{d}x_3/4}{2y_3/8} = 2\,\omega_3,$$

so $c_\iota = 2$.

(4) Show that $(\iota\circ\psi\circ\varphi)^* x_1 = [2]^* x_1$, so that $\iota\circ\psi\circ\varphi = \pm[2]$. Use (2) and (3) to decide the sign, and conclude that $\hat\varphi = \iota\circ\psi$, that $c_\varphi c_{\hat\varphi} = \deg\varphi$, and that $\ker\hat\varphi = \varphi(E_1[2](\bar k))$.

Solution. **The sign is $+$: $\iota\circ\psi\circ\varphi = [2]$, and $\hat\varphi = \iota\circ\psi$.** We have $\psi^* x_3 = y_2^2/x_2^2 = x_2 + 4/x_2$ and $\iota^* x_1 = x_3/4$, so

$$(\iota\circ\psi\circ\varphi)^* x_1 = \frac{1}{4}\left(\frac{x_1^2-1}{x_1} + \frac{4x_1}{x_1^2-1}\right) = \frac{(x_1^2+1)^2}{4(x_1^3 - x_1)},$$

which is $[2]^* x_1$ by (8.3). Two isogenies with the same pullback of x_1 agree up to sign. Now $c_{\iota\circ\psi\circ\varphi} = 2\cdot 1\cdot 1 = 2 = c_{[2]}$ by Problem 12(6), while $c_{[-2]} = -2 \neq 2$. So $\iota\circ\psi\circ\varphi = [2]$, and by uniqueness of the dual, $\hat\varphi = \iota\circ\psi$, with $c_{\hat\varphi} = 2$ and $c_\varphi c_{\hat\varphi} = 2 = \deg\varphi$. Finally $\ker\hat\varphi = \ker\psi = \{O_2, (0,0)\} = \varphi(E_1[2](\bar k))$ by (1).

(5) Is there an isogeny $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ of degree 2? If so, does it have a dual?

Solution. **Yes, $[2]$ and $[-2]$. Their only candidate duals, $[\pm 1]$, have degree 1.** The endomorphisms of $\mathbb{G}_\mathrm{m}$ are the maps $[m](t) = t^m$, $m \in \mathbb{Z}$, and $\deg[m] = |m|$, so the isogenies $\mathbb{G}_\mathrm{m} \to \mathbb{G}_\mathrm{m}$ of degree 2 are $[2]$ and $[-2]$, both with kernel $\mu_2(\bar k) = \{\pm 1\}$. In the sense of the defining identity $\hat\alpha\circ\alpha = [\deg\alpha]$, each has a unique dual: $[n]\circ[\pm 2] = [2]$ means $\pm 2n = 2$, so $n = \pm 1$. And in general the only $[n]$ with $[n]\circ[m] = [\deg[m]]$ is $[\operatorname{sign} m]$, which also satisfies $[m]\circ[\operatorname{sign} m] = [\deg[m]]$ and is compatible with composition. But for $|m| \geqslant 2$ it fails the other properties of the dual isogeny [Sil09, Theorem III.6.2]: its degree is 1, not $\deg[m]$; the dual of the dual is $[\operatorname{sign} m]$, not $[m]$; and $[m] \mapsto [\operatorname{sign} m]$ is not additive, since $\operatorname{sign}(1+1) \neq \operatorname{sign} 1 + \operatorname{sign} 1$.

Remark. All of this fails because $\deg[m] = |m|$, not m^2.

11. The Tate module

We first describe the Tate module of $\mathbb{G}_{\mathrm{m}}$, the familiar case, and then repeat each step for E. What links the two is the Weil pairing, which you may use without proof.

Proposition 11.1 (The Weil pairing [Sil09, Proposition III.8.1])**.** *Let $m \geqslant 1$ with $\operatorname{char} k \nmid m$. There is a pairing*

$$e_m \colon E[m](k^{\mathrm{sep}}) \times E[m](k^{\mathrm{sep}}) \to \mu_m(k^{\mathrm{sep}})$$

that is bilinear, alternating ($e_m(P,P) = 1$), non-degenerate, and Galois equivariant: $\sigma\big(e_m(P,Q)\big) = e_m(\sigma(P),\sigma(Q))$ for every $\sigma \in \mathrm{Gal}_k$. Here $E[m](k^{\mathrm{sep}}) = E[m](\overline{k})$, since $\operatorname{char} k \nmid m$. It follows that if (P,Q) is a basis of $E[m](k^{\mathrm{sep}})$ over $\mathbb{Z}/m\mathbb{Z}$, then $e_m(P,Q)$ is a primitive m-th root of unity, and if σ acts on this basis by the matrix $A \in \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$, then $\sigma\big(e_m(P,Q)\big) = e_m(P,Q)^{\det A}$.

Problem 14 (The 2-power torsion of $y^2 = x^3 - x$)**.**
Regard E as a curve over $\mathbb{Q}$, fix $\overline{\mathbb{Q}} \subset \mathbb{C}$, and recall that $E[m](\overline{\mathbb{Q}})$ is the kernel of $[m]$ on $E(\overline{\mathbb{Q}})$. Let σ be complex conjugation, acting on $\overline{\mathbb{Q}}$ and on $E(\overline{\mathbb{Q}})$ coordinatewise.

(1) Recall that $T_2(\mathbb{G}_{\mathrm{m}}) \coloneqq \varprojlim_n \mu_{2^n}(\overline{\mathbb{Q}})$, with transition maps $\zeta \mapsto \zeta^2$. Find a $\mathbb{Z}_2$-basis of $T_2(\mathbb{G}_{\mathrm{m}})$, and write down its first three terms. How does σ act on $\mu_4(\overline{\mathbb{Q}})$, and on $T_2(\mathbb{G}_{\mathrm{m}})$?

Solution. **$(-1, i, \zeta_8, \ldots)$. Also σ is $i \mapsto i^3$ on $\mu_4(\overline{\mathbb{Q}})$ and -1 on $T_2(\mathbb{G}_{\mathrm{m}})$.** The map $\zeta \mapsto \zeta^2$ sends the primitive 2^{n+1}-th roots of unity onto the primitive 2^n-th ones, so there are compatible sequences $(\zeta_{2^n})_n$ of primitive roots with $\zeta_{2^{n+1}}^2 = \zeta_{2^n}$. Since ζ_{2^n} generates the cyclic group $\mu_{2^n}(\overline{\mathbb{Q}})$, the maps $\mathbb{Z}/2^n\mathbb{Z} \to \mu_{2^n}(\overline{\mathbb{Q}})$, $a \mapsto \zeta_{2^n}^a$, are isomorphisms compatible with reduction on the left and squaring on the right, and in the limit $\mathbb{Z}_2 \xrightarrow{\sim} T_2(\mathbb{G}_{\mathrm{m}})$. So $(\zeta_{2^n})_n$ is a $\mathbb{Z}_2$-basis, for instance $(-1, i, \zeta_8, \ldots)$ with $\zeta_8 = (1+i)/\sqrt{2} = e^{2\pi i/8}$. Complex conjugation inverts every root of unity: on $\mu_4(\overline{\mathbb{Q}})$ it is $i \mapsto -i = i^3$, and on $T_2(\mathbb{G}_{\mathrm{m}})$ it is multiplication by $-1 \in \mathbb{Z}_2^\times$.

(2) List $E[2](\overline{\mathbb{Q}})$. Using Problem 12(6), find the x-coordinates of all $P \in E(\overline{\mathbb{Q}})$ with $[2]P \in E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$, and conclude that $\#E[4](\overline{\mathbb{Q}}) = 16$.

Solution. **$E[2](\overline{\mathbb{Q}}) = \{O, (0,0), (\pm 1, 0)\}$. The x_P are $\pm i$, $1 \pm \sqrt{2}$, $-1 \pm \sqrt{2}$.** A point $P \in E(\overline{\mathbb{Q}})$ with $[2]P \in E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$ is not in $E[2](\overline{\mathbb{Q}})$, so $4(x_P^3 - x_P) = 4y_P^2 \neq 0$. For such P, evaluating Problem 12(6) at P shows that $x([2]P) = ([2]^* x)(P) = e$, that is $[2]P = (e, 0)$, exactly when $g_e(x_P) = 0$: so $x_P = \pm i$ for $e = 0$, $x_P = 1 \pm \sqrt{2}$ for $e = 1$ and $x_P = -1 \pm \sqrt{2}$ for $e = -1$. None of these is a root of $x^3 - x$, so each gives two points $(x_P, \pm y_P) \in E(\overline{\mathbb{Q}})$ outside $E[2](\overline{\mathbb{Q}})$, twelve in all. With $E[2](\overline{\mathbb{Q}})$ this makes 16 points of $E(\overline{\mathbb{Q}})$. They are all of $E[4](\overline{\mathbb{Q}})$, since a point of $E[4](\overline{\mathbb{Q}})$ not in $E[2](\overline{\mathbb{Q}})$ doubles into $E[2](\overline{\mathbb{Q}}) \smallsetminus \{O\}$.

(3) Show that $P_2 \coloneqq (i, 1-i)$ and $Q_2 \coloneqq (1+\sqrt{2}, 2+\sqrt{2})$ are points of $E(\overline{\mathbb{Q}})$, with $[2]P_2 = (0,0)$ and $[2]Q_2 = (1,0)$. Show that (P_2, Q_2) is a basis of $E[4](\overline{\mathbb{Q}})$ over $\mathbb{Z}/4\mathbb{Z}$, and that $E[4](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q}(\zeta_8))$. Compare with $\mu_8(\overline{\mathbb{Q}})$ and Proposition 11.1.

Solution. For P_2, $x_{P_2}^3 - x_{P_2} = i^3 - i = -2i = (1-i)^2 = y_{P_2}^2$. For Q_2, $x_{Q_2}^2 = (1+\sqrt{2})^2 = 3 + 2\sqrt{2}$, so

$$x_{Q_2}^3 - x_{Q_2} = x_{Q_2}(x_{Q_2}^2 - 1) = 2x_{Q_2}^2 = 6 + 4\sqrt{2} = (2+\sqrt{2})^2 = y_{Q_2}^2.$$

By (2), $x([2]P_2) = 0$ and $x([2]Q_2) = 1$, and a point of $E(\overline{\mathbb{Q}})$ with x-coordinate 0 or 1 has y-coordinate 0. If $[a]P_2 + [b]Q_2 = O$, doubling gives $[a](0,0) + [b](1,0) = O$, so $a = 2a'$ and $b = 2b'$ are even. Then $[a'](0,0) + [b'](1,0) = O$ makes a', b' even too. So $(\mathbb{Z}/4\mathbb{Z})^2 \to E[4](\overline{\mathbb{Q}})$, $(a,b) \mapsto [a]P_2 + [b]Q_2$, is injective, hence bijective. Every point of $E[4](\overline{\mathbb{Q}})$ thus lies in $E(\mathbb{Q}(i, \sqrt{2}))$, and $\mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$, since $\zeta_8 = (1+i)/\sqrt{2}$. This is exactly $\mathbb{Q}(\mu_8(\overline{\mathbb{Q}}))$. Writing $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$

for the field generated by the coordinates of the points of $E[4](\overline{\mathbb{Q}})$, Proposition 11.1 forces $\mathbb{Q}(E[4](\overline{\mathbb{Q}})) \supseteq \mathbb{Q}(\mu_4(\overline{\mathbb{Q}}))$: the values of e_4 lie in $\mathbb{Q}(E[4](\overline{\mathbb{Q}}))$ and include a primitive fourth root of unity.

(4) Show that σ restricts to a group automorphism of $E[4](\overline{\mathbb{Q}})$. Compute $P_2 + (1,0)$, and write $\sigma(P_2)$ and $\sigma(Q_2)$ in the basis (P_2, Q_2). Compare the determinant of the resulting matrix in $\mathrm{GL}_2(\mathbb{Z}/4\mathbb{Z})$ with the action of σ on $\mu_4(\overline{\mathbb{Q}})$ found in (1), and explain the agreement using Proposition 11.1.

SOLUTION. **$\sigma(P_2) = [3]P_2 + [2]Q_2$, $\sigma(Q_2) = Q_2$, determinant $3 \equiv -1$.** The group law on $E(\overline{\mathbb{Q}})$ is given by rational functions with coefficients in $\mathbb{Q}$, so $\sigma(P+Q) = \sigma(P) + \sigma(Q)$. In particular $[4]\sigma(P) = \sigma([4]P)$, and σ maps $E[4](\overline{\mathbb{Q}})$ to itself, with inverse σ. Let $R := P_2 + (1,0)$. The line through $(i, 1-i)$ and $(1,0)$ has slope $\lambda = -1$, so

$$x_R = \lambda^2 - i - 1 = -i, \qquad y_R = -\big((1-i) + \lambda(x_R - i)\big) = -1 - i,$$

that is, $R = (-i, -1-i)$. Hence $\sigma(P_2) = (-i, 1+i) = -R = -P_2 - [2]Q_2$. Since P_2 and Q_2 have order 4, this is $[3]P_2 + [2]Q_2$. And $\sigma(Q_2) = Q_2$, since its coordinates are real. In the basis (P_2, Q_2), $\sigma|_{E[4](\overline{\mathbb{Q}})}$ has matrix

$$\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}, \qquad \det = 3 \equiv -1 \pmod 4,$$

and by (1), σ acts on $\mu_4(\overline{\mathbb{Q}})$ by the same exponent 3. This is no accident: by Proposition 11.1, $e_4(P_2, Q_2)$ is a primitive fourth root of unity and $\sigma\big(e_4(P_2,Q_2)\big) = e_4(P_2,Q_2)^{\det} = e_4(P_2,Q_2)^3$.

(5) Let $(P_n)_n$ and $(Q_n)_n$ be compatible sequences, $[2]P_{n+1} = P_n$ and $[2]Q_{n+1} = Q_n$, starting with $P_1 = (0,0)$, $Q_1 = (1,0)$ and P_2, Q_2 as above. Explain why they form a $\mathbb{Z}_2$-basis of $T_2(E)$, and compare with (1).

SOLUTION. The map $(\mathbb{Z}/2^n\mathbb{Z})^2 \to E[2^n](\overline{\mathbb{Q}})$, $(a,b) \mapsto [a]P_n + [b]Q_n$, is an isomorphism for every n. It is injective by the argument of (3), by induction on n: if $[a]P_n + [b]Q_n = O$, doubling gives $[a]P_{n-1} + [b]Q_{n-1} = O$, so $a = 2^{n-1}a'$ and $b = 2^{n-1}b'$ by induction. Then $[a']P_1 + [b']Q_1 = O$ makes a', b' even. It is surjective because $\#E[2^n](\overline{\mathbb{Q}}) = 4^n$, as $E[2^n](\overline{\mathbb{Q}}) \cong (\mathbb{Z}/2^n\mathbb{Z})^2$ [Sil09, Corollary III.6.4]. For $n = 2$ this is the count of (2). These isomorphisms are compatible with reduction on the left and $[2]$ on the right. Passing to the limit, $\mathbb{Z}_2^2 \xrightarrow{\sim} T_2(E)$. This is (1) in rank two: $\mu_{2^n}(\overline{\mathbb{Q}})$ is free of rank one over $\mathbb{Z}/2^n\mathbb{Z}$, while $E[2^n](\overline{\mathbb{Q}})$ is free of rank two.

12. MAGMA EXAMPLES

Example 12.1 (The isogenies of Problem 13)**.** We check in the function field of E_1 that φ and ψ land where they should, that $\iota \circ \psi \circ \varphi = [2]$, and compare with Magma's own 2-isogeny.

```
E := EllipticCurve([-1,0]);                    // y^2 = x^3 - x
Kx<x> := FunctionField(Rationals());
R<t> := PolynomialRing(Kx);
F<y> := ext<Kx | t^2 - (x^3 - x)>;                          // x = x_1, y = y_1
x2 := F!((x^2 - 1)/x);  y2 := y*(x^2 + 1)/x^2;              // phi^*x_2, phi^*y_2
y2^2 eq x2^3 + 4*x2;                                        // true
x3 := y2^2/x2^2;  y3 := y2*(x2^2 - 4)/x2^2;                 // (psi o phi)^*x_3, ^*y_3
y3^2 eq x3^3 - 16*x3;                                       // true
lam := (3*x^2 - 1)/(2*y);  xd := lam^2 - 2*x;  yd := -(y + lam*(xd - x));
x3/4 eq xd and y3/8 eq yd;                                  // true: [2]
```

```
phi := TwoIsogeny(E![0,0]);
aInvariants(Codomain(phi));                              // [ 0, 0, 0, 4, 0 ]
phi(E![1,0]);                                            // (0 : 0 : 1)
DualIsogeny(phi);
// ... taking (x : y : 1) to ((1/4*x^2 + 1) / x : (1/8*x^2*y - 1/2*y) / x^2 : 1)
```

Magma's dual is $\iota \circ \psi$ written out in the coordinates x_2, y_2 of E_2: $\big((x_2^2+4)/(4x_2),\ y_2(x_2^2-4)/(8x_2^2)\big)$.

Example 12.2 (Division polynomials and $E[4](\overline{\mathbb{Q}})$)**.** For $m \geqslant 2$, `DivisionPolynomial(E,m)` vanishes exactly at the x-coordinates of $E[m](\overline{\mathbb{Q}}) \smallsetminus \{O\}$. Its degree is $(m^2-1)/2$ for odd m and $(m^2+2)/2$ for even m, and its factors for $m=4$ are x, $x \pm 1$ and the g_e of Problem 12(6).

```
[ Degree(DivisionPolynomial(E,m)) : m in [2..8] ];
// [ 3, 4, 9, 12, 19, 24, 33 ]
Factorization(DivisionPolynomial(E,4));
// x - 1, x, x + 1, x^2 - 2*x - 1, x^2 + 1, x^2 + 2*x - 1
Q8<z> := CyclotomicField(8);  i := z^2;  s := z + z^7;  // s^2 = 2
E8 := BaseChange(E, Q8);
P2 := E8![i, 1-i];  Q2 := E8![1+s, 2+s];
2*P2, 2*Q2;                          // (0 : 0 : 1) (1 : 0 : 1)
#DivisionPoints(E8!0, 4);            // 16
E8![-i, 1+i] eq 3*P2 + 2*Q2;         // true: complex conjugation
```

Example 12.3 (Characteristic p)**.** Reducing modulo p, the p-division polynomial collapses, because $\#E[p](\overline{\mathbb{F}}_p) \in \{1, p\}$.

```
for p in [3,5,7,13] do
  Ep  := ChangeRing(E, GF(p));
  psi := DivisionPolynomial(Ep, p);
  print p, IsSupersingular(Ep), Degree(psi), Factorization(psi);
end for;
//  3 true   0  []
//  5 false 10  [ <x^2 + 2, 5> ]
//  7 true   0  []
// 13 false 78  [ <x^6 + 11*x^4 + 5*x^2 + 11, 13> ]
```

For $p = 3, 7$ the curve is supersingular and $E[p](\overline{\mathbb{F}}_p) = 0$. In general, $y^2 = x^3 - x$ is supersingular exactly when $p \equiv 3 \pmod 4$. For $p = 5, 13$ the division polynomial is the p-th power of a polynomial of degree $(p-1)/2$, whose roots are the x-coordinates of the $p-1$ non-zero points of $E[p](\overline{\mathbb{F}}_p) \cong \mathbb{Z}/p\mathbb{Z}$. The exponent p is the inseparable degree of $[p]$. The same computation for $\mathbb{G}_{\mathrm{m}}$ is $t^p - 1 = (t-1)^p$.

Looking ahead. The Weil pairing of Proposition 11.1 also gives $\det \varphi_\ell = \deg \varphi$ for the map φ_ℓ induced by an isogeny on Tate modules ([Sil09, Proposition III.8.6]). Formal groups make c_φ an honest derivative. And Frobenius factors $[p]$ in characteristic p.

Part 4. The zeta function of an elliptic curve

Section 13 defines the zeta function of a curve over $\mathbb{F}_q$ as a generating function for its numbers of rational points, and shows that it is also an Euler product over the closed points of the curve. Section 14 computes the zeta function of an elliptic curve E from the action of Frobenius on the Tate module, proves the Riemann hypothesis for it, and deduces Hasse's bound $|\#E(\mathbb{F}_q) - q - 1| \leqslant 2\sqrt{q}$.

Section 15 shows that the zeta function, equivalently the single number $\#E(\mathbb{F}_q)$, is a complete isogeny invariant.

Reading. [Sil09, §V.2], with [Sil09, §§III.7–III.8] for the Tate module, and [Poo06, §§2.2.1 and 3.4–3.5] for closed points and the Euler product. Three of the problems are exercises from [Sil09, Chapter V]: Problem 16 is Exercise 5.3, Problem 19 is Exercise 5.13 and Problem 21 is Exercise 5.4.

`Magma`. If you have no local installation, every block below runs in the free online calculator.

Standing notation.

- A power $q = p^r$ of a prime p, and the Galois group $\mathrm{Gal}_{\mathbb{F}_q} = \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$.
- The automorphism $\sigma \in \mathrm{Gal}_{\mathbb{F}_q}$, $t \mapsto t^q$, of $\overline{\mathbb{F}}_q$, which generates $\mathrm{Gal}_{\mathbb{F}_q}$ topologically. Its fixed field in $\overline{\mathbb{F}}_q$ is $\mathbb{F}_q$, and that of σ^n is $\mathbb{F}_{q^n}$.
- In Sections 14 and 15, an elliptic curve E over $\mathbb{F}_q$, given by a Weierstrass equation with coefficients in $\mathbb{F}_q$.
- The ring $\mathrm{End}(E)$ of the endomorphisms of E *defined over* $\mathbb{F}_q$.
- The q-power Frobenius endomorphism
$$\mathsf{F}\colon E \to E, \qquad (x, y) \mapsto (x^q, y^q).$$
- The integers
$$a := q + 1 - \#E(\mathbb{F}_q), \qquad a_n := q^n + 1 - \#E(\mathbb{F}_{q^n}) \quad (n \geqslant 1), \qquad a_0 := 2,$$
so that $a_1 = a$.
- For a prime $\ell \neq p$, the Tate module $T_\ell(E) = \varprojlim_n E[\ell^n](\overline{\mathbb{F}}_q)$, a free $\mathbb{Z}_\ell$-module of rank 2, and the map $\psi_\ell \in \mathrm{End}\big(T_\ell(E)\big)$ induced by $\psi \in \mathrm{End}(E)$.

13. Zeta functions and closed points

Let C be a smooth projective curve over $\mathbb{F}_q$. The zeta function of C (Definition 13.2) packages the numbers of rational points over all the extensions of $\mathbb{F}_q$ into one power series. The exponential in it looks odd: the more obvious series is $\sum_n \#C(\mathbb{F}_{q^n})T^n$. Problem 15 explains the choice. It is [Poo06, Proposition 3.4.3 and Exercise 3.7]. In [Poo06] the ground field is perfect throughout, as $\mathbb{F}_q$ is.

Definition 13.1 (Closed points [Poo06, Definition 2.2.1])**.** A closed point P of C is an integral subvariety of dimension 0. Its residue field $\mathbb{F}_q(P)$ is a finite extension of $\mathbb{F}_q$, and its degree is $\deg P := [\mathbb{F}_q(P) : \mathbb{F}_q]$.

Definition 13.2 (The zeta function)**.** The zeta function of C is
$$Z(C/\mathbb{F}_q; T) := \exp\Big(\sum_{n \geqslant 1} \#C(\mathbb{F}_{q^n})\frac{T^n}{n}\Big) \in \mathbb{Q}[[T]],$$
so that $\#C(\mathbb{F}_{q^n})$ is the coefficient of T^{n-1} in $\frac{d}{dT}\log Z(C/\mathbb{F}_q; T)$.

Proposition 13.3 (Closed points are Galois orbits [Poo06, Remark 2.2.2])**.** *Sending a point of $C(\overline{\mathbb{F}}_q)$ to its image is a surjection from $C(\overline{\mathbb{F}}_q)$ onto the set of closed points of C, whose fibers are the $\mathrm{Gal}_{\mathbb{F}_q}$-orbits. The orbit over a closed point P has $\deg P$ elements: a point of $C(\overline{\mathbb{F}}_q)$ over P is an $\mathbb{F}_q$-embedding $\mathbb{F}_q(P) \hookrightarrow \overline{\mathbb{F}}_q$, and there are $\deg P$ of these.*

Problem 15 (The Euler product)**.**
For $d \geqslant 1$, let N_d be the number of closed points of C of degree d.

(1) Show that a point of $C(\overline{\mathbb{F}}_q)$ lies in $C(\mathbb{F}_{q^n})$ if and only if σ^n fixes it. Deduce that a $\mathrm{Gal}_{\mathbb{F}_q}$-orbit of size d is contained in $C(\mathbb{F}_{q^n})$ if $d \mid n$, and is disjoint from it otherwise.

Solution. Embed C in $\mathbb{P}^N$ over $\mathbb{F}_q$ and write a point of $C(\overline{\mathbb{F}}_q)$ as $Q = [x_0 : \cdots : x_N]$ with some $x_i = 1$. Then $\sigma^n(Q) = [x_0^{q^n} : \cdots : x_N^{q^n}]$, again with i-th coordinate 1, so $\sigma^n(Q) = Q$ if and only if every x_j lies in $\mathbb{F}_{q^n}$, the fixed field of σ^n, that is, $Q \in C(\mathbb{F}_{q^n})$. The coordinates of Q lie in some finite field $\mathbb{F}_{q^m}$, so the action of $\mathrm{Gal}_{\mathbb{F}_q}$ on the orbit of Q factors through $\mathrm{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q)$, which is generated by the image of σ. Hence the orbit is $\{Q, \sigma(Q), \ldots, \sigma^{e-1}(Q)\}$, where σ^e generates the stabilizer of Q in $\langle\sigma\rangle$, and it has e elements. So $e = d$. Thus $\sigma^n(Q) = Q$ if and only if $d \mid n$, and since $\langle\sigma\rangle$ is abelian, every element of the orbit has the same stabilizer.

(2) Show that N_d is finite, and that

$$\#C(\mathbb{F}_{q^n}) = \sum_{d|n} dN_d \qquad \text{for every } n \geqslant 1.$$

Solution. By Proposition 13.3, a closed point of degree d is an orbit of size d, which by (1) lies in the finite set $C(\mathbb{F}_{q^d})$. So $N_d \leqslant \#C(\mathbb{F}_{q^d}) < \infty$. The set $C(\mathbb{F}_{q^n})$ is a union of orbits, namely of the orbits of size d for $d \mid n$, by (1). There are N_d of these, each with d elements, so $\#C(\mathbb{F}_{q^n}) = \sum_{d|n} dN_d$.

(3) Deduce the identity of formal power series

$$Z(C/\mathbb{F}_q; T) = \prod_P \bigl(1 - T^{\deg P}\bigr)^{-1} = \prod_{d \geqslant 1} (1 - T^d)^{-N_d},$$

the first product running over the closed points P of C. Why is the product well defined? Conclude that $Z(C/\mathbb{F}_q; T) \in 1 + T\mathbb{Z}[[T]]$, which is not obvious from the exponential.

Solution. Since N_d is finite, only finitely many factors of the product are $\not\equiv 1 \pmod{T^m}$ for each m, namely those with $\deg P < m$. So the product converges in $\mathbb{Z}[[T]]$, and each factor $(1 - T^d)^{-1} = 1 + T^d + T^{2d} + \cdots$ has integer coefficients. Taking logarithms and using $-\log(1 - x) = \sum_{k \geqslant 1} x^k/k$,

$$\log \prod_{d \geqslant 1} (1 - T^d)^{-N_d} = \sum_{d \geqslant 1} N_d \sum_{k \geqslant 1} \frac{T^{dk}}{k} = \sum_{n \geqslant 1} \Bigl(\sum_{d|n} dN_d \Bigr) \frac{T^n}{n} = \sum_{n \geqslant 1} \#C(\mathbb{F}_{q^n}) \frac{T^n}{n},$$

with $n = dk$, so that $1/k = d/n$. Exponentiating gives the claim, and the product has coefficients in $\mathbb{Z}$ and constant term 1.

(4) Invert (2): show that $N_d = \frac{1}{d} \sum_{e|d} \mu(d/e)\, \#C(\mathbb{F}_{q^e})$, where μ is the Möbius function. So the numbers of rational points over the extensions of $\mathbb{F}_q$ and the numbers of closed points of each degree determine each other.

Solution. Put $f(n) = \#C(\mathbb{F}_{q^n})$ and $g(d) = dN_d$. Then $f(n) = \sum_{d|n} g(d)$, and Möbius inversion gives $g(d) = \sum_{e|d} \mu(d/e) f(e)$.

(5) Take $C = \mathbb{P}^1$. Show that $Z(\mathbb{P}^1/\mathbb{F}_q; T) = 1/\bigl((1 - T)(1 - qT)\bigr)$. The closed points of $\mathbb{P}^1$ are the point at infinity, of degree 1, and the closed points of $\mathbb{A}^1$, which are the monic irreducible polynomials $f \in \mathbb{F}_q[t]$, with $\deg P = \deg f$. Deduce that the number of monic irreducible polynomials of degree d over $\mathbb{F}_q$ is $\frac{1}{d} \sum_{e|d} \mu(d/e) q^e$. How many are there of degree 2 over $\mathbb{F}_2$, and which are they?

SOLUTION. **One: t^2+t+1.** $\#\mathbb{P}^1(\mathbb{F}_{q^n}) = q^n+1$, so

$$\log Z(\mathbb{P}^1/\mathbb{F}_q;T) = \sum_n (1+q^n)T^n/n = -\log(1-T) - \log(1-qT).$$

By (4) applied to $\mathbb{P}^1$, for $d>1$ the point at infinity (the only closed point of $\mathbb{P}^1$ outside $\mathbb{A}^1$, of degree 1) does not count, and the extra $+1$ in $\#\mathbb{P}^1(\mathbb{F}_{q^e}) = q^e+1$ contributes $\frac{1}{d}\sum_{e|d}\mu(d/e) = 0$. So for $d>1$, and directly for $d=1$, the number of monic irreducible polynomials of degree d is $\frac{1}{d}\sum_{e|d}\mu(d/e)q^e$. For $q=2$ and $d=2$: $\frac{1}{2}(4-2)=1$, the polynomial t^2+t+1.

Remark. It is one closed point of degree 2, the orbit $\{\omega,\omega^2\}$ of a primitive cube root of unity in $\mathbb{F}_4$.

(6) Put $\zeta_C(s) \coloneqq Z(C/\mathbb{F}_q;q^{-s})$ and $NP \coloneqq \#\mathbb{F}_q(P)$. Show that $\zeta_C(s) = \prod_P \big(1-NP^{-s}\big)^{-1}$, and compare with the Euler product $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ of the Riemann zeta function.

SOLUTION. Substitute $T=q^{-s}$: $T^{\deg P} = (q^{\deg P})^{-s} = NP^{-s}$, since $\mathbb{F}_q(P)$ has $q^{\deg P}$ elements. The closed points play the role of the primes, and NP that of $p=\#\mathbb{F}_p$.

Remark 13.4 (The Weil conjectures for curves)**.** Let C be a smooth projective curve of genus g over $\mathbb{F}_q$. The Weil conjectures (proved for curves by Weil) say:

(1) (Rationality) $Z(C/\mathbb{F}_q;T) = P(T)/\big((1-T)(1-qT)\big)$ with $P\in\mathbb{Z}[T]$ of degree $2g$ and $P(0)=1$.
(2) (Functional equation) $Z\big(C/\mathbb{F}_q;1/(qT)\big) = q^{1-g}T^{2-2g}Z(C/\mathbb{F}_q;T)$.
(3) (Riemann hypothesis) $P(T) = \prod_{i=1}^{2g}(1-\alpha_i T)$ with $|\alpha_i| = \sqrt{q}$ for every i.

For $C=\mathbb{P}^1$, of genus 0, Problem 15(5) gives all three with $P=1$. We prove them for $g=1$ in Section 14.

14. The zeta function of an elliptic curve

Problem 16 is [Sil09, Exercise 5.3].

Problem 16 (Traces and determinants)**.**
Let A be a square matrix over a field K of characteristic 0. We prove, in two ways, that

$$\exp\Big(\sum_{n\geqslant 1}\operatorname{tr}(A^n)\frac{T^n}{n}\Big) = \frac{1}{\det(1-AT)} \qquad \text{in } K[[T]]. \tag{14.1}$$

(1) Let $f,g\in 1+TK[[T]]$. Show that f and g are units of $K[[T]]$, so that their logarithmic derivatives f'/f and g'/g are defined, and that if $f'/f = g'/g$, then $f=g$. Where do you use that K has characteristic 0? Give a counterexample in characteristic p.

SOLUTION. **Only to cancel n in $nh_n=0$.** A power series with constant term 1 is a unit: if $f=1-u$ with $u\in TK[[T]]$, then $f^{-1} = \sum_{k\geqslant 0}u^k$, which converges in $K[[T]]$ because $u^k\in T^kK[[T]]$. Put $h=f/g\in 1+TK[[T]]$. By the quotient rule,

$$h' = \frac{f'g-fg'}{g^2} = h\Big(\frac{f'}{f}-\frac{g'}{g}\Big) = 0.$$

Writing $h=\sum_n h_nT^n$, this says $nh_n=0$ for every $n\geqslant 1$. Since K has characteristic 0, n is invertible in K, so $h_n=0$ for $n\geqslant 1$, and $h=h_0=1$. This is the only use of characteristic 0, and it is essential: in characteristic p, $f=1+T^p$ and $g=1$ have $f'=pT^{p-1}=0=g'$, so both logarithmic derivatives vanish, but $f\neq g$.

(2) Show that both sides of (14.1) lie in $1+TK[[T]]$, compute their logarithmic derivatives, and conclude with (1).

Solution. For $F \in TK[[T]]$, $\exp F$ has constant term 1, and the chain rule gives $(\exp F)' = F' \exp F$. So the left side lies in $1+TK[[T]]$, with logarithmic derivative $\sum_{n\geqslant 1} \operatorname{tr}(A^n)T^{n-1}$. The right side has constant term $1/\det(1) = 1$. The matrix $M(T) = 1-AT$ is invertible over $K[[T]]$, with inverse $\sum_{k\geqslant 0} A^k T^k$, and $M'(T) = -A$. So the logarithmic derivative of $\det(1-AT)^{-1}$ is

$$-\frac{\frac{d}{dT}\det(1-AT)}{\det(1-AT)} = -\operatorname{tr}\big((1-AT)^{-1}(-A)\big) = \operatorname{tr}\Big(\sum_{k\geqslant 0} A^{k+1}T^k\Big) = \sum_{n\geqslant 1}\operatorname{tr}(A^n)T^{n-1}.$$

The two agree, so the two sides are equal by (1).

(3) Give a second proof: triangularize A over an algebraic closure $\bar{K}$ of K, reduce (14.1) to the case of a 1×1 matrix $A = (\lambda)$, and prove that case with (1), applied over $\bar{K}$.

Solution. Over $\bar{K}$, $A = BUB^{-1}$ with U upper triangular, with diagonal entries the eigenvalues $\lambda_1, \dots, \lambda_m$ of A. Then $A^n = BU^nB^{-1}$, and U^n is upper triangular with diagonal entries λ_i^n, so $\operatorname{tr}(A^n) = \sum_i \lambda_i^n$. Likewise $1-AT = B(1-UT)B^{-1}$, so $\det(1-AT) = \prod_i (1-\lambda_i T)$. Since $\exp(F+G) = \exp F \exp G$, both sides of (14.1) are products over i of the corresponding sides for the 1×1 matrix (λ_i). For $A = (\lambda)$, both $\exp\big(\sum_n \lambda^n T^n/n\big)$ and $(1-\lambda T)^{-1}$ lie in $1+T\bar{K}[[T]]$, with logarithmic derivatives $\sum_{n\geqslant 1}\lambda^n T^{n-1}$ and $\lambda/(1-\lambda T) = \sum_{n\geqslant 1}\lambda^n T^{n-1}$. So they are equal by (1) over $\bar{K}$, which also has characteristic 0. Both sides of (14.1) have coefficients in K, so the identity in $\bar{K}[[T]]$ is the one we want.

Problem 17 is the proof of [Sil09, Theorems V.2.3.1 and V.2.4], arranged around Problem 16. In the following results, E is an elliptic curve over $\mathbb{F}_q$ and $\ell \neq p$ is a prime.

Proposition 14.1 (Frobenius is purely inseparable [Sil09, Proposition II.2.11])**.** *The Frobenius endomorphism* F *is purely inseparable of degree* q*.*

Corollary 14.2 ($1-\mathsf{F}$ is separable [Sil09, Corollary III.5.5])**.** *The endomorphism* $1-\mathsf{F}$ *is separable.*

Theorem 14.3 (Kernels of separable isogenies [Sil09, Theorem III.4.10(c)])**.** *A nonzero separable isogeny* ψ *satisfies* $\#\ker\psi = \deg\psi$*, where* $\ker\psi$ *is taken in* $E(\overline{\mathbb{F}}_q)$*.*

Proposition 14.4 (Determinant and trace on $T_\ell(E)$ [Sil09, Proposition III.8.6])**.** *For* $\psi \in \operatorname{End}(E)$*,* $\det\psi_\ell = \deg\psi$ *and* $\operatorname{tr}\psi_\ell = 1 + \deg\psi - \deg(1-\psi)$*.*

Theorem 14.5 (Endomorphisms act faithfully on $T_\ell(E)$ [Sil09, Theorem III.7.4])**.** *The map* $\psi \mapsto \psi_\ell$ *is an injective ring homomorphism* $\operatorname{End}(E) \to \operatorname{End}\big(T_\ell(E)\big)$*.*

Problem 17 (The zeta function of E)**.**

(1) Show that $\sigma(Q) = \mathsf{F}(Q)$ for every $Q \in E(\overline{\mathbb{F}}_q)$. Deduce that the closed points of E of degree d are the F-orbits of size d in $E(\overline{\mathbb{F}}_q)$, and that $E(\mathbb{F}_{q^n}) = \ker(1-\mathsf{F}^n)$.

Solution. For $Q = (x_0, y_0)$ with $x_0, y_0 \in \overline{\mathbb{F}}_q$, $\mathsf{F}(Q) = (x_0^q, y_0^q) = \sigma(Q)$, and both fix O. So the $\operatorname{Gal}_{\mathbb{F}_q}$-orbits are the F-orbits (Problem 15(1): the orbit of Q is $\{Q, \sigma(Q), \sigma^2(Q), \dots\}$), and Proposition 13.3 identifies them with the closed points. By Problem 15(1), $Q \in E(\mathbb{F}_{q^n})$ if and only if $\mathsf{F}^n(Q) = \sigma^n(Q) = Q$, that is, $Q \in \ker(1-\mathsf{F}^n)$.

(2) Show that F^n is the q^n-power Frobenius endomorphism of E over $\mathbb{F}_{q^n}$, and deduce that

$$\#E(\mathbb{F}_{q^n}) = \deg(1-\mathsf{F}^n) = \det(1-\mathsf{F}_\ell^n) = 1 - \operatorname{tr}(\mathsf{F}_\ell^n) + q^n.$$

Where does the argument use that $1 - \mathsf{F}^n$ is separable?

SOLUTION. **In $\#\ker(1-\mathsf{F}^n) = \deg(1-\mathsf{F}^n)$.** $\mathsf{F}^n(x,y) = (x^{q^n}, y^{q^n})$, and E is defined over $\mathbb{F}_{q^n}$. So Corollary 14.2, applied to E over $\mathbb{F}_{q^n}$, says that $1-\mathsf{F}^n$ is separable. It is nonzero, since its kernel $E(\mathbb{F}_{q^n})$ is finite while $E(\overline{\mathbb{F}}_q)$ is infinite. By Theorem 14.3 and (1), $\#E(\mathbb{F}_{q^n}) = \#\ker(1-\mathsf{F}^n) = \deg(1-\mathsf{F}^n)$. This is where separability is used, since for an inseparable isogeny $\#\ker$ is only the separable degree. By Proposition 14.4 and Theorem 14.5,

$$\deg(1-\mathsf{F}^n) = \det\big((1-\mathsf{F}^n)_\ell\big) = \det(1-\mathsf{F}_\ell^n).$$

For a 2×2 matrix M, $\det(1-M) = 1 - \operatorname{tr} M + \det M$, and $\det \mathsf{F}_\ell^n = (\det \mathsf{F}_\ell)^n = (\deg \mathsf{F})^n = q^n$ by Proposition 14.1.

(3) Taking $n=1$, show that the characteristic polynomial of F_ℓ is $T^2 - aT + q$. In particular it has integer coefficients and does not depend on ℓ.

SOLUTION. For a 2×2 matrix the characteristic polynomial is $T^2 - (\operatorname{tr} M)T + \det M$. By (2) with $n=1$, $\operatorname{tr}\mathsf{F}_\ell = 1 + q - \#E(\mathbb{F}_q) = a$, and $\det \mathsf{F}_\ell = q$.

(4) Use Problem 16 to show that

$$Z(E/\mathbb{F}_q;T) = \frac{\det(1-\mathsf{F}_\ell T)}{(1-T)(1-qT)} = \frac{1-aT+qT^2}{(1-T)(1-qT)}.$$

By Problem 15, this is also $\prod_P (1-T^{\deg P})^{-1}$ over the closed points P of E.

SOLUTION. By (2), $\#E(\mathbb{F}_{q^n}) = 1 + q^n - \operatorname{tr}(\mathsf{F}_\ell^n)$, so, working in $\mathbb{Q}_\ell[[T]]$,

$$Z(E/\mathbb{F}_q;T) = \exp\Big(\sum_n \frac{T^n}{n}\Big)\exp\Big(\sum_n \frac{(qT)^n}{n}\Big)\exp\Big(-\sum_n \operatorname{tr}(\mathsf{F}_\ell^n)\frac{T^n}{n}\Big) = \frac{\det(1-\mathsf{F}_\ell T)}{(1-T)(1-qT)},$$

using Problem 16 over $K = \mathbb{Q}_\ell$ for the last factor. By (3),

$$\det(1-\mathsf{F}_\ell T) = T^2\det(T^{-1} - \mathsf{F}_\ell) = 1 - aT + qT^2.$$

(5) Verify the functional equation $Z\big(E/\mathbb{F}_q;1/(qT)\big) = Z(E/\mathbb{F}_q;T)$ of Remark 13.4.

SOLUTION. Substituting $T \mapsto 1/(qT)$ and multiplying numerator and denominator by qT^2:

$$\frac{1 - a/(qT) + 1/(qT^2)}{\big(1-1/(qT)\big)\big(1-1/T\big)} = \frac{qT^2 - aT + 1}{(qT-1)(T-1)} = Z(E/\mathbb{F}_q;T),$$

which is Remark 13.4(2) with $g=1$.

Problem 18 (The Riemann hypothesis and Hasse's bound)**.**
Let $\alpha, \beta \in \mathbb{C}$ be the roots of $T^2 - aT + q$, so that $1 - aT + qT^2 = (1-\alpha T)(1-\beta T)$.

(1) For $r, s \in \mathbb{Z}$ with $s \neq 0$, show that $\det(r/s - \mathsf{F}_\ell) = \deg(r - s\mathsf{F})/s^2 \geqslant 0$. Deduce that $T^2 - aT + q$ is nonnegative on $\mathbb{R}$, and hence that $\beta = \bar{\alpha}$ and $|\alpha| = |\beta| = \sqrt{q}$. This is the Riemann hypothesis of Remark 13.4. With ζ_E as in Problem 15(6), show that every zero of $\zeta_E(s)$ satisfies $\operatorname{Re}(s) = 1/2$, and that $\zeta_E(s) = \zeta_E(1-s)$.

SOLUTION. By Theorem 14.5, $(r - s\mathsf{F})_\ell = r - s\mathsf{F}_\ell$, so by Proposition 14.4

$$s^2\det(r/s - \mathsf{F}_\ell) = \det(r - s\mathsf{F}_\ell) = \deg(r - s\mathsf{F}) \geqslant 0.$$

So the real polynomial $T^2 - aT + q = \det(T - \mathsf{F}_\ell)$ (Problem 17(3)) is $\geqslant 0$ on $\mathbb{Q}$, hence on $\mathbb{R}$ by continuity, and its discriminant $a^2 - 4q$ is $\leqslant 0$. Either the roots are non-real and complex

conjugate, or there is a double real root. In both cases $\beta = \bar{\alpha}$, and $|\alpha|^2 = \alpha\beta = q$. The zeros of ζ_E are the s with $q^{-s} \in \{1/\alpha, 1/\beta\}$ (the denominator does not vanish there), so $q^{-\mathrm{Re}(s)} = |q^{-s}| = q^{-1/2}$. The substitution $s \mapsto 1-s$ sends q^{-s} to $1/(q \cdot q^{-s})$, so $\zeta_E(1-s) = \zeta_E(s)$ is Problem 17(5).

Remark. Compare [Sil09, Remark V.2.5].

(2) Show that $a_n = \alpha^n + \beta^n$ for every $n \geqslant 0$. Deduce Hasse's bound

$$|\#E(\mathbb{F}_q) - q - 1| \leqslant 2\sqrt{q},$$

and more generally $|a_n| \leqslant 2q^{n/2}$ for every $n \geqslant 1$.

SOLUTION. Taking logarithmic derivatives in Problem 17(4), written as

$$Z = \frac{(1-\alpha T)(1-\beta T)}{(1-T)(1-qT)},$$

gives $\#E(\mathbb{F}_{q^n}) = 1 + q^n - \alpha^n - \beta^n$, that is, $a_n = \alpha^n + \beta^n$ for $n \geqslant 1$. And $a_0 = 2 = \alpha^0 + \beta^0$. Then $|a_n| \leqslant |\alpha|^n + |\beta|^n = 2q^{n/2}$ by (1), and $n = 1$ is Hasse's bound.

(3) Conversely, suppose only that there is a constant c with $|a_n| \leqslant c\, q^{n/2}$ for all $n \geqslant 1$. Show that the power series $\sum_{n \geqslant 1} a_n T^n = \frac{\alpha T}{1-\alpha T} + \frac{\beta T}{1-\beta T}$ converges for $|T| < q^{-1/2}$, and deduce that $|\alpha| = |\beta| = \sqrt{q}$. So the Riemann hypothesis is equivalent to a Hasse bound over every extension of $\mathbb{F}_q$, even with a worse constant.

SOLUTION. The series has coefficients bounded by $c\, q^{n/2}$, so it converges for $|T| < q^{-1/2}$ and defines a holomorphic function there. As formal power series, $\sum_n (\alpha^n + \beta^n) T^n = \alpha T/(1-\alpha T) + \beta T/(1-\beta T)$. If $\alpha \neq \beta$, the right side has poles at $1/\alpha$ and $1/\beta$, which do not cancel. If $\alpha = \beta$, it is $2\alpha T/(1-\alpha T)$, with a pole at $1/\alpha$. A holomorphic function on the disc $|T| < q^{-1/2}$ has no poles there, so $|1/\alpha|, |1/\beta| \geqslant q^{-1/2}$, that is, $|\alpha|, |\beta| \leqslant \sqrt{q}$. Since $|\alpha||\beta| = q$, both are equalities.

Problem 19 is [Sil09, Exercise 5.13].

Problem 19 (A recurrence)**.**

(1) Show that the single number $\#E(\mathbb{F}_q)$ determines $\#E(\mathbb{F}_{q^n})$ for every $n \geqslant 1$.

SOLUTION. $\#E(\mathbb{F}_q)$ determines a, hence $Z(E/\mathbb{F}_q; T)$ by Problem 17(4), hence every $\#E(\mathbb{F}_{q^n})$. Explicitly, it determines α and β, and $a_n = \alpha^n + \beta^n$ by Problem 18(2).

(2) Prove that

$$a_{n+2} = a_1 a_{n+1} - q a_n \qquad \text{for all } n \geqslant 0.$$

(Recall the convention $a_0 = 2$.)

SOLUTION. By Problem 18(2), $a_n = \alpha^n + \beta^n$ for $n \geqslant 0$. Since α is a root of $T^2 - aT + q$, $\alpha^{n+2} = a\alpha^{n+1} - q\alpha^n$, and likewise for β. Adding, $a_{n+2} = a\, a_{n+1} - q\, a_n$, with $a = a_1$.

Remark. In particular every a_n is an integer, computed from a_1 alone.

(3) Here is the recurrence on $y^2 = x^3 + x + 1$ over $\mathbb{F}_{101}$:

```
q := 101;  E := EllipticCurve([GF(q) | 1, 1]);          // y^2 = x^3 + x + 1
a := [TraceOfFrobenius(ChangeRing(E, GF(q^n))) : n in [1..6]];
a;                                  // [ -3, -193, 882, 16847, -139623, -1282678 ]
b := [2, a[1]];
for n in [1..5] do Append(~b, a[1]*b[n+1] - q*b[n]); end for;
```

```
b[2..7] eq a;                    // true
```

Here `TraceOfFrobenius(E)` is a. Compute a_2 by hand from $a_1 = -3$. Check Hasse's bound for a_6 over $\mathbb{F}_{101^6}$.

SOLUTION. **$a_2 = -193$. Hasse's bound holds for a_6.** $a_2 = a_1^2 - 2q = 9 - 202 = -193$, as Magma says, and the recurrence reproduces all six traces. For a_6:

$$2\sqrt{101^6} = 2 \cdot 101^3 = 2\,060\,602 \geqslant 1\,282\,678.$$

Problem 20 (A zeta function by hand)**.**
Let E be the elliptic curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_2$ (its discriminant is 1).

(1) List the rational points of $E(\mathbb{F}_2)$, and find a.

SOLUTION. **$E(\mathbb{F}_2) = \{O, (0,1), (1,0), (1,1)\}$. So $a = -1$.** For $x = 0$: $y^2 = 1$, so $y = 1$. For $x = 1$: $y^2 + y = 0$, so $y \in \{0, 1\}$. With O, $E(\mathbb{F}_2) = \{O, (0,1), (1,0), (1,1)\}$, so $\#E(\mathbb{F}_2) = 4$ and $a = 2 + 1 - 4 = -1$.

(2) Write down $Z(E/\mathbb{F}_2; T)$, and check the Riemann hypothesis for it directly.

SOLUTION. By Problem 17(4),

$$Z(E/\mathbb{F}_2; T) = \frac{1 + T + 2T^2}{(1 - T)(1 - 2T)}.$$

The roots of $T^2 + T + 2$ are $\alpha, \beta = (-1 \pm \sqrt{-7})/2$, with $|\alpha|^2 = (1 + 7)/4 = 2 = q$.

(3) Use Problem 19 to predict $\#E(\mathbb{F}_4)$ and $\#E(\mathbb{F}_8)$. Check the first by listing the rational points of $E(\mathbb{F}_4)$, where $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$ with $\omega^2 = \omega + 1$.

SOLUTION. **$\#E(\mathbb{F}_4) = 8$ and $\#E(\mathbb{F}_8) = 4$.** With $a_0 = 2$, $a_1 = -1$ and $q = 2$:

$$a_2 = (-1)(-1) - 2 \cdot 2 = -3, \qquad a_3 = (-1)(-3) - 2(-1) = 5.$$

So $\#E(\mathbb{F}_4) = 4 + 1 + 3 = 8$ and $\#E(\mathbb{F}_8) = 8 + 1 - 5 = 4$. To check the first: for $x = 0$, $y^2 = 1$ gives $y = 1$ only. For $x \neq 0$ put $y = xz$. Dividing by x^2, the equation becomes $z^2 + z = x + x^{-2}$. For $x = 1$ the right side is 0. For $x = \omega$ it is $\omega + \omega^{-2} = \omega + \omega = 0$ (as $\omega^3 = 1$). For $x = \omega^2$ it is $\omega^2 + \omega^{-4} = \omega^2 + \omega^2 = 0$. So in each case $z \in \{0, 1\}$, giving two rational points each, and $\#E(\mathbb{F}_4) = 1 + 1 + 3 \cdot 2 = 8$.

(4) Use Problem 15(4) to find the numbers N_1, N_2, N_3 of closed points of E of degree 1, 2 and 3. Deduce that every $\mathbb{F}_8$-rational point of E is $\mathbb{F}_2$-rational. Is $\#E(\mathbb{F}_8)$ close to the edge of the Hasse interval?

SOLUTION. **$N_1 = 4$, $N_2 = 2$, $N_3 = 0$. So $\#E(\mathbb{F}_8) = 4$ is the least Hasse allows.** $N_1 = \#E(\mathbb{F}_2) = 4$, $N_2 = \frac{1}{2}(8 - 4) = 2$ and $N_3 = \frac{1}{3}(4 - 4) = 0$. By Problem 15(2), $E(\mathbb{F}_8)$ is the union of the orbits of size 1 and 3, and there are none of size 3. So $E(\mathbb{F}_8) = E(\mathbb{F}_2)$. The Hasse interval over $\mathbb{F}_8$ is $|a_3| \leqslant 2\sqrt{8} \approx 5.66$, and $a_3 = 5$ is as large as it can be: $E(\mathbb{F}_8)$ is as small as an elliptic curve over $\mathbb{F}_8$ allows.

(5) Check the Euler product of Problem 15(3) against (2) up to the term in T^2.

Solution. **Both are $1+4T+12T^2$ modulo T^3.** Modulo T^3, the factors with $\deg P \geqslant 3$ are 1, so

$$\prod_P (1-T^{\deg P})^{-1} \equiv (1-T)^{-4}(1-T^2)^{-2} \equiv (1+4T+10T^2)(1+2T^2) \equiv 1+4T+12T^2,$$

while $(1+T+2T^2)(1+3T+7T^2) \equiv 1+4T+12T^2$, using $1/\big((1-T)(1-2T)\big) = \sum_n (2^{n+1}-1)T^n$.

(6) Check, where `Places(E, d)` lists the closed points of degree d:

```
E := EllipticCurve([GF(2) | 1, 0, 0, 0, 1]);            // y^2 + xy = x^3 + 1
[#ChangeRing(E, GF(2^n)) : n in [1..8]];     // [ 4, 8, 4, 16, 44, 56, 116, 288 ]
[#Places(E, d) : d in [1..6]];               // [ 4, 2, 0, 2, 8, 8 ]
ZetaFunction(E);              // (2*$.1^2 + $.1 + 1)/(2*$.1^2 - 3*$.1 + 1)
```

Solution. **They agree.** The counts agree with $2^n + 1 - a_n$ for the recurrence of Problem 19, and the numbers of closed points with (4) and, beyond it, with Problem 15(4): for instance $N_4 = \frac{1}{4}(16-8) = 2$. `Magma`'s zeta function is ours, with both polynomials written in decreasing powers of its indeterminate, our T, and the denominator expanded: $2T^2 - 3T + 1 = (1-T)(1-2T)$.

15. Isogenous curves

Problem 21 is [Sil09, Exercise 5.4]. Let E_1 and E_2 be elliptic curves over $\mathbb{F}_q$, with q-power Frobenius endomorphisms F_1 and F_2 and traces $a(E_1)$, $a(E_2)$.

Definition 15.1 (Isogenies and Galois-equivariant maps)**.** We write $\mathrm{Hom}(E_1, E_2)$ for the isogenies $E_1 \to E_2$ *defined over* $\mathbb{F}_q$. The Galois group $\mathrm{Gal}_{\mathbb{F}_q}$ acts on each Tate module $T_\ell(E_i)$, and we write $\mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}\big(T_\ell(E_1), T_\ell(E_2)\big)$ for the $\mathbb{Z}_\ell$-linear maps that commute with this action.

Theorem 15.2 (Tate's isogeny theorem [Sil09, Theorem III.7.7(a)])**.** *Let $\ell \neq p$ be prime. The natural map*

$$\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \to \mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}\big(T_\ell(E_1), T_\ell(E_2)\big)$$

is an isomorphism.

Corollary 15.3 (Isogenies form a free module [Sil09, Corollary III.7.5])**.** *The group* $\mathrm{Hom}(E_1, E_2)$ *is a free $\mathbb{Z}$-module of finite rank, being a subgroup of the free module of.*

Problem 21 (Isogenous curves have the same number of rational points)**.**

(1) Let $\varphi\colon E_1 \to E_2$ be a nonzero isogeny defined over $\mathbb{F}_q$. Show that $\varphi \circ \mathsf{F}_1 = \mathsf{F}_2 \circ \varphi$, and deduce that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

Solution. The isogeny φ is given by rational functions with coefficients in $\mathbb{F}_q$, and raising to the q-th power fixes those coefficients: $\varphi(x^q, y^q) = \varphi(x,y)^q$ coordinatewise, that is, $\varphi \circ \mathsf{F}_1 = \mathsf{F}_2 \circ \varphi$. Hence $\varphi \circ (1 - \mathsf{F}_1) = (1 - \mathsf{F}_2) \circ \varphi$, and taking degrees, which are multiplicative,

$$\deg \varphi \cdot \deg(1 - \mathsf{F}_1) = \deg(1 - \mathsf{F}_2) \cdot \deg \varphi.$$

Since $\deg \varphi \neq 0$, Problem 17(2) gives $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$.

(2) Deduce that $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$.

Solution. The zeta function depends only on q and $a = q + 1 - \#E(\mathbb{F}_q)$, by Problem 17(4).

For the converse, suppose that $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$, so that $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$ have the same characteristic polynomial $f(T) = T^2 - aT + q$ (Problem 17(3)). Fix a prime $\ell \neq p$ and put $V_i \coloneqq T_\ell(E_i) \otimes \mathbb{Q}_\ell$.

(3) Show that a $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ commutes with the action of $\mathrm{Gal}_{\mathbb{F}_q}$ if and only if it intertwines $\mathsf{F}_{1,\ell}$ and $\mathsf{F}_{2,\ell}$.

Solution. For $Q \in E_i(\overline{\mathbb{F}}_q)$ we have $\sigma(Q) = \mathsf{F}_i(Q)$, as in Problem 17(1). So σ acts on $T_\ell(E_i)$ by $\mathsf{F}_{i,\ell}$. A map commuting with $\mathrm{Gal}_{\mathbb{F}_q}$ therefore intertwines the Frobenius maps. Conversely, if λ commutes with σ, it commutes with every element of the subgroup $\langle\sigma\rangle$. The action of $\mathrm{Gal}_{\mathbb{F}_q}$ on each $E_i[\ell^n](\overline{\mathbb{F}}_q)$ factors through a finite quotient $\mathrm{Gal}(F/\mathbb{F}_q)$ for a finite extension F, which is generated by the image of σ. So λ commutes with the whole group modulo ℓ^n for every n, and hence on T_ℓ.

(4) Suppose $a^2 \neq 4q$. Show that $\mathsf{F}_{1,\ell}$ is not a scalar, that V_1 has a basis $v, \mathsf{F}_{1,\ell}v$, and that $V_1 \cong \mathbb{Q}_\ell[T]/(f)$ as $\mathbb{Q}_\ell[T]$-modules, with T acting by $\mathsf{F}_{1,\ell}$. Deduce that there is a nonzero $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ intertwining the two Frobenius maps.

Solution. If $\mathsf{F}_{1,\ell} = c$ were a scalar, its characteristic polynomial would be $(T-c)^2$, with discriminant $a^2 - 4q = 0$. So some $v \in V_1$ is not an eigenvector, and then v and $\mathsf{F}_{1,\ell}v$ are linearly independent, hence a basis of the 2-dimensional space V_1. The map $\mathbb{Q}_\ell[T] \to V_1$, $g \mapsto g(\mathsf{F}_{1,\ell})v$, is therefore surjective. Its kernel contains f by Cayley–Hamilton, and $\mathbb{Q}_\ell[T]/(f)$ has dimension 2, so $V_1 \cong \mathbb{Q}_\ell[T]/(f)$. The same holds for V_2, so there is a $\mathbb{Q}_\ell[T]$-isomorphism $\lambda\colon V_1 \to V_2$, that is, a linear isomorphism with $\lambda \circ \mathsf{F}_{1,\ell} = \mathsf{F}_{2,\ell} \circ \lambda$. The image $\lambda(T_\ell(E_1))$ is a finitely generated $\mathbb{Z}_\ell$-submodule of V_2, so $\ell^k\lambda$ maps $T_\ell(E_1)$ into $T_\ell(E_2)$ for k large, and $\ell^k\lambda$ still intertwines and is nonzero.

(5) Suppose $a^2 = 4q$. Show that q is a square, that $a/2 \in \mathbb{Z}$, and that $\mathsf{F}_i = [a/2]$ in $\mathrm{End}(E_i)$ for $i = 1, 2$. Deduce that every $\mathbb{Z}_\ell$-linear map $T_\ell(E_1) \to T_\ell(E_2)$ intertwines the two Frobenius maps.

Solution. Since $a^2 = 4q$ is even, so is a. Thus $a/2 \in \mathbb{Z}$, and $q = (a/2)^2$ is the square of the integer $a/2 = \pm\sqrt{q}$. Now $f(T) = (T - a/2)^2$, and by Proposition 14.4

$$\deg\bigl(\mathsf{F}_i - [a/2]\bigr) = \det\bigl(\mathsf{F}_{i,\ell} - a/2\bigr) = f(a/2) = 0,$$

so $\mathsf{F}_i = [a/2]$. Then $\mathsf{F}_{i,\ell}$ is multiplication by $a/2$ on $T_\ell(E_i)$, and every $\mathbb{Z}_\ell$-linear map commutes with multiplication by a scalar.

(6) Use Theorem 15.2 to conclude that E_1 and E_2 are isogenous over $\mathbb{F}_q$.

Solution. In either case, by (3), there is a nonzero element of $\mathrm{Hom}_{\mathrm{Gal}_{\mathbb{F}_q}}(T_\ell(E_1), T_\ell(E_2))$, so by Theorem 15.2 $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \neq 0$. As $\mathrm{Hom}(E_1, E_2) \cong \mathbb{Z}^\rho$ is free (Corollary 15.3), $\mathrm{Hom}(E_1, E_2) \otimes \mathbb{Z}_\ell \cong \mathbb{Z}_\ell^\rho$, so $\rho \geqslant 1$ and there is a nonzero isogeny $E_1 \to E_2$ defined over $\mathbb{F}_q$.

Remark 15.4. Together with Problem 17(4), Problem 21 says that for elliptic curves E_1, E_2 over $\mathbb{F}_q$ the following are equivalent: E_1 and E_2 are isogenous over $\mathbb{F}_q$; $\#E_1(\mathbb{F}_q) = \#E_2(\mathbb{F}_q)$; $Z(E_1/\mathbb{F}_q; T) = Z(E_2/\mathbb{F}_q; T)$; and E_1 and E_2 have the same number of closed points of each degree (Problem 15(4)).

Looking ahead. By Problem 18(1), $\alpha = \sqrt{q}\,e^{i\theta}$ for a unique $\theta \in [0, \pi]$, and $a = 2\sqrt{q}\cos\theta$. Fix an elliptic curve over $\mathbb{Q}$, reduce it modulo every prime p of good reduction, and let θ_p be the angle of the reduction over $\mathbb{F}_p$. How are the θ_p distributed in $[0, \pi]$ as p varies? That is the Sato–Tate conjecture. The product over the primes of good reduction of the factors $(1 - a_p p^{-s} + p^{1-2s})^{-1}$, the

reciprocals of the numerators of the ζ-functions of the reductions, is, up to finitely many factors, the L-function of the curve, the subject of the Birch and Swinnerton-Dyer conjecture.

Part 5. Supersingular curves and the trace of Frobenius

Section 16 proves that an elliptic curve over $\mathbb{F}_q$ is supersingular if and only if p divides its trace of Frobenius, and works out what this says over $\mathbb{F}_p$, over $\mathbb{F}_2$ and $\mathbb{F}_3$, and over $\mathbb{F}_8$. Section 17 shows that $E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$ with $m \mid q-1$, and pins down m for a supersingular curve over $\mathbb{F}_p$. Section 18 shows that an elliptic curve over $\mathbb{Q}$ is ordinary modulo infinitely many primes.

Reading. [Sil09, §§V.3–V.4], with [Sil09, §V.2] for the trace of Frobenius, [Sil09, §III.6] for the torsion subgroups and [Sil09, §III.8] for the Weil pairing. The three problems are exercises from [Sil09, Chapter V]: Problem 22 is Exercise 5.10, Problem 23 is Exercise 5.6 and Problem 24 is Exercise 5.11.

`Magma`**.** If you have no local installation, every block below runs in the free online calculator.

Standing notation.

- A power $q = p^e$ of a prime p.
- The p^r-power Frobenius morphism $\mathsf{F}^r \colon C \to C^{(p^r)}$ of a curve C over a field of characteristic p, for $r \geqslant 1$, which raises coordinates to the p^r-th power. For an elliptic curve E, its dual isogeny $\mathsf{V}^r \colon E^{(p^r)} \to E$ is the Verschiebung.
- The q-power Frobenius *endomorphism* of an elliptic curve E over $\mathbb{F}_q$. Since $E^{(q)} = E$, the morphism
$$\mathsf{F}^e \colon E \to E, \qquad (x, y) \mapsto (x^q, y^q),$$
is an endomorphism of E, with dual $\mathsf{V}^e \in \mathrm{End}(E)$. Here $\mathrm{End}(E)$ consists of the endomorphisms defined over $\mathbb{F}_q$.
- The group $E[m](\overline{\mathbb{F}}_q)$ of m-torsion points of E with coordinates in $\overline{\mathbb{F}}_q$, for $m \geqslant 1$.
- The supersingular elliptic curves E over $\mathbb{F}_q$: those with $E[p^r](\overline{\mathbb{F}}_q) = 0$ for one, equivalently every, $r \geqslant 1$ [Sil09, Theorem V.3.1 and the definition after it]. The others are ordinary. Silverman also says that a supersingular curve has "Hasse invariant 0" and an ordinary one "Hasse invariant 1". That is the phrasing of Exercise 5.11.

16. Supersingularity and the trace of Frobenius

This section is [Sil09, Exercise 5.10]. Part (5) is a little stronger than [Sil09, Exercise 5.10(e)], and part (6) needs the stronger form. By Deuring's theorem [Sil09, Theorem V.3.1], a supersingular curve is one whose Verschiebung is purely inseparable, whose endomorphism ring over $\overline{\mathbb{F}}_q$ is an order in a quaternion algebra, and whose formal group has height 2. None of these is easy to read off from a Weierstrass equation. Over a finite field there is a criterion that is: count the rational points.

Definition 16.1 (Trace of Frobenius)**.** Let E be an elliptic curve over $\mathbb{F}_q$. The trace of Frobenius of E is
$$a \coloneqq q + 1 - \#E(\mathbb{F}_q).$$

Definition 16.2 (Separable degree)**.** For an isogeny ψ, $\deg_s \psi$ denotes its separable degree.

In the following results, E is an elliptic curve over $\mathbb{F}_q$, with $q = p^e$, and a is its trace of Frobenius.

Proposition 16.3 (Frobenius is purely inseparable [Sil09, Proposition II.2.11])**.** *For $r \geqslant 1$, the Frobenius morphism F^r is purely inseparable of degree p^r.*

Corollary 16.4 (Purely inseparable isogenies [Sil09, Corollary II.2.12]). *A purely inseparable isogeny $\psi\colon E \to E'$ of degree p^r factors as $\psi = \lambda \circ \mathsf{F}^r$ with $\lambda\colon E^{(p^r)} \to E'$ an isomorphism.*

Theorem 16.5 (Kernels and separable degree [Sil09, Theorem III.4.10(a)]). *A nonzero isogeny ψ satisfies $\#\ker\psi = \deg_s\psi$, the kernel taken on points with coordinates in $\overline{\mathbb{F}}_q$.*

Corollary 16.6 (Separability of $[m]+[n]\mathsf{F}^e$ [Sil09, Corollary III.5.5]). *For $m, n \in \mathbb{Z}$, the endomorphism $[m]+[n]\mathsf{F}^e$ is separable if and only if $p \nmid m$.*

Theorem 16.7 (Dual isogenies [Sil09, Theorem III.6.2(a)]). *Every isogeny ψ satisfies $\hat{\psi}\circ\psi = [\deg\psi]$ and $\psi\circ\hat{\psi} = [\deg\psi]$. So $\mathsf{V}^e \circ \mathsf{F}^e = \mathsf{F}^e \circ \mathsf{V}^e = [q]$.*

Corollary 16.8 (The degree of $[m]$ [Sil09, Corollary III.6.4(a)]). *For $m \in \mathbb{Z}$, the multiplication-by-m map has degree $\deg[m] = m^2$.*

Corollary 16.9 (The p-power torsion [Sil09, Corollary III.6.4(c)]). *Either $E[p^r](\overline{\mathbb{F}}_q) = 0$ for every $r \geqslant 1$, or $E[p^r](\overline{\mathbb{F}}_q) \cong \mathbb{Z}/p^r\mathbb{Z}$ for every $r \geqslant 1$.*

Theorem 16.10 (The characteristic polynomial of Frobenius [Sil09, Theorem V.2.3.1(b)]). *The Frobenius endomorphism satisfies $(\mathsf{F}^e)^2 - [a]\mathsf{F}^e + [q] = 0$ in $\mathrm{End}(E)$.*

Theorem 16.11 (Hasse's bound [Sil09, Theorem V.1.1]). *The trace of Frobenius satisfies $|a| \leqslant 2\sqrt{q}$.*

Problem 22 (Supersingularity and the trace of Frobenius).
Let E be an elliptic curve over $\mathbb{F}_q$.
(1) Show that $\mathsf{V}^e = [a] - \mathsf{F}^e$ in $\mathrm{End}(E)$, and deduce that V^e is separable if and only if $p \nmid a$. Show that $\ker\mathsf{V}^e$ has $\#E[q](\overline{\mathbb{F}}_q)$ elements, so that $\deg_s\mathsf{V}^e \in \{1, q\}$. Conclude that

$$E \text{ is supersingular} \iff a \equiv 0 \pmod{p}.$$

Solution. By Theorems 16.7 and 16.10,

$$0 = (\mathsf{F}^e)^2 - [a]\mathsf{F}^e + \mathsf{V}^e \circ \mathsf{F}^e = \left(\mathsf{F}^e - [a] + \mathsf{V}^e\right) \circ \mathsf{F}^e.$$

The map F^e is surjective on points with coordinates in $\overline{\mathbb{F}}_q$, since $(x, y) = \mathsf{F}^e(x^{1/q}, y^{1/q})$. So the left factor vanishes on all of $E(\overline{\mathbb{F}}_q)$, that is, $\mathsf{V}^e = [a] - \mathsf{F}^e$. By Corollary 16.6 with $m = a$ and $n = -1$, V^e is separable if and only if $p \nmid a$.
For the kernel: $\mathsf{V}^e \circ \mathsf{F}^e = [q]$ maps $E[q](\overline{\mathbb{F}}_q)$ to O, so F^e maps $E[q](\overline{\mathbb{F}}_q)$ into $\ker\mathsf{V}^e$. Conversely a point Q of $E(\overline{\mathbb{F}}_q)$ in $\ker\mathsf{V}^e$ is $\mathsf{F}^e(P)$ for some $P \in E(\overline{\mathbb{F}}_q)$, and then $[q]P = O$. So $\ker\mathsf{V}^e = \mathsf{F}^e\big(E[q](\overline{\mathbb{F}}_q)\big)$, and F^e is injective on $E(\overline{\mathbb{F}}_q)$, because $t \mapsto t^q$ is injective on the field $\overline{\mathbb{F}}_q$. Hence $\#\ker\mathsf{V}^e = \#E[q](\overline{\mathbb{F}}_q)$, which is 1 or q by Corollary 16.9 with $r = e$. By Theorem 16.5, $\deg_s\mathsf{V}^e \in \{1, q\}$. And $\deg\mathsf{V}^e = q$, since $\deg\mathsf{V}^e \cdot \deg\mathsf{F}^e = \deg[q] = q^2$ by Theorem 16.7, Corollary 16.8, and Proposition 16.3. So V^e is either separable ($\deg_s = q$) or purely inseparable ($\deg_s = 1$), and

$$E \text{ supersingular} \iff E[q](\overline{\mathbb{F}}_q) = 0 \iff \deg_s\mathsf{V}^e = 1 \iff \mathsf{V}^e \text{ not separable} \iff p \mid a.$$

The first equivalence is the definition, with $r = e$. The last is the first paragraph.

(2) Suppose $q = p \geqslant 5$. Show that E is supersingular if and only if $\#E(\mathbb{F}_p) = p + 1$. Where is $p \geqslant 5$ used?

Solution. **To make $a = 0$ the only multiple of p in the Hasse interval.** For $q = p$, part (1) says that E is supersingular if and only if $p \mid a$. By Theorem 16.11, $|a| \leqslant 2\sqrt{p}$, and $2\sqrt{p} < p$ exactly when $p > 4$. So for $p \geqslant 5$ the only multiple of p in the Hasse interval is $a = 0$,

that is, $\#E(\mathbb{F}_p) = p+1$. For $p = 2$ and $p = 3$ the interval also contains $a = \pm p$, and parts (3) and (4) show that these values occur.

(3) Every elliptic curve over $\mathbb{F}_3$ is isomorphic to one given by $y^2 = x^3 + a_2x^2 + a_4x + a_6$. `Magma` lists the isomorphism classes, with $\#E(\mathbb{F}_3)$, a and whether E is supersingular:

```
K := GF(3);  reps := [];
for a2, a4, a6 in K do
  ok, E := IsEllipticCurve([K | 0, a2, 0, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
  end if;
end for;
#reps;                                                          // 8
for E in reps do
  printf "%o  %o  %o  %o\n", aInvariants(E), #E,
         TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 0, 1, 0 ]  4  0  true
// [ 0, 0, 0, 2, 0 ]  4  0  true
// [ 0, 0, 0, 2, 1 ]  7  -3  true
// [ 0, 0, 0, 2, 2 ]  1  3  true
// [ 0, 1, 0, 0, 1 ]  6  -2  false
// [ 0, 1, 0, 0, 2 ]  3  1  false
// [ 0, 2, 0, 0, 1 ]  5  -1  false
// [ 0, 2, 0, 0, 2 ]  2  2  false
```

Which column decides supersingularity, by part (1)? Count the rational points of $y^2 = x^3 - x + 1$ and of $y^2 = x^3 - x - 1$ by hand, and explain why part (2) fails for $p = 3$.

Solution. **The third column, a. The two curves have 7 and 1 rational points.** E is supersingular exactly when $3 \mid a$, that is, when $\#E(\mathbb{F}_3) \equiv 1 \pmod 3$. The table shows the four classes with $a_2 = 0$ (these are the curves with $j = 0$) and the four with $a_2 \neq 0$. Over $\mathbb{F}_3$ every x satisfies $x^3 = x$, so $x^3 - x + 1 = 1$ for all $x \in \mathbb{F}_3$: the equation $y^2 = x^3 - x + 1$ has the two solutions $y = \pm 1$ for each of the three values of x, and with O, $\#E(\mathbb{F}_3) = 7$, $a = -3$. Likewise $x^3 - x - 1 = -1 = 2$ is not a square in $\mathbb{F}_3$, so $y^2 = x^3 - x - 1$ has no affine rational points and $\#E(\mathbb{F}_3) = 1$, $a = 3$. Both curves are supersingular with $\#E(\mathbb{F}_3) \neq 4$: part (2) fails for $p = 3$ because $a = \pm 3$ lies in the Hasse interval $|a| \leqslant 2\sqrt{3} \approx 3.46$.

(4) The same over $\mathbb{F}_2$, where a Weierstrass equation needs all five coefficients:

```
K := GF(2);  reps := [];
for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([K | a1, a2, a3, a4, a6]);
  if ok and not exists{F : F in reps | IsIsomorphic(E, F)} then
    Append(~reps, E);
  end if;
end for;
#reps;                                                          // 5
for E in reps do
  printf "%o  %o  %o  %o\n", aInvariants(E), #E,
         TraceOfFrobenius(E), IsSupersingular(E);
end for;
// [ 0, 0, 1, 0, 0 ]  3  0  true
```

```
// [ 0, 0, 1, 1, 0 ]  5  -2  true
// [ 0, 0, 1, 1, 1 ]  1  2   true
// [ 1, 0, 0, 0, 1 ]  4  -1  false
// [ 1, 0, 1, 0, 1 ]  2  1   false
```

Count the rational points of the three supersingular curves by hand. Which coefficient decides supersingularity over $\mathbb{F}_2$, and what does the analogue of part (2) say for $p = 2$?

SOLUTION. **3, 5 and 1 rational points. Supersingular means $a_1 = 0$, not $a = 0$.** Over $\mathbb{F}_2$, E is supersingular exactly when a is even, that is, when $\#E(\mathbb{F}_2)$ is odd. In the table this is the condition $a_1 = 0$. For the three supersingular curves, $y^2 + y$ takes the value 0 at $y = 0, 1$ and never the value 1:

- $y^2 + y = x^3$: two rational points over $x = 0$, none over $x = 1$. With O, $\#E(\mathbb{F}_2) = 3$ and $a = 0$.
- $y^2 + y = x^3 + x$: the right side is 0 for both x, so four affine rational points, $\#E(\mathbb{F}_2) = 5$ and $a = -2$.
- $y^2 + y = x^3 + x + 1$: the right side is 1 for both x, so no affine rational points, $\#E(\mathbb{F}_2) = 1$ and $a = 2$.

The analogue of part (2) would say $\#E(\mathbb{F}_2) = 3$, and two of the three supersingular curves violate it: $a = \pm 2$ lies in the Hasse interval $|a| \leqslant 2\sqrt{2} \approx 2.83$.

Remark. Again $j = 0$. Compare [Sil09, Exercise 5.7].

(5) Suppose $p \mid a$. Use Corollary 16.4 to write $\mathsf{V}^e = \lambda \circ \mathsf{F}^e$ with $\lambda \in \mathrm{Aut}(E)$, deduce that $[a] = ([1] + \lambda) \circ \mathsf{F}^e$, and conclude that $q \mid a^2$. So

$$a \equiv 0 \pmod{p} \iff q \mid a^2 \iff p^{\lceil e/2 \rceil} \mid a.$$

SOLUTION. If $p \mid a$, then by part (1) V^e is purely inseparable, of degree $q = p^e$. By Corollary 16.4, $\mathsf{V}^e = \lambda \circ \mathsf{F}^e$ with $\lambda \colon E^{(q)} \to E$ an isomorphism. Since $E^{(q)} = E$, $\lambda \in \mathrm{Aut}(E)$. Then

$$[a] = \mathsf{F}^e + \mathsf{V}^e = \mathsf{F}^e + \lambda \circ \mathsf{F}^e = ([1] + \lambda) \circ \mathsf{F}^e,$$

and taking degrees, which are multiplicative, $a^2 = \deg([1] + \lambda) \cdot q$ by Corollary 16.8 and Proposition 16.3. If $[1] + \lambda = 0$ this reads $a = 0$. So $q \mid a^2$. Conversely $q \mid a^2$ implies $p \mid a$. Finally, writing $a = p^k b$ with $p \nmid b$, $p^e \mid a^2 = p^{2k} b^2$ if and only if $2k \geqslant e$, that is, $k \geqslant \lceil e/2 \rceil$. For $q = p^{2i}$ this is $p^i \mid a$. For $q = p^{2i+1}$ it is $p^{i+1} \mid a$.

(6) Show that no elliptic curve over $\mathbb{F}_8$ has exactly 7 or exactly 11 rational points. `Magma` collects the values of $\#E(\mathbb{F}_8)$ over all Weierstrass equations, and the corresponding traces:

```
K := GF(8);  counts := {};
for a1, a2, a3, a4, a6 in K do
  ok, E := IsEllipticCurve([a1, a2, a3, a4, a6]);
  if ok then Include(~counts, #E); end if;
end for;
counts;                          // { 4, 5, 6, 8, 9, 10, 12, 13, 14 }
{ 9 - n : n in counts };         // { -5, -4, -3, -1, 0, 1, 3, 4, 5 }
```

Which integers in the Hasse interval are missing, and which traces belong to supersingular curves? Predict both from parts (1) and (5) before you look.

SOLUTION. **Missing: $a = \pm 2$. Supersingular: $a \in \{0, \pm 4\}$.** Over $\mathbb{F}_8$, $\#E(\mathbb{F}_8) = 7$ or 11 means $a = 2$ or $a = -2$. Then $2 \mid a$, so by part (5) $8 \mid a^2 = 4$, which is false. The Hasse interval

is $|a| \leqslant 2\sqrt{8} \approx 5.66$, so $a \in [-5, 5]$ and $\#E(\mathbb{F}_8) \in [4, 14]$. The values 7 and 11 are the only ones missing from the `Magma` set. The supersingular traces are the even a with $8 \mid a^2$, that is, $4 \mid a$: $a \in \{0, \pm 4\}$, and $\#E(\mathbb{F}_8) \in \{5, 9, 13\}$. All three occur.

Remark 16.12 (The traces of supersingular curves)**.** The identity $a^2 = \deg([1] + \lambda) \cdot q$ of Problem 22(5) says more than $q \mid a^2$. Since $a^2 \leqslant 4q$, the integer $d := \deg([1] + \lambda)$ is at most 4, so a supersingular curve over $\mathbb{F}_q$ has

$$a^2 \in \{0,\ q,\ 2q,\ 3q,\ 4q\}\,,$$

and only the values with dq a perfect square can occur. Over $\mathbb{F}_8$ these are $a^2 \in \{0, 16\}$, as found above; over $\mathbb{F}_p$ with $p \geqslant 5$ only $a = 0$; over $\mathbb{F}_3$ also $a^2 = 9 = 3q$, and over $\mathbb{F}_2$ also $a^2 = 4 = 2q$, the cases of parts (3) and (4). Which of the five values occur, over which q, is a theorem of Waterhouse. We do not need it.

17. The group structure of $E(\mathbb{F}_q)$

This section is [Sil09, Exercise 5.6].

In the following results, E is an elliptic curve over a field K and $m \geqslant 1$ is an integer with $\operatorname{char} K \nmid m$.

Corollary 17.1 (The m-torsion points [Sil09, Corollary III.6.4(b)])**.** *The group $E[m](\overline{K})$ is isomorphic to $\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$.*

Corollary 17.2 (Rational torsion and roots of unity [Sil09, Corollary III.8.1.1])**.** *If every m-torsion point of $E(\overline{K})$ is K-rational, then K contains the m-th roots of unity.*

Problem 23 (The group structure of $E(\mathbb{F}_q)$)**.**
Let E be an elliptic curve over $\mathbb{F}_q$.

(1) Show that there are integers $m, n \geqslant 1$ with $p \nmid m$ and $E(\mathbb{F}_q) \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$.

Solution. Put $G := E(\mathbb{F}_q)$, a finite abelian group, and $N := \#G$. Then G is the direct sum of its ℓ-primary parts G_ℓ over the primes $\ell \mid N$, and $G_\ell \cong \bigoplus_i \mathbb{Z}/\ell^{k_i}\mathbb{Z}$ for some exponents $k_1 \leqslant \ldots \leqslant k_s$, where the number s of cyclic factors is the dimension of the $\mathbb{F}_\ell$-vector space $G[\ell] = G_\ell[\ell]$. Now $G[\ell]$ is a subgroup of $E[\ell](\overline{\mathbb{F}}_q)$, which is $(\mathbb{Z}/\ell\mathbb{Z})^2$ for $\ell \neq p$ by Corollary 17.1, and 0 or $\mathbb{Z}/p\mathbb{Z}$ for $\ell = p$ by Corollary 16.9. So $s \leqslant 2$ for $\ell \neq p$, and $s \leqslant 1$ for $\ell = p$: write

$$G_\ell \cong \mathbb{Z}/\ell^{i_\ell}\mathbb{Z} \times \mathbb{Z}/\ell^{j_\ell}\mathbb{Z}$$

with $0 \leqslant i_\ell \leqslant j_\ell$ and $i_p = 0$. Put

$$m := \prod_\ell \ell^{i_\ell}, \qquad mn := \prod_\ell \ell^{j_\ell},$$

so that $m \mid mn$ and $p \nmid m$. By the Chinese remainder theorem, $G \cong \bigoplus_\ell G_\ell \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$.

(2) With m as in part (1), show that $q \equiv 1 \pmod{m}$.

Solution. Since $m \mid m$ and $m \mid mn$, the subgroup $G[m]$ of $G \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/mn\mathbb{Z}$ is $(\mathbb{Z}/m\mathbb{Z})^2$, with m^2 elements. It is contained in $E[m](\overline{\mathbb{F}}_q)$, which also has m^2 elements because $p \nmid m$ (Corollary 17.1). So every m-torsion point of $E(\overline{\mathbb{F}}_q)$ is $\mathbb{F}_q$-rational, and by Corollary 17.2 the group $\mathbb{F}_q^\times$ contains the m-th roots of unity, which form a cyclic group of order m (as $p \nmid m$). Since $\mathbb{F}_q^\times$ has order $q - 1$, $m \mid q - 1$.

(3) Suppose $q = p \geqslant 5$ and E is supersingular. Show that $m = 1$ or $m = 2$, and that $m = 1$ if $p \equiv 1 \pmod 4$.

Solution. By Problem 22(2), $\#E(\mathbb{F}_p) = p + 1$, so $m^2 n = p + 1$ and in particular $m^2 \mid p + 1$. By part (2), $m \mid p - 1$. Hence m divides $\gcd(p - 1, p + 1) = 2$, so $m \in \{1, 2\}$. If $m = 2$, then $4 = m^2$ divides $p + 1$, so $p \equiv 3 \pmod 4$. Therefore $p \equiv 1 \pmod 4$ forces $m = 1$, and $E(\mathbb{F}_p)$ is cyclic of order $p + 1$.

(4) Both values of m occur when $p \equiv 3 \pmod 4$. Over $\mathbb{F}_7$ the curves $y^2 = x^3 + x$ and $y^2 = x^3 - x$, both with $j = 1728$, are supersingular. Here `Invariants(AbelianGroup(E))` returns $[m, mn]$, or $[mn]$ when $m = 1$:

```
E1 := EllipticCurve([GF(7) | 1, 0]);  E2 := EllipticCurve([GF(7) | -1, 0]);
IsSupersingular(E1), IsSupersingular(E2);                    // true true
Invariants(AbelianGroup(E1));                                // [ 8 ]
Invariants(AbelianGroup(E2));                                // [ 2, 4 ]
[ j : j in GF(13) | IsSupersingular(EllipticCurveFromjInvariant(j)) ];  // [ 5 ]
E3 := EllipticCurveFromjInvariant(GF(13)!5);  aInvariants(E3);   // [ 1, 0, 0, 7, 2 ]
Invariants(AbelianGroup(E3));                                // [ 14 ]
Invariants(AbelianGroup(QuadraticTwist(E3)));                // [ 14 ]
```

Explain the two values of m over $\mathbb{F}_7$ from the rational points of order 2 of each curve, and the value over $\mathbb{F}_{13}$ from part (3).

Solution. **$m = 1$ for E_1 and $m = 2$ for E_2 over $\mathbb{F}_7$. Over $\mathbb{F}_{13}$, $m = 1$.** Both curves over $\mathbb{F}_7$ have $j = 1728$ and $7 \equiv 3 \pmod 4$, and `Magma` confirms that they are supersingular, with 8 rational points each. The rational points of order 2 are $(x_0, 0)$ with $x_0 \in \mathbb{F}_7$ a root of the cubic. For $y^2 = x^3 + x = x(x^2 + 1)$: since $7 \equiv 3 \pmod 4$, -1 is not a square in $\mathbb{F}_7$, so $x^2 + 1$ is irreducible and $x = 0$ is the only root. So $E_1(\mathbb{F}_7)$ has a single rational point of order 2, so $E_1(\mathbb{F}_7)[2] \neq (\mathbb{Z}/2\mathbb{Z})^2$, $m = 1$, and $E_1(\mathbb{F}_7) \cong \mathbb{Z}/8\mathbb{Z}$. For $y^2 = x^3 - x = x(x - 1)(x + 1)$ all three roots are rational, so $E_2(\mathbb{F}_7)[2] = (\mathbb{Z}/2\mathbb{Z})^2$ and $m = 2$: $E_2(\mathbb{F}_7) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$. Over $\mathbb{F}_{13}$, with $13 \equiv 1 \pmod 4$, the only supersingular j-invariant is $j = 5$, and part (3) predicts $m = 1$ for every supersingular curve, which is what `Magma` finds: both curves with $j = 5$ (the second is the quadratic twist of the first) have $E(\mathbb{F}_{13}) \cong \mathbb{Z}/14\mathbb{Z}$.

18. Infinitely many ordinary primes

This section is [Sil09, Exercise 5.11], with "ordinary" for "Hasse invariant 1". Silverman's exercise asks for infinitely many ordinary primes. The curve of part (4) and its supersingular primes are [Sil09, Example V.4.6]. The proof uses results that we have not proved.

Definition 18.1 (Reduction modulo p)**.** Let E be an elliptic curve over $\mathbb{Q}$, given by a Weierstrass equation with coefficients in $\mathbb{Z}$ and discriminant $\Delta \in \mathbb{Z}$. For a prime $p \nmid \Delta$, reducing the coefficients modulo p gives a Weierstrass equation over $\mathbb{F}_p$ with nonzero discriminant, hence an elliptic curve E_p over $\mathbb{F}_p$.

Definition 18.2 (Supersingular and ordinary primes)**.** A prime $p \nmid \Delta$ is supersingular or ordinary for E according as E_p is.

Definition 18.3 (Primes that split completely)**.** Let $L/\mathbb{Q}$ be a finite Galois extension. A prime p splits completely in L if $p\mathcal{O}_L$ is a product of $[L : \mathbb{Q}]$ distinct prime ideals of the ring of integers $\mathcal{O}_L$.

In the following results, E, Δ and E_p are as in Definition 18.1, and $L/\mathbb{Q}$ is a finite Galois extension.

Proposition 18.4 (Reduction modulo p [Sil09, Proposition VII.2.1])**.** *Let $p \nmid \Delta$. Every point of $E(\mathbb{Q}_p)$ can be written $[X : Y : Z]$ with $X, Y, Z \in \mathbb{Z}_p$ not all divisible by p, and reducing the coordinates modulo p gives a group homomorphism $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$.*

Proposition 18.5 (Reduction of torsion [Sil09, Proposition VII.3.1(b)])**.** *Let $p \nmid \Delta$ and let $m \geqslant 1$ with $p \nmid m$. The restriction of the reduction homomorphism $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ to the m-torsion points of $E(\mathbb{Q}_p)$ is injective.*

Proposition 18.6 (Completions at a completely split prime [Neu99])**.** *Let p be a prime that splits completely in L. For each prime $\mathfrak{p}$ of L above p the completion $L_\mathfrak{p}$ is $\mathbb{Q}_p$, so that L embeds in $\mathbb{Q}_p$.*

Theorem 18.7 (Chebotarev density theorem [Neu99])**.** *Infinitely many primes split completely in L: they have density $1/[L : \mathbb{Q}]$ among all primes.*

Problem 24 (Infinitely many ordinary primes)**.**
Fix a prime $\ell \geqslant 3$, and let $L \coloneqq \mathbb{Q}(E[\ell])$ be the field generated over $\mathbb{Q}$ by the affine coordinates of the nonzero ℓ-torsion points of $E(\overline{\mathbb{Q}})$.

(1) Show that L is a finite Galois extension of $\mathbb{Q}$, and that L contains the ℓ-th roots of unity.

Solution. The group $E[\ell](\overline{\mathbb{Q}})$ is finite (it has ℓ^2 elements by Corollary 17.1), so the coordinates of its points are algebraic numbers, each with finitely many conjugates, and L is a finite extension of $\mathbb{Q}$. If $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and P is an ℓ-torsion point, then $\sigma(P)$ is again an ℓ-torsion point: the multiplication-by-ℓ map is given by rational functions with coefficients in $\mathbb{Q}$, so $[\ell]\sigma(P) = \sigma([\ell]P) = O$. Hence σ permutes the finite set of coordinates generating L, and $\sigma(L) = L$. So $L/\mathbb{Q}$ is normal, hence Galois. The ℓ-torsion points of $E(\overline{\mathbb{Q}})$ all have coordinates in L, so Corollary 17.2, applied to E over $K = L$, gives $\mu_\ell \subset L$.

(2) Let $p \nmid \ell\Delta$ be a prime that splits completely in L. Show that $\ell^2 \mid \#E_p(\mathbb{F}_p)$ and that $p \equiv 1 \pmod{\ell}$.

Solution. By Proposition 18.6, choose an embedding $L \hookrightarrow \mathbb{Q}_p$. Every ℓ-torsion point of $E(\overline{\mathbb{Q}})$ has coordinates in L, so $E(\mathbb{Q}_p)$ contains a copy of $E[\ell](\overline{\mathbb{Q}}) \cong (\mathbb{Z}/\ell\mathbb{Z})^2$. By Propositions 18.4 and 18.5 with $m = \ell$, reduction maps this group injectively into $E_p(\mathbb{F}_p)$, so $E_p(\mathbb{F}_p)$ has a subgroup of order ℓ^2, and $\ell^2 \mid \#E_p(\mathbb{F}_p)$.
For the congruence, let $\zeta \in L \subset \mathbb{Q}_p$ be a primitive ℓ-th root of unity. It is a root of $x^\ell - 1$, so it lies in $\mathbb{Z}_p$, and its reduction $\bar{\zeta}$ is a root of $x^\ell - 1$ in $\mathbb{F}_p$. The ℓ roots $1, \zeta, \ldots, \zeta^{\ell-1}$ reduce to ℓ distinct elements of $\mathbb{F}_p$: their reductions are the roots of $x^\ell - 1 = \prod_i (x - \zeta^i)$ read modulo p, and $x^\ell - 1$ is separable over $\mathbb{F}_p$ because its derivative $\ell x^{\ell-1}$ does not vanish at a root, as $p \neq \ell$. Hence $\bar{\zeta} \in \mathbb{F}_p^\times$ has order exactly ℓ, and $\ell \mid p - 1$.

(3) Show that every prime $p \geqslant 5$ with $p \nmid \ell\Delta$ that splits completely in L is ordinary for E, and conclude. Where did you use $\ell \geqslant 3$?

Solution. **The hypothesis $\ell \geqslant 3$ is used exactly at $\ell \mid 2$.** Suppose $p \geqslant 5$ splits completely in L, with $p \nmid \ell\Delta$, and that E_p is supersingular. By Problem 22(2), $\#E_p(\mathbb{F}_p) = p+1$, so part (2) gives $\ell \mid p + 1$ and $\ell \mid p - 1$, hence $\ell \mid 2$, contradicting $\ell \geqslant 3$. So every such p is ordinary. By Theorem 18.7 there are infinitely many primes that split completely in L, and removing the finitely many primes dividing $6\ell\Delta$ leaves infinitely many ordinary primes. For $\ell = 2$ the two congruences say only that p is odd, and the argument gives nothing.

(4) Take the curve $E\colon y^2 + y = x^3 - x^2 - 10x - 20$, of discriminant -11^5, and $\ell = 3$. By part (2), a prime $p \neq 3, 11$ that splits completely in $\mathbb{Q}(E[3])$ has all nine 3-torsion points of

$E_p(\overline{\mathbb{F}}_p)$ rational. Magma lists the supersingular primes below 100 and the primes below 3000 with that property:

```
E := EllipticCurve([0, -1, 1, -10, -20]);          // y^2 + y = x^3 - x^2 - 10x - 20
Discriminant(E);                                    // -161051
good := [p : p in PrimesUpTo(3000) | p ne 11];
[p : p in good | p lt 100 and IsSupersingular(ChangeRing(E, GF(p)))];
// [ 2, 19, 29 ]
full := function(p, l)       // are all the l-torsion points of E_p rational?
  I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
  return #I eq 2 and I[1] mod l eq 0;
end function;
split := [p : p in good | p ne 3 and full(p, 3)];  split;
// [ 337, 523, 1087, 2437, 2719, 2749 ]
[p mod 3 : p in split];                                    // [ 1, 1, 1, 1, 1, 1 ]
[IsSupersingular(ChangeRing(E, GF(p))) : p in split];
// [ false, false, false, false, false, false ]
[p : p in good | p lt 300 and p ne 5 and full(p, 5)];      // and the same for l = 5:
// [ 31, 41, 61, 71, 101, 131, 151, 181, 191, 211, 241, 251, 271, 281 ]
```

Check the output against parts (2) and (3), and compare the list of supersingular primes with the list in the example cited before this problem. The last line runs the same test for $\ell = 5$: which primes below 300 appear, and what does that suggest about the field $\mathbb{Q}(E[5])$? (The rational point $(5, 5)$ of E has order 5.)

Solution. **The good primes $p \equiv 1 \pmod 5$. That $\mathbb{Q}(E[5]) = \mathbb{Q}(\mu_5)$.** The supersingular primes below 100 are 2, 19 and 29, as in [Sil09, Example V.4.6] once the errata add $p = 2$. The primes $p < 3000$ for which all nine 3-torsion points of E_p are rational are all $\equiv 1 \pmod 3$ and all ordinary, as parts (2) and (3) predict for the primes splitting completely in $\mathbb{Q}(E[3])$.
For $\ell = 5$ the list is every prime $p < 300$ with $p \nmid 5\Delta$ and $p \equiv 1 \pmod 5$ (the bad prime 11 is also $\equiv 1 \pmod 5$, but the listing leaves it out, as part (2) says nothing about it), which suggests that $\mathbb{Q}(E[5])$ is the field $\mathbb{Q}(\mu_5)$ of 5-th roots of unity: the primes splitting completely in $\mathbb{Q}(\mu_5)$ are exactly those with $p \equiv 1 \pmod 5$.

Remark. That is right for this curve. The rational point $(5, 5)$ generates a subgroup $\mathbb{Z}/5\mathbb{Z}$ of $E[5](\overline{\mathbb{Q}})$ on which the Galois group acts trivially, and pairing with it under the Weil pairing, which is nondegenerate and Galois invariant [Sil09, Proposition III.8.1(c),(d)], identifies the quotient $E[5](\overline{\mathbb{Q}})/\langle(5, 5)\rangle$ with $\mu_5(\overline{\mathbb{Q}})$, compatibly with the Galois action. That $\mathbb{Q}(E[5])$ is no larger than $\mathbb{Q}(\mu_5)$ is a further fact about this curve that we do not prove.
For $\ell = 3$, the primes found and the primes splitting completely in $\mathbb{Q}(E[3])$ are in fact the same: for $p \nmid 3\Delta$, the Frobenius at a prime of $\mathbb{Q}(E[3])$ above p acts on the 3-torsion points as F^1 acts on $E_p[3](\overline{\mathbb{F}}_p)$, so p splits completely if and only if $E_p[3](\overline{\mathbb{F}}_p)$ is rational. We do not prove this here.

Remark 18.8 (Supersingular primes)**.** The other direction is much harder. For the curve of Problem 24(4), D. H. Lehmer found exactly 27 odd supersingular primes below 31500 [Sil09, Example V.4.6]. For an elliptic curve over $\mathbb{Q}$ without complex multiplication, the supersingular primes have density 0 (Serre, Elkies), and yet there are infinitely many of them (Elkies) [Sil09, Theorems V.4.7 and V.4.9]. The Lang–Trotter conjecture predicts that about $c\sqrt{x}/\log x$ of them lie below x [Sil09, Conjecture V.4.8]. For a curve with complex multiplication, such as $y^2 = x^3 + x$, the supersingular primes have density 1/2 [Sil09, Examples V.4.4 and V.4.5].

Part 6. Reduction modulo p and a bound on the torsion

The aim is a `Magma` script, written from scratch, that bounds the torsion subgroup of an elliptic curve over $\mathbb{Q}$ using only the local theory of [Sil09, §§VII.1–VII.3]. Section 19 shows that reduction modulo an odd prime of good reduction is injective on the rational torsion points, Section 20 turns this into two bounds, and Section 21 tests them against the LMFDB. Each problem prints a piece of the script with its output. You prove the mathematics behind it.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by a Weierstrass equation with coefficients $a_1, \dots, a_6 \in \mathbb{Z}$ and discriminant Δ.
- The torsion subgroup $T \coloneqq E(\mathbb{Q})_{\mathrm{tors}}$.
- A prime p is good if $p \nmid \Delta$. Then reducing the equation modulo p gives an elliptic curve E_p over $\mathbb{F}_p$.
- For an abelian group G, $G[m]$ is the subgroup killed by m.
- A finite abelian group G is isomorphic to $\mathbb{Z}/n_1\mathbb{Z} \times \mathbb{Z}/n_2\mathbb{Z}$ for at most one pair of integers $1 \leqslant n_1 \mid n_2$. When it is, we write $n_1(G)$ and $n_2(G)$ for them.

19. Reduction is injective on torsion

Everything in this section is in [Sil09, §§VII.1–VII.3]. The curve of Problem 25 is from Application VII.3.5 there.

In the following definition and results, p is a good prime for E.

Definition 19.1 (Reduction modulo p)**.** We regard $E(\mathbb{Q})$ as a subgroup of $E(\mathbb{Q}_p)$, and write $P \mapsto \bar{P}$ for the reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ and $E_1(\mathbb{Q}_p)$ for its kernel.

Proposition 19.2 (Minimality [Sil09, Remark VII.1.1])**.** *The equation is minimal at p, because $v_p(\Delta) = 0 < 12$. So E_p is the reduction of E modulo p.*

Proposition 19.3 (Reduction is a homomorphism [Sil09, Proposition VII.2.1 and the discussion after it])**.** *The reduction map $E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$ is a surjective group homomorphism. Its kernel $E_1(\mathbb{Q}_p)$ consists of O and the $(x, y) \in E(\mathbb{Q}_p)$ with $v_p(x) < 0$.*

Proposition 19.4 (Torsion prime to p [Sil09, Proposition VII.3.1])**.** *If $p \nmid m$, then $E_1(\mathbb{Q}_p)[m] = \{O\}$, and the reduction map above restricts to an injection $E(\mathbb{Q}_p)[m] \hookrightarrow E_p(\mathbb{F}_p)$.*

Theorem 19.5 (Torsion of p-power order [Sil09, Theorem VII.3.4(b)])**.** *Let $P \in E(\mathbb{Q}_p)$ have exact order p^n with $n \geqslant 1$, and let $r \coloneqq \lfloor 1/(p^n - p^{n-1}) \rfloor$. Then $p^{2r}x(P) \in \mathbb{Z}_p$. In particular $x(P) \in \mathbb{Z}_p$ unless $p = 2$ and $n = 1$.*

Problem 25 (Injectivity of reduction on torsion)**.**

(1) Let $p \geqslant 3$ be a good prime. Show that the reduction map

$$E(\mathbb{Q}) \hookrightarrow E(\mathbb{Q}_p) \to E_p(\mathbb{F}_p)$$

is injective on T. In particular T is finite, and isomorphic to a subgroup of $E_p(\mathbb{F}_p)$.

Solution. By Proposition 19.3 it suffices to show that $T \cap E_1(\mathbb{Q}_p) = \{O\}$. Suppose $P \in T \cap E_1(\mathbb{Q}_p)$ has order $m \geqslant 2$, let ℓ be a prime dividing m, and put $Q \coloneqq (m/\ell)P$, which lies in $E_1(\mathbb{Q}_p)$ and has order ℓ. If $\ell \neq p$, this contradicts Proposition 19.4. If $\ell = p$, then Theorem 19.5 with $n = 1$ gives $x(Q) \in \mathbb{Z}_p$, since $p \geqslant 3$, whereas $v_p(x(Q)) < 0$ by Proposition 19.3. This is a contradiction. So reduction is injective on T, and T embeds in the finite group $E_p(\mathbb{F}_p)$.

Remark. No global input, such as the Mordell–Weil theorem, was used.

(2) The prime 2 is different: even if $p = 2$ is good, $T \to E_2(\mathbb{F}_2)$ need not be injective. Take for instance the curve $E\colon y^2 + xy = x^3 + 4x + 1$:

```
E := EllipticCurve([1, 0, 0, 4, 1]);            // y^2 + xy = x^3 + 4x + 1
Factorization(Integers()!Discriminant(E));      // [ <5, 2>, <13, 2> ]
P := E![-1/4, 1/8];  2*P eq E!0;                // true
```

So 2 is a good prime and $P = (-1/4, 1/8)$ is a rational point of order 2. Show that $\bar{P} = O$ in $E_2(\mathbb{F}_2)$. Which hypothesis of Proposition 19.4 and Theorem 19.5 fails, and what does Proposition 19.4 still give at $p = 2$?

Solution. **The hypothesis $p \nmid m$ fails. Torsion of odd order still injects.** Clearing denominators, $P = [-2 : 1 : 8]$, which reduces to $[0 : 1 : 0] = O$ modulo 2. So the distinct torsion points O and P have the same reduction. Proposition 19.4 needs $p \nmid m$, but $p = m = 2$. For $p = 2$ and $n = 1$, Theorem 19.5 has $r = 1$ and only gives $4x(P) \in \mathbb{Z}_2$ (which is true for $x(P) = -1/4$). Proposition 19.4 still shows that the torsion points of odd order inject into $E_2(\mathbb{F}_2)$.

Remark. The script skips $p = 2$.

20. Two bounds

This section is [Sil09, Application VII.3.2], and its three curves are [Sil09, Examples VII.3.3.1–VII.3.3.3].

Problem 26 (The order bound)**.**
The first piece of the script computes the greatest common divisor b of the orders $\#E_p(\mathbb{F}_p)$ over the good odd primes $p \leqslant N$:

```
// b = gcd of #E_p(F_p) over the good odd primes p <= N (there must be one)
OrderBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  return GCD([#ChangeRing(E, GF(p)) : p in good]);
end function;
E1 := EllipticCurve([0, 0, 1, -1, 1]);          // y^2 + y = x^3 - x + 1
Discriminant(E1);                               // -611
[#ChangeRing(E1, GF(p)) : p in [3, 5, 7]];      // [ 1, 8, 6 ]
OrderBound(E1, 30);                             // 1
E2 := EllipticCurve([0, 3]);                    // y^2 = x^3 + 3
Discriminant(E2);                               // -3888
[#ChangeRing(E2, GF(p)) : p in [5, 7, 11, 13]]; // [ 6, 13, 12, 9 ]
OrderBound(E2, 30);                             // 1
```

(1) Deduce that both curves have trivial torsion, and that $y^2 = x^3 + 3$ has infinitely many rational points.

Solution. By Problem 25(1), $\#T$ divides $\#E_p(\mathbb{F}_p)$ for every good odd prime p, hence divides b. Both outputs are $b = 1$: for $y^2 + y = x^3 - x + 1$ already $\#E_3(\mathbb{F}_3) = 1$, and for $y^2 = x^3 + 3$ the orders 6 and 13 at $p = 5, 7$ are coprime. So $T = \{O\}$ for both curves. The rational point $(1, 2)$ is not O, so it has infinite order, and its multiples are infinitely many distinct rational points.

(2) `OrderBound` skips the primes dividing the discriminant of the equation it is given. Compare two equations for the first curve, with $N = 7$:

```
E4 := EllipticCurve([0, 0, 27, -81, 729]);      // y^2 + 27y = x^3 - 81x + 729
Discriminant(E1), Discriminant(E4);             // -611 -324710451
IsIsomorphic(E1, E4);                           // true
OrderBound(E1, 7), OrderBound(E4, 7);           // 1 2
```

Here $-324710451 = -3^{12} \cdot 611$. Find a change of variables between the two equations. Explain why a non-minimal equation can make the script skip a prime it could have used, but never makes it use one it should not.

SOLUTION. $\boldsymbol{x = 9x'}$, $\boldsymbol{y = 27y'}$. Substituting $x = 9x'$, $y = 27y'$ in $y^2 + 27y = x^3 - 81x + 729$ gives

$$3^6(y'^2 + y') = 3^6(x'^3 - x' + 1), \quad \text{that is,} \quad y'^2 + y' = x'^3 - x' + 1.$$

This change of variables has $u = 3$, and the two discriminants differ by $u^{12} = 3^{12}$ [Sil09, §III.1]. So 3 is a good prime for $y^2 + y = x^3 - x + 1$, although it divides the discriminant of the other equation. For any equation with integer coefficients, a prime not dividing its discriminant is good, and Proposition 19.2 makes the equation minimal there, so Problem 25(1) applies. A non-minimal equation only has more primes dividing its discriminant, which the script skips: the bound can get weaker, never wrong. Here, with $N = 7$, the minimal equation uses $p = 3, 5, 7$ and $\#E_3(\mathbb{F}_3) = 1$ gives $b = 1$, while the other equation only uses $p = 5, 7$ and gives $b = \gcd(8, 6) = 2$.

Corollary 20.1 (The m-torsion [Sil09, Corollary III.6.4])**.** *Let E be an elliptic curve over a field K, and $m \geqslant 1$. If $\operatorname{char} K \nmid m$, then $E[m](\overline{K}) \cong (\mathbb{Z}/m\mathbb{Z})^2$. If $\operatorname{char} K = p > 0$, then $E[p^e](\overline{K})$ is cyclic for every $e \geqslant 1$.*

Problem 27 (The structure bound)**.**
The curve $y^2 = x^3 + x$ shows that the orders are not the whole story. `Invariants` returns $[n_1(G), n_2(G)]$ for $G = E_p(\mathbb{F}_p)$, omitting $n_1(G)$ when it is 1, and returns [] when G is trivial:

```
E3 := EllipticCurve([1, 0]);                    // y^2 = x^3 + x
Discriminant(E3);                               // -64
OrderBound(E3, 30);                             // 4
[Invariants(AbelianGroup(ChangeRing(E3, GF(p)))) : p in [3, 5, 7, 11, 13]];
// [ [ 4 ], [ 2, 2 ], [ 8 ], [ 12 ], [ 2, 10 ] ]
```

(1) Let G be a finite subgroup of $E(\overline{K})$, for an elliptic curve E over a field K. Show that $n_1(G)$ and $n_2(G)$ are defined, and that $n_2(G)$ is the exponent of G. Show that $\#E[n_1(G)](\overline{K}) = n_1(G)^2$, and that an integer $m \geqslant 1$ with $\#E[m](\overline{K}) = m^2$ satisfies $E[m](\overline{K}) \subseteq G$ if and only if $m \mid n_1(G)$. Deduce that $n_1(H) \mid n_1(G)$ and $n_2(H) \mid n_2(G)$ for every subgroup $H \subseteq G$.

SOLUTION. By the structure theorem,

$$G \cong \mathbb{Z}/d_1\mathbb{Z} \times \cdots \times \mathbb{Z}/d_k\mathbb{Z}, \qquad 1 \leqslant d_1 \mid \cdots \mid d_k,$$

for some integers $d_1, \ldots, d_k$ with $k \geqslant 2$. Take k as small as possible. If $k > 2$, then $d_1 > 1$, since otherwise the factor $\mathbb{Z}/d_1\mathbb{Z}$ could be dropped. For a prime $\ell \mid d_1$, the subgroup $G[\ell] \cong (\mathbb{Z}/\ell\mathbb{Z})^k$

lies in $E[\ell](\overline{K})$, which has at most ℓ^2 elements by Corollary 20.1, a contradiction. So $k = 2$, and $n_1(G) = d_1$, $n_2(G) = d_2$ are defined. The exponent of $\mathbb{Z}/n_1(G)\mathbb{Z} \times \mathbb{Z}/n_2(G)\mathbb{Z}$ is $n_2(G)$.
Put $n_1 := n_1(G)$, so that $G[n_1] \cong (\mathbb{Z}/n_1\mathbb{Z})^2$. If $\operatorname{char} K = p > 0$ and $p \mid n_1$, then $G[p] \cong (\mathbb{Z}/p\mathbb{Z})^2$ would lie in the cyclic group $E[p](\overline{K})$, which is impossible. So $\operatorname{char} K \nmid n_1$, and $E[n_1](\overline{K}) \cong (\mathbb{Z}/n_1\mathbb{Z})^2$ has n_1^2 elements. It contains $G[n_1]$, which has as many, so $E[n_1](\overline{K}) = G[n_1] \subseteq G$. Hence $m \mid n_1$ implies $E[m](\overline{K}) \subseteq E[n_1](\overline{K}) \subseteq G$. Conversely, if $\#E[m](\overline{K}) = m^2$ and $E[m](\overline{K}) \subseteq G$, then $E[m](\overline{K}) = G[m]$, and

$$m^2 = \#G[m] = \gcd(m, n_1)\gcd(m, n_2(G)) \leqslant \gcd(m, n_1)\, m.$$

So $\gcd(m, n_1) = m$, that is, $m \mid n_1$.
Finally, let $H \subseteq G$. Then $E[n_1(H)](\overline{K}) \subseteq H \subseteq G$ has $n_1(H)^2$ elements, so $n_1(H) \mid n_1(G)$. And $n_2(G)$ kills H, so $n_2(H) \mid n_2(G)$.

(2) Let b_1 and b_2 be the greatest common divisors of $n_1(E_p(\mathbb{F}_p))$ and of $n_2(E_p(\mathbb{F}_p))$ over the good odd primes $p \leqslant N$. Show that $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $b_1 b_2 \mid b$. This is the second piece of the script:

```
// [b1, b2]: n1(T) | b1 and n2(T) | b2
StructureBound := function(E, N)
  Delta := Integers()!Discriminant(E);
  good := [p : p in PrimesUpTo(N) | p gt 2 and Delta mod p ne 0];
  b1 := 0;  b2 := 0;
  for p in good do
    I := Invariants(AbelianGroup(ChangeRing(E, GF(p))));
    b1 := GCD(b1, #I eq 2 select I[1] else 1);
    b2 := GCD(b2, #I eq 0 select 1 else I[#I]);
  end for;
  return [b1, b2];
end function;
StructureBound(E3, 30);                          // [ 1, 2 ]
P := E3![0, 0];  2*P eq E3!0;                    // true
```

Conclude that $T = \{O, (0,0)\} \cong \mathbb{Z}/2\mathbb{Z}$ for $y^2 = x^3 + x$. Why does b stay at 4 however large N is?

Solution. **Because $4 \mid \#E_p(\mathbb{F}_p)$ for every prime $p \geqslant 3$.** By Problem 25(1), T is isomorphic to a subgroup of $E_p(\mathbb{F}_p) \subseteq E_p(\overline{\mathbb{F}}_p)$ for each good odd prime p, so part (1) gives $n_i(T) \mid n_i(E_p(\mathbb{F}_p))$ for $i = 1, 2$, so $n_i(T) \mid b_i$. Also $b_1 b_2$ divides $n_1(E_p(\mathbb{F}_p))\, n_2(E_p(\mathbb{F}_p)) = \#E_p(\mathbb{F}_p)$ for each p, hence divides b: the structure bound is never worse than the order bound. For $y^2 = x^3 + x$ the output $[b_1, b_2] = [1, 2]$ makes T cyclic of order dividing 2, and $(0,0)$ has order 2. The order bound stays at 4 because $4 \mid \#E_p(\mathbb{F}_p)$ for every prime $p \geqslant 3$ [Sil09, Exercise 5.12]: the orders cannot tell $\mathbb{Z}/4\mathbb{Z}$ from $(\mathbb{Z}/2\mathbb{Z})^2$, but the group structures can.

Remark. Already $p = 3$ and $p = 5$ suffice, with $E_3(\mathbb{F}_3) \cong \mathbb{Z}/4\mathbb{Z}$ and $E_5(\mathbb{F}_5) \cong (\mathbb{Z}/2\mathbb{Z})^2$. This is Silverman's argument.

(3) Show that a rational point P is a torsion point if and only if $b_2 P = O$, and that rational torsion points generating a subgroup of order $b_1 b_2$ generate T.

Solution. If $P \in T$, its order divides the exponent $n_2(T)$, which divides b_2. Conversely, $b_2 P = O$ says that P has finite order. If rational torsion points generate $H \subseteq T$ with $\#H = b_1 b_2$, then $\#T = n_1(T) n_2(T) \leqslant b_1 b_2 = \#H$, so $H = T$.

21. How good are the bounds? The LMFDB

The L-functions and modular forms database [LMF25] lists millions of elliptic curves over $\mathbb{Q}$ with their torsion subgroups, computed by methods beyond [Sil09, Chapter VII]. We use it as the answer key.

Proposition 21.1 (Reduction of isogenies [Mil20, proof of Theorem V.4.1])**.** *An isogeny over $\mathbb{Q}$ reduces to an isogeny over $\mathbb{F}_p$ at every good prime p of both equations.*

Proposition 21.2 (Isogenous curves over $\mathbb{F}_p$ [Sil09, Exercise 5.4(a)])**.** *Isogenous curves over $\mathbb{F}_p$ have the same number of rational points.*

Problem 28 (Testing the bounds)**.**

(1) Here are the curves of smallest conductor with each torsion structure that occurs over $\mathbb{Q}$, with their LMFDB labels, a-invariants and torsion structures ([1] for the trivial group):

```
curves := [<"11.a1", [0, -1, 1, -7820, -263580], [1]>,
  <"14.a2", [1, 0, 1, -171, -874], [2]>,
  <"19.a2", [0, 1, 1, -9, -15], [3]>, <"15.a4", [1, 1, 1, -80, 242], [4]>,
  <"11.a2", [0, -1, 1, -10, -20], [5]>, <"14.a6", [1, 0, 1, 4, -6], [6]>,
  <"26.b2", [1, -1, 1, -3, 3], [7]>, <"15.a8", [1, 1, 1, 35, -28], [8]>,
  <"54.b2", [1, -1, 1, -14, 29], [9]>, <"66.c3", [1, 0, 0, -45, 81], [10]>,
  <"90.c7", [1, -1, 1, -122, 1721], [12]>, <"15.a2", [1, 1, 1, -135, -660], [2, 2]>,
  <"15.a5", [1, 1, 1, -10, -10], [2, 4]>, <"30.a6", [1, 0, 1, -19, 26], [2, 6]>,
  <"210.e6", [1, 0, 0, -1070, 7812], [2, 8]>];
for c in curves do
  E := EllipticCurve(c[2]);
  printf "%-7o LMFDB %-9o order bound %-3o structure bound %o\n",
    c[1], c[3], OrderBound(E, 50), StructureBound(E, 50);
end for;
// 11.a1   LMFDB [ 1 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a2   LMFDB [ 2 ]     order bound 6   structure bound [ 1, 6 ]
// 19.a2   LMFDB [ 3 ]     order bound 3   structure bound [ 1, 3 ]
// 15.a4   LMFDB [ 4 ]     order bound 8   structure bound [ 1, 4 ]
// 11.a2   LMFDB [ 5 ]     order bound 5   structure bound [ 1, 5 ]
// 14.a6   LMFDB [ 6 ]     order bound 6   structure bound [ 1, 6 ]
// 26.b2   LMFDB [ 7 ]     order bound 7   structure bound [ 1, 7 ]
// 15.a8   LMFDB [ 8 ]     order bound 8   structure bound [ 1, 8 ]
// 54.b2   LMFDB [ 9 ]     order bound 9   structure bound [ 1, 9 ]
// 66.c3   LMFDB [ 10 ]    order bound 10  structure bound [ 1, 10 ]
// 90.c7   LMFDB [ 12 ]    order bound 12  structure bound [ 1, 12 ]
// 15.a2   LMFDB [ 2, 2 ]  order bound 8   structure bound [ 2, 4 ]
// 15.a5   LMFDB [ 2, 4 ]  order bound 8   structure bound [ 2, 4 ]
// 30.a6   LMFDB [ 2, 6 ]  order bound 12  structure bound [ 2, 6 ]
// 210.e6  LMFDB [ 2, 8 ]  order bound 16  structure bound [ 2, 8 ]
```

Check that no bound contradicts the LMFDB. Where is each bound attained?

Solution. **No contradiction. The bounds are attained eleven and twelve times.** In every row $n_1(T) \mid b_1$, $n_2(T) \mid b_2$ and $\#T \mid b$. The order bound equals $\#T$ for eleven curves, all but 11.a1, 14.a2, 15.a4 and 15.a2. The structure bound equals $[n_1(T), n_2(T)]$ for twelve, all but 11.a1, 14.a2 and 15.a2.

(2) The curves 15.a4 and 15.a8 are isogenous:

```
E := EllipticCurve([1, 1, 1, -80, 242]);        // 15.a4
F := EllipticCurve([1, 1, 1, 35, -28]);          // 15.a8
good := [p : p in PrimesUpTo(50) | p ne 3 and p ne 5];
[#ChangeRing(E, GF(p)) : p in good] eq [#ChangeRing(F, GF(p)) : p in good];
// true
[<p, Invariants(AbelianGroup(ChangeRing(E, GF(p)))),
     Invariants(AbelianGroup(ChangeRing(F, GF(p))))> : p in [7, 11, 13]];
// [ <7, [ 2, 4 ], [ 8 ]>, <11, [ 2, 8 ], [ 16 ]>, <13, [ 16 ], [ 2, 8 ]> ]
```

Using Propositions 21.1 and 21.2, explain why the order bound cannot separate isogenous curves, and why the structure bound separates these two.

Solution. If E and E' are isogenous over $\mathbb{Q}$, then E_p and E'_p are isogenous over $\mathbb{F}_p$ at each good prime of both equations, so $\#E_p(\mathbb{F}_p) = \#E'_p(\mathbb{F}_p)$ [Sil09, Exercise 5.4(a)], and the order bounds agree. The group structures need not: at $p = 7$ the groups are $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$ for 15.a4 and $\mathbb{Z}/8\mathbb{Z}$ for 15.a8, so $b_2 = 4$ for the first and $b_2 = 8$ for the second, and the structure bound is sharp for both.

(3) The LMFDB lists $(5, -2)$ as a torsion generator of 15.a4:

```
E := EllipticCurve([1, 1, 1, -80, 242]);  P := E![5, -2];        // 15.a4
[k*P eq E!0 : k in [1..4]];                        // [ false, false, false, true ]
```

Prove that $T \cong \mathbb{Z}/4\mathbb{Z}$ for 15.a4 without trusting the LMFDB. Can you do the same for 11.a1, 14.a2 or 15.a2?

Solution. **Not for 11.a1, 14.a2 or 15.a2: no choice of primes helps.** The output shows that P has order 4, so $\langle P \rangle$ is a subgroup of T of order $4 = b_1 b_2$, and $T = \langle P \rangle$ by Problem 27(3). For 11.a1 the bound $[1, 5]$ allows $T = \{O\}$ or $\mathbb{Z}/5\mathbb{Z}$. To prove $T = \{O\}$ we would have to rule out a rational point of order 5, and the local theory gives no finite list of candidates: it makes the coordinates of a torsion point of order at least 3 integral, and $4x(P)$ integral for a point P of order 2 [Sil09, Application VII.3.5], but it does not bound them. Likewise 14.a2 needs a rational point of order 3 ruled out, and 15.a2 one of order 4. No choice of primes helps. The isogenous curves 11.a2, 14.a6 and 15.a5 have rational torsion of orders 5, 6 and 8, and the same good primes as their partners: the discriminants are -11 and -11^5, $-2^{18} \cdot 7$ and $-2^6 \cdot 7^3$, $3^8 \cdot 5^2$ and $3^4 \cdot 5^4$ (in general, isogenous curves have good reduction at the same primes [Sil09, Corollary VII.7.2]). So by part (2) these orders divide $\#E_p(\mathbb{F}_p)$ at every good odd prime p of 11.a1, 14.a2 and 15.a2, and b never drops below them. Nor does b_2: a prime dividing the order of a finite abelian group divides its exponent, and $8 \mid n_1 n_2$ with $n_1 \mid n_2$ forces $4 \mid n_2$.

Remark. Remark 21.3 says what is missing.

(4) Find these curves yourself: on https://www.lmfdb.org/EllipticCurve/Q/, search by "torsion structure", such as `[2,4]`, and sort by conductor. On a curve's page, find its equation, discriminant, torsion subgroup and bad primes. Pick a curve of your own and compute both bounds.

Solution. The section "Invariants" of a curve's page gives the equation, the a-invariants and the discriminant; "Mordell–Weil group" gives the torsion structure and generators; the table "Local data" lists the bad primes. The LMFDB's torsion must satisfy both bounds, and its generators can be checked as in part (3).

Remark 21.3 (Beyond the local theory)**.** Over $\mathbb{Q}$ the bounds complete to an algorithm. For $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$, a rational torsion point $P \neq O$ has integer coordinates, and either $2P = O$ or $y(P)^2 \mid 4A^3 + 27B^2$ (Nagell–Lutz, [Sil09, Corollary VIII.7.2]). This leaves finitely many candidates, and the test $b_2 P = O$ decides which are torsion points. Mazur proved that $E(\mathbb{Q})_{\mathrm{tors}}$ is one of the fifteen groups in the listing [Sil09, Theorem VIII.7.5].

Part 7. Good and bad reduction: examples

An elliptic curve over $\mathbb{Q}_p$ has good, multiplicative or additive reduction, and multiplicative reduction is split or nonsplit [Sil09, §VII.5]. The problems below run through concrete curves of each kind: Problem 29 reads the type off an equation, Problem 30 counts the rational points of the singular reduction, and Problem 31 changes the type by twisting the curve or extending the field. Problem 32 ($\star$) goes further: the curves Frey attached to solutions of $A + B = C$ have no additive reduction at all.

Standing notation.

- A prime p, with $v = v_p$ its valuation on $\mathbb{Q}_p$, extended to $\overline{\mathbb{Q}}_p$.
- An elliptic curve E over $\mathbb{Q}_p$, given by a Weierstrass equation with coefficients in $\mathbb{Z}_p$, with discriminant Δ and invariants c_4, c_6 [Sil09, §III.1].
- When the equation is minimal, E_p is its reduction modulo p, a Weierstrass cubic over $\mathbb{F}_p$.
- We write
$$a_p \coloneqq p + 1 - \#E_p(\mathbb{F}_p),$$
counting the singular point too.

22. Reading off the type

The following definition and results hold for every prime p, and are used throughout. In them, K is $\mathbb{Q}_p$ or a finite extension of it, with normalized valuation v_K, ring of integers $\mathcal{O}_K$ and residue field k, and E is an elliptic curve over K given by a Weierstrass equation with coefficients in $\mathcal{O}_K$.

Definition 22.1 (Reduction types [Sil09, §VII.5])**.** Suppose that the equation is minimal, and consider its reduction, a Weierstrass cubic over k. Then E has good reduction if the reduction is nonsingular, multiplicative reduction if the reduction has a node, and additive reduction if it has a cusp. Multiplicative reduction is split if the slopes of the tangent lines at the node lie in k, and nonsplit otherwise.

Proposition 22.2 (Discriminant and c_4 [Sil09, §III.1])**.** *For an equation $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (that is, $a_1 = a_3 = 0$),*

$$\Delta = -16\big(4a_2^3a_6 - a_2^2a_4^2 - 18a_2a_4a_6 + 4a_4^3 + 27a_6^2\big), \qquad c_4 = 16\big(a_2^2 - 3a_4\big), \tag{22.1}$$

which follow from the formulas of.

Proposition 22.3 (Minimal equations [Sil09, Remark VII.1.1])**.** *Any other Weierstrass equation for E has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$ for some $u \in K^\times$. So if $v_K(\Delta) < 12$ or $v_K(c_4) < 4$, the equation is minimal.*

Proposition 22.4 (Singular Weierstrass cubics [Sil09, Proposition III.1.4(a)])**.** *A Weierstrass cubic over a field with $\Delta = 0$ has exactly one singular point over an algebraic closure. It is a node, with two distinct tangent lines, if $c_4 \neq 0$, and a cusp, with one tangent line, if $c_4 = 0$.*

Proposition 22.5 (Criteria for the reduction type [Sil09, Proposition VII.5.1])**.** *Suppose that the equation is minimal. Then E has good reduction if $v_K(\Delta) = 0$, multiplicative reduction if $v_K(\Delta) > 0$ and $v_K(c_4) = 0$, and additive reduction if $v_K(\Delta) > 0$ and $v_K(c_4) > 0$.*

Proposition 22.6 (Extending the field [Sil09, Proposition VII.5.4(b)])**.** *Let L be a finite extension of K with ramification index e, so that $v_L = e\, v_K$ on K. If E has good, respectively multiplicative, reduction, then so does E_L.*

Proposition 22.7 (Potential good reduction [Sil09, Proposition VII.5.5])**.** *There is a finite extension L of K such that E_L has good reduction if and only if $j(E) \in \mathcal{O}_K$.*

Problem 29 (Reading off the type)**.**
Let $p \geqslant 5$, and let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p. For each equation, decide whether it is minimal at p, and find the reduction type of the curve over $\mathbb{Q}_p$: good, split multiplicative, nonsplit multiplicative, or additive.

Solution. Here $a_4 = 0$, so (22.1) reads $\Delta = -16a_6(4a_2^3 + 27a_6)$ and $c_4 = 16a_2^2$. Since $p \geqslant 5$, the factor 16 is a unit.

(1) $y^2 = x^3 + x^2 + p$.

Solution. **Minimal. Split multiplicative.**

$$\Delta = -16p(27p + 4), \qquad v(\Delta) = 1 < 12, \qquad c_4 = 16,$$

so the equation is minimal (Proposition 22.3), and c_4 is a unit: multiplicative reduction (Proposition 22.5). The reduction $y^2 = x^3+x^2 = x^2(x+1)$ has its node at $(0,0)$, where the lowest-degree terms $y^2 - x^2$ give the tangent lines $y = \pm x$. Their slopes lie in $\mathbb{F}_p$: split.

(2) $y^2 = x^3 + \varepsilon x^2 + p$.

Solution. **Minimal. Nonsplit multiplicative.**

$$\Delta = -16p(4\varepsilon^3 + 27p), \qquad v(\Delta) = 1, \qquad c_4 = 16\varepsilon^2,$$

using $p \nmid \varepsilon$. So c_4 is a unit: multiplicative. The tangent lines to $y^2 = x^2(x + \varepsilon)$ at $(0,0)$ are $y^2 = \varepsilon x^2$, with slopes $\pm\sqrt{\varepsilon} \notin \mathbb{F}_p$: nonsplit.

(3) $y^2 = x^3 + p$.

Solution. **Minimal. Additive.** $\Delta = -432p^2$ has $v(\Delta) = 2$, so the equation is minimal, and $c_4 = 0$: additive. The reduction $y^2 = x^3$ has a cusp at $(0,0)$.

(4) $y^2 = x^3 + p^6$.

Solution. **Not minimal. Good reduction.** Now $\Delta = -432p^{12}$ and $c_4 = 0$, so Proposition 22.3 does not apply, and the reduction $y^2 = x^3$ of this equation has a cusp. But the equation is not minimal: substituting $x = p^2x'$, $y = p^3y'$ and dividing by p^6 gives

$$y'^2 = x'^3 + 1, \qquad \Delta = -432,$$

a unit. So the curve has good reduction, and the cusp was an artifact of a bad model.

Remark. This is why reduction types are defined with a minimal equation.

(5) $y^2 = x^3 + x^2 + p^{12}$.

Solution. **Minimal. Split multiplicative.** $\Delta = -16p^{12}(27p^{12} + 4)$ has $v(\Delta) = 12$, but $c_4 = 16$ is a unit, so the equation is minimal by Proposition 22.3: a scaling as in (4) would divide c_4 by u^4 and leave $\mathbb{Z}_p$. The reduction is the same as in (1).

Remark. Large $v(\Delta)$ alone does not make an equation non-minimal.

(6) Here is Magma's opinion at $p = 7$ and $\varepsilon = 3$. For each curve it prints the valuation of the minimal discriminant, the Kodaira symbol of the reduction (I_0 for good reduction, I_n with $n \geqslant 1$ for multiplicative, anything else for additive), and the polynomial $1 - a_7T$, or $1 - a_7T + 7T^2$ in the case of good reduction.

```
R<T> := PolynomialRing(Integers());
p := 7; eps := 3;                          // 3 is not a square modulo 7
Info := function(a)
  E := EllipticCurve(a); L := LocalInformation(E, p);
  return <L[2], L[5], R!EulerFactor(E, p)>;
end function;
Info([0, 1, 0, 0, p]);                     // <1, I1, -T + 1>
Info([0, eps, 0, 0, p]);                   // <1, I1, T + 1>
Info([0, 0, 0, 0, p]);                     // <2, II, 1>
Info([0, 0, 0, 0, p^6]);                   // <0, I0, 7*T^2 + 4*T + 1>
Info([0, 1, 0, 0, p^12]);                  // <12, I12, -T + 1>
```

Check that it agrees with your answers. Note the values of a_7 at the curves with bad reduction. Problem 30 explains them.

Solution. **They agree.** I_1, I_1 and I_{12} are multiplicative, II is additive, and Magma finds the good reduction of (4), with $v(\Delta) = 0$ for its minimal equation:

	(1)	(2)	(3)	(4)	(5)
$v(\Delta)$, minimal	1	1	2	0	12
Kodaira symbol	I_1	I_1	II	I_0	I_{12}
reduction	split	nonsplit	additive	good	split
a_7	1	-1	0	-4	1

23. Counting rational points on the singular reduction

Definition 23.1 (The nonsingular locus)**.** When the equation of E is minimal, the nonsingular locus E_p^{ns} is E_p minus its singular point, if it has one.

Problem 30 (Counting with lines through the singular point)**.**
Let E be an elliptic curve over $\mathbb{Q}_p$ with bad reduction, and let S be the singular point of its reduction E_p.

(1) Show that $S \in E_p(\mathbb{F}_p)$, and that $S \neq O$.

Solution. By Proposition 22.5 and Proposition 22.4, E_p has exactly one singular point S over $\overline{\mathbb{F}}_p$. The Frobenius automorphism σ of $\overline{\mathbb{F}}_p$ maps singular points of E_p to singular points, because E_p and its partial derivatives have coefficients in $\mathbb{F}_p$. So $\sigma(S) = S$, and $S \in E_p(\mathbb{F}_p)$. The rational point $O = [0 : 1 : 0]$ is nonsingular: in the chart $Y = 1$ the equation is

$$z + a_1xz + a_3z^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3,$$

whose linear part z does not vanish.

(2) Show that a change of variables over $\mathbb{F}_p$ moves S to $(0, 0)$ and puts E_p in the form

$$y^2 + a_1xy = x^3 + a_2x^2,$$

and that the tangent lines at S are the lines $y = \alpha x$ with $g(\alpha) = 0$, where $g(t) \coloneqq t^2 + a_1t - a_2$.

Solution. Write $S = (x_0, y_0)$ and substitute $x + x_0$ for x and $y + y_0$ for y. This keeps the Weierstrass form and has coefficients in $\mathbb{F}_p$. In the new equation

$$f(x, y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6 = 0,$$

the rational point $(0, 0)$ lies on the curve, so $a_6 = 0$, and is singular, so

$$a_4 = -\frac{\partial f}{\partial x}(0, 0) = 0, \qquad a_3 = \frac{\partial f}{\partial y}(0, 0) = 0.$$

The tangent lines at $(0, 0)$ are given by the terms of lowest degree, $y^2 + a_1xy - a_2x^2$, which factor as $(y - \alpha x)(y - \alpha' x)$ with α, α' the roots of g.

(3) Let $t \in \mathbb{F}_p$. Show that the line $y = tx$ meets E_p at S and, if $g(t) \neq 0$, at exactly one more $\mathbb{F}_p$-rational point $P_t = \big(g(t), t\,g(t)\big)$, and nowhere else in the affine plane. Show that $t \mapsto P_t$ is a bijection from $\{t \in \mathbb{F}_p : g(t) \neq 0\}$ onto $E_p^{\mathrm{ns}}(\mathbb{F}_p) - \{O\}$.

Solution. Substituting $y = tx$ gives

$$t^2x^2 + a_1tx^2 = x^3 + a_2x^2, \quad \text{that is,} \quad x^2\big(g(t) - x\big) = 0.$$

So the affine points of E_p on the line are S and, if $g(t) \neq 0$, the point P_t. If $g(t) = 0$, the line is tangent at S and meets E_p only there. P_t is nonsingular, since $P_t \neq S$. Conversely, a nonsingular affine point (x, y) has $x \neq 0$, since $x = 0$ forces $y^2 = 0$, which is S. So it lies on the line of slope $t = y/x$ through S, with $g(t) \neq 0$, and equals P_t. The two maps are inverse to each other.

(4) Deduce that $\#E_p^{\mathrm{ns}}(\mathbb{F}_p) = p + 1 - r$, where r is the number of roots of g in $\mathbb{F}_p$, and hence that

$$a_p = \begin{cases} 1 & \text{for split multiplicative reduction,} \\ -1 & \text{for nonsplit multiplicative reduction,} \\ 0 & \text{for additive reduction.} \end{cases}$$

Compare with the output in Problem 29(6).

Solution. By (3), $\#E_p^{\mathrm{ns}}(\mathbb{F}_p) = 1 + (p - r)$, counting O. Adding S,

$$\#E_p(\mathbb{F}_p) = p + 2 - r, \qquad \text{so} \qquad a_p = r - 1.$$

By Definition 22.1 and (2), $r = 2$ for split multiplicative reduction, $r = 0$ for nonsplit, and $r = 1$ for additive reduction, where g is a square. So $a_p = 1, -1, 0$. This matches Problem 29(6).

Remark. The three values are the numbers of $\mathbb{F}_p$-points of $\mathbb{G}_{\mathrm{m}}$, of a twisted multiplicative group and of $\mathbb{G}_{\mathrm{a}}$: E_p^{ns} is isomorphic to one of these as a group [Sil09, Proposition III.2.5], [Poo08, §6.3].

(5) Check (4) by hand for $y^2 = x^3 + x^2$ and $y^2 = x^3 + 2x^2$ over $\mathbb{F}_5$.

Solution. **$a_5 = 1$ and $a_5 = -1$.** For $y^2 = x^3 + x^2$ over $\mathbb{F}_5$, $g(t) = t^2 - 1$ has the roots ± 1, and $P_t = (t^2 - 1, t^3 - t)$ for $t = 0, 2, 3$ gives $(4, 0)$, $(3, 1)$ and $(3, 4)$. With O and S there are 5 rational points, so $a_5 = 1$. For $y^2 = x^3 + 2x^2$, $g(t) = t^2 - 2$ has no roots, since 2 is not a square modulo 5, and $t = 0, \ldots, 4$ gives $(3, 0)$, $(4, 4)$, $(2, 4)$, $(2, 1)$ and $(4, 1)$: with O and S, 7 rational points, so $a_5 = -1$.

24. Changing the curve, changing the field

Problem 31 (Twists and extensions)**.**
Let $p \geqslant 5$. For $d \in \mathbb{Q}_p{}^\times$ and $E\colon y^2 = x^3 + a_2x^2 + a_4x + a_6$, the quadratic twist of E by d is

$$E^d\colon y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6.$$

(1) Show that $(x, y) \mapsto (dx, d^{3/2}y)$ is an isomorphism $E \to E^d$ over $\mathbb{Q}_p(\sqrt{d})$, and that $\Delta(E^d) = d^6\Delta(E)$, $c_4(E^d) = d^2c_4(E)$ and $j(E^d) = j(E)$.

Solution. If $y^2 = x^3 + a_2x^2 + a_4x + a_6$, then

$$(d^{3/2}y)^2 = d^3y^2 = (dx)^3 + da_2(dx)^2 + d^2a_4(dx) + d^3a_6,$$

so the image lies on E^d, and $(x, y) \mapsto (x/d, d^{-3/2}y)$ is the inverse. This is a change of variables with $u = d^{-1/2}$, which multiplies Δ by $u^{-12} = d^6$ and c_4 by $u^{-4} = d^2$ [Sil09, Table 3.1]. Hence $j = c_4^3/\Delta$ is unchanged.

Remark. Or use (22.1): every term of Δ is multiplied by d^6, and every term of c_4 by d^2.

From now on, $E\colon y^2 = x^3 + x^2 + p$ is the curve of Problem 29(1), with split multiplicative reduction.

(2) Let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p. Show that E^ε has nonsplit multiplicative reduction, and that $a_p(E^\varepsilon) = -a_p(E)$. Since $\mathbb{Q}_p(\sqrt{\varepsilon})$ is an unramified extension of $\mathbb{Q}_p$, what does this say about which features of the reduction are preserved by unramified extensions?

Solution. **Unramified extensions preserve the reduction type, but not split versus nonsplit.** $E^\varepsilon\colon y^2 = x^3 + \varepsilon x^2 + \varepsilon^3 p$ has $v(\Delta) = v(\Delta(E)) = 1$ and $v(c_4) = 0$, so it is minimal with multiplicative reduction, and its reduction $y^2 = x^2(x + \varepsilon)$ has tangent slopes $\pm\sqrt{\varepsilon} \notin \mathbb{F}_p$: nonsplit. By Problem 30, $a_p(E) = 1$ and $a_p(E^\varepsilon) = -1$. Over the unramified extension $\mathbb{Q}_p(\sqrt{\varepsilon})$, whose residue field is $\mathbb{F}_{p^2}$, the two curves become isomorphic, and both have split multiplicative reduction there.

(3) Show that $E^p\colon y^2 = x^3 + px^2 + p^4$ has additive reduction. Over which quadratic extension of $\mathbb{Q}_p$ does it acquire multiplicative reduction? Show that it has good reduction over no finite extension of $\mathbb{Q}_p$.

Solution. **Additive. Multiplicative over $\mathbb{Q}_p(\sqrt{p})$. Never good.** By (1),

$$v(\Delta(E^p)) = 6 + 1 = 7 < 12, \qquad v(c_4(E^p)) = v(16p^2) = 2 > 0,$$

so the equation is minimal, with additive reduction. Its reduction is $y^2 = x^3$. Over $\mathbb{Q}_p(\sqrt{p})$, $E^p \cong E$ by (1), and E keeps its multiplicative reduction by Proposition 22.6. Finally $v(j) = 3v(c_4) - v(\Delta) = -1$ for E, and hence for E^p, so $j(E^p) \notin \mathbb{Z}_p$ and E^p has good reduction over no finite extension (Proposition 22.7).

(4) Let $E_3\colon y^2 = x^3 + p$, and let L be a finite extension of $\mathbb{Q}_p$ with ramification index e. Show that E_3 has good reduction over L if and only if $6 \mid e$.

Solution. Over L, $v_L(\Delta) = e\,v(-432p^2) = 2e$. If E_3 has good reduction over L, a minimal equation over L has $v_L(\Delta') = 0$, and $\Delta' = u^{-12}\Delta$ by Proposition 22.3, so $2e = 12v_L(u)$ and $6 \mid e$. Conversely, if $6 \mid e$, write $p = \varpi^e w$ and substitute $x = \varpi^{e/3}x'$, $y = \varpi^{e/2}y'$. Dividing by ϖ^e gives

$$y'^2 = x'^3 + w, \qquad \Delta = -432w^2,$$

a unit: good reduction. For instance $L = \mathbb{Q}_p(p^{1/6})$ works.

(5) Show that $y^2 = x^3 + px$ has additive reduction, and good reduction over L as in (4) if and only if $4 \mid e$.

SOLUTION. **Additive. Good over L if and only if $4 \mid e$.** Here $\Delta = -64p^3$ and $c_4 = -48p$, so $v(\Delta) = 3$ and $v(c_4) = 1$: minimal, with additive reduction. Over L, $v_L(\Delta) = 3e$, and the argument of (4) gives good reduction if and only if $12 \mid 3e$, that is, $4 \mid e$. For "if", substitute $x = \varpi^{e/2}x'$, $y = \varpi^{3e/4}y'$ to get

$$y'^2 = x'^3 + wx', \qquad \Delta = -64w^3,$$

a unit.

Remark. Both E_3 and this curve have $j \in \mathbb{Z}_p$ (0 and 1728), as Proposition 22.7 requires.

25. ⋆ Frey curves

This section is [Sil09, Exercise 8.23], with the conductor of [Mil20, Aside II.3.5]. The LMFDB [LMF25] is the answer key.

Problem 32 (Frey curves are semistable)**.**
An elliptic curve over $\mathbb{Q}$ is semistable if it has good or multiplicative reduction at every prime. Let A, B, C be nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$. Let

$$E\colon y^2 = x(x + A)(x - B).$$

(1) Show that $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + AB + B^2)$.

SOLUTION. The roots of $x(x+A)(x-B)$ are 0, $-A$ and B, with differences A, B and $A+B = C$. For $y^2 = (x - e_1)(x - e_2)(x - e_3)$,

$$\Delta = 16\prod_{i<j}(e_i - e_j)^2, \qquad c_4 = 16\big(e_1^2 + e_2^2 + e_3^2 - e_1e_2 - e_1e_3 - e_2e_3\big),$$

which follow from (22.1) with $a_2 = -(e_1 + e_2 + e_3)$, $a_4 = e_1e_2 + e_1e_3 + e_2e_3$ and $a_6 = -e_1e_2e_3$. Here they give $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + B^2 + AB)$.

(2) Let p be an odd prime dividing ABC. Show that the equation is minimal at p and that E has multiplicative reduction at p. Show that the reduction is split if and only if $-B$, A or B is a square modulo p, according as p divides A, B or C.

SOLUTION. Since A, B, C are pairwise coprime ($A + B = C$ and $\gcd(A, B, C) = 1$), p divides exactly one of them. If $p \mid A$, then $c_4 \equiv 16B^2$. If $p \mid B$, then $c_4 \equiv 16A^2$. If $p \mid C$, then $B \equiv -A$ and $c_4 \equiv 16A^2$. In each case c_4 is a unit, so the equation is minimal (Proposition 22.3) and the reduction is multiplicative (Proposition 22.5). The reduction and the tangent lines at its node are:

- $p \mid A$: $y^2 = x^2(x - B)$, node $(0, 0)$, tangents $y^2 = -Bx^2$;
- $p \mid B$: $y^2 = x^2(x + A)$, node $(0, 0)$, tangents $y^2 = Ax^2$;
- $p \mid C$: $y^2 = x(x + A)^2$, node $(-A, 0)$, tangents $y^2 = -A(x + A)^2$, and $-A \equiv B$.

So the slopes lie in $\mathbb{F}_p$ exactly when $-B$, A, respectively B, is a square modulo p.

(3) Suppose that $A \equiv 0 \pmod{16}$ and $B \equiv 3 \pmod 4$. Show that the substitution $x = 4x'$, $y = 8y' + 4x'$ gives an equation with coefficients in $\mathbb{Z}$,

$$y'^2 + x'y' = x'^3 + \frac{A - B - 1}{4}x'^2 - \frac{AB}{16}x',$$

with discriminant $(ABC)^2/2^8$ and $c_4 = A^2 + AB + B^2$. Deduce that E has good reduction at 2 if $v_2(A) = 4$, and multiplicative reduction at 2 if $v_2(A) \geqslant 5$. So E is semistable.

Solution. Substituting and dividing by 64, $(8y'+4x')^2 = 4x'(4x'+A)(4x'-B)$ becomes the displayed equation of the statement. The coefficients are integers because $A-B-1 \equiv 0-3-1 \equiv 0 \pmod 4$ and $16 \mid A$. This is a change of variables with $u=2$, so (Proposition 22.3)

$$\Delta' = 2^{-12}\Delta = (ABC)^2/2^8, \qquad c_4' = 2^{-4}c_4 = A^2+AB+B^2.$$

Since A is even and B odd, c_4' is odd, so the new equation is minimal at 2. Since B and C are odd, $v_2(\Delta') = 2v_2(A) - 8$: good reduction if $v_2(A) = 4$, and multiplicative reduction if $v_2(A) \geqslant 5$ (Proposition 22.5). With (2), and good reduction at the odd primes not dividing ABC, E is semistable. (At odd p, the substitution has $u = 2$, a unit, so it does not change the reduction type there.)

(4) The conductor of E is $N = \prod_p p^{f_p}$, where $f_p = 0$ if E has good reduction at p, $f_p = 1$ if it has multiplicative reduction, and $f_p \geqslant 2$ if it has additive reduction. Under the hypotheses of part (3), compute N in terms of A, B, C. Find the curves for $(A,B,C) = (16,-1,15)$ and $(32,-5,27)$ in the LMFDB, and check their conductors and their split and nonsplit primes against your answers.

Solution. **N is the product of the odd primes dividing ABC, times 2 if $32 \mid A$.** Every f_p is 0 or 1, and $f_p = 1$ exactly at the odd primes dividing ABC, and at 2 when $32 \mid A$. For $(16,-1,15)$, $N = 15$: the curve is 15a3 in Cremona's labels, with split multiplicative reduction at 5 ($B = -1$ is a square modulo 5) and nonsplit at 3 (-1 is not a square modulo 3). For $(32,-5,27)$, $N = 30$: the curve is 30a2, split at 3 ($B = -5 \equiv 1$), nonsplit at 5 ($A = 32 \equiv 2$ is not a square modulo 5), and multiplicative at 2.

(5) Suppose that $a^\ell + b^\ell = c^\ell$ with $\ell \geqslant 5$ prime and a, b, c nonzero coprime integers. Show that, after reordering and changing signs, $(A,B,C) = (a^\ell, b^\ell, c^\ell)$ satisfies the hypotheses of part (3), and that the resulting curve has conductor $\prod_{p \mid abc} p$ and minimal discriminant $(abc)^{2\ell}/2^8$.

Solution. Exactly one of a, b, c is even, since they are coprime and $a^\ell + b^\ell = c^\ell$. Since ℓ is odd, $x \mapsto x^\ell$ commutes with signs, so the relation can be rewritten as $A + B = C$ with A the ℓ-th power of the even one: $a^\ell + b^\ell = c^\ell$ itself if a is even, and $c^\ell + (-b)^\ell = a^\ell$ if c is even. Then $v_2(A) \geqslant \ell \geqslant 5$. If $B \equiv 1 \pmod 4$, replace (A,B,C) by $(-A,-B,-C)$. By (4) and (3), the conductor is the product of all primes dividing ABC, which is $\prod_{p \mid abc} p$, and the minimal discriminant is $(ABC)^2/2^8 = (abc)^{2\ell}/2^8$, an ℓ-th power up to 2^8.

Remark. Frey observed that such a curve would be very strange. Ribet proved that, if it were modular, its ℓ-torsion would come from a modular form of level 2, and there is none. Wiles and Taylor–Wiles proved that semistable elliptic curves over $\mathbb{Q}$ are modular, so the curve, and the solution, cannot exist [Mil20, §V.9].

Part 8. Compactness and inertia

Let E be an elliptic curve over a local field K. The rational points of E with nonsingular reduction form a subgroup $E_0(K)$ of finite index in $E(K)$ [Sil09, Theorem VII.6.1 and Corollary VII.6.2], and E has good reduction if and only if the inertia group acts trivially on its torsion, by the criterion of Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]. Problem 33 proves the finiteness of $E(K)/E_0(K)$ by a compactness argument. Problem 34 studies how the inertia group acts on the torsion of a curve with potential good reduction.

Standing notation.

- A finite extension K of $\mathbb{Q}_p$, with normalized valuation v, ring of integers $\mathcal{O}_K$, uniformizer π and residue field $k = \mathcal{O}_K/\pi\mathcal{O}_K$, a finite field of characteristic p. (Silverman's Chapter VII assumes that K and k are perfect, and here both are.)
- A fixed algebraic closure $\overline{K}$ of K.
- A bar denotes reduction modulo π.
- An elliptic curve E over K, given by a minimal Weierstrass equation $F(X,Y,Z) = 0$, with $F \in \mathcal{O}_K[X,Y,Z]$ homogeneous of degree 3. It embeds E in $\mathbb{P}^2$ as $V_+(F)$.
- The primitive coordinates $[X : Y : Z]$ of $P \in \mathbb{P}^2(K)$: coordinates in $\mathcal{O}_K$, not all in $\pi\mathcal{O}_K$. Every P has them, and they are unique up to a factor in $\mathcal{O}_K^\times$.
- The reduction $\bar{P} \coloneqq [\bar{X} : \bar{Y} : \bar{Z}] \in \mathbb{P}^2(k)$ of $P \in \mathbb{P}^2(K)$, computed from primitive coordinates, hence well defined.
- The reduced curve E_π, defined by $\bar{F} = 0$. If $P \in E(K)$, then $\bar{P} \in E_\pi(k)$.

26. Finiteness by compactness

This section is [Sil09, Exercise 7.6]. The argument uses that k is finite, so it does not apply, for instance, to the completion of the maximal unramified extension of $\mathbb{Q}_p$: a field complete with respect to a discrete valuation, with residue field $\overline{\mathbb{F}}_p$ (item (6)). Yet this is the case that the criterion of Néron–Ogg–Shafarevich needs: Silverman's proof of it uses the finiteness of $E(K^{\mathrm{unr}})/E_0(K^{\mathrm{unr}})$, where K^{unr} is the maximal unramified extension of K, whose residue field is an algebraic closure of k [Sil09, proof of Theorem VII.7.1].

Definition 26.1 (The topology of K)**.** We give K the topology defined by v: the sets $a + \pi^n\mathcal{O}_K$, for $a \in K$ and $n \in \mathbb{Z}$, form a basis, and each of them is both open and closed. With it, K is a Hausdorff topological field.

Definition 26.2 (The subgroup $E_0(K)$)**.** Let E_π^{ns} be E_π minus its singular point, if it has one. Then

$$E_0(K) \coloneqq \{P \in E(K) : \bar{P} \in E_\pi^{\mathrm{ns}}(k)\}.$$

Definition 26.3 (Regular functions [Poo06, Definition 1.8.3])**.** A function $\psi \in K(E)$ is regular at $Q \in E(\overline{K})$ if $\psi = g/h$ with $g, h \in K[X,Y,Z]$ homogeneous of the same degree and $h(Q) \neq 0$. Then $\psi(Q) \coloneqq g(Q)/h(Q)$, an element of K if $Q \in E(K)$. The functions regular at Q form a discrete valuation ring with fraction field $K(E)$.

Proposition 26.4 ($E_0(K)$ is a subgroup [Sil09, Proposition VII.2.1])**.** *The set $E_0(K)$ is a subgroup of $E(K)$.*

Proposition 26.5 (Compactness [Poo17, §1.1.2])**.** *The ring $\mathcal{O}_K$ is compact: it is complete, and for every n it is covered by the $(\#k)^n$ translates of $\pi^n\mathcal{O}_K$. Equivalently, K is locally compact.*

Proposition 26.6 (Morphisms $E \to E$ over K)**.**

(1) The morphisms $\alpha\colon E \to E$ defined over K are exactly the points $\alpha = [\psi_0 : \psi_1 : \psi_2] \in E(K(E))$, that is, $\psi_0, \psi_1, \psi_2 \in K(E)$, not all zero, with $F(\psi_0, \psi_1, \psi_2) = 0$ [Poo17, Proposition 3.6.5(b)]*.*
(2) If $Q \in E(\overline{K})$ and ψ_0, ψ_1, ψ_2 are regular at Q and not all zero there, then $\alpha(Q) = [\psi_0(Q) : \psi_1(Q) : \psi_2(Q)]$ [Sil09, §I.3]*.*
(3) Replacing each ψ_i by $\phi\psi_i$, for $\phi \in K(E)^\times$, gives the same point of $E(K(E))$, and for every $Q \in E(\overline{K})$ some ϕ makes $\phi\psi_0, \phi\psi_1, \phi\psi_2$ regular at Q and not all zero there: a power of a uniformizer of the discrete valuation ring of Definition 26.3 will do [Sil09, proof of Proposition II.2.1]*.*

Proposition 26.7 (Translations [Sil09, Theorem III.3.6 and Example III.4.7])**.** *Let $P \in E(K)$. The translation $\tau_P\colon E \to E$, $Q \mapsto Q + P$, is an isomorphism defined over K, with inverse τ_{-P}.*

Problem 33 ($E(K)/E_0(K)$ is finite)**.**
Let $q\colon K^3-\{0\}\to\mathbb{P}^2(K)$ be the quotient map. Give $K^3-\{0\}$ the product topology and $\mathbb{P}^2(K)$ the quotient topology: $U\subseteq\mathbb{P}^2(K)$ is open if and only if $q^{-1}(U)$ is open. Let

$$S\coloneqq\{\mathbf{x}=(x_0,x_1,x_2)\in\mathcal{O}_K^3 : x_i\in\mathcal{O}_K^\times \text{ for some } i\}$$

be the set of primitive coordinate vectors.

(1) Show that S is compact and open in $K^3-\{0\}$, and that $q(S)=\mathbb{P}^2(K)$. Deduce that $\mathbb{P}^2(K)$ is compact. Show that $U\subseteq\mathbb{P}^2(K)$ is open if and only if $q^{-1}(U)\cap S$ is open in S.

Solution. $S=\mathcal{O}_K^3-(\pi\mathcal{O}_K)^3$. By Proposition 26.5 and Tychonoff's theorem, $\mathcal{O}_K^3$ is compact, and $(\pi\mathcal{O}_K)^3$ is open, so S is closed in a compact space, hence compact. It is also open: S is the union over i of the sets $\mathcal{O}_K\times\mathcal{O}_K\times\mathcal{O}_K$ with the i-th factor replaced by $\mathcal{O}_K^\times$, and $\mathcal{O}_K^\times$ is open, being the union of the balls $u+\pi\mathcal{O}_K$ with u a unit. If $\mathbf{x}\in K^3-\{0\}$ and $m=\min_i v(x_i)$, then $\pi^{-m}\mathbf{x}\in S$ and $q(\pi^{-m}\mathbf{x})=q(\mathbf{x})$, so $q(S)=\mathbb{P}^2(K)$. As q is continuous, $\mathbb{P}^2(K)=q(S)$ is compact.
If U is open, then $q^{-1}(U)\cap S$ is open in S. Conversely, suppose that $V\coloneqq q^{-1}(U)\cap S$ is open in S, hence in $K^3-\{0\}$, since S is open. Every $\mathbf{x}\in K^3-\{0\}$ is $c\,\mathbf{s}$ with $c\in K^\times$ and $\mathbf{s}\in S$, and $q^{-1}(U)$ is stable under $K^\times$, so

$$q^{-1}(U)=\bigcup_{c\in K^\times} cV.$$

Multiplication by c is a homeomorphism of $K^3-\{0\}$, so this is a union of open sets, and U is open.

(2) Show that $E(K)=V_+(F)(K)$ is closed in $\mathbb{P}^2(K)$, and deduce that it is compact.

Solution. The set $q^{-1}(E(K))=\{\mathbf{x}\in K^3-\{0\} : F(\mathbf{x})=0\}$ is closed, because F is a continuous function $K^3\to K$ (K is a topological field) and $\{0\}$ is closed in the Hausdorff space K. So $E(K)$ is closed in the compact space $\mathbb{P}^2(K)$, hence compact.

(3) Let $g,h\in K[X,Y,Z]$ be homogeneous of the same degree. Show that $W\coloneqq\{Q\in\mathbb{P}^2(K) : h(Q)\neq 0\}$ is open, and that $W\to K$, $Q\mapsto g(Q)/h(Q)$, is continuous. Deduce from Propositions 26.6 and 26.7 that every translation $\tau_P\colon E(K)\to E(K)$ is a homeomorphism.

Solution. As in (2), the set where h vanishes is closed, so W is open. The function $G(\mathbf{x})\coloneqq g(\mathbf{x})/h(\mathbf{x})$ is continuous on the open set $q^{-1}(W)$, and $G(c\mathbf{x})=G(\mathbf{x})$ for $c\in K^\times$, since g and h have the same degree. So $G=\theta\circ q$ for a function $\theta\colon W\to K$. If $V\subseteq K$ is open, then $q^{-1}(\theta^{-1}(V))=G^{-1}(V)$ is open in $q^{-1}(W)$, which is open in $K^3-\{0\}$. So $\theta^{-1}(V)$ is open, and θ is continuous.
Let $P,Q\in E(K)$. By Proposition 26.7, τ_P is a morphism defined over K. Write $\tau_P=[\psi_0:\psi_1:\psi_2]$ as in Proposition 26.6, with the ψ_i regular at Q and $\psi_j(Q)\neq 0$ for some j (part (3)). Write $\psi_i=g_i/h_i$ with $h_i(Q)\neq 0$ (Definition 26.3). On the open neighborhood $U\coloneqq E(K)\cap\{h_0h_1h_2\neq 0\}$ of Q in $E(K)$, each ψ_i is regular and, by the first part, continuous. So $U'\coloneqq\{Q'\in U : \psi_j(Q')\neq 0\}$ is an open neighborhood of Q. On U', by part (2), τ_P is the composition of the continuous maps

$$U'\to K^3-\{0\},\ Q'\mapsto\big(\psi_0(Q'),\psi_1(Q'),\psi_2(Q')\big),\qquad\text{and}\qquad q\colon K^3-\{0\}\to\mathbb{P}^2(K).$$

So τ_P is continuous on each set of an open cover of $E(K)$, hence continuous. Its inverse τ_{-P} is continuous for the same reason, so τ_P is a homeomorphism.

(4) Show that the reduction map $\mathbb{P}^2(K)\to\mathbb{P}^2(k)$, $P\mapsto\bar{P}$, is locally constant. Deduce that $E_0(K)$ is open in $E(K)$, and also closed.

Solution. For $\mathbf{x} \in S$, the open neighborhood $\mathbf{x} + (\pi\mathcal{O}_K)^3$ of $\mathbf{x}$ lies in S, and every $\mathbf{x}'$ in it has $\bar{\mathbf{x}}' = \bar{\mathbf{x}}$. So $\mathbf{x} \mapsto \overline{q(\mathbf{x})}$ is locally constant on S. Hence for every subset $T \subseteq \mathbb{P}^2(k)$, the set $q^{-1}(\{P : \bar{P} \in T\}) \cap S$ is a union of such neighborhoods, so open in S, and $\{P : \bar{P} \in T\}$ is open by (1). Its complement is the set for $\mathbb{P}^2(k) - T$, so it is also closed. With $T = E_\pi^{\mathrm{ns}}(k)$, $E_0(K)$ is the intersection of $E(K)$ with an open and closed subset of $\mathbb{P}^2(K)$.

(5) Prove that $E(K)/E_0(K)$ is finite.

Solution. By Proposition 26.4, the cosets $P + E_0(K) = \tau_P(E_0(K))$ partition $E(K)$, and each is open by (3) and (4). By (2), $E(K)$ is compact, so finitely many cosets cover it. Since the cosets are disjoint and nonempty, each one must appear in this finite subcover, so there are finitely many.

(6) Let K' be a field complete with respect to a discrete valuation whose residue field k' is infinite, for instance algebraically closed. Show that its ring of integers is not compact. Which step of the argument above fails for an elliptic curve over K'?

Solution. **The first step, (1): $\mathbb{P}^2(K')$ is not compact.** Choose lifts $a_\alpha \in \mathcal{O}_{K'}$ of the elements $\alpha \in k'$. The balls $a_\alpha + \pi\mathcal{O}_{K'}$ are open, pairwise disjoint and cover $\mathcal{O}_{K'}$, and there are infinitely many of them. Since none can be removed, $\mathcal{O}_{K'}$ is not compact. So the first step fails: $\mathcal{O}_{K'}^3$, S and $\mathbb{P}^2(K')$ are not compact, and for an elliptic curve E' over K', the cover of $E'(K')$ by the cosets of $E'_0(K')$ need have no finite subcover.

Remark. The finiteness of $E'(K')/E'_0(K')$ is still true, but it needs the theory of Néron models [Sil09, Corollary VII.6.2].

Compare [Poo17, §2.6.2 and Proposition 2.6.1], which define the same kind of topology on $X(K)$ for every variety X over K, and show that $X(K)$ is compact when X is proper and K is a local field.

27. Inertia acting on torsion

This section is [Sil09, Exercise 7.9], with the hypothesis $m \geqslant 3$ added to its first two parts (item (5) explains why) and a `Magma` check. We use the notation of [Sil09, §VII.4].

Definition 27.1 (Reduction types)**.** The curve E has good reduction if E_π is nonsingular, that is, if $v(\Delta) = 0$, and bad reduction otherwise. It has potential good reduction if it has good reduction over some finite extension of K.

Definition 27.2 (Inertia)**.** Let $\mathrm{Gal}_K := \mathrm{Gal}(\overline{K}/K)$, $K^{\mathrm{unr}} \subseteq \overline{K}$ the maximal unramified extension of K, and $I_v := \mathrm{Gal}(\overline{K}/K^{\mathrm{unr}})$ the inertia group. A set on which Gal_K acts is unramified if I_v acts trivially on it.

Definition 27.3 (Ramification of finite extensions)**.** For an extension L of K inside $\overline{K}$, let $\mathrm{Gal}_L := \mathrm{Gal}(\overline{K}/L)$. If L/K is finite and Galois, the inertia group of L/K is the image of I_v in $\mathrm{Gal}(L/K)$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_L)$. The extension L/K is unramified if its inertia group is trivial, and tamely ramified if its order is prime to p.

Definition 27.4 (Torsion fields and Tate modules [Sil09, §III.7])**.** For $m \geqslant 1$ with $p \nmid m$, $K(E[m])$ is the field generated over K by the coordinates of the points of $E[m](\overline{K})$, so $\mathrm{Gal}_{K(E[m])}$ is the kernel of the action of Gal_K on $E[m](\overline{K})$. For a prime $\ell \neq p$, the Tate module $T_\ell(E)$ is the inverse limit of the $E[\ell^n](\overline{K})$ along the surjections $[\ell]$, and $\rho_\ell \colon \mathrm{Gal}_K \to \mathrm{Aut}\, T_\ell(E)$ is the action of Gal_K on it.

In the following results, $\ell \neq p$ is a prime and $m \geqslant 1$ is an integer with $p \nmid m$.

Proposition 27.5 (Minimal equations [Sil09, Remark VII.1.1 and §III.1])**.** *A Weierstrass equation with coefficients in $\mathcal{O}_K$ and $v(\Delta) < 12$ is minimal. For $y^2 = x^3 + Ax + B$, $\Delta = -16(4A^3 + 27B^2)$.*

Corollary 27.6 (Torsion [Sil09, Corollary III.6.4(b)]). *The group $E[m](\overline{K})$ is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^2$.*

Proposition 27.7 (The Tate module [Sil09, Proposition III.7.1(a)]). *The Tate module $T_\ell(E)$ is isomorphic to $\mathbb{Z}_\ell^2$. Hence each projection $T_\ell(E) \to E[\ell^n](\overline{K})$ is a surjective homomorphism of Gal_K-modules, and its kernel is $\ell^n T_\ell(E)$, since both have index ℓ^{2n}.*

Proposition 27.8 (Good reduction [Sil09, Proposition VII.4.1]). *If E has good reduction, then $E[m](\overline{K})$ and $T_\ell(E)$ are unramified.*

Theorem 27.9 (Néron–Ogg–Shafarevich [Sil09, Theorem VII.7.1]). *If $T_\ell(E)$ is unramified, then E has good reduction.*

Corollary 27.10 (Potential good reduction [Sil09, Corollary VII.7.3]). *The curve E has potential good reduction if and only if $\rho_\ell(I_v)$ is finite.*

Proposition 27.11 (Inertia groups of finite extensions [Sil09, proof of Corollary VII.7.3]). *If $H \subseteq I_v$ is a closed normal subgroup of finite index, there is a finite extension K' of K inside $\overline{K}$ whose inertia group $\mathrm{Gal}(\overline{K}/K'K^{\mathrm{unr}}) = I_v \cap \mathrm{Gal}_{K'}$ is H.*

Problem 34 (The inertia group of $K(E[m])/K$).
Suppose that E has potential good reduction, and let J be the subgroup of the elements of I_v that act trivially on $E[m](\overline{K})$ for every $m \geqslant 1$ with $p \nmid m$.

(1) Let ℓ be a prime and $r \geqslant 1$ with $\ell^r \geqslant 3$, and let $g \in \mathrm{GL}_2(\mathbb{Z}_\ell)$ be an element of finite order with $g \equiv I \pmod{\ell^r}$. Show that $g = I$.

Solution. Suppose that $g \neq I$. Some power of g has prime order q, and it is still congruent to I modulo ℓ^r, so we may assume that g has order q. Write $g = I + \ell^s h$ with $h \in M_2(\mathbb{Z}_\ell)$ not divisible by ℓ. Then $s \geqslant r$, so $\ell^s \geqslant 3$. Expanding,

$$0 = g^q - I = q\ell^s h + \sum_{i=2}^{q} \binom{q}{i} \ell^{is} h^i.$$

If $q \neq \ell$, every term of the sum is divisible by ℓ^{2s}, so $q\ell^s h \equiv 0 \pmod{\ell^{s+1}}$ and, q being a unit, $h \equiv 0 \pmod{\ell}$: a contradiction. If $q = \ell$, then ℓ divides $\binom{\ell}{i}$ for $2 \leqslant i < \ell$, so those terms are divisible by ℓ^{2s+1}, hence by ℓ^{s+2}. The last term is divisible by $\ell^{\ell s}$, and $\ell s \geqslant s+2$ because $s(\ell-1) \geqslant 2$: this is where $\ell^s \geqslant 3$ is needed. So $\ell^{s+1} h \equiv 0 \pmod{\ell^{s+2}}$, and again $h \equiv 0 \pmod{\ell}$.
Remark. For $\ell^r = 2$ the statement is false: $g = -I$ has order 2 and $-I \equiv I \pmod 2$.

(2) Show that $J = I_v \cap \ker \rho_\ell$ for every prime $\ell \neq p$, and that J has finite index in I_v.

Solution. Let $J_\ell := I_v \cap \ker \rho_\ell$. An element of J acts trivially on every $E[\ell^n](\overline{K})$, hence on their inverse limit $T_\ell(E)$, so $J \subseteq J_\ell$. Conversely, $\rho_\ell(I_v)$ is finite by Corollary 27.10, so J_ℓ, the kernel of $I_v \to \rho_\ell(I_v)$, has finite index. It is normal, and closed, being the intersection of the kernels of the actions on the finite sets $E[\ell^n](\overline{K})$. By Proposition 27.11, J_ℓ is the inertia group of a finite extension K' of K, again a finite extension of $\mathbb{Q}_p$. Over K', $T_\ell(E)$ is unramified, so E has good reduction over K' (Theorem 27.9). Then $E[m](\overline{K})$ is unramified over K' for every m with $p \nmid m$ (Proposition 27.8), that is, J_ℓ acts trivially on it. So $J_\ell \subseteq J$.

(3) Let $m \geqslant 3$ with $p \nmid m$. Show that $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and deduce that the inertia group of $K(E[m])/K$ is isomorphic to I_v/J, independently of m.

Solution. By definition, $J \subseteq I_v \cap \mathrm{Gal}_{K(E[m])}$. Conversely, since $m \geqslant 3$, it has a divisor $\ell^r \geqslant 3$ with ℓ prime: an odd prime factor of m, or 4 if m is a power of 2. Then $\ell \neq p$, as $p \nmid m$, and $E[\ell^r] \subseteq E[m]$. Let $\sigma \in I_v \cap \mathrm{Gal}_{K(E[m])}$. Then σ acts trivially on $E[\ell^r](\overline{K}) \cong T_\ell(E)/\ell^r T_\ell(E)$

(Proposition 27.7), so the matrix of $\rho_\ell(\sigma)$ in a $\mathbb{Z}_\ell$-basis of $T_\ell(E)$ is congruent to I modulo ℓ^r. It has finite order, since $\rho_\ell(I_v)$ is finite (Corollary 27.10), so $\rho_\ell(\sigma) = I$ by (1), and $\sigma \in J$ by (2). So $J = I_v \cap \mathrm{Gal}_{K(E[m])}$, and the inertia group of $K(E[m])/K$ is isomorphic to $I_v/(I_v \cap \mathrm{Gal}_{K(E[m])}) = I_v/J$.

(4) Let $m \geqslant 3$ with $p \nmid m$. Show that $K(E[m])/K$ is unramified if and only if E has good reduction. Which direction holds for every $m \geqslant 1$ with $p \nmid m$, and for every elliptic curve over K?

Solution. **Good reduction implies unramified, for all such m and E.** If E has good reduction, then $E[m](\overline{K})$ is unramified by Proposition 27.8, so $K(E[m])/K$ is unramified. This holds for every m with $p \nmid m$ and every elliptic curve over K. Conversely, suppose that $m \geqslant 3$ and that $K(E[m])/K$ is unramified. Then I_v/J is trivial by (3), so $I_v = J = I_v \cap \ker \rho_\ell$ for a prime $\ell \neq p$ by (2). So $T_\ell(E)$ is unramified, and E has good reduction by Theorem 27.9.

(5) Let $p \geqslant 3$ and $E\colon y^2 = x^3 - \pi^2 x$.

(a) Show that E has bad reduction, potential good reduction, and $K(E[2]) = K$.

Solution. Here

$$\Delta = -16 \cdot 4(-\pi^2)^3 = 64\pi^6, \qquad v(\Delta) = 6 < 12,$$

since p is odd. So the equation is minimal (Proposition 27.5) and E has bad reduction. Over $K(\sqrt{\pi})$, E is isomorphic to $E_1\colon y_1^2 = x_1^3 - x_1$, as shown in (5)(b), and E_1 has $\Delta = 64$, a unit: E has potential good reduction. And $x^3 - \pi^2 x = x(x-\pi)(x+\pi)$, so the points of order two $(0,0)$, $(\pm\pi, 0)$ are rational, and $K(E[2]) = K$.

Remark. Compare: $j(E) = 1728 \in \mathcal{O}_K$, and [Sil09, Proposition VII.5.5].

(b) Show that I_v/J has order 2, and that its nontrivial element acts on every $T_\ell(E)$ as -1.

Solution. Let $E_1\colon y_1^2 = x_1^3 - x_1$, with $\Delta = 64$, a unit: it has good reduction over every finite extension of K. Fix a square root $\sqrt{\pi} \in \overline{K}$. The map $\phi\colon E_1 \to E$, $(x_1, y_1) \mapsto (\pi x_1, \pi\sqrt{\pi}\, y_1)$, is an isomorphism, since

$$(\pi\sqrt{\pi}\, y_1)^2 = \pi^3(x_1^3 - x_1) = (\pi x_1)^3 - \pi^2(\pi x_1).$$

Let $\sigma \in \mathrm{Gal}_K$ and $\chi(\sigma) \coloneqq \sigma(\sqrt{\pi})/\sqrt{\pi} \in \{\pm 1\}$. Then $\sigma(\phi(P)) = \phi(\chi(\sigma)\,\sigma(P))$ for $P \in E_1(\overline{K})$, because $(x_1, y_1) \mapsto (x_1, -y_1)$ is $[-1]$. If $\sigma \in I_v$, then σ fixes $E_1[m](\overline{K})$ for every m with $p \nmid m$ (Proposition 27.8), so σ acts on $E[m](\overline{K}) = \phi(E_1[m](\overline{K}))$, and on every $T_\ell(E)$, as multiplication by $\chi(\sigma)$. Since $t^2 - \pi$ is Eisenstein, $K(\sqrt{\pi})/K$ is ramified, so χ is nontrivial on I_v. Hence $J = I_v \cap \ker \chi$ has index 2, and the other coset acts as -1.

(c) So (3) and (4) fail for $m = 2$. Where did your proofs use $m \geqslant 3$?

Solution. **In (3), to find a prime power $\ell^r \geqslant 3$ dividing m.** For $m = 2$, $K(E[2]) = K$, so $I_v \cap \mathrm{Gal}_{K(E[2])} = I_v \neq J$, and (3) fails. And $K(E[2])/K$ is unramified although E has bad reduction, so (4) fails. The proof of (3) needed a prime power $\ell^r \geqslant 3$ dividing m, which does not exist for $m = 2$: the element $-I \in \mathrm{GL}_2(\mathbb{Z}_2)$ has order 2 and is congruent to I modulo 2, but not modulo 4.

Remark. This is why Silverman's hint works modulo 4 when $\ell = 2$.

(6) Suppose that $p \geqslant 5$. Show that $K(E[m])/K$ is tamely ramified for every $m \geqslant 1$ with $p \nmid m$.

Solution. Since $p \geqslant 5$, (3) applies with $m = 3$: I_v/J is isomorphic to the inertia group of $K(E[3])/K$, a subgroup of $\mathrm{Gal}(K(E[3])/K)$, which acts faithfully on $E[3](\overline{K}) \cong (\mathbb{Z}/3\mathbb{Z})^2$ (Corollary 27.6). So I_v/J embeds in $\mathrm{GL}_2(\mathbb{F}_3)$, of order $48 = 2^4 \cdot 3$, and its order is prime to p.

For every m with $p \nmid m$, J acts trivially on $E[m](\overline{K})$, so $J \subseteq I_v \cap \mathrm{Gal}_{K(E[m])}$, and the inertia group of $K(E[m])/K$, isomorphic to $I_v/(I_v \cap \mathrm{Gal}_{K(E[m])})$, is a quotient of I_v/J. So its order is prime to p too.

(7) `Magma` computes $\#(I_v/J)$ for elliptic curves over $\mathbb{Q}_7$:

```
p := 7;
Inertia := func< A, B |
  #InertiaGroup(GaloisRepresentation(EllipticCurve([A, B]), p)) >;
Inertia(-1, 0);                    // 1
Inertia(-p^2, 0);                  // 2
Inertia(0, p);                     // 6
```

Check the first two values against your answers, and the third against (6).

SOLUTION. **They agree.** The first curve has $\Delta = 64$, a unit at 7: good reduction, so $I_v = J$ by (4) and (3). The second is the curve of (5) with $\pi = 7$, and $\#(I_v/J) = 2$. The third, $y^2 = x^3 + 7$, has $\#(I_v/J) = 6$, which divides 48. It acquires good reduction over $L = \mathbb{Q}_7(\varpi)$ with $\varpi^6 = 7$: the substitution $x = \varpi^2 x_1$, $y = \varpi^3 y_1$ gives $y_1^2 = x_1^3 + 1$. As $\mu_6(\overline{\mathbb{Q}}_7) \subseteq \mathbb{Q}_7$, $L/\mathbb{Q}_7$ is Galois, and it is totally ramified of degree 6. Its inertia group $I_v \cap \mathrm{Gal}_L$ lies in J, so I_v/J is a quotient of the inertia group of $L/\mathbb{Q}_7$, of order 6. `Magma` says that it is all of it.

Part 9. Hilbert 90 and Kummer theory

Hilbert's Theorem 90 says that $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$. Applied to the Kummer sequence $1 \to \mu_m \to \mathbb{G}_{\mathrm{m}} \xrightarrow{m} \mathbb{G}_{\mathrm{m}} \to 1$, it gives an isomorphism $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ with an explicit formula, and, when k contains the m-th roots of unity, a classification of the abelian extensions of k of exponent dividing m. The problems below prove all of this from the definitions in [Sil09, Appendix B]. Problem 35 computes cohomology groups by hand, Problem 36 proves Hilbert 90, Problem 37 unwinds the Kummer sequence and Problem 38 derives Kummer theory. Problems 39 and 40 compute the groups $k^\times/k^{\times m}$ for finite, local and global fields, and cut out a finite subgroup by conditions at the primes. Later, the same sequence for an elliptic curve E will give an injection $E(k)/m \hookrightarrow H^1(k, E[m])$. Then there is no Hilbert 90, and its failure is measured by $H^1(k, E)$.

Standing notation.

- A field k, not assumed perfect, a separable closure k^{sep} of k, and the absolute Galois group $\mathrm{Gal}_k := \mathrm{Gal}(k^{\mathrm{sep}}/k)$. For an extension L of k inside k^{sep}, $\mathrm{Gal}_L := \mathrm{Gal}(k^{\mathrm{sep}}/L)$. (Silverman's Appendix B assumes k perfect and works with an algebraic closure. Nothing in it changes with k^{sep} in its place.)
- A profinite group G, often finite, and a discrete G-module A: an abelian group with a left action of G, written $a \mapsto ga$, such that $G \times A \to A$ is continuous for the discrete topology on A. (Silverman writes right actions, $a \mapsto a^\sigma$.)
- Elements g, h of G, and elements σ, τ, γ of a Galois group such as Gal_k or $\mathrm{Gal}(L/k)$. Where σ is a generator of a cyclic group, it keeps that meaning.
- An integer $m \geqslant 1$ with $\mathrm{char}\, k \nmid m$, and the algebraic group μ_m of m-th roots of unity, the kernel of $x \mapsto x^m$ on $\mathbb{G}_{\mathrm{m}}$. For a field L containing k, $\mu_m(L)$ is the group of m-th roots of unity in L.
- **An abuse of notation.** For a commutative algebraic group A over k, such as $\mathbb{G}_{\mathrm{m}}$, μ_m or an elliptic curve, we write $H^i(k, A) := H^i(\mathrm{Gal}_k, A(k^{\mathrm{sep}}))$, the cohomology of the Gal_k-module of

k^{sep}-points of A. For instance,

$$H^1(k,\mathbb{G}_{\mathrm{m}}) = H^1(\mathrm{Gal}_k,(k^{\mathrm{sep}})^\times), \qquad H^1(k,\mu_m) = H^1(\mathrm{Gal}_k,\mu_m(k^{\mathrm{sep}})).$$

This is the only place where A stands for $A(k^{\mathrm{sep}})$.

- The quotient $B/m := B/mB$ of an abelian group B. For the multiplicative group of a field L it is $L^\times/L^{\times m}$, where $L^{\times m}$ is the subgroup of m-th powers.

28. Cocycles by hand

This section uses the definitions of [Sil09, §§B.1–B.2] and [Poo02, §§2–3]. Its last item is [Sil09, Exercise B.2(a)].

Definition 28.1 (Cohomology in degrees 0 and 1 [Sil09, §§B.1–B.2], [Poo02, §3])**.** Let A be a discrete G-module. Then $H^0(G,A) := A^G$, the subgroup of the $a \in A$ with $ga = a$ for every $g \in G$. A left 1-cocycle, or simply 1-cocycle, is a continuous map $\xi\colon G \to A$, $g \mapsto \xi_g$, such that

$$\xi_{gh} = \xi_g + g\xi_h \qquad \text{for all } g,h \in G.$$

A left 1-coboundary, or simply 1-coboundary, is a map of the form $g \mapsto ga - a$, for some $a \in A$. Both are *left*: they are written for the left action $a \mapsto ga$. Silverman's are the right ones, for $a \mapsto a^\sigma$: $\xi_{\sigma\tau} = \xi_\sigma^\tau + \xi_\tau$ and $\xi_\sigma = a^\sigma - a$. The 1-cocycles form a group $Z^1(G,A)$ under addition of values, the 1-coboundaries a subgroup $B^1(G,A)$, and $H^1(G,A) := Z^1(G,A)/B^1(G,A)$. If G is finite, every map $G \to A$ is continuous. When the group law of A is written multiplicatively, as for $L^\times$ or $\mu_m(L)$, the conditions read $\xi_{gh} = \xi_g \cdot g(\xi_h)$ and $\xi_g = g(a)/a$.

Proposition 28.2 (Trivial action [Sil09, Remarks B.1.1 and B.2.2], [Poo02, §3])**.** *If G acts trivially on A, then $H^0(G,A) = A$ and $H^1(G,A) = \mathrm{Hom}_{\mathrm{conts}}(G,A)$, the group of continuous homomorphisms.*

Problem 35 (Computing H^1 from the definition)**.**
In this problem G is finite.

(1) Let $G = \{1,\sigma\}$ have order 2, acting on $\mathbb{Z}$ by $\sigma a = -a$. Compute $H^0(G,\mathbb{Z})$ and $H^1(G,\mathbb{Z})$. Compare with the trivial action of G on $\mathbb{Z}$.

Solution. **$H^0 = 0$ and $H^1 \cong \mathbb{Z}/2\mathbb{Z}$. For the trivial action, $H^0 = \mathbb{Z}$ and $H^1 = 0$.** Here $\mathbb{Z}^G = \{a : -a = a\} = 0$. Every 1-cocycle has $\xi_1 = 0$, since $\xi_1 = \xi_{1\cdot 1} = \xi_1 + \xi_1$. So it is determined by $x := \xi_\sigma$, and the only other condition, from $\sigma^2 = 1$, is $0 = \xi_1 = \xi_\sigma + \sigma\xi_\sigma = x - x$, which always holds. So $Z^1(G,\mathbb{Z}) \cong \mathbb{Z}$, via $\xi \mapsto \xi_\sigma$. The coboundary of a has $\xi_\sigma = \sigma a - a = -2a$, so $B^1(G,\mathbb{Z}) \cong 2\mathbb{Z}$ and $H^1(G,\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$. For the trivial action, Proposition 28.2 gives $H^0 = \mathbb{Z}$ and $H^1 = \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z},\mathbb{Z}) = 0$, as $\mathbb{Z}$ has no element of order 2.

(2) Let $G = \langle\sigma\rangle$ be cyclic of order n, and $N := 1 + \sigma + \cdots + \sigma^{n-1}$, acting on A. Show that $\xi \mapsto \xi_\sigma$ induces an isomorphism

$$H^1(G,A) \cong \ker(N\colon A \to A)/(\sigma - 1)A.$$

Solution. Let ξ be a 1-cocycle and $x := \xi_\sigma$. By induction on i, using $\xi_{\sigma^{i+1}} = \xi_\sigma + \sigma\xi_{\sigma^i}$,

$$\xi_{\sigma^i} = (1 + \sigma + \cdots + \sigma^{i-1})x \qquad (i \geqslant 1). \tag{28.1}$$

So ξ is determined by x, and $Nx = \xi_{\sigma^n} = \xi_1 = 0$. Conversely, let $x \in \ker N$, and define ξ_{σ^i} by (28.1) for every $i \geqslant 0$ (the empty sum for $i = 0$). Replacing i by $i + n$ adds $\sigma^i Nx = 0$, so ξ is well defined on G. It is a 1-cocycle, because

$$\xi_{\sigma^{i+j}} = (1 + \cdots + \sigma^{i-1})x + \sigma^i(1 + \cdots + \sigma^{j-1})x = \xi_{\sigma^i} + \sigma^i\xi_{\sigma^j}.$$

So $\xi \mapsto \xi_\sigma$ is an isomorphism $Z^1(G,A) \to \ker N$. It maps the coboundary of a to $\sigma a - a$, so $B^1(G,A)$ maps onto $(\sigma - 1)A$.

(3) Let $G = \mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{1, c\}$. Use (2) to compute $H^1(G, \mathbb{C}^\times)$ and $H^1(G, \mu_m(\mathbb{C}))$. Compare the order of the second group with that of $\mathbb{R}^\times/\mathbb{R}^{\times m}$.

SOLUTION. **$H^1(G, \mathbb{C}^\times) = 1$, and $\#H^1(G, \mu_m(\mathbb{C})) = \gcd(2, m) = \#\mathbb{R}^\times/\mathbb{R}^{\times m}$.** In multiplicative notation, $Nz = z\bar{z} = |z|^2$ and $(c-1)A$ is the set of quotients $\bar{z}/z$. For $A = \mathbb{C}^\times$, $\ker N$ is the unit circle. Every w on it is a quotient: if $w \neq -1$, then $z := 1 + \bar{w}$ is nonzero and $\bar{z}/z = (1+w)/(1+w^{-1}) = w$, and $-1 = \bar{\imath}/\imath$. So $H^1(G, \mathbb{C}^\times) = 1$.
For $A = \mu_m(\mathbb{C})$, complex conjugation is inversion, so $N\zeta = \zeta\zeta^{-1} = 1$ and $\ker N = \mu_m(\mathbb{C})$, while $\bar{\zeta}/\zeta = \zeta^{-2}$ and $(c-1)\mu_m(\mathbb{C}) = \mu_m(\mathbb{C})^2$. The cyclic group $\mu_m(\mathbb{C})$ of order m has $\mu_m(\mathbb{C})/\mu_m(\mathbb{C})^2$ of order $\gcd(2,m)$. On the other side, every positive real number is an m-th power, and -1 is an m-th power in $\mathbb{R}$ exactly when m is odd. So $\mathbb{R}^\times/\mathbb{R}^{\times m}$ also has order $\gcd(2,m)$.
Remark. This is no accident: Problem 37 shows that $H^1(k, \mu_m) \cong k^\times/k^{\times m}$ for every field k, and the first computation is Hilbert 90 for $\mathbb{C}/\mathbb{R}$.

(4) Suppose that G has order n. Show that $nH^1(G,A) = 0$.

SOLUTION. Let ξ be a 1-cocycle and $a := \sum_{h\in G} \xi_h$. As h runs over G, so does gh, so summing the cocycle condition over h gives $a = n\xi_g + ga$. Hence $n\xi_g = g(-a) - (-a)$, and $n\xi$ is the coboundary of $-a$.

29. Hilbert 90

This section is [Poo17, Proposition 1.3.15(ii) and Exercises 1.10–1.11], with the proof of [Mil20, Proposition IV.1.3] (also [Mil22, Theorem 5.23]). [Sil09, Proposition B.2.5] states the result and refers to Serre for the proof. Item (2) is the form in which Hilbert stated it.

Theorem 29.1 (Dedekind [Mil22, Theorem 5.14])**.** *Let F be a field and H a group. If $\chi_1, \dots, \chi_r$ are distinct homomorphisms $H \to F^\times$, then they are linearly independent over F: if $a_1\chi_1 + \cdots + a_r\chi_r = 0$ as a function $H \to F$, with $a_i \in F$, then $a_1 = \cdots = a_r = 0$.*

Definition 29.2 (Norm and trace)**.** Let L/k be a finite Galois extension with group G. The norm and the trace of $a \in L$ are $N_{L/k}(a) := \prod_{\sigma\in G} \sigma(a)$ and $\mathrm{Tr}_{L/k}(a) := \sum_{\sigma\in G} \sigma(a)$. Both lie in k.

Proposition 29.3 (Infinite Galois theory)**.**
(1) The subgroups Gal_L, for L running over the finite Galois extensions of k inside k^{sep}, are open and normal in Gal_k, and they form a basis of open neighborhoods of 1 [Sil09, §B.2].
(2) The maps $H \mapsto (k^{\mathrm{sep}})^H$ and $L \mapsto \mathrm{Gal}_L$ are inverse bijections between the closed subgroups of Gal_k and the extensions of k inside k^{sep}. The subgroup H is open if and only if $(k^{\mathrm{sep}})^H$ is finite over k, and then $[\mathrm{Gal}_k : H] = [(k^{\mathrm{sep}})^H : k]$. It is normal if and only if $(k^{\mathrm{sep}})^H$ is Galois over k, and then $\mathrm{Gal}((k^{\mathrm{sep}})^H/k) \cong \mathrm{Gal}_k/H$ by restriction. In particular, $(k^{\mathrm{sep}})^{\mathrm{Gal}_k} = k$ [Mil22, Theorem 7.13].

Definition 29.4 (Restriction and inflation [Sil09, §B.2], [Poo02, §4])**.** Let A be a discrete G-module and $H \subseteq G$ a closed subgroup. The restriction $\mathrm{Res}\colon H^1(G,A) \to H^1(H,A)$ maps the class of ξ to the class of $\xi|_H$. If H is normal, the inflation $\mathrm{Inf}\colon H^1(G/H, A^H) \to H^1(G,A)$ maps the class of ξ to the class of the composition $G \to G/H \xrightarrow{\xi} A^H \subseteq A$. A 1-cocycle of G of this form is inflated from G/H.

Problem 36 (Hilbert's Theorem 90)**.**

(1) Let L/k be a finite Galois extension with group G, and $\xi\colon G \to L^\times$ a 1-cocycle. Show that there is a $c \in L^\times$ with

$$b \coloneqq \sum_{\tau \in G} \xi_\tau \cdot \tau(c) \neq 0,$$

and that then $\xi_\sigma = b/\sigma(b)$ for every $\sigma \in G$. Deduce that $H^1(G, L^\times) = 1$.

Solution. The elements of G restrict to distinct homomorphisms $L^\times \to L^\times$, and the coefficients ξ_τ are nonzero. By Theorem 29.1, $\sum_\tau \xi_\tau \cdot \tau$ is not the zero function on $L^\times$, so some $c \in L^\times$ has $b \neq 0$. For $\sigma \in G$, the cocycle condition gives $\sigma(\xi_\tau) = \xi_\sigma^{-1}\xi_{\sigma\tau}$, so

$$\sigma(b) = \sum_{\tau \in G} \sigma(\xi_\tau) \cdot (\sigma\tau)(c) = \xi_\sigma^{-1} \sum_{\tau \in G} \xi_{\sigma\tau} \cdot (\sigma\tau)(c) = \xi_\sigma^{-1} b,$$

since $\sigma\tau$ runs over G with τ. So $\xi_\sigma = b/\sigma(b) = \sigma(b^{-1})/b^{-1}$, the coboundary of b^{-1}.

(2) Suppose moreover that $G = \langle\sigma\rangle$ is cyclic. Show that if $a \in L^\times$ has $N_{L/k}(a) = 1$, then $a = \sigma(b)/b$ for some $b \in L^\times$.

Solution. By Problem 35(2), in multiplicative notation, $H^1(G, L^\times)$ is the group of the a with $N_{L/k}(a) = 1$ modulo the quotients $\sigma(b)/b$. It is trivial by (1).

(3) Let (x, y) be a rational point of the circle $x^2 + y^2 = 1$. Apply (2) to $\mathbb{Q}(i)/\mathbb{Q}$ to find $u, v \in \mathbb{Q}$, not both 0, with

$$(x, y) = \Big(\frac{u^2 - v^2}{u^2 + v^2}, \frac{2uv}{u^2 + v^2}\Big).$$

Solution. Let σ be complex conjugation, which generates $\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})$. The element $a \coloneqq x + iy$ has $N(a) = x^2 + y^2 = 1$, so $a = \sigma(\beta)/\beta$ for some $\beta \in \mathbb{Q}(i)^\times$. Write $\beta = u - iv$ with $u, v \in \mathbb{Q}$ not both 0. Then

$$x + iy = \frac{u + iv}{u - iv} = \frac{(u + iv)^2}{u^2 + v^2} = \frac{u^2 - v^2}{u^2 + v^2} + i\,\frac{2uv}{u^2 + v^2}.$$

(4) Let $\xi\colon \mathrm{Gal}_k \to (k^{\mathrm{sep}})^\times$ be a continuous 1-cocycle.

(a) Show that there is a finite Galois extension L of k inside k^{sep} such that ξ takes values in $L^\times$ and $\xi_{\sigma\tau} = \xi_\sigma$ for all $\sigma \in \mathrm{Gal}_k$ and $\tau \in \mathrm{Gal}_L$. So ξ is inflated from a 1-cocycle $\mathrm{Gal}(L/k) \to L^\times$.

Solution. Since ξ is continuous and $(k^{\mathrm{sep}})^\times$ is discrete, $\xi^{-1}(1)$ is open, and it contains 1, because $\xi_1 = \xi_1 \cdot \xi_1$. By part (1) of Proposition 29.3, it contains Gal_{L_0} for some finite Galois extension L_0 of k. The fibers of ξ are open and cover the compact group Gal_k, so ξ takes finitely many values, and they lie in some finite extension of k inside k^{sep}. Let L be the Galois closure of the compositum of this extension and L_0, a finite Galois extension of k with $\mathrm{Gal}_L \subseteq \mathrm{Gal}_{L_0}$. For $\tau \in \mathrm{Gal}_L$, $\xi_\tau = 1$, so $\xi_{\sigma\tau} = \xi_\sigma \cdot \sigma(\xi_\tau) = \xi_\sigma$. So ξ factors through $\mathrm{Gal}_k \,/\, \mathrm{Gal}_L \cong \mathrm{Gal}(L/k)$ (Proposition 29.3), with values in $L^\times = ((k^{\mathrm{sep}})^\times)^{\mathrm{Gal}_L}$, and the cocycle condition for ξ is the one for the induced map $\mathrm{Gal}(L/k) \to L^\times$.

(b) Deduce that $H^1(k, \mathbb{G}_{\mathrm{m}}) = 1$.

Solution. By (4)(a) and (1), there is $b \in L^\times$ with $\xi_\sigma = b/\sigma(b)$ for every $\sigma \in \mathrm{Gal}_k$ (the action of σ on L is that of its image in $\mathrm{Gal}(L/k)$). So ξ is the coboundary of b^{-1}.

Remark. The same argument shows that $H^1(\mathrm{Gal}_k, A) = \varinjlim_L H^1(\mathrm{Gal}(L/k), A^{\mathrm{Gal}_L})$ for every discrete Gal_k-module A [Sil09, Exercise B.4].

(5) ($\star$) Let L/k be finite Galois with group G, and $\xi\colon G \to L$ a 1-cocycle for the additive group of L. Show that $H^1(G, L) = 0$, by a variant of (1).

Solution. By Theorem 29.1 with all coefficients 1, $\mathrm{Tr}_{L/k} = \sum_\tau \tau$ is not the zero function, so some $c \in L$ has $t \coloneqq \mathrm{Tr}_{L/k}(c) \neq 0$, and $t \in k$. Let $b \coloneqq t^{-1} \sum_\tau \xi_\tau \cdot \tau(c)$. Then $\sigma(\xi_\tau) = \xi_{\sigma\tau} - \xi_\sigma$, so

$$\sigma(b) = t^{-1} \sum_{\tau \in G} (\xi_{\sigma\tau} - \xi_\sigma) \cdot (\sigma\tau)(c) = b - \xi_\sigma t^{-1} \sum_{\tau \in G} (\sigma\tau)(c) = b - \xi_\sigma.$$

So $\xi_\sigma = \sigma(-b) - (-b)$.

Remark. [Poo17, Proposition 1.3.15(i)] deduces this, and the vanishing of all the higher groups, from the normal basis theorem. As in (4), it gives $H^1(k, \mathbb{G}_\mathrm{a}) = 0$.

30. The Kummer sequence

This section is the exercise in [Poo02, §3], with the explicit formula of [Sil09, Propositions B.2.5(c) and VIII.2.2]. Hilbert's Theorem 90, $H^1(k, \mathbb{G}_\mathrm{m}) = 1$, is Problem 36(4).

Proposition 30.1 (Finite multiplicative groups [Mil22, Exercise 1-3])**.** *Every finite subgroup of the multiplicative group of a field is cyclic.*

Proposition 30.2 (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3], [Poo02, Theorem 1])**.** *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete G-modules. There is an exact sequence*

$$0 \to A^G \to B^G \to C^G \xrightarrow{\delta} H^1(G, A) \to H^1(G, B) \to H^1(G, C),$$

where, for $c \in C^G$, $\delta(c)$ is the class of $g \mapsto gb - b$ for any $b \in B$ with $\psi(b) = c$.

Proposition 30.3 (Inflation–restriction [Sil09, Proposition B.2.4])**.** *Let A be a discrete Gal_k-module and L/k a finite Galois extension inside k^sep. The sequence*

$$0 \to H^1(\mathrm{Gal}(L/k), A^{\mathrm{Gal}_L}) \xrightarrow{\mathrm{Inf}} H^1(\mathrm{Gal}_k, A) \xrightarrow{\mathrm{Res}} H^1(\mathrm{Gal}_L, A)$$

is exact.

Problem 37 (The Kummer sequence)**.**
Recall that $\mathrm{char}\, k \nmid m$, so $m \neq 0$ in k.

(1) Show that $\mu_m(k^\mathrm{sep})$ is cyclic of order m, and that

$$1 \to \mu_m(k^\mathrm{sep}) \to (k^\mathrm{sep})^\times \xrightarrow{x \mapsto x^m} (k^\mathrm{sep})^\times \to 1$$

is an exact sequence of discrete Gal_k-modules. Deduce from Proposition 30.2 that its connecting map induces an isomorphism $\delta\colon k^\times/k^{\times m} \to H^1(k, \mu_m)$.

Solution. For $a \in k^\times$, or more generally $a \in (k^\mathrm{sep})^\times$, the polynomial $x^m - a$ is separable: its derivative mx^{m-1} vanishes only at 0, since $m \neq 0$ in k, and 0 is not a root. So it has m distinct roots, all in k^sep. For $a = 1$ this shows that $\mu_m(k^\mathrm{sep})$ has order m. It is cyclic by Proposition 30.1. For general a it shows that $x \mapsto x^m$ is surjective on $(k^\mathrm{sep})^\times$. Its kernel is $\mu_m(k^\mathrm{sep})$, and it commutes with the action of Gal_k, as $\sigma(x^m) = (\sigma(x))^m$. The long exact sequence reads

$$k^\times \xrightarrow{m} k^\times \xrightarrow{\delta} H^1(k, \mu_m) \to H^1(k, \mathbb{G}_\mathrm{m}) = 1,$$

so δ is surjective with kernel $k^{\times m}$.

(2) Let $a \in k^\times$ and $\alpha \in k^{\text{sep}}$ with $\alpha^m = a$. Check directly, without Proposition 30.2, that $\xi_\sigma := \sigma(\alpha)/\alpha$ is a continuous 1-cocycle with values in $\mu_m(k^{\text{sep}})$, whose class does not depend on the choice of α. Show that $a \mapsto [\xi]$ is a homomorphism $k^\times \to H^1(k, \mu_m)$ with kernel $k^{\times m}$. This is the map δ of (1).

Solution. Since $(\sigma(\alpha))^m = \sigma(a) = a$, $\xi_\sigma \in \mu_m(k^{\text{sep}})$. The cocycle condition holds: $\xi_{\sigma\tau} = (\sigma\tau)(\alpha)/\alpha = (\sigma(\alpha)/\alpha) \cdot \sigma(\tau(\alpha)/\alpha) = \xi_\sigma \cdot \sigma(\xi_\tau)$. And $\xi_{\sigma\tau} = \xi_\sigma$ for τ in the open subgroup $\text{Gal}_{k(\alpha)}$, so ξ is locally constant, that is, continuous. The other choices of α are the $\zeta\alpha$ with $\zeta \in \mu_m(k^{\text{sep}})$, and they replace ξ by $\xi \cdot (\sigma \mapsto \sigma(\zeta)/\zeta)$, which differs from it by a coboundary. If $\beta^m = b$, then $(\alpha\beta)^m = ab$ and $\sigma(\alpha\beta)/(\alpha\beta) = \xi_\sigma \cdot \sigma(\beta)/\beta$, so the map is a homomorphism. Its kernel: if $\sigma(\alpha)/\alpha = \sigma(\zeta)/\zeta$ for some $\zeta \in \mu_m(k^{\text{sep}})$ and every σ, then $\alpha/\zeta \in (k^{\text{sep}})^{\text{Gal}_k} = k$ (Proposition 29.3), and $a = (\alpha/\zeta)^m \in k^{\times m}$. Conversely, if $a = c^m$ with $c \in k^\times$, choose $\alpha = c$, and $\xi = 1$. Finally, α lifts a along $x \mapsto x^m$, so by Proposition 30.2 this is the formula for δ.

(3) Show directly from Hilbert 90 that every continuous 1-cocycle $\xi\colon \text{Gal}_k \to \mu_m(k^{\text{sep}})$ is cohomologous to one of the form of (2).

Solution. View ξ as a 1-cocycle with values in $(k^{\text{sep}})^\times$. By Hilbert 90 it is a coboundary there: $\xi_\sigma = \sigma(\beta)/\beta$ for some $\beta \in (k^{\text{sep}})^\times$. Then $\sigma(\beta^m)/\beta^m = \xi_\sigma^m = 1$ for every σ, so $a := \beta^m \in k^\times$, and ξ is the cocycle of (2) for a and $\alpha = \beta$.

Remark. So only surjectivity needs Hilbert 90. In the analogous sequence for an elliptic curve, $H^1(k, E)$ replaces $H^1(k, \mathbb{G}_{\text{m}})$, and it is not zero in general.

(4) Let $m = 2$, with $\text{char}\, k \neq 2$. Show that $H^1(k, \mu_2) = \text{Hom}_{\text{conts}}(\text{Gal}_k, \{\pm 1\})$, and that for $d \in k^\times - k^{\times 2}$, $\delta(d)$ is the homomorphism $\chi_d\colon \sigma \mapsto \sigma(\sqrt{d})/\sqrt{d}$, with kernel $\text{Gal}_{k(\sqrt{d})}$. Deduce that $d \mapsto k(\sqrt{d})$ is a bijection from the nontrivial elements of $k^\times/k^{\times 2}$ to the quadratic extensions of k inside k^{sep}.

Solution. The action of Gal_k on $\mu_2(k^{\text{sep}}) = \{\pm 1\} \subseteq k$ is trivial, so the first claim is Proposition 28.2. The formula for $\delta(d)$ is (2), and $\chi_d(\sigma) = 1$ exactly when σ fixes $\sqrt{d}$, that is, when $\sigma \in \text{Gal}_{k(\sqrt{d})}$. A nontrivial continuous homomorphism $\text{Gal}_k \to \{\pm 1\}$ is determined by its kernel, an open subgroup of index 2, and every such subgroup is the kernel of exactly one. By Proposition 29.3, these subgroups correspond to the extensions of degree 2 inside k^{sep}, all Galois. So

$$k^\times/k^{\times 2} - \{1\} \xrightarrow{\delta} \text{Hom}_{\text{conts}}(\text{Gal}_k, \{\pm 1\}) - \{1\} \to \{\text{quadratic extensions}\}$$

is a composition of bijections, and it maps d to the fixed field of $\text{Gal}_{k(\sqrt{d})}$, which is $k(\sqrt{d})$. For $k = \mathbb{Q}$, the quadratic fields are the $\mathbb{Q}(\sqrt{d})$ with d a squarefree integer other than 1, each obtained once.

(5) Let L be an extension of k inside k^{sep}. Show that the restriction $H^1(k, \mu_m) \to H^1(L, \mu_m)$ corresponds, under δ, to the map $k^\times/k^{\times m} \to L^\times/L^{\times m}$ induced by inclusion, and that its kernel corresponds to $(k^\times \cap L^{\times m})/k^{\times m}$. Compute the kernel for $\mathbb{Q}(i)/\mathbb{Q}$ and $m = 2$, and compare it with Proposition 30.3.

Solution. **$\{\pm 1\}$, of order 2, as Proposition 30.3 predicts.** The field k^{sep} is also a separable closure of L, with $\text{Gal}_L \subseteq \text{Gal}_k$. For $a \in k^\times$ and $\alpha^m = a$, the restriction of $\sigma \mapsto \sigma(\alpha)/\alpha$ to Gal_L is the same formula, which is $\delta_L(a)$. So $\text{Res} \circ \delta_k = \delta_L$ on $k^\times$, and since δ_L is injective, the kernel of Res is the image of the classes a with $a \in L^{\times m}$.
For $k = \mathbb{Q}$, $L = \mathbb{Q}(i)$ and $m = 2$: if $a \in \mathbb{Q}^\times$ equals $(x + iy)^2$ with $x, y \in \mathbb{Q}$, then $2xy = 0$, so $a = x^2$ or $a = -y^2$. Conversely $-1 = i^2$. So the kernel is $\{\pm 1\}\mathbb{Q}^{\times 2}/\mathbb{Q}^{\times 2}$, of order 2. By

Proposition 30.3, it is $H^1(\mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q}), \mu_2(\mathbb{Q}(i))) = \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \{\pm 1\})$ (Proposition 28.2), also of order 2.

31. Kummer theory

This section is [Sil09, Exercise 8.4], in the form of [Mil22, Theorem 5.30 and Remark 5.32], together with an example that shows why the roots of unity are needed. In it, $\mu_m(k^{\mathrm{sep}}) \subseteq k$, except in item (4).

Proposition 31.1 (Characters of a finite abelian group [Mil22, discussion before Theorem 5.30])**.** *If B is a finite abelian group of exponent dividing m, then $\mathrm{Hom}(B, \mu_m(k^{\mathrm{sep}}))$ has the same order as B.*

Problem 38 (Abelian extensions of exponent dividing m)**.**

(1) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that δ is an isomorphism

$$\delta\colon k^\times/k^{\times m} \to \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}})).$$

For $a \in k^\times$ and $\alpha^m = a$, show that $\ker \delta(a) = \mathrm{Gal}_{k(\alpha)}$, and deduce that $k(\alpha)/k$ is Galois and cyclic, of degree the order of a in $k^\times/k^{\times m}$.

Solution. Now Gal_k acts trivially on $\mu_m(k^{\mathrm{sep}})$, so

$$H^1(k, \mu_m) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_k, \mu_m(k^{\mathrm{sep}}))$$

by Proposition 28.2, and Problem 37(1) gives the isomorphism. The homomorphism $\chi := \delta(a)$, $\sigma \mapsto \sigma(\alpha)/\alpha$, has kernel the σ fixing α, that is, $\mathrm{Gal}_{k(\alpha)}$. This kernel is normal, so $k(\alpha)/k$ is Galois with group $\mathrm{Gal}_k \,/\, \mathrm{Gal}_{k(\alpha)} \cong \chi(\mathrm{Gal}_k)$ (Proposition 29.3), a subgroup of the cyclic group $\mu_m(k^{\mathrm{sep}})$. A homomorphism into a cyclic group has order equal to the order of its image, and δ is an isomorphism, so $[k(\alpha) : k] = \#\chi(\mathrm{Gal}_k)$ is the order of a.

(2) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Let Δ be a subgroup of $k^\times$ containing $k^{\times m}$, with $\Delta/k^{\times m}$ finite, and $L := k(\Delta^{1/m})$ the extension of k inside k^{sep} generated by the m-th roots of the elements of Δ. Show that L/k is finite and Galois, and that

$$\langle\ ,\ \rangle\colon \mathrm{Gal}(L/k) \times \Delta/k^{\times m} \to \mu_m(k^{\mathrm{sep}}), \qquad \langle \sigma, a\rangle := \sigma(\alpha)/\alpha \quad (\alpha^m = a),$$

is a well defined bilinear pairing, and that it is perfect: the induced maps

$$\mathrm{Gal}(L/k) \to \mathrm{Hom}(\Delta/k^{\times m}, \mu_m(k^{\mathrm{sep}})), \qquad \Delta/k^{\times m} \to \mathrm{Hom}(\mathrm{Gal}(L/k), \mu_m(k^{\mathrm{sep}}))$$

are isomorphisms. Deduce that $\mathrm{Gal}(L/k)$ is abelian of exponent dividing m and that $[L : k] = (\Delta : k^{\times m})$.

Solution. Let $a_1, \dots, a_r$ represent generators of $\Delta/k^{\times m}$. Every $a \in \Delta$ is $a_1^{e_1} \cdots a_r^{e_r} c^m$ with $c \in k^\times$, so L is generated by the roots of the separable polynomials $x^m - a_i$. All the roots of $x^m - a_i$ are $\zeta\alpha_i$ with $\zeta \in \mu_m(k^{\mathrm{sep}}) \subseteq k$, so L is the splitting field of $\prod_i (x^m - a_i)$: finite and Galois.

The value $\sigma(\alpha)/\alpha$ lies in $\mu_m(k^{\mathrm{sep}})$, and does not depend on α, since σ fixes $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Replacing a by ac^m replaces α by αc, with the same value. So the pairing is well defined. It is multiplicative in a, as in Problem 37(2), and in σ, since $(\sigma\tau)(\alpha)/\alpha = \sigma(\tau(\alpha)/\alpha) \cdot \sigma(\alpha)/\alpha$ and $\tau(\alpha)/\alpha \in k$ is fixed by σ.

If $\langle \sigma, a\rangle = 1$ for every a, then σ fixes all the generators of L, so $\sigma = 1$. If $\langle \sigma, a\rangle = 1$ for every σ, then $\alpha \in L^{\mathrm{Gal}(L/k)} = k$, so $a \in k^{\times m}$. So both induced maps are injective. The first

embeds $\mathrm{Gal}(L/k)$ in an abelian group of exponent dividing m, so $\mathrm{Gal}(L/k)$ is one too. By Proposition 31.1, the two injections give $[L:k] \leqslant (\Delta : k^{\times m}) \leqslant [L:k]$. So equality holds, and both injections are bijections.

(3) Suppose that $\mu_m(k^{\mathrm{sep}}) \subseteq k$. Show that $\Delta \mapsto k(\Delta^{1/m})$ is a bijection from the subgroups Δ as in (2) to the finite abelian extensions of k inside k^{sep} of exponent dividing m, with inverse $L \mapsto \Delta_L := k^\times \cap L^{\times m}$.

Solution. Let L be a finite abelian extension of exponent dividing m. By Problem 37(5) and Proposition 30.3, with $\mu_m(k^{\mathrm{sep}})^{\mathrm{Gal}_L} = \mu_m(L) = \mu_m(k^{\mathrm{sep}})$,

$$\Delta_L/k^{\times m} \cong \ker\big(H^1(k,\mu_m) \to H^1(L,\mu_m)\big) \cong H^1(\mathrm{Gal}(L/k), \mu_m(L)) = \mathrm{Hom}(\mathrm{Gal}(L/k), \mu_m(L)),$$

which has order $[L:k]$ by Proposition 31.1. Every element of Δ_L is an m-th power in L, and then all its m-th roots lie in L, as $\mu_m(k^{\mathrm{sep}}) \subseteq k$. So $L' := k(\Delta_L^{1/m}) \subseteq L$, and $[L':k] = (\Delta_L : k^{\times m}) = [L:k]$ by (2). Hence $L' = L$. Conversely, let $L = k(\Delta^{1/m})$. Then $\Delta \subseteq \Delta_L$, and both have index $[L:k]$ over $k^{\times m}$, by (2) and the computation above. So $\Delta = \Delta_L$.

Remark. For $k = \mathbb{Q}$ and $m = 2$: the finite abelian extensions of $\mathbb{Q}$ of exponent 2 are the $\mathbb{Q}(\sqrt{d_1}, \dots, \sqrt{d_r})$, indexed by the finite subgroups of $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ [Mil22, Example 5.31].

(4) ($\star$) Let $k = \mathbb{Q}$, $m = 3$, $\zeta \in \overline{\mathbb{Q}}$ a primitive cube root of unity, and $F := \mathbb{Q}(\zeta)$. Here $\mu_3(\overline{\mathbb{Q}}) \not\subseteq \mathbb{Q}$.

(a) Show that $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not Galois, although $2 \notin \mathbb{Q}^{\times 3}$. Which step of (1) fails?

Solution. **Proposition 28.2 fails: $\mathrm{Gal}_\mathbb{Q}$ acts nontrivially on $\mu_3(\overline{\mathbb{Q}})$.** The real field $\mathbb{Q}(\sqrt[3]{2})$ does not contain the other roots $\zeta\sqrt[3]{2}$, $\zeta^2\sqrt[3]{2}$ of the irreducible polynomial $x^3 - 2$, so it is not normal. In (1), Proposition 28.2 no longer applies: complex conjugation inverts ζ. The set $\{\sigma : \sigma(\alpha) = \alpha\} = \mathrm{Gal}_{\mathbb{Q}(\alpha)}$, for $\alpha = \sqrt[3]{2}$, is still the set where the cocycle $\xi_\sigma = \sigma(\alpha)/\alpha$ is 1, and it is still a subgroup, but no longer the kernel of a homomorphism, and it is not normal.

(b) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 3} \to F^\times/F^{\times 3}$ is injective.

Solution. By Problem 37(5) and Proposition 30.3, the kernel is isomorphic to

$$H^1(\mathrm{Gal}(F/\mathbb{Q}), \mu_3(F)).$$

This group is killed by $\#\,\mathrm{Gal}(F/\mathbb{Q}) = 2$ (Problem 35(4)), and by 3, since the cocycles take values in $\mu_3(F)$. So it is trivial.

Remark. An elementary proof: if $a = \beta^3$ with $\beta \in F$, then $a^2 = N_{F/\mathbb{Q}}(a) = N_{F/\mathbb{Q}}(\beta)^3$, so $a = a^3/a^2 = (a/N_{F/\mathbb{Q}}(\beta))^3 \in \mathbb{Q}^{\times 3}$.

(c) Let $a \in \mathbb{Q}^\times - \mathbb{Q}^{\times 3}$, $\alpha^3 = a$, and $L := F(\alpha)$. Show that $L/\mathbb{Q}$ is Galois of degree 6, and that conjugation by any $\tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\tau(\zeta) = \zeta^{-1}$ acts on $\mathrm{Gal}(L/F) \cong \mu_3(F)$, the isomorphism of (1) over F, as inversion. Deduce that $\mathrm{Gal}(L/\mathbb{Q}) \cong S_3$. Check with `Magma`:

```
R<x> := PolynomialRing(Rationals());
G := GaloisGroup(x^3 - 2);
#G, IsAbelian(G);                    // 6 false
F<zeta> := CyclotomicField(3);
S<y> := PolynomialRing(F);
#GaloisGroup(y^3 - 2);               // 3
```

SOLUTION. The field L is generated by the roots $\alpha, \zeta\alpha, \zeta^2\alpha$ of $x^3 - a$, so it is its splitting field, Galois over $\mathbb{Q}$. By (4)(b), a is not a cube in F, so $[L : F] = 3$ by (1) over F, and $[L : \mathbb{Q}] = 6$. Let $\gamma \in \mathrm{Gal}(L/F)$ and $\omega := \gamma(\alpha)/\alpha \in \mu_3(F)$. The element $\beta := \tau^{-1}\alpha$ is another cube root of a, so $\gamma(\beta)/\beta = \omega$ too, as γ fixes $\mu_3(F) \subseteq F$. Then

$$(\tau\gamma\tau^{-1})(\alpha) = \tau(\gamma(\beta)) = \tau(\omega\beta) = \tau(\omega)\,\alpha = \omega^{-1}\alpha.$$

So $\tau\gamma\tau^{-1}$ corresponds to ω^{-1}. Hence $\mathrm{Gal}(L/\mathbb{Q})$, of order 6, is nonabelian, so it is S_3. `Magma` agrees for $a = 2$: the group has order 6 over $\mathbb{Q}$, is not abelian, and has order 3 over F.

Remark. So $\mathrm{Gal}(L/F) \cong \mu_3(F)$ is an isomorphism of $\mathrm{Gal}(F/\mathbb{Q})$-modules, for the action by conjugation on the left and the natural action on the right, and it is the nontrivial action on $\mu_3(F)$ that makes $L/\mathbb{Q}$ nonabelian. In cohomological terms, (4)(b) says that Res: $H^1(\mathbb{Q}, \mu_3) \to H^1(F, \mu_3)$ is injective, and its image lies in the $\mathrm{Gal}(F/\mathbb{Q})$-invariants (compare [Sil09, Exercise B.3], for finite groups).

32. Computing $k^\times/k^{\times m}$

This section computes $k^\times/k^{\times m} \cong H^1(k, \mu_m)$ for finite fields, $\mathbb{Q}_p$ and $\mathbb{Q}$, and then cuts out a finite subgroup by conditions at the primes, as in [Poo02, §§9–10]. The last item is [Sil09, proof of Proposition VIII.1.6], with a `Magma` check.

Lemma 32.1 (Hensel's lemma [Sil09, Lemma IV.1.2])**.** *Let $F \in \mathbb{Z}_p[w]$ and $a \in \mathbb{Z}_p$ with $F(a) \in p^n\mathbb{Z}_p$ for some $n \geqslant 1$ and $F'(a) \in \mathbb{Z}_p^\times$. Then F has a root $b \in \mathbb{Z}_p$ with $b \equiv a \pmod{p^n}$.*

Definition 32.2 (Unramified classes [Poo02, §§4 and 9])**.** Let p be a prime. Embed $\overline{\mathbb{Q}}$ in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} := \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_{\mathbb{Q}}$. Let ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ be the maximal unramified extension of $\mathbb{Q}_p$, the union of its finite unramified extensions. A finite extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}}_p$ is unramified if and only if it lies in ${\mathbb{Q}_p}^{\mathrm{unr}}$. The inertia group is $I_p := \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}}) \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$. A class $\xi \in H^1(\mathbb{Q}, A)$, for a commutative algebraic group A over $\mathbb{Q}$, is unramified at p if its image in $H^1(I_p, A(\overline{\mathbb{Q}}_p))$ is trivial. For a finite set S of primes, $H^1(\mathbb{Q}, A; S)$ is the subgroup of the classes unramified at every prime $p \notin S$.

Proposition 32.3 (Unramified Kummer extensions [Sil09, proof of Proposition VIII.1.6])**.** *Let p be a prime with $p \nmid m$, $a \in {\mathbb{Q}_p}^\times$ and $\alpha \in \overline{\mathbb{Q}}_p$ with $\alpha^m = a$. Then $\mathbb{Q}_p(\alpha)/\mathbb{Q}_p$ is unramified if and only if $v_p(a) \equiv 0 \pmod{m}$.*

Problem 39 (Finite and local fields)**.**

(1) Let q be a power of a prime not dividing m. Show that $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ is cyclic of order $\gcd(m, q-1)$. If $m \mid q-1$, so that $\mu_m(\overline{\mathbb{F}}_q) \subseteq \mathbb{F}_q$, compare the subgroups of $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ with the extensions of $\mathbb{F}_q$ of degree dividing m, as Problem 38(3) predicts.

SOLUTION. **Both are indexed by the divisors of m.** By Proposition 30.1, $\mathbb{F}_q^\times$ is cyclic of order $q-1$. In a cyclic group C of order n, written additively as $\mathbb{Z}/n\mathbb{Z}$, the subgroup mC is generated by m, of order $n/\gcd(m,n)$, so C/mC is cyclic of order $\gcd(m,n)$. If $m \mid q-1$, the group $\mathbb{F}_q^\times/\mathbb{F}_q^{\times m}$ is cyclic of order m, and its subgroups are indexed by the divisors d of m, with the subgroup of order d. On the other side, $\mathbb{F}_q$ has exactly one extension of each degree d inside an algebraic closure, $\mathbb{F}_{q^d}$, cyclic over $\mathbb{F}_q$, and it has exponent dividing m exactly when $d \mid m$. The bijection of Problem 38(3) matches the subgroup of order d with $\mathbb{F}_{q^d}$.

(2) Let p be odd and $u \in \mathbb{Z}_p^\times$ a unit whose reduction is not a square in $\mathbb{F}_p$. Show that ${\mathbb{Q}_p}^\times/{\mathbb{Q}_p}^{\times 2}$ has order 4, with representatives $1, u, p, up$. List the quadratic extensions of $\mathbb{Q}_p$, and say which one is unramified.

SOLUTION. **$\mathbb{Q}_p(\sqrt{u})$, $\mathbb{Q}_p(\sqrt{p})$, $\mathbb{Q}_p(\sqrt{up})$. Only $\mathbb{Q}_p(\sqrt{u})$ is unramified.** If $v \in \mathbb{Z}_p^\times$ reduces to a square $\bar{c}^2$ in $\mathbb{F}_p$, with $c \in \mathbb{Z}_p^\times$, then $F(w) := w^2 - v$ has $F(c) \in p\mathbb{Z}_p$ and $F'(c) = 2c \in \mathbb{Z}_p^\times$, so v is a square by Lemma 32.1. Conversely, squares of units reduce to squares. So $\mathbb{Z}_p^\times/\mathbb{Z}_p^{\times 2} \cong \mathbb{F}_p^\times/\mathbb{F}_p^{\times 2}$, of order 2 by (1), with representatives $1, u$. As ${\mathbb{Q}_p}^\times = p^{\mathbb{Z}} \times \mathbb{Z}_p^\times$ and p^2 is a square, ${\mathbb{Q}_p}^\times/{\mathbb{Q}_p}^{\times 2}$ has order 4, with representatives $1, u, p, up$. By Problem 37(4), the quadratic extensions are $\mathbb{Q}_p(\sqrt{d})$ for $d = u, p, up$. By Proposition 32.3, the unramified one is $\mathbb{Q}_p(\sqrt{u})$, as $v_p(u) = 0$ is even.

(3) Show that a unit $v \in \mathbb{Z}_2^\times$ is a square if and only if $v \equiv 1 \pmod 8$. Deduce that $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$ has order 8, with representatives $\pm 1, \pm 2, \pm 5, \pm 10$. `Magma` counts the quadratic extensions of $\mathbb{Q}_p$:

```
#AllExtensions(pAdicRing(3, 20), 2);    // 3
#AllExtensions(pAdicRing(5, 20), 2);    // 3
#AllExtensions(pAdicRing(2, 20), 2);    // 7
```

Check these against your answers.

SOLUTION. **They agree: 3 for odd p, and $8 - 1 = 7$ for $p = 2$.** An odd integer has square $\equiv 1 \pmod 8$, and the same holds in $\mathbb{Z}_2$, by continuity, so squares of units are $\equiv 1 \pmod 8$. Conversely, let $v \equiv 1 \pmod 8$ and $c := (v-1)/4 \in 2\mathbb{Z}_2$. Then $F(w) = w^2 + w - c$ has $F(0) = -c \in 2\mathbb{Z}_2$ and $F'(0) = 1$, so it has a root $b \in \mathbb{Z}_2$ by Lemma 32.1, and $(2b+1)^2 = 4(b^2+b) + 1 = 4c + 1 = v$. Hence $\mathbb{Z}_2^\times/\mathbb{Z}_2^{\times 2} \cong (\mathbb{Z}/8\mathbb{Z})^\times = \{\pm 1, \pm 5\}$, and $\mathbb{Q}_2^\times/\mathbb{Q}_2^{\times 2}$, with $\mathbb{Q}_2^\times = 2^{\mathbb{Z}} \times \mathbb{Z}_2^\times$, has the eight representatives $\pm 1, \pm 5, \pm 2, \pm 10$. By Problem 37(4), the quadratic extensions of $\mathbb{Q}_p$ correspond to the nontrivial classes: 3 for odd p, by (2), and 7 for $p = 2$.

Remark. Proposition 32.3 does not apply to $p = 2$ and $m = 2$. Of the seven quadratic extensions of $\mathbb{Q}_2$, only $\mathbb{Q}_2(\sqrt{5}) = \mathbb{Q}_2(\sqrt{-3})$ is unramified.

Definition 32.4 (The group $k(S, m)$ [Sil09, Theorem X.1.1(c)])**.** Let k be a number field, S a finite set of places of k containing the archimedean ones, and

$$k(S, m) := \{a \in k^\times/k^{\times m} : v(a) \equiv 0 \pmod m \text{ for all } v \notin S\},$$

where v runs over the normalized valuations of k. Let $\mathcal{O}_S$ be the ring of S-integers, the $a \in k$ with $v(a) \geqslant 0$ for all $v \notin S$, and $\mathrm{Cl}(\mathcal{O}_S)$ its ideal class group. For $k = \mathbb{Q}$ we list only the primes in S, and in [Poo02, §10], $\mathbb{Q}(S, 2)$ is written $\langle -1, S\rangle$.

Proposition 32.5 (Finiteness [Sil09, proof of Proposition VIII.1.6])**.** *The group $\mathrm{Cl}(\mathcal{O}_S)$ is finite, and $\mathcal{O}_S^\times$ is finitely generated (Dirichlet's S-unit theorem).*

Problem 40 (Classes unramified outside S)**.**

(1) Show that $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ is an $\mathbb{F}_2$-vector space with basis the classes of -1 and of the primes. In particular, it is infinite.

Solution. Every $a \in \mathbb{Q}^\times$ is $\pm\prod_p p^{v_p(a)}$, uniquely. Modulo squares only the sign and the parities of the $v_p(a)$ matter, so $a \mapsto (\operatorname{sign} a, (v_p(a) \bmod 2)_p)$ is an isomorphism from $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ to $\{\pm 1\} \times \bigoplus_p \mathbb{Z}/2\mathbb{Z}$, where the sum is over the primes.

(2) Let S be a finite set of primes. Show that $\mathbb{Q}(S,2)$ is the subgroup generated by -1 and the primes in S, of order $2^{\#S+1}$.

Solution. In the coordinates of (1), $\mathbb{Q}(S,2)$ is the set of classes whose coordinates at the primes outside S vanish, which is the span of -1 and the primes in S.

(3) Let p be an odd prime and $a \in \mathbb{Q}^\times$. Show that $\delta(a) \in H^1(\mathbb{Q}, \mu_2)$ is unramified at p if and only if $v_p(a)$ is even. Deduce that $\mathbb{Q}(S,2) = \delta^{-1}H^1(\mathbb{Q}, \mu_2; S)$ for every finite set S of primes containing 2.

Solution. The action of I_p on $\mu_2(\overline{\mathbb{Q}}_p) = \{\pm 1\}$ is trivial, so $H^1(I_p, \mu_2(\overline{\mathbb{Q}}_p)) = \operatorname{Hom}_{\text{conts}}(I_p, \{\pm 1\})$ (Proposition 28.2), and the restriction of $\delta(a)$ is the homomorphism $\sigma \mapsto \sigma(\sqrt{a})/\sqrt{a}$. It is trivial if and only if $I_p = \operatorname{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\text{unr}})$ fixes $\sqrt{a}$, that is, if and only if $\sqrt{a} \in {\mathbb{Q}_p}^{\text{unr}}$ (Proposition 29.3 over $\mathbb{Q}_p$), that is, by Definition 32.2, if and only if $\mathbb{Q}_p(\sqrt{a})/\mathbb{Q}_p$ is unramified. By Proposition 32.3, with $p \nmid 2$, this happens exactly when $v_p(a)$ is even. So for $S \ni 2$, $\delta(a)$ is unramified at every $p \notin S$ exactly when $v_p(a)$ is even for every $p \notin S$, that is, when $a \in \mathbb{Q}(S,2)$.

Remark. This is the computation behind 2-descent: [Poo02, §10] identifies $H^1(\mathbb{Q}, E[2]; S)$ with $\langle -1, S\rangle^2$ when $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$.

(4) ($\star$) Let k be a number field and S as in Definition 32.4. Construct an exact sequence

$$1 \to \mathcal{O}_S^\times/\mathcal{O}_S^{\times m} \to k(S,m) \to \operatorname{Cl}(\mathcal{O}_S)[m] \to 0,$$

and deduce that $k(S,m)$ is finite. For $k = \mathbb{Q}(\sqrt{-5})$, $m = 2$ and S the archimedean place, show that $k(S,2) = \langle -1, 2\rangle$, of order 4, and check with `Magma`:

```
K<s> := QuadraticField(-5);
OK := Integers(K);
#ClassGroup(OK);                    // 2
G, m := pSelmerGroup(2, {PowerIdeal(OK)|});
#G;                                 // 4
[K | G.i @@ m : i in [1..Ngens(G)]];  // [ -1, 2 ]
```

Solution. **$a \mapsto [\mathfrak{a}]$, where $a\mathcal{O}_S = \mathfrak{a}^m$.** If a represents an element of $k(S,m)$, its fractional ideal $a\mathcal{O}_S$ has all exponents divisible by m, so it is $\mathfrak{a}^m$ for a unique fractional ideal $\mathfrak{a}$. Replacing a by ac^m replaces $\mathfrak{a}$ by $c\mathfrak{a}$, so $a \mapsto [\mathfrak{a}]$ is a well defined homomorphism to $\operatorname{Cl}(\mathcal{O}_S)$, with values in $\operatorname{Cl}(\mathcal{O}_S)[m]$, as $\mathfrak{a}^m$ is principal. It is surjective: if $\mathfrak{a}^m = a\mathcal{O}_S$, then a represents an element of $k(S,m)$ mapping to $[\mathfrak{a}]$. Its kernel: if $\mathfrak{a} = c\mathcal{O}_S$, then $a/c^m \in \mathcal{O}_S^\times$, so the kernel is the image of $\mathcal{O}_S^\times$. Finally, $u \in \mathcal{O}_S^\times$ maps to 1 in $k^\times/k^{\times m}$ if $u = c^m$ with $c \in k^\times$, and then $c \in \mathcal{O}_S^\times$, since $mv(c) = v(u) = 0$ for $v \notin S$. So $\mathcal{O}_S^\times/\mathcal{O}_S^{\times m}$ injects. Both ends are finite by Proposition 32.5, so $k(S,m)$ is finite.

For $k = \mathbb{Q}(\sqrt{-5})$ and S archimedean, $\mathcal{O}_S = \mathbb{Z}[\sqrt{-5}]$, with unit group $\{\pm 1\}$, so $\mathcal{O}_S^\times/\mathcal{O}_S^{\times 2}$ is generated by -1. The class group has order 2 (`Magma`), generated by $\mathfrak{p} := (2, 1+\sqrt{-5})$, which is not principal, as no element of $\mathbb{Z}[\sqrt{-5}]$ has norm 2. And $\mathfrak{p}^2 = (2)$. So 2 represents an element of $k(S,2)$ mapping to $[\mathfrak{p}]$, and $k(S,2) = \langle -1, 2\rangle$, of order 4, as `Magma` says.

Remark. In the notation of Definition 32.2, extended to k, $k(S,m) = H^1(k, \mu_m; S)$ when S contains the places above m, and the sequence is the Kummer sequence for the étale cohomology of $\operatorname{Spec}\mathcal{O}_S$, with

$\mathrm{Cl}(\mathcal{O}_S) = \operatorname{Pic}\mathcal{O}_S$ in place of $H^1(k, \mathbb{G}_m) = 1$. Compare the analogy between $\mathrm{Cl}(k)$ and the Shafarevich–Tate group in [Poo02, §7].

Part 10. The Kummer pairing and weak Mordell–Weil

The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite for an elliptic curve E over a number field K. The proof in [Sil09, §VIII.1] has three steps. The Kummer pairing turns the finiteness of $E(K)/mE(K)$ into the finiteness of the field L generated by the m-th divisors of the rational points. This field is abelian of exponent m and unramified outside a finite set of places S. And Kummer theory shows that such fields are finite. This worksheet runs the argument for $m = 2$ and $K = \mathbb{Q}$, on curves whose points of order two are rational. Then the class group and the S-unit theorem disappear, and the injectivity of reduction on the 2-torsion becomes elementary. Problem 41 sets up the pairing. Problem 42, the heart of the worksheet, proves that L is unramified outside S and deduces a bound for $E(\mathbb{Q})/2E(\mathbb{Q})$. Problem 43 tests the bound on $y^2 = x^3 - x$.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by
$$y^2 = (x - e_1)(x - e_2)(x - e_3)$$
with distinct $e_1, e_2, e_3 \in \mathbb{Z}$, and its points of order two $T_i := (e_i, 0)$. So $E[2](\overline{\mathbb{Q}}) = \{O, T_1, T_2, T_3\} \subseteq E(\mathbb{Q})$.
- The finite set of primes $S := \{2\} \cup \{p : p \mid (e_1 - e_2)(e_1 - e_3)(e_2 - e_3)\}$, and $s := \#S$.
- The absolute Galois group $\mathrm{Gal}_{\mathbb{Q}} := \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, with elements σ, τ, γ. For an extension F of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$, $\mathrm{Gal}_F := \mathrm{Gal}(\overline{\mathbb{Q}}/F)$.
- For $Q \in E(\overline{\mathbb{Q}})$, the field $\mathbb{Q}(Q)$ generated by the coordinates of Q, and the point $\sigma(Q)$, obtained by applying $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$ to the coordinates.
- The compositum L of the fields $\mathbb{Q}(Q)$ for $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$. Silverman writes $L = \mathbb{Q}([2]^{-1}E(\mathbb{Q}))$.
- The quotient $B/2 := B/2B$ of an abelian group B, and the group $\mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ of classes of rational numbers modulo squares.

33. The Kummer pairing

This section is [Sil09, Proposition VIII.1.2], for $m = 2$ and $K = \mathbb{Q}$, and its reinterpretation in [Sil09, §VIII.2], with cohomology as in [Sil09, §§B.1–B.2].

Definition 33.1 (The Kummer pairing [Sil09, §VIII.1])**.** The Kummer pairing is the map
$$\kappa\colon E(\mathbb{Q}) \times \mathrm{Gal}_{\mathbb{Q}} \to E[2](\overline{\mathbb{Q}})$$
defined as follows. Let $P \in E(\mathbb{Q})$ and $\sigma \in \mathrm{Gal}_{\mathbb{Q}}$. Choose $Q \in E(\overline{\mathbb{Q}})$ with $2Q = P$, and set $\kappa(P, \sigma) := \sigma(Q) - Q$. That this lands in $E[2](\overline{\mathbb{Q}})$, and does not depend on Q, is Problem 41(1).

Proposition 33.2 (Infinite Galois theory [Mil22, Theorem 7.13])**.** *For a closed subgroup H of $\mathrm{Gal}_{\mathbb{Q}}$, let $\overline{\mathbb{Q}}^H$ be its fixed field. The maps $H \mapsto \overline{\mathbb{Q}}^H$ and $F \mapsto \mathrm{Gal}_F$ are inverse bijections between the closed subgroups of $\mathrm{Gal}_{\mathbb{Q}}$ and the extensions of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$. The subgroup H is normal if and only if $\overline{\mathbb{Q}}^H$ is Galois over $\mathbb{Q}$, and then $\mathrm{Gal}(\overline{\mathbb{Q}}^H/\mathbb{Q}) \cong \mathrm{Gal}_{\mathbb{Q}}/H$ by restriction. In particular, $\overline{\mathbb{Q}}^{\mathrm{Gal}_{\mathbb{Q}}} = \mathbb{Q}$.*

Definition 33.3 (Cohomology in degree 1 [Sil09, §§B.1–B.2])**.** Let A be a discrete $\mathrm{Gal}_{\mathbb{Q}}$-module. A 1-cocycle is a continuous map $\xi\colon \mathrm{Gal}_{\mathbb{Q}} \to A$, $\sigma \mapsto \xi_\sigma$, with $\xi_{\sigma\tau} = \xi_\sigma + \sigma\xi_\tau$ for all σ, τ. For $a \in A$, the map $\sigma \mapsto \sigma a - a$ is a 1-cocycle, and the 1-cocycles of this form are the 1-coboundaries. Then $H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ is the group of 1-cocycles modulo the 1-coboundaries. For an elliptic curve or its 2-torsion, we abuse notation and write $H^1(\mathbb{Q}, E[2]) \coloneqq H^1(\mathrm{Gal}_{\mathbb{Q}}, E[2](\overline{\mathbb{Q}}))$.

Proposition 33.4 (The long exact sequence [Sil09, Propositions B.1.2 and B.2.3])**.** *Let $0 \to A \to B \xrightarrow{\psi} C \to 0$ be an exact sequence of discrete $\mathrm{Gal}_{\mathbb{Q}}$-modules. For $c \in C^{\mathrm{Gal}_{\mathbb{Q}}}$, choose $b \in B$ with $\psi(b) = c$, and let $\delta(c) \in H^1(\mathrm{Gal}_{\mathbb{Q}}, A)$ be the class of $\sigma \mapsto \sigma b - b$. This does not depend on b, and the sequence*

$$0 \to A^{\mathrm{Gal}_{\mathbb{Q}}} \to B^{\mathrm{Gal}_{\mathbb{Q}}} \to C^{\mathrm{Gal}_{\mathbb{Q}}} \xrightarrow{\delta} H^1(\mathrm{Gal}_{\mathbb{Q}}, A) \to H^1(\mathrm{Gal}_{\mathbb{Q}}, B)$$

is exact.

Proposition 33.5 (Trivial action [Sil09, Remark B.2.2])**.** *If $\mathrm{Gal}_{\mathbb{Q}}$ acts trivially on A, then*

$$H^1(\mathrm{Gal}_{\mathbb{Q}}, A) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_{\mathbb{Q}}, A),$$

the group of continuous homomorphisms.

Problem 41 (The Kummer pairing)**.**

(1) Show that $\kappa(P, \sigma) \in E[2](\overline{\mathbb{Q}})$, and that it does not depend on the choice of Q.

Solution. The multiplication map $[2]$ is given by rational functions with coefficients in $\mathbb{Q}$, so it commutes with σ. Hence

$$2\big(\sigma(Q) - Q\big) = \sigma(2Q) - 2Q = \sigma(P) - P = O,$$

as P is a rational point. Any other choice is $Q + T$ with $T \in E[2](\overline{\mathbb{Q}})$, and $\sigma(T) = T$ because $T \in E(\mathbb{Q})$. So $\sigma(Q+T) - (Q+T) = \sigma(Q) - Q$.

(2) Let $P, P' \in E(\mathbb{Q})$ and $\sigma, \tau \in \mathrm{Gal}_{\mathbb{Q}}$. Show that $\kappa(P + P', \sigma) = \kappa(P, \sigma) + \kappa(P', \sigma)$ and

$$\kappa(P, \sigma\tau) = \kappa(P, \sigma) + \sigma\big(\kappa(P, \tau)\big).$$

Deduce that κ is bilinear. Where does the hypothesis $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$ enter?

Solution. **In the independence of Q, and to remove σ from $\sigma(\kappa(P,\tau))$.** The addition law is given by rational functions over $\mathbb{Q}$, so σ acts on $E(\overline{\mathbb{Q}})$ by group automorphisms. If $2Q = P$ and $2Q' = P'$, then $2(Q+Q') = P+P'$, and $\sigma(Q+Q') - (Q+Q') = (\sigma(Q) - Q) + (\sigma(Q') - Q')$. For the second identity,

$$(\sigma\tau)(Q) - Q = \big(\sigma(Q) - Q\big) + \sigma\big(\tau(Q) - Q\big).$$

This is the cocycle identity of Definition 33.3, and it holds for every E. Here $\kappa(P, \tau) \in E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$ is fixed by σ, so $\kappa(P, \sigma\tau) = \kappa(P, \sigma) + \kappa(P, \tau)$, and $\kappa(P, \cdot)$ is a homomorphism. The hypothesis is used twice: for the independence of Q in (1), and here.

(3) Show that $\kappa(P, \cdot) = O$ if and only if $P \in 2E(\mathbb{Q})$, and that $\kappa(\cdot, \sigma) = O$ if and only if $\sigma \in \mathrm{Gal}_L$. Deduce that $L/\mathbb{Q}$ is Galois and that κ induces a perfect pairing

$$E(\mathbb{Q})/2 \times \mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}}),$$

that is, both $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}(L/\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$ and $\mathrm{Gal}(L/\mathbb{Q}) \to \mathrm{Hom}(E(\mathbb{Q})/2, E[2](\overline{\mathbb{Q}}))$ are injective. Conclude that $E(\mathbb{Q})/2$ is finite if and only if $L/\mathbb{Q}$ is finite.

Solution. If $P = 2Q$ with $Q \in E(\mathbb{Q})$, then $\sigma(Q) = Q$ for all σ. Conversely, if $\sigma(Q) = Q$ for all σ, then the coordinates of Q lie in $\overline{\mathbb{Q}}^{\mathrm{Gal}_\mathbb{Q}} = \mathbb{Q}$ (Proposition 33.2), so $P = 2Q \in 2E(\mathbb{Q})$. Next, if $\sigma \in \mathrm{Gal}_L$ and $2Q \in E(\mathbb{Q})$, then $Q \in E(L)$, so $\sigma(Q) = Q$. Conversely, if $\kappa(\cdot, \sigma) = O$, then $\sigma(Q) = Q$ for every Q with $2Q \in E(\mathbb{Q})$, so σ fixes each $\mathbb{Q}(Q)$, hence their compositum L, and $\sigma \in \mathrm{Gal}_L$ by Proposition 33.2.
By (2), $\sigma \mapsto \kappa(\cdot, \sigma)$ is a homomorphism $\mathrm{Gal}_\mathbb{Q} \to \mathrm{Hom}(E(\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$ with kernel Gal_L, so Gal_L is normal and $L/\mathbb{Q}$ is Galois (Proposition 33.2). The pairing factors through $E(\mathbb{Q})/2$ on the left and through $\mathrm{Gal}_\mathbb{Q} / \mathrm{Gal}_L \cong \mathrm{Gal}(L/\mathbb{Q})$ on the right, and the two kernels computed above say that both induced maps are injective. If $E(\mathbb{Q})/2$ is finite, then $\mathrm{Gal}(L/\mathbb{Q})$ injects into the finite group $\mathrm{Hom}(E(\mathbb{Q})/2, E[2](\overline{\mathbb{Q}}))$. If $L/\mathbb{Q}$ is finite, then $E(\mathbb{Q})/2$ injects into the finite group $\mathrm{Hom}(\mathrm{Gal}(L/\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$.

(4) Show that $0 \to E[2](\overline{\mathbb{Q}}) \to E(\overline{\mathbb{Q}}) \xrightarrow{2} E(\overline{\mathbb{Q}}) \to 0$ is an exact sequence of discrete $\mathrm{Gal}_\mathbb{Q}$-modules, and that its connecting map $\delta\colon E(\mathbb{Q}) \to H^1(\mathbb{Q}, E[2])$ sends P to the class of $\kappa(P, \cdot)$. Show that $H^1(\mathbb{Q}, E[2]) = \mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_\mathbb{Q}, E[2](\overline{\mathbb{Q}}))$, and recover from Proposition 33.4 the injectivity of $E(\mathbb{Q})/2 \to \mathrm{Hom}(\mathrm{Gal}_\mathbb{Q}, E[2](\overline{\mathbb{Q}}))$.

Solution. The map $[2]$ is a nonconstant morphism of curves, so it is surjective on $\overline{\mathbb{Q}}$-points [Sil09, Proposition II.2.3], and its kernel is $E[2](\overline{\mathbb{Q}})$ by definition. Every point of $E(\overline{\mathbb{Q}})$ has coordinates in a finite extension F of $\mathbb{Q}$, so its stabilizer contains the open subgroup Gal_F, and the module is discrete. By Proposition 33.4, $\delta(P)$ is the class of $\sigma \mapsto \sigma(Q) - Q$ with $2Q = P$, which is $\kappa(P, \cdot)$. As $\mathrm{Gal}_\mathbb{Q}$ acts trivially on $E[2](\overline{\mathbb{Q}})$, Proposition 33.5 identifies $H^1(\mathbb{Q}, E[2])$ with $\mathrm{Hom}_{\mathrm{conts}}(\mathrm{Gal}_\mathbb{Q}, E[2](\overline{\mathbb{Q}}))$, a group in which every coboundary is 0. The exact sequence of Proposition 33.4 reads

$$E(\mathbb{Q}) \xrightarrow{2} E(\mathbb{Q}) \xrightarrow{\delta} H^1(\mathbb{Q}, E[2]),$$

so $\ker \delta = 2E(\mathbb{Q})$, which is the first half of (3).

Remark. Without the hypothesis $E[2](\overline{\mathbb{Q}}) \subseteq E(\mathbb{Q})$, the map δ is still defined and injective on $E(\mathbb{Q})/2$, but $H^1(\mathbb{Q}, E[2])$ is no longer a group of homomorphisms. Silverman reduces to the case of rational torsion by [Sil09, Lemma VIII.1.1.1], or by inflation–restriction [Sil09, §VIII.2]. The next group in the long exact sequence, $H^1(\mathbb{Q}, E)$, measures the failure of δ to be surjective, and leads to the Selmer and Shafarevich–Tate groups [Sil09, §X.4].

34. Unramified outside S

This section is [Sil09, Proposition VIII.1.5], for $m = 2$ and $K = \mathbb{Q}$. Its argument is that of [Sil09, Proposition VII.4.1(a)], applied to the halves of rational points instead of torsion points. The last item adds Kummer theory, as in [Sil09, proof of Proposition VIII.1.6]. In this section p is a prime, and $\overline{\mathbb{Q}}$ is embedded in an algebraic closure $\overline{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, so that restriction is an injection $\mathrm{Gal}_{\mathbb{Q}_p} := \mathrm{Gal}(\overline{\mathbb{Q}}_p/\mathbb{Q}_p) \hookrightarrow \mathrm{Gal}_\mathbb{Q}$. The residue field of the valuation ring of $\overline{\mathbb{Q}}_p$ is an algebraic closure $\overline{\mathbb{F}}_p$ of $\mathbb{F}_p$, and a bar denotes reduction modulo its maximal ideal.

Definition 34.1 (Inertia [Sil09, §VII.4])**.** An element $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$ preserves the valuation ring of $\overline{\mathbb{Q}}_p$, so it induces an automorphism $\bar{\sigma}$ of $\overline{\mathbb{F}}_p$. The inertia group $I_p \subseteq \mathrm{Gal}_{\mathbb{Q}_p}$ is the subgroup of the σ with $\bar{\sigma} = 1$. If ${\mathbb{Q}_p}^{\mathrm{unr}} \subseteq \overline{\mathbb{Q}}_p$ is the maximal unramified extension of $\mathbb{Q}_p$, then $I_p = \mathrm{Gal}(\overline{\mathbb{Q}}_p/{\mathbb{Q}_p}^{\mathrm{unr}})$. A Galois extension F of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ is unramified at p if $I_p \subseteq \mathrm{Gal}_F$, that is, if I_p fixes F.

Proposition 34.2 (Good reduction [Sil09, §III.1 and Remark VIII.1.3])**.** *The Weierstrass equation of E has discriminant*

$$\Delta = 16(e_1 - e_2)^2(e_1 - e_3)^2(e_2 - e_3)^2.$$

If $p \nmid \Delta$, the equation is minimal at p and its reduction E_p is an elliptic curve over $\mathbb{F}_p$.

Proposition 34.3 (Reduction [Sil09, Proposition VII.2.1])**.** *Suppose that $p \nmid \Delta$. Every $Q \in E(\overline{\mathbb{Q}}_p)$ has projective coordinates $[X : Y : Z]$ in the valuation ring of $\overline{\mathbb{Q}}_p$, not all in its maximal ideal, and $\bar{Q} := [\bar{X} : \bar{Y} : \bar{Z}] \in E_p(\overline{\mathbb{F}}_p)$. The reduction map $E(\overline{\mathbb{Q}}_p) \to E_p(\overline{\mathbb{F}}_p)$, $Q \mapsto \bar{Q}$, is a homomorphism, and $\overline{\sigma(Q)} = \bar{\sigma}(\bar{Q})$ for $\sigma \in \mathrm{Gal}_{\mathbb{Q}_p}$.*

Proposition 34.4 (Kummer theory [Sil09, Exercise 8.4])**.** *Let F be a Galois extension of $\mathbb{Q}$ inside $\overline{\mathbb{Q}}$ with $\mathrm{Gal}(F/\mathbb{Q})$ abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in F. Then $F = \mathbb{Q}(\sqrt{d} : d \in D)$.*

Proposition 34.5 (Unramified quadratic extensions [Sil09, proof of Proposition VIII.1.6])**.** *Let p be an odd prime and $d \in \mathbb{Q}^\times$. Then $\mathbb{Q}(\sqrt{d})$ is unramified at p if and only if $v_p(d)$ is even.*

Definition 34.6 (The group $\mathbb{Q}(S, 2)$ [Sil09, Theorem X.1.1(c)])**.** The subgroup $\mathbb{Q}(S, 2) \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ consists of the classes of the $d \in \mathbb{Q}^\times$ with $v_p(d)$ even for every prime $p \notin S$. It is generated by the classes of -1 and of the primes in S, so it has order 2^{s+1}.

Problem 42 ($L/\mathbb{Q}$ is unramified outside S)**.**

(1) Show that $\mathrm{Gal}(L/\mathbb{Q})$ is abelian of exponent dividing 2. Let $D \subseteq \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$ be the subgroup of the classes of the $d \in \mathbb{Q}^\times$ that are squares in L. Deduce that $L = \mathbb{Q}(\sqrt{d} : d \in D)$.

Solution. By Problem 41(2) and (3), $\sigma \mapsto \kappa(\cdot, \sigma)$ is an injective homomorphism $\mathrm{Gal}(L/\mathbb{Q}) \hookrightarrow \mathrm{Hom}(E(\mathbb{Q}), E[2](\overline{\mathbb{Q}}))$. The target is abelian and killed by 2, so the same holds for $\mathrm{Gal}(L/\mathbb{Q})$. As $L/\mathbb{Q}$ is Galois, Proposition 34.4 applies to $F = L$.

(2) Let $p \notin S$. Show that E has good reduction at p, and that O, T_1, T_2, T_3 reduce to four distinct points of $E_p(\mathbb{F}_p)$. Deduce that reduction is injective on $E[2](\overline{\mathbb{Q}})$.

Solution. The prime divisors of Δ are 2 and the primes dividing some $e_i - e_j$, that is, the primes in S. So $p \nmid \Delta$, and E_p is an elliptic curve by Proposition 34.2. The reductions $\bar{T}_i = (\bar{e}_i, 0)$ are affine, so they differ from $\bar{O} = [0 : 1 : 0]$, and they are distinct, because $p \nmid e_i - e_j$ for $i \neq j$. So reduction is injective on the four points of $E[2](\overline{\mathbb{Q}})$.

Remark. Both conditions defining S are needed. If $p \mid e_i - e_j$, then $\bar{T}_i = \bar{T}_j$, so $T_i - T_j$, the third point of order two, reduces to $\bar{O}$. At $p = 2$, injectivity fails for every model of E: an elliptic curve over a field of characteristic 2 has at most two 2-torsion points over an algebraic closure [Sil09, Corollary III.6.4(c)]. Problem 43 shows that L can indeed ramify at 2.

(3) Let $p \notin S$, $Q \in E(\overline{\mathbb{Q}})$ with $2Q \in E(\mathbb{Q})$, and $\sigma \in I_p$. Show that $\sigma(Q) - Q$ reduces to $\bar{O}$, and deduce that $\sigma(Q) = Q$. Conclude that $L/\mathbb{Q}$ is unramified at p.

Solution. By Proposition 34.3 and $\bar{\sigma} = 1$,

$$\overline{\sigma(Q) - Q} = \bar{\sigma}(\bar{Q}) - \bar{Q} = \bar{Q} - \bar{Q} = \bar{O}.$$

By Problem 41(1), $\sigma(Q) - Q \in E[2](\overline{\mathbb{Q}})$, so $\sigma(Q) = Q$ by (2). So σ fixes the coordinates of Q, hence $\mathbb{Q}(Q)$. As this holds for every such Q, the element σ fixes their compositum L. Hence $I_p \subseteq \mathrm{Gal}_L$.

Remark. In the language of Problem 41(4), the homomorphism $\kappa(P, \cdot)$ is trivial on I_p: the class $\delta(P)$ is unramified at p. For general m and K the same argument works, with the injectivity of reduction on $E[m]$ for $v(m) = 0$ taken from the formal group [Sil09, Proposition VII.3.1(b)]. Item (2) is where rational 2-torsion makes this elementary.

(4) Show that $D \subseteq \mathbb{Q}(S, 2)$. Deduce that $\#\,\mathrm{Gal}(L/\mathbb{Q}) \leqslant 2^{s+1}$ and $\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 4^{s+1}$.

Solution. **$\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 4^{s+1}$.** Let d represent a class of D, and let $p \notin S$, so p is odd, as $2 \in S$. Then $\sqrt{d} \in L$, and I_p fixes L by (3), so $\mathbb{Q}(\sqrt{d})$ is unramified at p, and $v_p(d)$ is even by Proposition 34.5. So $D \subseteq \mathbb{Q}(S,2)$, an $\mathbb{F}_2$-vector space of dimension $s+1$ (Definition 34.6). If $d_1, \dots, d_n$, with $n \leqslant s+1$, represent a basis of D, then $L = \mathbb{Q}(\sqrt{d_1}, \dots, \sqrt{d_n})$, and $[L:\mathbb{Q}] \leqslant 2^n \leqslant 2^{s+1}$. So $\mathrm{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/2\mathbb{Z})^t$ with $t \leqslant s+1$, and by Problem 41(3),

$$\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant \#\operatorname{Hom}\big((\mathbb{Z}/2\mathbb{Z})^t, (\mathbb{Z}/2\mathbb{Z})^2\big) = 4^t \leqslant 4^{s+1}.$$

Remark. This is the weak Mordell–Weil theorem for these curves. With the Mordell–Weil theorem, $E(\mathbb{Q}) \cong \mathbb{Z}^r \times E(\mathbb{Q})_{\mathrm{tors}}$, and $E(\mathbb{Q})_{\mathrm{tors}}/2$ has the same order as $E(\mathbb{Q})_{\mathrm{tors}}[2] = E[2](\overline{\mathbb{Q}})$, namely 4. So $\#E(\mathbb{Q})/2E(\mathbb{Q}) = 2^{r+2}$, and the bound says $r \leqslant 2s$.

35. An example

This section runs Problem 42 on $y^2 = x^3 - x$, that is, $(e_1, e_2, e_3) = (0, 1, -1)$. The images of its points of order two under the explicit form of the Kummer map are [Sil09, Proposition X.1.4].

Problem 43 (The curve $y^2 = x^3 - x$)**.**
In this problem, E is the curve $y^2 = x^3 - x$, with $T_1 = (0,0)$, $T_2 = (1,0)$ and $T_3 = (-1,0)$. `Magma` finds its rank and its torsion subgroup, and checks that the halves of each T_i are defined over $\mathbb{Q}(\zeta_8)$:

```
E := EllipticCurve([-1, 0]);
Rank(E);                          // 0 true
TorsionSubgroup(E);               // Z/2 + Z/2
K<z> := CyclotomicField(8);
EK := ChangeRing(E, K);
[#DivisionPoints(EK ! [e, 0], 2) : e in [0, 1, -1]];   // [ 4, 4, 4 ]
```

So $E(\mathbb{Q}) = E[2](\overline{\mathbb{Q}})$.

(1) Compute S and the bound of Problem 42(4).

Solution. **$S = \{2\}$, so $\#E(\mathbb{Q})/2E(\mathbb{Q}) \leqslant 16$.** The differences $e_i - e_j$ are ± 1 and ± 2, so $S = \{2\}$ and $s = 1$. Here $\mathbb{Q}(S,2) = \langle -1, 2\rangle$, so $L \subseteq \mathbb{Q}(i, \sqrt{2})$, and the bound is $4^2 = 16$.

(2) For $i = 1, 2, 3$, find the four $Q \in E(\overline{\mathbb{Q}})$ with $2Q = T_i$, and show that $L = \mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$.

Solution. **The halves of T_1, T_2, T_3 generate $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\zeta_8)$.** A half $Q = (x, y)$ of a point of order two has order 4, so $y \neq 0$, and $y^2 = x^3 - x$. For T_1, $x(2Q) = 0$ gives $x^2 = -1$. Then $y^2 = x(x^2 - 1) = -2x$, and

$$Q = (i, \pm(1-i)), \quad (-i, \pm(1+i)),$$

as $(1-i)^2 = -2i$ and $(1+i)^2 = 2i$. For T_2, $x(2Q) = 1$ gives $(x^2+1)^2 = 4(x^3 - x)$, that is, $(x^2 - 2x - 1)^2 = 0$, so $x = 1 \pm \sqrt{2}$. Then $x^2 - 1 = 2x$, so $y^2 = 2x^2$ and $Q = (x, \pm\sqrt{2}\,x)$. For T_3, $x(2Q) = -1$ gives $(x^2 + 2x - 1)^2 = 0$, so $x = -1 \pm \sqrt{2}$. Then $x^2 - 1 = -2x$, so $y^2 = -2x^2$ and $Q = (x, \pm\sqrt{-2}\,x)$. As $E(\mathbb{Q}) = \{O, T_1, T_2, T_3\}$, the field L is the compositum of $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{2}, \sqrt{-2})$, which is $\mathbb{Q}(i, \sqrt{2}) = \mathbb{Q}(\zeta_8)$, as $\zeta_8 = (1+i)/\sqrt{2}$. It is ramified at 2 only, as it should be.

(3) Let $\sigma, \tau \in \mathrm{Gal}(L/\mathbb{Q})$ with $\sigma(i) = -i$, $\sigma(\sqrt{2}) = \sqrt{2}$, $\tau(i) = i$ and $\tau(\sqrt{2}) = -\sqrt{2}$. Compute $\kappa(T_i, \sigma)$ and $\kappa(T_i, \tau)$ for $i = 1, 2, 3$, and deduce that no T_i lies in $2E(\mathbb{Q})$. Compute $\#E(\mathbb{Q})/2E(\mathbb{Q})$ and compare it with the bound of (1).

Solution. **$\#E(\mathbb{Q})/2E(\mathbb{Q}) = 4$, against the bound 16.** By bilinearity it is enough to treat T_1 and T_2, as $T_3 = T_1 + T_2$. Take the half $Q = (i, 1-i)$ of T_1. It is fixed by τ, so $\kappa(T_1, \tau) = O$. And $\sigma(Q) = (-i, 1+i)$, while $-Q = (i, i-1)$. The line through these two points has slope $(i-1-1-i)/(2i) = i$, so $\sigma(Q) - Q$ has x-coordinate $i^2 - (-i) - i = -1$, and $\kappa(T_1, \sigma) = T_3$. Take the half $Q = (1+\sqrt{2}, 2+\sqrt{2})$ of T_2. It is fixed by σ, so $\kappa(T_2, \sigma) = O$. And $\tau(Q) = (1-\sqrt{2}, 2-\sqrt{2})$, while $-Q = (1+\sqrt{2}, -2-\sqrt{2})$. The slope is $-4/(2\sqrt{2}) = -\sqrt{2}$, so $\tau(Q) - Q$ has x-coordinate $2 - (1-\sqrt{2}) - (1+\sqrt{2}) = 0$, and $\kappa(T_2, \tau) = T_1$. In all,

	σ	τ
T_1	T_3	O
T_2	O	T_1
T_3	T_3	T_1

No row vanishes, so no T_i lies in $2E(\mathbb{Q})$, by Problem 41(3). As $E(\mathbb{Q}) = E[2](\overline{\mathbb{Q}})$, also $2E(\mathbb{Q}) = \{O\}$ directly. Hence $\#E(\mathbb{Q})/2E(\mathbb{Q}) = 4$. The pairing is perfect, as it should be: $\mathrm{Gal}(L/\mathbb{Q})$ has order 4, and only 4 of the 16 homomorphisms $\mathrm{Gal}(L/\mathbb{Q}) \to E[2](\overline{\mathbb{Q}})$ come from rational points.

Remark. The explicit Kummer map of [Sil09, Proposition X.1.4] sends $P = (x, y) \notin \{T_1, T_2\}$ to $(x - e_1, x - e_2) \in \mathbb{Q}(S, 2)^2$, with a modified formula at T_1 and T_2. Here it gives $T_1 \mapsto (-1, -1)$, $T_2 \mapsto (1, 2)$ and $T_3 \mapsto (-1, -2)$. Each coordinate records the action of $\mathrm{Gal}(L/\mathbb{Q})$ on a square root, which matches the table: $\kappa(T_1, \cdot)$ sees only $\sqrt{-1}$, $\kappa(T_2, \cdot)$ only $\sqrt{2}$. Deciding which of the 16 classes of $\mathbb{Q}(S, 2)^2$ come from rational points is the descent problem of [Sil09, §X.1].

Part 11. Heights on abelian groups

The Mordell–Weil theorem has two steps. The weak Mordell–Weil theorem says that $E(K)/mE(K)$ is finite, and a height function turns this into finite generation [Sil09, §VIII.3]. The second step does not use the curve: it works for any abelian group with a height that is quadratic up to a bounded error. This worksheet is about such heights on an abstract abelian group: which functions are heights of which kind, what their $O(1)$ constants are, and how a quadratic height behaves. Problem 44 sets up the definitions and the first estimates. Problem 45, the heart of the worksheet, proves the descent theorem and shows, on $\mathbb{Q}$ and on a direct sum of copies of $\mathbb{Z}$, that its hypotheses are needed. Problem 46 is the algebra of quadratic heights.

Standing notation.

- An abelian group A, written additively, with elements P, Q, R, torsion subgroup A_{tors}, and, for an integer $m \geqslant 2$, the quotient $A/m := A/mA$.
- For real functions f, g on a set X, the notation $f = g + O(1)$, meaning that $f - g$ is bounded on X. An inequality $f \leqslant g + c$ that holds for one constant c holds for every larger one, so the constants that bound heights are taken non-negative, and a lower bound is written $-c$ with $c \geqslant 0$.

36. Heights and the parallelogram law

This section is the vocabulary of [Sil09, §VIII.3], and the estimates [Sil09, Corollary VIII.6.4] that the heights on an elliptic curve satisfy, proved here for any height that satisfies the parallelogram law up to a bounded error [HS00, §B.4].

Definition 36.1 (Heights [Sil09, §§VIII.3, VIII.6])**.** A height on a set X is a function $h\colon X \to \mathbb{R}$. Two heights h, h' on X are equivalent, written $h = h' + O(1)$, if $h - h'$ is bounded on X. A height h has the Northcott property if $\{x \in X : h(x) \leqslant C\}$ is finite for every $C \in \mathbb{R}$. The defect of a height h on A is the function

$$\delta_h(P, Q) \coloneqq h(P + Q) + h(P - Q) - 2h(P) - 2h(Q)$$

on $A \times A$. The height h is quasi-quadratic if δ_h is bounded, and quadratic if $\delta_h = 0$ (the parallelogram law).

Problem 44 (Heights and the parallelogram law)**.**

(1) Show that equivalence of heights on a set X is an equivalence relation. Show that the Northcott property depends only on the equivalence class, and that a height with the Northcott property is bounded below. Give a height on $\mathbb{Z}$ that is bounded below and does not have the Northcott property.

Solution. **If $|h - h'| \leqslant c$, then $\{h' \leqslant C\} \subseteq \{h \leqslant C + c\}$.** The difference $h - h$ is zero, $h' - h$ is bounded when $h - h'$ is, and if $|h - h'| \leqslant c$ and $|h' - h''| \leqslant c'$, then $|h - h''| \leqslant c + c'$. If $|h - h'| \leqslant c$ and $h'(x) \leqslant C$, then $h(x) \leqslant C + c$, which gives the inclusion of the answer. So if h has the Northcott property, the set $\{h' \leqslant C\}$ is contained in a finite set. For the lower bound, let $x_0 \in X$. The set $\{h \leqslant h(x_0)\}$ is finite and contains x_0, so it has a minimum, and that minimum is a lower bound for h on X. The zero height on $\mathbb{Z}$ is bounded below, and $\{h \leqslant 0\} = \mathbb{Z}$. A more interesting example is the number $\omega(n)$ of distinct prime divisors of $n \neq 0$, with $\omega(0) \coloneqq 0$. It is non-negative, and the set $\{\omega \leqslant 1\}$ contains every prime power, so it is infinite.

(2) Let h be a quasi-quadratic height on A, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that $|h(0)| \leqslant c/2$, that $|h(-P) - h(P)| \leqslant 2c$, and that $|h(2P) - 4h(P)| \leqslant 3c/2$ for every $P \in A$.

Solution. By definition, $\delta_h(0, 0) = -2h(0)$, $\delta_h(0, P) = h(-P) - h(P) - 2h(0)$ and $\delta_h(P, P) = h(2P) - 4h(P) + h(0)$, using $h(P - 0) = h(P)$ and $h(P - P) = h(0)$. The first gives $|h(0)| \leqslant c/2$. The second gives $|h(-P) - h(P)| \leqslant c + 2|h(0)| \leqslant 2c$, and the third gives $|h(2P) - 4h(P)| \leqslant c + |h(0)| \leqslant 3c/2$.

(3) Let h be a quasi-quadratic height on A, and let $c \geqslant 0$ be a constant with $|\delta_h| \leqslant c$. Show that for every $n \in \mathbb{Z}$ there is a constant $d_n \geqslant 0$ with $|h(nP) - n^2 h(P)| \leqslant d_n$ for all $P \in A$, and give a recursion for d_n in terms of c. Suppose moreover that $h \geqslant -c'$ for a constant $c' \geqslant 0$. Show that for every $Q \in A$ there is a constant $C(Q) \geqslant 0$ with $h(P + Q) \leqslant 2h(P) + C(Q)$ for all $P \in A$, and give $C(Q)$ explicitly.

Solution. **$d_{n+1} = 2d_n + d_{n-1} + c$, $d_0 = c/2$, $d_1 = 0$, $C(Q) = 2h(Q) + 2c' + c$.** Fix $P \in A$, and let $e_n \coloneqq h(nP) - n^2 h(P)$ for $n \in \mathbb{Z}$. By (2), $|e_0| = |h(0)| \leqslant c/2$, and $e_1 = 0$. So $d_0 = c/2$ and $d_1 = 0$ work.
Let $n \geqslant 1$. Since $(n + 1)^2 + (n - 1)^2 = 2n^2 + 2$, the defect at (nP, P) is

$$\delta_h(nP, P) = h((n + 1)P) + h((n - 1)P) - 2h(nP) - 2h(P) = e_{n+1} + e_{n-1} - 2e_n.$$

So $e_{n+1} = 2e_n - e_{n-1} + \delta_h(nP, P)$, and $|e_{n+1}| \leqslant 2d_n + d_{n-1} + c$. By induction on n, the recursion of the answer gives a constant d_n for every $n \geqslant 0$. For negative multiples, (2) applied to nP gives $|h(-nP) - h(nP)| \leqslant 2c$, so $|e_{-n}| \leqslant |e_n| + 2c$, and $d_{-n} \coloneqq d_n + 2c$ works. None of these bounds depends on P.
For the last claim, the defect at (P, Q) gives

$$h(P + Q) = 2h(P) + 2h(Q) - h(P - Q) + \delta_h(P, Q) \leqslant 2h(P) + 2h(Q) + c' + c,$$

since $h(P-Q) \geqslant -c'$ and $\delta_h \leqslant c$. The constant $2h(Q)+c'+c$ can be negative. Enlarging it to $C(Q) := 2h(Q)+2c'+c$ makes it at least $c \geqslant 0$, since $h(Q) \geqslant -c'$.

Remark. The recursion gives $d_2 = 3c/2$, $d_3 = 4c$ and $d_4 = 21c/2$, and in general d_n grows like $(1+\sqrt{2})^n$. A bound of the order of n^2 is true, and comes with the canonical height, in a later worksheet.

(4) Let $f\colon A \to \mathbb{R}$ be a homomorphism. Show that the height f^2 on A is quadratic. Show that a sum of quadratic heights on A is quadratic.

SOLUTION. **$\delta_{f^2} = 0$, and $\delta_{h+h'} = \delta_h + \delta_{h'}$.** Let $a := f(P)$ and $b := f(Q)$. Then $f(P \pm Q) = a \pm b$, and

$$\delta_{f^2}(P,Q) = (a+b)^2 + (a-b)^2 - 2a^2 - 2b^2 = 0,$$

which is the parallelogram law in $\mathbb{R}$. Each term of the defect is linear in the height, so $\delta_{h+h'} = \delta_h + \delta_{h'}$, and a sum of heights with zero defect has zero defect.

(5) Let $A = \mathbb{Z}^2$, $q(x,y) := x^2+y^2$ and $q'(x,y) := x^2+(x+y)^2$. Show that q and q' are quadratic heights with the Northcott property, that $q \leqslant 3q'$ and $q' \leqslant 3q$, and that they are not equivalent.

SOLUTION. **$q' - q = x(x+2y)$ is unbounded.** By (4), applied to the homomorphisms x, y and $x+y$, both q and q' are quadratic. If $q \leqslant C$ then $|x|, |y| \leqslant \sqrt{C}$, so q has the Northcott property. From $(x+y)^2 \leqslant 2x^2+2y^2$ we get $q' \leqslant 3x^2+2y^2 \leqslant 3q$, and from $y^2 = ((x+y)-x)^2 \leqslant 2(x+y)^2+2x^2$ we get $q \leqslant 3q'$. So q' has the Northcott property as well. But $q'(n,n) - q(n,n) = 3n^2$ is unbounded.

Remark. Two heights with $q \leqslant 3q'$ and $q' \leqslant 3q$ are comparable, which is weaker than equivalent. So a quadratic height is far from determined by its equivalence class, although an equivalence class contains at most one quadratic height, as a later worksheet shows.

37. The descent theorem

This section is [Sil09, Theorem VIII.3.1] and its proof, for an abstract abelian group, and the examples of groups that fail one of its hypotheses. The count of rational numbers of bounded height is [HS00, Theorem B.2.3].

Theorem 37.1 (Descent theorem [Sil09, Theorem VIII.3.1])**.** *Let h be a height on A with the following properties.*

(1) For every $Q \in A$ there is a constant $C_1(Q)$ with $h(P+Q) \leqslant 2h(P) + C_1(Q)$ for all $P \in A$.
(2) There are an integer $m \geqslant 2$ and a constant C_2 with $h(mP) \geqslant m^2 h(P) - C_2$ for all $P \in A$.
(3) The height h has the Northcott property.

If A/m is finite for the m of (2), then A is finitely generated.

Problem 45 (The descent theorem)**.**

(1) Let h be a quasi-quadratic height on A with the Northcott property. Show that h satisfies the hypotheses of Theorem 37.1 for every $m \geqslant 2$.

SOLUTION. **$C_1(Q) = C(Q)$ and $C_2 = d_m$, from Problem 44(3).** By Problem 44(1), $h \geqslant -c'$ for some $c' \geqslant 0$. Then Problem 44(3) gives $h(P+Q) \leqslant 2h(P) + C(Q)$, which is (1), and $h(mP) \geqslant m^2 h(P) - d_m$, which is (2), for every $m \geqslant 2$.

(2) Let h be a height on A with the Northcott property, let $m \geqslant 2$ be an integer, and let $0 < a < b$ be real numbers. Suppose that for every $Q \in A$ there is a constant $C_1(Q) \geqslant 0$ with $h(P+Q) \leqslant a\,h(P) + C_1(Q)$ for all P, and that there is a constant $C_2 \geqslant 0$ with $h(mP) \geqslant b\,h(P) - C_2$ for all P. Suppose also that A/m is finite, let $Q_1, \dots, Q_s$ represent its cosets, and

let $C := \max_i C_1(-Q_i) + C_2$. Show that A is generated by $Q_1, \dots, Q_s$ and the finitely many $P \in A$ with $h(P) \leqslant 1 + C/(b-a)$.

SOLUTION. **If $P_{j-1} = mP_j + Q_{i_j}$, then $h(P_j) \leqslant \rho\, h(P_{j-1}) + C/b$ with $\rho := a/b < 1$.** Given $P_0 := P$, choose P_j and i_j with $P_{j-1} = mP_j + Q_{i_j}$. Then

$$b\, h(P_j) \leqslant h(mP_j) + C_2 = h(P_{j-1} - Q_{i_j}) + C_2 \leqslant a\, h(P_{j-1}) + C.$$

Since $\rho > 0$, induction gives

$$h(P_n) \leqslant \rho^n h(P) + \frac{C}{b}\big(1 + \rho + \cdots + \rho^{n-1}\big) \leqslant \rho^n h(P) + \frac{C}{b-a},$$

where the last step uses $C \geqslant 0$. For n large, $\rho^n h(P) \leqslant 1$, so $h(P_n) \leqslant 1 + C/(b-a)$, and $P = m^n P_n + \sum_{j=1}^{n} m^{j-1} Q_{i_j}$. The set $\{h \leqslant 1 + C/(b-a)\}$ is finite by the Northcott property. No lower bound on h is needed.

Remark. Silverman's case is $a = 2$ and $b = m^2 \geqslant 4$, where $C/(b-a) \leqslant C/2$ recovers his bound $1 + \frac{1}{2}(C_1 + C_2)$ [Sil09, proof of Theorem VIII.3.1].

(3) Let $A = \mathbb{Q}$, as an additive group, with the logarithmic Weil height

$$h(u/w) := \log \max(|u|, |w|),$$

for u/w in lowest terms. Show that h has the Northcott property, that $h(x+y) \leqslant h(x) + h(y) + \log 2$, with equality for $x = y = 1$, that $h(mx) \leqslant h(x) + \log m$ and $\mathbb{Q}/m = 0$ for every integer $m \geqslant 2$. Which hypothesis of (2) fails, for every $m \geqslant 2$ and every $b > 1$? Is h quasi-quadratic?

SOLUTION. **The second, since $h(mx) \leqslant h(x) + \log m$. Not quasi-quadratic: $\delta_h(n,n) = \log 2 - 3 \log n$.** There are at most $(2e^C + 1)^2$ pairs (u, w) of integers with $|u|, |w| \leqslant e^C$, so $\{h \leqslant C\}$ is finite. From $u/w + u'/w' = (uw' + u'w)/(ww')$, before canceling, $\max(|uw' + u'w|, |ww'|) \leqslant 2 \max(|u|, |w|) \max(|u'|, |w'|)$, and canceling only lowers the height. For $x = y = 1$ both sides are $\log 2$. So the first hypothesis holds with $a = 1$ and $C_1(Q) = h(Q) + \log 2$, since $h \geqslant 0$. Likewise $h(mu/w) \leqslant \log(m \max(|u|, |w|))$. Every rational number is m times a rational number, so $\mathbb{Q}/m = 0$. But $\mathbb{Q}$ is not finitely generated: a finitely generated subgroup of $\mathbb{Q}$ has a common denominator. So the second hypothesis fails, and directly: if $h(mx) \geqslant b\, h(x) - C_2$ with $b > 1$, then $(b-1)h(x) \leqslant C_2 + \log m$ for every x, contradicting the Northcott property on the infinite set $\mathbb{Q}$. Finally $\delta_h(n,n) = h(2n) + h(0) - 4h(n) = \log(2n) - 4 \log n$ for $n \geqslant 1$, which is unbounded.

(4) Let $A = \bigoplus_{i \geqslant 1} \mathbb{Z}$, let $w_1, w_2, \dots$ be positive real numbers, and $h(x) := \sum_i w_i x_i^2$. Show that h is quadratic. For which weights w_i does h have the Northcott property? For such weights, which hypothesis of Theorem 37.1 fails for A?

SOLUTION. **Northcott if and only if $w_i \to \infty$. The quotient A/m is infinite.** Each map $x \mapsto \sqrt{w_i}\, x_i$ is a homomorphism $A \to \mathbb{R}$, so each $w_i x_i^2$ satisfies the parallelogram law, by Problem 44(4), and the sums are finite. Suppose $w_i \to \infty$ and let $C \in \mathbb{R}$. Only finitely many i have $w_i \leqslant C$. If $h(x) \leqslant C$ and $x_i \neq 0$, then $w_i \leqslant w_i x_i^2 \leqslant C$, so x is supported on those i, with $|x_i| \leqslant \sqrt{C/w_i}$. So $\{h \leqslant C\}$ is finite. If w_i does not tend to infinity, then $w_i \leqslant C$ for infinitely many i, and the corresponding basis vectors e_i have $h(e_i) = w_i \leqslant C$. For $w_i = i$, the height h satisfies every hypothesis on h, by (1), but $A/m \cong \bigoplus_{i \geqslant 1} \mathbb{Z}/m\mathbb{Z}$ is infinite, and A is not finitely generated.

Remark. Conversely, let A be finitely generated, and choose an isomorphism $A \cong A_{\text{tors}} \times \mathbb{Z}^r$. The sum of the squares of the r coordinates is a quadratic height on A, and $\{h \leqslant C\}$ is the preimage of finitely many elements of $\mathbb{Z}^r$, each preimage a coset of the finite group A_{tors}. So A is finitely generated if and only if $A/2$ is finite and A has a quasi-quadratic height with the Northcott property.

38. Quadratic heights and their pairing

This section is [HS00, Lemma B.5.2], the algebra of a height that satisfies the parallelogram law exactly, which [Sil09, Theorem VIII.9.3(c)] states for the canonical height on an elliptic curve.

Definition 38.1 (The pairing of a quadratic height [HS00, §B.5])**.** Let q be a quadratic height on A. Its pairing is

$$\langle P, Q\rangle_q := q(P+Q) - q(P) - q(Q).$$

Problem 46 (Quadratic heights and their pairing)**.**
Let q be a quadratic height on A.

(1) Show that $q(0) = 0$, $q(-P) = q(P)$ and $q(2P) = 4q(P)$, and that $\langle\ ,\ \rangle_q$ is symmetric and bilinear. Deduce that $\langle P, P\rangle_q = 2q(P)$ and $q(nP) = n^2 q(P)$ for every $n \in \mathbb{Z}$. Conversely, show that for a symmetric bilinear map $B\colon A \times A \to \mathbb{R}$, the function $P \mapsto \frac{1}{2}B(P,P)$ is a quadratic height with pairing B.

Solution. **The four laws combine to $\langle P+Q, R\rangle_q = \langle P, R\rangle_q + \langle Q, R\rangle_q$.** The parallelogram law with $P = Q = 0$ gives $2q(0) = 4q(0)$, with $P = 0$ it gives $q(Q) + q(-Q) = 2q(Q)$, and with $Q = P$ it gives $q(2P) = 4q(P)$. The pairing is symmetric by definition. The parallelogram law for the pairs $(X, Y) = (P+R, Q)$, $(P, R-Q)$, $(P+Q, R)$ and (R, Q) gives

$$q(P+R+Q) + q(P+R-Q) = 2q(P+R) + 2q(Q), \tag{38.1}$$

$$q(P+R-Q) + q(P-R+Q) = 2q(P) + 2q(R-Q), \tag{38.2}$$

$$q(P+Q+R) + q(P+Q-R) = 2q(P+Q) + 2q(R), \tag{38.3}$$

$$q(R+Q) + q(R-Q) = 2q(R) + 2q(Q). \tag{38.4}$$

Take

$$(38.1) - (38.2) + (38.3) - 2\,(38.4).$$

On the left, the two terms $q(P+R-Q)$ cancel, $q(P-R+Q)$ cancels against $q(P+Q-R)$, and $q(P+R+Q)$ and $q(P+Q+R)$ add up. The term $2q(R-Q)$ appears on both sides. Dividing by 2 gives

$$q(P+Q+R) - q(P+Q) - q(P+R) - q(Q+R) + q(P) + q(Q) + q(R) = 0,$$

which says that $\langle P+Q, R\rangle_q = \langle P, R\rangle_q + \langle Q, R\rangle_q$. Then $\langle P, P\rangle_q = q(2P) - 2q(P) = 2q(P)$, and by bilinearity $q(nP) = \frac{1}{2}\langle nP, nP\rangle_q = n^2 q(P)$. Conversely, for $q(P) := \frac{1}{2}B(P,P)$, expanding $B(P \pm Q, P \pm Q)$ gives $q(P+Q) + q(P-Q) = B(P,P) + B(Q,Q) = 2q(P) + 2q(Q)$ and $q(P+Q) - q(P) - q(Q) = B(P,Q)$.

Remark. So $q \mapsto \langle\ ,\ \rangle_q$ and $B \mapsto \frac{1}{2}B(P,P)$ are inverse bijections between quadratic heights on A and symmetric bilinear maps $A \times A \to \mathbb{R}$.

(2) Let $T \in A_{\mathrm{tors}}$. Show that $q(T) = 0$, that $\langle P, T\rangle_q = 0$ for every $P \in A$, and that $q(P+T) = q(P)$. So q and its pairing factor through A/A_{tors}.

Solution. $\boldsymbol{n\langle P, T\rangle_q = \langle P, nT\rangle_q = \langle P, 0\rangle_q = 0}$**.** Let $n \geqslant 1$ with $nT = 0$. By (1), $n^2 q(T) = q(nT) = q(0) = 0$, and $\langle P, 0\rangle_q = q(P) - q(P) - q(0) = 0$, which gives the answer. Then $q(P+T) = q(P) + q(T) + \langle P, T\rangle_q = q(P)$.

(3) Let $A = \mathbb{Z}^r$ with standard basis $e_1, \dots, e_r$. Show that every quadratic height on A is $q(x) = \frac{1}{2}x^{\mathsf{T}}Gx$ for a unique real symmetric $r \times r$ matrix G, and that every symmetric G arises. Show that if G is positive definite, then q has the Northcott property.

SOLUTION. **$G_{ij} = \langle e_i, e_j\rangle_q$, and $q(x) \geqslant \frac{1}{2}\lambda|x|^2$ for the least eigenvalue λ of G.** By bilinearity, $\langle x, y\rangle_q = \sum_{i,j} x_i y_j \langle e_i, e_j\rangle_q = x^{\mathsf{T}} G y$ with $G_{ij} \coloneqq \langle e_i, e_j\rangle_q$, which is symmetric, and $q(x) = \frac{1}{2}\langle x, x\rangle_q$. The matrix is unique because G_{ij} is determined by q. Conversely, $(x, y) \mapsto x^{\mathsf{T}} G y$ is symmetric and bilinear for every symmetric G, so (1) applies. If G is positive definite, then $x^{\mathsf{T}} G x \geqslant \lambda|x|^2$, so $\{q \leqslant C\}$ is contained in the finite set $\{x \in \mathbb{Z}^r : |x|^2 \leqslant 2C/\lambda\}$.

Remark. The converse holds: a quadratic height on $\mathbb{Z}^r$ with the Northcott property is positive definite [Sil09, Lemma VIII.9.5]. Its proof needs Minkowski's theorem, and [Sil09, Exercise 8.16] shows why: $q(x, y) = (x - \sqrt{2}\, y)^2$ is positive on $\mathbb{Z}^2 \setminus \{0\}$ without being positive definite. That is for next time. Note that a quadratic height can be negative, as $G = -1$ on $\mathbb{Z}$ shows.

Part 12. Heights on projective space

To apply the descent theorem to $E(K)$, one needs a height on $E(K)$ with the Northcott property. Silverman builds it from a height on projective space [Sil09, §VIII.5]: a point of $\mathbb{P}^N(\overline{\mathbb{Q}})$ is large when its coordinates are large at the archimedean places or have large denominators at the non-archimedean ones. This worksheet works out the first properties of this height. Problem 47, the heart of the worksheet, proves Kronecker's theorem: the algebraic numbers of height 1 are 0 and the roots of unity. Problem 48 proves the inequalities for products and sums, and the behavior of the height under the Segre and d-uple embeddings. Problem 49 computes the height through the ideal generated by the coordinates, and shows on $\mathbb{Q}(\sqrt{-5})$ why the recipe of coprime integer coordinates does not extend to every number field.

Standing notation.

- An algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, and its subring $\overline{\mathbb{Z}}$ of algebraic integers.
- A number field K, its ring of integers $\mathcal{O}_K$, its set of places $\mathcal{P}_K$, and the subsets $\mathcal{P}_K^\infty$ and $\mathcal{P}_K^0$ of archimedean and non-archimedean places.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v.
- For a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$ and $v \in \mathcal{P}_K$, the number $|P|_v \coloneqq \max_i |x_i|_v$. It depends on the choice of coordinates.

39. Kronecker's theorem

This section is [Sil09, Exercise 8.6], with the proof of [HS00, Corollary B.2.3.1]. The definitions and the first properties of the height are [Sil09, §VIII.5]. The last two items compare Kronecker's theorem with the enumeration of polynomials that proves it in small degree, and with algebraic numbers of small height.

Proposition 39.1 (Extension and product formulas [Sil09, (VIII.5.2), (VIII.5.3)])**.** *(1) Let L be a finite extension of K, and let $v \in \mathcal{P}_K$. Then $\sum_{w|v} n_w = [L : K]\, n_v$, the sum over the places $w \in \mathcal{P}_L$ that restrict to v.*
(2) Let $x \in K^\times$. Then $\prod_{v \in \mathcal{P}_K} |x|_v^{n_v} = 1$.

Definition 39.2 (Heights on projective space [Sil09, §§VIII.5, VIII.6])**.** The height relative to K of a point $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$ is

$$H_K(P) \coloneqq \prod_{v \in \mathcal{P}_K} |P|_v^{n_v}.$$

Let $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$, and let K be a number field with $P \in \mathbb{P}^N(K)$. The height of P is $H(P) \coloneqq H_K(P)^{1/[K:\mathbb{Q}]}$, and its logarithmic height is $h(P) \coloneqq \log H(P)$. For $x \in \overline{\mathbb{Q}}$, let $H(x) \coloneqq H([x:1])$ and $h(x) \coloneqq \log H(x)$.

Proposition 39.3 (Properties of the height [Sil09, Proposition VIII.5.4])**.** *Let $P \in \mathbb{P}^N(K)$.*

(1) The height $H_K(P)$ does not depend on the choice of coordinates of P.
(2) The height satisfies $H_K(P) \geqslant 1$.
(3) Let L be a finite extension of K. Then $H_L(P) = H_K(P)^{[L:K]}$.

In particular, $H(P)$ does not depend on the choice of K, and $H(P) \geqslant 1$.

Theorem 39.4 (Northcott's theorem [Sil09, Theorem VIII.5.11])**.** *Let C and d be real numbers. Then the set of points $P \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(P) \leqslant C$ and $[\mathbb{Q}(P) : \mathbb{Q}] \leqslant d$ is finite, where $\mathbb{Q}(P)$ is the field generated by the ratios x_i/x_j with $x_j \neq 0$ of the coordinates of P. In particular, the set $\{P \in \mathbb{P}^N(K) : H_K(P) \leqslant C\}$ is finite.*

Problem 47 (Kronecker's theorem)**.**
This problem proves Kronecker's theorem: a nonzero algebraic number $x \in \overline{\mathbb{Q}}^\times$ has $H(x) = 1$ if and only if x is a root of unity. By (2), it is equivalent to Kronecker's original form: a nonzero algebraic integer $x \in \overline{\mathbb{Z}}$ with $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon \mathbb{Q}(x) \to \mathbb{C}$ is a root of unity.

(1) Show that every root of unity has height 1. Show that $H(0) = 1$, so the hypothesis $x \neq 0$ cannot be dropped.

Solution. **If $\zeta^n = 1$, then $|\zeta|_v^n = 1$ for every place v.** Let ζ be a root of unity, with $\zeta^n = 1$ for some $n \geqslant 1$, and let $K \coloneqq \mathbb{Q}(\zeta)$. For every $v \in \mathcal{P}_K$, $|\zeta|_v^n = 1$ and $|\zeta|_v > 0$, so $|\zeta|_v = 1$. So every factor $\max(|\zeta|_v, 1)^{n_v}$ of $H_K([\zeta:1])$ is 1. Likewise every factor of $H_{\mathbb{Q}}([0:1])$ is $\max(0,1) = 1$. But 0 is not a root of unity.

(2) Let $x \in \overline{\mathbb{Q}}$ and $K \coloneqq \mathbb{Q}(x)$. Show that $H(x) = 1$ if and only if $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K$, and that this holds if and only if $x \in \overline{\mathbb{Z}}$ and $|\tau(x)| \leqslant 1$ for every embedding $\tau\colon K \to \mathbb{C}$. Show that $x = (3+4i)/5$ satisfies $|\tau(x)| = 1$ for both embeddings of $\mathbb{Q}(i)$, and compute $H(x)$.

Solution. **$H(x) = \sqrt{5}$.** Every factor $\max(|x|_v, 1)^{n_v}$ of $H_K([x:1])$ is at least 1, so the product is 1 if and only if every factor is 1, that is, $|x|_v \leqslant 1$ for every v. The elements of K with $|x|_v \leqslant 1$ for every $v \in \mathcal{P}_K^0$ form $\mathcal{O}_K$, and the archimedean absolute values are $|x|_v = |\tau(x)|$ for the embeddings $\tau\colon K \to \mathbb{C}$, each place coming from one embedding or from a pair of complex conjugate ones [HS00, Proposition B.1.3]. This gives the second equivalence.
For $x = (3+4i)/5$, the two embeddings send x to $(3 \pm 4i)/5$, of absolute value $\sqrt{9+16}/5 = 1$. In $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$ and $3+4i = (2+i)^2$, so

$$x = \frac{(2+i)^2}{(2+i)(2-i)} = \frac{2+i}{2-i}.$$

So $x \notin \overline{\mathbb{Z}}$. At the place v of the prime $(2-i)$, of residue field $\mathbb{F}_5$, we have $n_v = 1$ and $|x|_v = 5$. At every other non-archimedean place, $|x|_v \leqslant 1$, and at the complex place $|x|_v = 1$. So $H_K([x:1]) = 5$ and $H(x) = 5^{1/2}$.

Remark. So the integrality in Kronecker's original statement (an algebraic integer all of whose conjugates lie on the unit circle is a root of unity) cannot be dropped. Problem 49 computes such heights in one step.

(3) Let $x \in \overline{\mathbb{Q}}$. Show that $H(x^n) = H(x)^n$ for every integer $n \geqslant 1$, and that $H(1/x) = H(x)$ if $x \neq 0$. Prove Kronecker's theorem.

SOLUTION. **The powers x^n lie in $\mathbb{Q}(x)$ and have height 1, so $x^r = x^s$ for some $r < s$.** Let $K := \mathbb{Q}(x)$. For every $v \in \mathcal{P}_K$, $\max(|x^n|_v, 1) = \max(|x|_v, 1)^n$, so $H_K([x^n : 1]) = H_K([x : 1])^n$. If $x \neq 0$, then $[1/x : 1] = [1 : x]$, and $|[1 : x]|_v = |[x : 1]|_v$, so $H(1/x) = H(x)$.

One direction of Kronecker's theorem is (1). Conversely, let $x \in \overline{\mathbb{Q}}^\times$ with $H(x) = 1$. Then $H_K([x^n : 1]) = 1$ for every $n \geqslant 1$, and the points $[x^n : 1]$ lie in $\mathbb{P}^1(K)$. By Theorem 39.4, they are finitely many, so $x^r = x^s$ for some $1 \leqslant r < s$. Since $x \neq 0$, $x^{s-r} = 1$.

(4) Let $f = t^d + a_1 t^{d-1} + \cdots + a_d \in \mathbb{Z}[t]$ be monic of degree $d \geqslant 1$, with all complex roots in the closed unit disc. Show that $|a_i| \leqslant \binom{d}{i}$ for every i. So there are finitely many such f of each degree. `Magma` lists those of degree 4 with $a_4 \neq 0$, and checks that each is a product of cyclotomic polynomials:

```
R<t> := PolynomialRing(Integers());
d := 4;
B := [[-Binomial(d, i)..Binomial(d, i)] : i in [1..d]];
S := [];
for c in CartesianProduct(B) do
  f := t^d + &+[c[i]*t^(d - i) : i in [1..d]];
  if c[d] ne 0 and forall{r : r in Roots(f, ComplexField(30)) |
                          Abs(r[1]) le 1 + 10^-10} then
    Append(~S, f);
  end if;
end for;
#CartesianProduct(B), #S;                          // 3159 24
forall{f : f in S | forall{g : g in Factorization(f) |
                           IsCyclotomicPolynomial(g[1])}};        // true
```

Use this output to prove Kronecker's theorem for every $x \in \overline{\mathbb{Q}}^\times$ with $[\mathbb{Q}(x) : \mathbb{Q}] \leqslant 4$, without Theorem 39.4.

SOLUTION. **The minimal polynomial g of x, times $(t-1)^{4-\deg g}$, is in the list.** Let $\alpha_1, \ldots, \alpha_d$ be the roots of f, with multiplicity. Then $a_i = (-1)^i e_i(\alpha_1, \ldots, \alpha_d)$, a sum of $\binom{d}{i}$ products of roots, each of absolute value at most 1. So $|a_i| \leqslant \binom{d}{i}$.

Let $x \in \overline{\mathbb{Q}}^\times$ with $H(x) = 1$ and $e := [\mathbb{Q}(x) : \mathbb{Q}] \leqslant 4$. By (2), $x \in \overline{\mathbb{Z}}$, and its conjugates lie in the closed unit disc. So its minimal polynomial $g \in \mathbb{Z}[t]$ is monic of degree e, with all roots in the disc, and $g(0) \neq 0$ since $x \neq 0$. Then $f := g \cdot (t-1)^{4-e}$ is monic of degree 4, with all roots in the disc and $f(0) \neq 0$. So f is one of the 24 polynomials of the list, and its irreducible factor g is cyclotomic. So x is a root of unity.

Remark. This is Kronecker's own proof: the minimal polynomials of the powers x^n of such an x all satisfy the same bound on their coefficients, so they are finitely many. The bound $|a_i| \leqslant \binom{d}{i}$ plays the role of [Sil09, Theorem VIII.5.9] in the proof of Theorem 39.4. The enumeration is exponential in d: for $d = 6$ it already takes minutes.

(5) Let $d \geqslant 1$, and let $2^{1/d}$ be the real root of $t^d - 2$. Show that $h(2^{1/d}) = (\log 2)/d$. So h takes arbitrarily small positive values on $\overline{\mathbb{Q}}^\times$. `Magma` checks this, and computes the logarithmic height of a root b of Lehmer's polynomial:

```
R<t> := PolynomialRing(Integers());
[RealField(10) | d*AbsoluteLogarithmicHeight(NumberField(t^d - 2).1)
                 : d in [2..5]];       // [ 0.6931471805, 0.6931471805, ... ]
```

```
RealField(10)!Log(2);                    // 0.6931471805
f := t^10 + t^9 - t^7 - t^6 - t^5 - t^4 - t^3 + t + 1;
IsIrreducible(f), IsCyclotomicPolynomial(f);          // true false
K<b> := NumberField(f);
RealField(10)!AbsoluteLogarithmicHeight(b);           // 0.01623576120
```

Explain why $h(b) > 0$, and compute $[\mathbb{Q}(b) : \mathbb{Q}]\, h(b)$.

Solution. **$h(2^{1/d}) = (\log 2)/d$, and $10\,h(b) \approx 0.1624$.** Let $\alpha := 2^{1/d}$ and $K := \mathbb{Q}(\alpha)$. The polynomial $t^d - 2$ is Eisenstein at 2, so $[K : \mathbb{Q}] = d$. Since $\alpha \in \mathcal{O}_K$, the non-archimedean factors of $H_K([\alpha : 1])$ are 1. The archimedean factors are $\max(|\tau(\alpha)|, 1)$, one for each of the d embeddings τ, a complex place contributing its two embeddings through $n_v = 2$. Each $\tau(\alpha)$ is $2^{1/d}$ times a root of unity, so $|\tau(\alpha)| = 2^{1/d} > 1$. So $H_K([\alpha : 1]) = (2^{1/d})^d = 2$, and $h(\alpha) = (\log 2)/d$.
The polynomial f is irreducible and not cyclotomic, so its root b is not a root of unity. By Kronecker's theorem, $h(b) > 0$. Here $[\mathbb{Q}(b) : \mathbb{Q}] = 10$, and $10\,h(b) \approx 0.1624$.

Remark. So there is no positive lower bound for h on the algebraic numbers that are not roots of unity. Lehmer asked whether there is one for $[\mathbb{Q}(x) : \mathbb{Q}]\, h(x)$, a question still open. The value $10\,h(b) = \log 1.17628\ldots$ of his polynomial is the smallest positive one known.

40. Basic properties of the height

This section is [Sil09, Exercise 8.8]. The last two items are [HS00, Proposition B.2.4(b), (c)], for the Segre embedding [Har77, Exercise I.2.14] and the d-uple embedding [Har77, Exercise I.2.12]. The heights are those of Definition 39.2.

Problem 48 (Basic properties of the height)**.**
In (1) and (2), let $N \geqslant 1$ and $x_1, \ldots, x_N \in \overline{\mathbb{Q}}$.

(1) Show that $H(x_1 x_2 \cdots x_N) \leqslant H(x_1) H(x_2) \cdots H(x_N)$. Show that the inequality can be strict.

Solution. **$\max(|xy|_v, 1) \leqslant \max(|x|_v, 1) \max(|y|_v, 1)$.** Let K be a number field containing $x_1, \ldots, x_N$. If $a, b \geqslant 0$, then $\max(ab, 1) \leqslant \max(a, 1) \max(b, 1)$, since the right side is at least ab and at least 1. By induction on N, for every $v \in \mathcal{P}_K$,

$$\max(|x_1 \cdots x_N|_v, 1) \leqslant \prod_{i=1}^{N} \max(|x_i|_v, 1).$$

Raise to the power n_v, multiply over v, and take the $[K : \mathbb{Q}]$-th root. For $x = 2$, the product $x \cdot x^{-1} = 1$ has height $1 < H(2)H(1/2) = 4$.

(2) Show that $H(x_1 + x_2 + \cdots + x_N) \leqslant N\, H(x_1) H(x_2) \cdots H(x_N)$, and that the factor N cannot be replaced by a smaller one.

Solution. **$\prod_v N^{\epsilon(v) n_v} = N^{[K:\mathbb{Q}]}$. Equality for $x_1 = \cdots = x_N = 1$.** Let K be a number field containing $x_1, \ldots, x_N$, and $v \in \mathcal{P}_K$. Let $\epsilon(v) := 1$ if v is archimedean and $\epsilon(v) := 0$ otherwise. The triangle inequality, or the ultrametric one, gives $|x_1 + \cdots + x_N|_v \leqslant N^{\epsilon(v)} \max_i |x_i|_v$. Since moreover $\max_i a_i \leqslant \prod_i \max(a_i, 1)$ for $a_i \geqslant 0$,

$$\max(|x_1 + \cdots + x_N|_v, 1) \leqslant N^{\epsilon(v)} \max\Big(\max_i |x_i|_v, 1\Big) \leqslant N^{\epsilon(v)} \prod_{i=1}^{N} \max(|x_i|_v, 1).$$

Raise to the power n_v and multiply over v. By part (1) of Proposition 39.1, for the extension $K/\mathbb{Q}$ and the archimedean place of $\mathbb{Q}$, $\sum_{v\in\mathcal{P}_K^\infty} n_v = [K:\mathbb{Q}]$. So the factors $N^{\epsilon(v)n_v}$ multiply to $N^{[K:\mathbb{Q}]}$, and the $[K:\mathbb{Q}]$-th root gives the claim. For $x_1 = \cdots = x_N = 1$, the left side is $H(N) = N$ and the right side is $N \cdot 1 = N$.

(3) Let $N, M \geqslant 0$, and let $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ and $Q = [y_0 : \cdots : y_M] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. The Segre embedding sends (P, Q) to the point

$$S(P,Q) \coloneqq [x_0y_0 : x_0y_1 : \cdots : x_iy_j : \cdots : x_Ny_M] \in \mathbb{P}^{MN+M+N}(\overline{\mathbb{Q}}),$$

with the products x_iy_j in lexicographic order of (i,j). Show that $S(P,Q)$ does not depend on the choice of coordinates, and that $H(S(P,Q)) = H(P)H(Q)$.

Solution. $\mathbf{\max_{i,j}|x_iy_j|_v = |P|_v|Q|_v}$. There are $(N+1)(M+1) = MN+M+N+1$ products. If some $x_i \neq 0$ and some $y_j \neq 0$, then $x_iy_j \neq 0$. Rescaling the coordinates of P by λ and those of Q by μ rescales every product by $\lambda\mu$. Let K be a number field containing all x_i and y_j, and $v \in \mathcal{P}_K$. Since $|x_iy_j|_v = |x_i|_v|y_j|_v$, the maximum over (i,j) is attained at a pair where both factors are maximal. Raise to the power n_v, multiply over v, and take the $[K:\mathbb{Q}]$-th root.

(4) Let $N \geqslant 0$ and $d \geqslant 1$, let $M \coloneqq \binom{N+d}{N} - 1$, and let $f_0, \ldots, f_M$ be the distinct monomials of degree d in $X_0, \ldots, X_N$. The d-uple embedding sends $P = [x_0 : \cdots : x_N] \in \mathbb{P}^N(\overline{\mathbb{Q}})$ to $P^{(d)} \coloneqq [f_0(P) : \cdots : f_M(P)] \in \mathbb{P}^M(\overline{\mathbb{Q}})$. Show that

$$H(P^{(d)}) = H(P)^d = H([x_0^d : \cdots : x_N^d]).$$

Solution. $\mathbf{\max_j|f_j(P)|_v = |P|_v^d = \max_i|x_i^d|_v}$. The monomials include $X_0^d, \ldots, X_N^d$, so $P^{(d)}$ has a nonzero coordinate, and rescaling P by λ rescales $P^{(d)}$ by λ^d. Let K be a number field containing $x_0, \ldots, x_N$, and $v \in \mathcal{P}_K$. Each $f_j(P)$ is a product of d coordinates of P, so $|f_j(P)|_v \leqslant |P|_v^d$, with equality for $f_j = X_i^d$ when $|x_i|_v = |P|_v$. The same holds for the coordinates x_i^d. No triangle inequality is involved, so no constant appears. Raise to the power n_v, multiply over v, and take the $[K:\mathbb{Q}]$-th root.

Remark. For $N = 1$ and $P = [x:1]$, the last equality is $H(x^d) = H(x)^d$, as in Problem 47(3). Compare [Sil09, Theorem VIII.5.6]: for a morphism F of degree d, only $H(F(P)) = H(P)^d$ up to bounded factors holds.

41. Heights and ideals

This section is [Sil09, Exercise 8.9], which extends [Sil09, Remark VIII.5.5] from $\mathbb{Q}$ to a number field. The description of the non-archimedean places is [HS00, Proposition B.1.3] and the discussion before it. For a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$, let $\operatorname{ord}_\mathfrak{p}\colon K^\times \to \mathbb{Z}$ be the exponent of $\mathfrak{p}$ in the factorization of the fractional ideal $x\mathcal{O}_K$, and $\operatorname{Nm}\mathfrak{p} \coloneqq \#\mathcal{O}_K/\mathfrak{p}$. The norm extends multiplicatively to fractional ideals: $\operatorname{Nm}_{K/\mathbb{Q}}\mathfrak{a} \coloneqq \prod_\mathfrak{p}(\operatorname{Nm}\mathfrak{p})^{\operatorname{ord}_\mathfrak{p}\mathfrak{a}}$.

Proposition 41.1 (Non-archimedean places [HS00, Proposition B.1.3])**.** *The non-archimedean places of K correspond to the nonzero prime ideals of $\mathcal{O}_K$. If $v \in \mathcal{P}_K^0$ corresponds to $\mathfrak{p}$, then $|x|_v^{n_v} = (\operatorname{Nm}\mathfrak{p})^{-\operatorname{ord}_\mathfrak{p}(x)}$ for every $x \in K^\times$.*

Problem 49 (Heights and ideals)**.**

(1) Let $x_0, \dots, x_N \in K$, not all zero, let $P := [x_0 : \cdots : x_N] \in \mathbb{P}^N(K)$, and let $\mathfrak{b}$ be the fractional ideal of K generated by $x_0, \dots, x_N$. Show that

$$H_K(P) = \left(\mathrm{Nm}_{K/\mathbb{Q}}\,\mathfrak{b}\right)^{-1} \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}.$$

SOLUTION. **$\mathrm{ord}_\mathfrak{p}\,\mathfrak{b} = \min_i \mathrm{ord}_\mathfrak{p}(x_i)$, over the i with $x_i \neq 0$.** A sum of fractional ideals has, at each $\mathfrak{p}$, the least exponent of its summands, so $\mathrm{ord}_\mathfrak{p}\,\mathfrak{b} = \min_i \mathrm{ord}_\mathfrak{p}(x_i)$, over the i with $x_i \neq 0$. Let $v \in \mathcal{P}_K^0$ correspond to $\mathfrak{p}$. By Proposition 41.1, $|x_i|_v^{n_v} = (\mathrm{Nm}\,\mathfrak{p})^{-\mathrm{ord}_\mathfrak{p}(x_i)}$, which is largest when $\mathrm{ord}_\mathfrak{p}(x_i)$ is least. So $|P|_v^{n_v} = (\mathrm{Nm}\,\mathfrak{p})^{-\mathrm{ord}_\mathfrak{p}\,\mathfrak{b}}$. The exponent is zero for all but finitely many $\mathfrak{p}$, and the product over $\mathcal{P}_K^0$ is $\prod_\mathfrak{p}(\mathrm{Nm}\,\mathfrak{p})^{-\mathrm{ord}_\mathfrak{p}\,\mathfrak{b}} = (\mathrm{Nm}_{K/\mathbb{Q}}\,\mathfrak{b})^{-1}$.

(2) Let $K = \mathbb{Q}$. Deduce that if $x_0, \dots, x_N$ are coprime integers, then $H_\mathbb{Q}(P) = \max_i |x_i|$. More generally, show that if $\mathfrak{b}$ is principal, then P has coordinates in $\mathcal{O}_K$ that generate the unit ideal, and for such coordinates $H_K(P) = \prod_{v \in \mathcal{P}_K^\infty} |P|_v^{n_v}$.

SOLUTION. **If $\mathfrak{b} = \lambda\mathcal{O}_K$, the coordinates x_i/λ generate $\mathcal{O}_K$.** If $\mathfrak{b} = \lambda\mathcal{O}_K$, then the elements x_i/λ generate $\lambda^{-1}\mathfrak{b} = \mathcal{O}_K$, so they lie in $\mathcal{O}_K$ and generate the unit ideal. For such coordinates the ideal of (1) is $\mathcal{O}_K$, of norm 1. Over $\mathbb{Q}$, every fractional ideal is principal, the ideal generated by coprime integers is $\mathbb{Z}$, and $\mathcal{P}_\mathbb{Q}^\infty$ is one place with $n_\infty = 1$.

(3) Let $K := \mathbb{Q}(\sqrt{-5})$, with $a := \sqrt{-5}$, and let $P := [2 : 1 + a] \in \mathbb{P}^1(K)$. `Magma` computes both sides of (1), and compares with its own logarithmic height of $(1 + a)/2$:

```
K<a> := QuadraticField(-5);
O := Integers(K);
xs := [K | 2, 1 + a];
b := ideal<O | xs>;
IsPrincipal(b), Norm(b);                                    // false 2
arch := &*[Max([Abs(Evaluate(x, v)) : x in xs])^LocalDegree(v)
           : v in InfinitePlaces(K)];
primes := &join[Support(x*O) : x in xs | x ne 0];
fin := &*[Norm(p)^(-Min([Valuation(x, p) : x in xs])) : p in primes];
RealField(10)!arch, fin;                                    // 6.000000000 1/2
[RealField(10) | AbsoluteLogarithmicHeight((1 + a)/2), Log(3)/2];
                                          // [ 0.5493061444, 0.5493061444 ]
```

Check the output by hand: show that $\mathfrak{b}$ has norm 2 and is not principal, and compute $H_K(P)$ and $H(P)$. Show that P has no coordinates in $\mathcal{O}_K$ that generate the unit ideal. Explain the last line of the output.

SOLUTION. **$H_K(P) = 6/2 = 3$, $H(P) = \sqrt{3}$, and $h((1 + a)/2) = (\log 3)/2$.** In $\mathcal{O}_K = \mathbb{Z}[a]$, $(1 + a)(1 - a) = 6$, and $\mathfrak{b}^2 = (4, 2 + 2a, -4 + 2a) = 2\,(2, 1 + a, -2 + a) = (2)$, since $(1 + a) - (-2 + a) = 3$ and $3 - 2 = 1$ lie in $(2, 1 + a, -2 + a)$. So $(\mathrm{Nm}\,\mathfrak{b})^2 = \mathrm{Nm}(2) = 4$, and $\mathrm{Nm}\,\mathfrak{b} = 2$. A generator of $\mathfrak{b}$ would be an element $u + wa$ of norm $u^2 + 5w^2 = 2$, and there is none. The field has one archimedean place, complex, with $n_v = 2$, where $|P|_v^2 = \max(|2|^2, |1 + a|^2) = \max(4, 6) = 6$. So $H_K(P) = 6/2 = 3$ and $H(P) = 3^{1/2}$.
Coordinates of P are $\lambda \cdot 2$ and $\lambda(1 + a)$ for some $\lambda \in K^\times$, and they generate $\lambda\mathfrak{b}$, which is not principal, hence not $\mathcal{O}_K$. In the output, `arch` $= 6$ is the archimedean product, and `fin` $= 1/2$ the non-archimedean one, which is $(\mathrm{Nm}\,\mathfrak{b})^{-1}$. Finally, $[(1 + a)/2 : 1] = [1 + a : 2]$, which is P with its coordinates swapped, so $h((1 + a)/2) = h(P) = (\log 3)/2$.

Remark. Choosing λ so that $\lambda\mathfrak{b}$ is one of finitely many representatives of the class group, (1) gives another proof of the second part of Theorem 39.4: the coordinates then lie in a fixed fractional ideal, with bounded absolute values at the archimedean places, and such elements are finitely many. Over $\mathbb{Q}$ the class group is trivial, which is why [Sil09, Example VIII.5.1] needs no product over the places.

Part 13. Heights and torsion on elliptic curves

Kronecker's theorem says that the elements of $\overline{\mathbb{Q}}^\times$ of height 1 are the roots of unity, the torsion of $\mathbb{G}_\mathrm{m}$. This worksheet proves its analogue for an elliptic curve, for the height h_x of the x-coordinate, assuming the duplication bound of [Sil09, Corollary VIII.6.4(b)]. The torsion points have bounded height, not height 0, and that is enough for finiteness. First come the valuations of a sum of several terms, and the shape of the coordinates of a rational point, over $\mathbb{Q}$ and over a number field: the first step of Silverman's proof of the Mordell–Weil theorem over $\mathbb{Q}$.

Standing notation.

- A number field k, elements $a, b \in k$ with $\Delta := -16(4a^3 + 27b^2) \neq 0$, and the elliptic curve $E_{a,b}\colon y^2 = x^3 + ax + b$ over k.
- The height H on $\mathbb{P}^N(\overline{\mathbb{Q}})$, and $h := \log H$. For $t \in \mathbb{Q}$, with $t = u/w$ in lowest terms, $H([t:1]) = \max(|u|, |w|)$. See [Sil09, §VIII.5].
- The x-coordinate $x\colon E_{a,b} \to \mathbb{P}^1$, with $x(O) = [1:0]$, and the height $h_x(P) := h(x(P))$ of a point $P \in E_{a,b}(\overline{\mathbb{Q}})$. So $h_x(O) = 0$.

42. Valuations of sums

The proofs below compare the valuations of the terms of a sum. This section makes the ultrametric inequality precise: when it is an equality, and what happens when it is not.

Definition 42.1 (Discrete valuations)**.** A discrete valuation on a field F is a surjective map $v\colon F \to \mathbb{Z} \cup \{\infty\}$ with $v(\alpha) = \infty$ only for $\alpha = 0$, $v(\alpha\beta) = v(\alpha) + v(\beta)$ and $v(\alpha + \beta) \geqslant \min(v(\alpha), v(\beta))$. Its valuation ring is $\mathcal{O}_v := \{\alpha \in F : v(\alpha) \geqslant 0\}$, a uniformizer is an element $\pi \in F$ with $v(\pi) = 1$, and the residue field is $\kappa := \mathcal{O}_v/\pi\mathcal{O}_v$, with reduction $\alpha \mapsto \bar{\alpha}$. Examples are v_p on $\mathbb{Q}$, and $v_\mathfrak{p}$ on a number field k, for a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$.

Problem 50 (The unique minimizer criterion)**.**
Let v be a discrete valuation on a field F, with uniformizer π and residue field κ. Let $\alpha_1, \dots, \alpha_n \in F$, not all 0, $m := \min_i v(\alpha_i)$, and $I := \{i : v(\alpha_i) = m\}$.

(1) Show that $v(\alpha_1 + \cdots + \alpha_n) \geqslant m$, with equality if I has one element.

Solution. **The other terms add up to an element of valuation $> m$.** The inequality follows from the definition, by induction on n. For the equality, first let $n = 2$ and $v(\alpha_1) < v(\alpha_2)$. If $v(\alpha_1 + \alpha_2) > v(\alpha_1)$, then

$$v(\alpha_1) = v\big((\alpha_1 + \alpha_2) - \alpha_2\big) \geqslant \min\big(v(\alpha_1 + \alpha_2), v(\alpha_2)\big) > v(\alpha_1),$$

a contradiction, since $v(-\alpha_2) = v(\alpha_2)$. In general, let $I = \{1\}$. Then $v(\alpha_2 + \cdots + \alpha_n) \geqslant \min_{i \geqslant 2} v(\alpha_i) > m$, and the case $n = 2$ applies to α_1 and $\alpha_2 + \cdots + \alpha_n$.

(2) Show that $v(\alpha_1 + \cdots + \alpha_n) = m$ if and only if $\sum_{i \in I} \overline{\alpha_i/\pi^m} \neq 0$ in κ. Give an example with $\#I = 2$ where the inequality is strict.

Solution. **Divide by π^m and reduce: the terms outside I vanish in κ.** Let $\beta_i := \alpha_i/\pi^m$, so that $v(\beta_i) \geqslant 0$, with equality exactly for $i \in I$. Then $v(\alpha_1 + \cdots + \alpha_n) = m + v(\beta_1 + \cdots + \beta_n)$, and $\beta_1 + \cdots + \beta_n \in \mathcal{O}_v$ has valuation 0 if and only if its reduction is nonzero. For $i \notin I$, $v(\beta_i) \geqslant 1$, so $\bar{\beta}_i = 0$, and the reduction is $\sum_{i \in I} \bar{\beta}_i$. For an example, take $v = v_p$ and $1 + (p-1) = p$: both terms have valuation 0, and their residues 1 and -1 cancel.

Remark. In particular, equality holds when I has one element, since then the sum of residues is a single nonzero residue. This is (1) again. The criterion is the non-archimedean form of the dominant term: for large $|x|$, a polynomial behaves like its leading term, up to a constant. Here the dominant term wins exactly.

43. The shape of a rational point

This section is the first step of the proof of [Sil09, Lemma VIII.4.2], which proves that h_x satisfies the hypotheses of the descent theorem over $\mathbb{Q}$. It turns the rational points of bounded height into a finite search over integers. The second part is the same argument over a number field, where ideals take the place of the integer d. The integrality of a, b is harmless: if $u^4 a, u^6 b$ are integral, then $(x, y) \mapsto (u^2 x, u^3 y)$ is an isomorphism onto $E_{u^4 a, u^6 b}$, and it changes h_x by at most $2 \log|u|$ for $u \in \mathbb{Z}$.

Definition 43.1 (Heights over a number field [Sil09, §VIII.5])**.** For $t \in k$,

$$H([t:1])^{[k:\mathbb{Q}]} = \prod_{v \in \mathcal{P}_k} \max(1, |t|_v)^{n_v},$$

with $n_v := [k_v : \mathbb{Q}_p]$ or $[k_v : \mathbb{R}]$. For the place of a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$, $|t|_v^{n_v} = N(\mathfrak{p})^{-v_\mathfrak{p}(t)}$, where $N(\mathfrak{p})$ is the order of $\mathcal{O}_k/\mathfrak{p}$. The denominator ideal of t is $\mathfrak{d}(t) := \prod_\mathfrak{p} \mathfrak{p}^{\max(0, -v_\mathfrak{p}(t))}$, an ideal of $\mathcal{O}_k$.

Problem 51 (The shape of a rational point)**.**

(1) Let $a, b \in \mathbb{Z}$, and $P \in E_{a,b}(\mathbb{Q}) \smallsetminus \{O\}$. Show that $x(P) = n/d^2$ and $y(P) = m/d^3$ with $n, m, d \in \mathbb{Z}$, $d \geqslant 1$ and $\gcd(n, d) = \gcd(m, d) = 1$, and that $H(x(P)) = \max(|n|, d^2)$.

Solution. **At a prime p with $v_p(x) < 0$, $3v_p(x) = 2v_p(y)$.** Write $x = x(P)$ and $y = y(P)$. Let p be a prime with $v_p(x) < 0$. Since $a, b \in \mathbb{Z}$, $v_p(ax) \geqslant v_p(x) > 3v_p(x)$ and $v_p(b) \geqslant 0 > 3v_p(x)$, so x^3 is the only term of smallest valuation in $x^3 + ax + b$, and $2v_p(y) = v_p(x^3 + ax + b) = 3v_p(x)$ by Problem 50(1). Then $v_p(x) = -2r$ and $v_p(y) = -3r$ for some integer $r \geqslant 1$. If $v_p(x) \geqslant 0$, then $v_p(y^2) = v_p(x^3 + ax + b) \geqslant 0$, so $v_p(y) \geqslant 0$. Let $d := \prod_p p^r$, the product over the primes with $v_p(x) < 0$. Then $n := d^2 x$ and $m := d^3 y$ are integers prime to d. As $\gcd(n, d^2) = 1$, the fraction n/d^2 is in lowest terms, and $H(x(P)) = \max(|n|, d^2)$.

Remark. So the rational points of $E_{a,b}$ with $H(x) \leqslant B$ have $|n| \leqslant B$ and $d \leqslant \sqrt{B}$, and $m^2 = n^3 + and^4 + bd^6$: a finite search. This is how the descent bound of [Sil09, Theorem VIII.3.1] becomes a list of generators of $E_{a,b}(\mathbb{Q})$.

(2) Let $a, b \in \mathcal{O}_k$, and $P \in E_{a,b}(k) \smallsetminus \{O\}$. Show that there is an ideal $\mathfrak{d}$ of $\mathcal{O}_k$ with $\mathfrak{d}(x(P)) = \mathfrak{d}^2$ and $\mathfrak{d}(y(P)) = \mathfrak{d}^3$, and that

$$H(x(P))^{[k:\mathbb{Q}]} = N(\mathfrak{d})^2 \prod_{v|\infty} \max(1, |x(P)|_v)^{n_v}.$$

Check that for $k = \mathbb{Q}$ this is (1).

Solution. **The argument of (1) works at each prime $\mathfrak{p}$.** Write $x = x(P)$ and $y = y(P)$. At a prime $\mathfrak{p}$ with $v_\mathfrak{p}(x) < 0$, the term x^3 is again the only one of smallest valuation, as $v_\mathfrak{p}(a), v_\mathfrak{p}(b) \geqslant 0$. So $v_\mathfrak{p}(x) = -2r_\mathfrak{p}$ and $v_\mathfrak{p}(y) = -3r_\mathfrak{p}$ with $r_\mathfrak{p} \geqslant 1$, and $v_\mathfrak{p}(y) \geqslant 0$ when $v_\mathfrak{p}(x) \geqslant 0$. Let $\mathfrak{d} := \prod_\mathfrak{p} \mathfrak{p}^{r_\mathfrak{p}}$, over the primes with $v_\mathfrak{p}(x) < 0$. Then $\mathfrak{d}(x) = \mathfrak{d}^2$ and $\mathfrak{d}(y) = \mathfrak{d}^3$. In Definition 43.1, the factor of a prime $\mathfrak{p}$ is $\max(1, N(\mathfrak{p})^{-v_\mathfrak{p}(x)}) = N(\mathfrak{p})^{\max(0,-v_\mathfrak{p}(x))}$, so the product of these factors is $N(\mathfrak{d}(x)) = N(\mathfrak{d})^2$. For $k = \mathbb{Q}$, $\mathfrak{d} = d\mathbb{Z}$ with $d \geqslant 1$, $N(\mathfrak{d}) = d$, and the one archimedean factor is $\max(1, |n|/d^2)$, so $H(x) = d^2 \max(1, |n|/d^2) = \max(|n|, d^2)$.

Remark. If $\mathfrak{d} = d\mathcal{O}_k$ is principal, then $x = n/d^2$ and $y = m/d^3$ with $n := d^2x$ and $m := d^3y$ in $\mathcal{O}_k$, as over $\mathbb{Q}$. In general $\mathfrak{d}$ need not be principal, and the class group is the obstruction to writing x this way. The height still bounds $N(\mathfrak{d})$, which leaves finitely many ideals $\mathfrak{d}$.

44. Kronecker's theorem for elliptic curves

Kronecker's theorem [Sil09, Exercise 8.6] says that $\alpha \in \overline{\mathbb{Q}}^\times$ has $H(\alpha) = 1$ if and only if α is a root of unity. This section proves the analogue for $E_{a,b}$ and h_x, from the results below.

Proposition 44.1 (The duplication formula [Sil09, Algorithm III.2.3(d)])**.** *Let $P \in E_{a,b}(\overline{\mathbb{Q}})$ with $2P \neq O$, and $x := x(P)$. Then*

$$x(2P) = \frac{x^4 - 2ax^2 - 8bx + a^2}{4(x^3 + ax + b)}.$$

Proposition 44.2 (The duplication bound [Sil09, Corollary VIII.6.4(b)])**.** *There is a constant $C \geqslant 0$, depending on a and b, with $|h_x(2P) - 4h_x(P)| \leqslant C$ for all $P \in E_{a,b}(\overline{\mathbb{Q}})$.*

Theorem 44.3 (Northcott [Sil09, Theorem VIII.5.11])**.** *For all constants B and D, the set of $R \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(R) \leqslant B$ and $[\mathbb{Q}(R) : \mathbb{Q}] \leqslant D$ is finite. Here $\mathbb{Q}(R)$ is the field of definition of R.*

Problem 52 (Kronecker for elliptic curves)**.**
Let C be the constant of Proposition 44.2. Only its lower bound, $h_x(2P) \geqslant 4h_x(P) - C$, is used.

(1) Show that every torsion point $T \in E_{a,b}(\overline{\mathbb{Q}})$ has $h_x(T) \leqslant C/3$.

Solution. **The largest height M on the group $\langle T \rangle$ has $4M - C \leqslant M$.** Let $M := \max\{h_x(P) : P \in \langle T \rangle\}$, attained at nT for some $n \in \mathbb{Z}$. Then $2nT \in \langle T \rangle$, so

$$4M - C = 4h_x(nT) - C \leqslant h_x(2nT) \leqslant M,$$

and $M \leqslant C/3$. In particular, $h_x(T) \leqslant M \leqslant C/3$.

Remark. The argument uses only two properties of $\langle T \rangle$: it is finite, and it contains the double of each of its elements. The orbit $\{2^jT : j \geqslant 0\}$ of T under doubling has both, and would do as well.

(2) Show that for every $D \geqslant 1$ there are only finitely many torsion points $T \in E_{a,b}(\overline{\mathbb{Q}})$ with $[\mathbb{Q}(T) : \mathbb{Q}] \leqslant D$. Deduce that the torsion subgroup of $E_{a,b}(K)$ is finite for every number field K containing k.

Solution. **Their x-coordinates have bounded height and bounded degree.** Let $T \neq O$ be a torsion point with $[\mathbb{Q}(T) : \mathbb{Q}] \leqslant D$. Then $x(T) \in \mathbb{Q}(T)$, so $[\mathbb{Q}(x(T)) : \mathbb{Q}] \leqslant D$, and $H(x(T)) \leqslant e^{C/3}$ by (1). By Theorem 44.3, only finitely many points of $\mathbb{P}^1(\overline{\mathbb{Q}})$ satisfy both conditions, and each is $x(T)$ for at most two points T. For a number field $K \supseteq k$, every $T \in E_{a,b}(K)$ has $\mathbb{Q}(T) \subseteq K$, so $D = [K : \mathbb{Q}]$ works.

Remark. For $\mathbb{G}_{\mathrm{m}}$ the same argument, with $h(\alpha^2) = 2h(\alpha)$ in place of Proposition 44.2, gives $h(\zeta) = 0$ for every root of unity ζ, and the finiteness of the roots of unity of bounded degree. On $E_{a,b}$, the error C leaves room for torsion points of positive height.

(3) Show that the naive analogue of Kronecker's theorem fails in both directions: a torsion point need not have $h_x = 0$, and a rational point with $h_x = 0$ need not be a torsion point.

(a) On $y^2 = x^3 + 1$, show that $T = (2, 3)$ is a torsion point of order 6, with $h_x(T) = \log 2$.

SOLUTION. **Here $x(2T) = 0$ and $x(4T) = 0$, so $4T = -2T$.** By Proposition 44.1 with $a = 0$ and $b = 1$, $x(2T) = (16 - 16)/(4 \cdot 9) = 0$, so $2T = (0, \pm 1)$, and $x(4T) = 0/4 = 0$. So $4T = \pm 2T$. As $y(T) \neq 0$, $2T \neq O$, so $4T \neq 2T$, and $4T = -2T$, that is, $6T = O$. The order is not 1, 2 or 3: $T \neq O$, $2T \neq O$, and $3T = O$ would mean $x(2T) = x(T) = 2$. So T has order 6, and $h_x(T) = \log H(2) = \log 2$.

(b) On $y^2 = x^3 + x + 1$, the rational point $P = (0, 1)$ has $h_x(P) = 0$. For this curve over $\mathbb{Q}$, one can take $C = \log 619$ in the lower bound $h_x(2P) \geqslant 4h_x(P) - C$. Show that P has infinite order.

SOLUTION. **Here $x(4P) = -287/1296$, and $1296 > 619^{1/3}$.** By Proposition 44.1 with $a = b = 1$, $x(2P) = 1/4$, and

$$x(4P) = \frac{(1/4)^4 - 2(1/4)^2 - 8(1/4) + 1}{4\big((1/4)^3 + 1/4 + 1\big)} = \frac{-287/256}{81/16} = -\frac{287}{1296},$$

in lowest terms, as $287 = 7 \cdot 41$. If P were a torsion point, so would be $4P$, and (1), which uses only the lower bound, would give $H(x(4P)) \leqslant 619^{1/3} < 9$. But $H(x(4P)) = 1296$. So P has infinite order.

Remark. The constant $\log 619$ comes from the explicit identities of [Sil09, Sublemma VIII.4.3], as in the proof of [Sil09, Lemma VIII.4.2]. The Lutz–Nagell theorem [Sil09, Corollary VIII.7.2] gives another proof: $x(2P) = 1/4$ is not an integer, so $2P$ is not a torsion point.

Remark. The canonical height $\hat{h}(P) := \lim_{n\to\infty} 4^{-n} h_x(2^n P)$ gives the exact analogue of Kronecker's theorem: $\hat{h}(mP) = m^2 \hat{h}(P)$, and $\hat{h}(P) = 0$ if and only if P is a torsion point [Sil09, Theorem VIII.9.3(b), (d)], where $\hat{h}$ is half of ours. Writing $\hat{h} - h_x$ as a telescoping series with terms $4^{-n-1}\big(h_x(2^{n+1}P) - 4h_x(2^n P)\big)$ gives $|\hat{h} - h_x| \leqslant C/3$. So (1) says that $h_x(T)$ is within $C/3$ of $\hat{h}(T) = 0$.

Part 14. The Kubert–Tate normal form and good reduction everywhere

This worksheet is about two sections of [Sil09]: torsion points (§VIII.7), and minimal equations and good reduction (§VIII.8). Mazur's theorem lists the fifteen groups that occur as the torsion subgroup of an elliptic curve over $\mathbb{Q}$. The Kubert–Tate normal form puts an elliptic curve with a k-point P of order $m \geqslant 4$ in a family with two parameters, in which m is cut out by a polynomial condition on the parameters. These conditions define curves, the modular curves $Y_1(m)$, whose k-points parametrize such pairs (E, P). Problem 53 derives the normal form, and finds the conditions for $m = 4, 5, 6, 7, 9, 11, 12$. For $m = 11$ the curve has genus one. Problem 54 shows that every elliptic curve over $\mathbb{Q}$ has bad reduction at some prime.

Standing notation.

- A field k, and an elliptic curve E over k given by a Weierstrass equation

$$y^2 + a_1 xy + a_3 y = x^3 + a_2 x^2 + a_4 x + a_6$$

with coefficients in k, discriminant Δ and point at infinity O [Sil09, §III.1].
- A change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$, with $u \in k^\times$ and $r, s, t \in k$. It gives a Weierstrass equation for E in x', y', with discriminant $u^{-12}\Delta$. Any two Weierstrass equations for E are related by a change of variables [Sil09, Proposition III.3.1(b)].
- Coordinates $x(Q)$ and $y(Q)$ of a $\overline{k}$-point $Q \neq O$ of E.

45. The Kubert–Tate normal form and Mazur's list

Problem 53 is [Sil09, Exercise 8.13]. Its item (1) is Exercise 8.13(a), the orders 6, 7, 9 and 12 are Exercise 8.13(b), and the order 11 is Exercise 8.13(c). The normal form, and the conditions for all the orders allowed by Theorem 45.1, are in [Kub76]. The elliptic curves with a k-point of order $m \geqslant 4$ are parametrized by the k-points of a modular curve [Sil09, Remark VIII.7.8].

Theorem 45.1 (Mazur [Sil09, Theorem VIII.7.5])**.** *Let E be an elliptic curve over $\mathbb{Q}$. Then $E(\mathbb{Q})_{\text{tors}}$ is isomorphic to one of the fifteen groups*

$$\mathbb{Z}/N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 10 \text{ or } N = 12, \qquad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2N\mathbb{Z} \text{ with } 1 \leqslant N \leqslant 4.$$

Each of them occurs as $E(\mathbb{Q})_{\text{tors}}$ for some elliptic curve E over $\mathbb{Q}$.

Proposition 45.2 (The group law [Sil09, Proposition III.2.2(a) and Algorithm III.2.3(a)])**.** *If a line in $\mathbb{P}^2$ meets E at the $\overline{k}$-points P_1, P_2, P_3, counted with multiplicity, then $P_1 + P_2 + P_3 = O$. The negative of a $\overline{k}$-point (x_0, y_0) is $(x_0, -y_0 - a_1x_0 - a_3)$.*

Definition 45.3 (Kubert–Tate normal form)**.** A Weierstrass equation is in Kubert–Tate normal form if it is

$$y^2 + (1 - c)xy - by = x^3 - bx^2 \tag{45.1}$$

for some $b, c \in k$.

Problem 53 (The Kubert–Tate normal form and Mazur's list)**.**
(1) Let $P \in E(k)$ be a k-point of order $m \geqslant 4$, possibly infinite. In three steps, show that a change of variables gives a Weierstrass equation for E in Kubert–Tate normal form, in which $P = (0, 0)$. After each change of variables, write again x, y for the coordinates and $a_1, \dots, a_6$ for the coefficients.

(a) Show that a change of variables with $u = 1$ and $s = 0$ moves P to $(0, 0)$, and that then $a_6 = 0$.

Solution. **Take $r = x(P)$ and $t = y(P)$.** Since $P \neq O$, it is an affine k-point $(x(P), y(P))$. The change of variables $x = x' + x(P)$, $y = y' + y(P)$ has $u = 1$ and $s = 0$, and gives P the coordinates $(0, 0)$. As $(0, 0)$ satisfies the new equation, its constant term a_6 is 0.

(b) Suppose that $P = (0, 0)$, so that $a_6 = 0$. Show that $a_3 \neq 0$, and that the change of variables with $u = 1$, $r = t = 0$ and $s = a_4/a_3$ makes $a_4 = 0$. Show that then the tangent line to E at P is $y = 0$.

Solution. **If $a_3 = 0$, the tangent line at P is $x = 0$, and then $2P = O$.** Put

$$F := y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x.$$

At $(0, 0)$, $\partial F/\partial x$ takes the value $-a_4$ and $\partial F/\partial y$ the value a_3, and they are not both 0, since E is nonsingular. So the tangent line at P is $a_3y = a_4x$. If $a_3 = 0$, it is the line $x = 0$. In projective coordinates it is $X = 0$, which passes through $O = [0 : 1 : 0]$. So it meets E at P twice and at O, and Proposition 45.2 gives $2P = O$, a contradiction. So $a_3 \neq 0$.

Substituting $y = y' + sx$ gives

$$y'^2 + (a_1 + 2s)xy' + a_3y' = x^3 + (a_2 - sa_1 - s^2)x^2 + (a_4 - sa_3)x,$$

whose coefficient of x is 0 for $s = a_4/a_3$. The new coordinates of P are still $(0,0)$, and the tangent line there is $a_3y' = 0$, that is, $y' = 0$.

(c) Suppose that $P = (0,0)$ and $a_4 = a_6 = 0$, with $a_3 \neq 0$. Show that $a_2 \neq 0$, and that the change of variables with $u = a_3/a_2$ and $r = s = t = 0$ gives a Weierstrass equation in Kubert–Tate normal form. Find b and c.

SOLUTION. **Here $b = -a_2^3/a_3^2$ and $c = 1 - a_1a_2/a_3$.** As in (1)(b), the tangent line at P is $a_3y = a_4x$, that is, $y = 0$. On it the equation becomes $x^3 + a_2x^2 = x^2(x + a_2) = 0$. If $a_2 = 0$, the line meets E at P three times, and Proposition 45.2 gives $3P = O$, a contradiction. So $a_2 \neq 0$, and $u := a_3/a_2$ lies in $k^\times$. Substituting $x = u^2x'$ and $y = u^3y'$, and dividing by u^6, gives

$$y'^2 + \frac{a_1}{u}x'y' + \frac{a_3}{u^3}y' = x'^3 + \frac{a_2}{u^2}x'^2.$$

Here $a_3/u^3 = a_2^3/a_3^2 = a_2/u^2$ and $a_1/u = a_1a_2/a_3$. So the equation is in Kubert–Tate normal form, with $b = -a_2^3/a_3^2$ and $c = 1 - a_1a_2/a_3$, and P is still $(0,0)$. The composite of the three changes of variables is again a change of variables, which proves (1).

(2) Conversely, suppose that E is given by an equation in Kubert–Tate normal form. Let $P := (0,0) \in E(k)$, and let m be its order, possibly infinite. Show that $b \neq 0$, that $2P = (b, bc)$ and $3P = (c, b - c)$, and that $m \geqslant 4$.

SOLUTION. **The tangent line at P is $y = 0$, and the line through P and $2P$ is $y = cx$.** If $b = 0$, every term of the equation has degree at least 2 in x and y, so $(0,0)$ is a singular point, which is impossible. So $b \neq 0$. As in (1)(b), the tangent line at P is $a_3y = a_4x$, here $-by = 0$, that is, $y = 0$. On it the equation becomes $x^3 - bx^2 = x^2(x - b) = 0$. So the line meets E at P twice and at $(b,0)$, and Proposition 45.2, with $a_1 = 1 - c$ and $a_3 = -b$, gives

$$2P = -(b,0) = \big(b, -(1-c)b + b\big) = (b, bc).$$

As $b \neq 0$, the line through P and $2P$ is $y = cx$. On it the equation becomes

$$c^2x^2 + (1-c)cx^2 - bcx = x^3 - bx^2, \quad \text{that is,} \quad x(x-b)(x-c) = 0.$$

So the line meets E at P, $2P$ and (c, c^2), counted with multiplicity (which matters when $c = 0$ or $c = b$), and

$$3P = -(c, c^2) = \big(c, -c^2 - (1-c)c + b\big) = (c, b - c).$$

The point at infinity O is none of P, $2P$ and $3P$, which are affine. So $m \geqslant 4$.

(3) With E and P as in (2), suppose that a change of variables gives another equation for E in Kubert–Tate normal form, with parameters b' and c', in which P is again $(0,0)$. Show that $b' = b$ and $c' = c$. So the pair (E, P) determines b and c.

SOLUTION. **The change of variables is the identity.** As P is $(0,0)$ in both coordinates, $r = t = 0$. Substituting $x = u^2x'$ and $y = u^3y' + su^2x'$ in (45.1), and dividing by u^6, gives as in (1)(b)

$$a_1' = \frac{a_1 + 2s}{u}, \quad a_2' = \frac{a_2 - sa_1 - s^2}{u^2}, \quad a_3' = \frac{a_3}{u^3}, \quad a_4' = \frac{a_4 - sa_3}{u^4}.$$

Both equations have $a_4 = a_4' = 0$, and $a_3 = -b \neq 0$ by (2). So $s = 0$. Both have $a_2 = a_3$ and $a_2' = a_3'$, so $a_2/u^2 = a_2/u^3$ with $a_2 = -b \neq 0$, and $u = 1$. Then $b' = -a_3' = b$, and $1 - c' = a_1' = a_1 = 1 - c$.

(4) With E, P and m as in (2), show that $m = 4$ if and only if $c = 0$, and $m = 5$ if and only if $b = c$.

SOLUTION. **Here $m = 4$ means $3P = -P$, and $m = 5$ means $3P = -2P$.** By (2), $m \geqslant 4$. So $m = 4$ if and only if $4P = O$, that is, $3P = -P$. By Proposition 45.2, $-P = (0, b)$, and $3P = (c, b - c)$ equals $(0, b)$ if and only if $c = 0$. Likewise, $m = 5$ if and only if $5P = O$, since 5 is prime, that is, $3P = -2P$. Here $-2P = (b, 0)$, the third k-point on the tangent line at P, and $(c, b - c) = (b, 0)$ if and only if $c = b$.

Remark. The formulas of [Sil09, §III.1] give the discriminant of (45.1):

$$\Delta = b^3\big(16b^2 - (8c^2 + 20c - 1)b + c(c - 1)^3\big).$$

For $c = 0$ it is $b^4(16b + 1)$, and for $b = c$ it is $b^5(b^2 - 11b - 1)$. So, up to a change of variables, the elliptic curves with a k-point of order 4 are the curves $y^2 + xy - by = x^3 - bx^2$ with $b(16b + 1) \neq 0$, and those with a k-point of order 5 are the curves $y^2 + (1 - b)xy - by = x^3 - bx^2$ with $b(b^2 - 11b - 1) \neq 0$. For $b = 1$ the second family gives $y^2 - y = x^3 - x^2$, with $\Delta = -11$, from [Sil09, Exercise 8.12(e)].

(5) With E, P and m as in (2), find the condition on b and c for $m = 6$.

SOLUTION. **Here $b = c + c^2$.** By (2), $m \geqslant 4$, so $m = 6$ if and only if $6P = O$, that is, $3P = -3P$. By Proposition 45.2, $-3P = (c, c^2)$, the third k-point on the line through P and $2P$. And $(c, b - c) = (c, c^2)$ if and only if $b = c + c^2$.

Remark. For $b = c + c^2$ the discriminant is $c^6(c + 1)^3(9c + 1)$. So, up to a change of variables, the elliptic curves with a k-point of order 6 are the curves

$$y^2 + (1 - c)xy - (c + c^2)y = x^3 - (c + c^2)x^2, \qquad c(c + 1)(9c + 1) \neq 0.$$

For $c = 1$ this is $y^2 - 2y = x^3 - 2x^2$, with $\Delta = 80$.

(6) With E, P and m as in (2), suppose that $m \geqslant 5$, so that $c \neq 0$, and put $d := b/c$. Show that

$$4P = \big(d^2 - d,\ d^2(c - d + 1)\big).$$

SOLUTION. **The line through P and $3P$ meets E again at $-4P = (d^2 - d, d(d - 1)^2)$.** Here $b = cd$ and $3P = (c, c(d - 1))$, so the line through P and $3P$ is $y = (d - 1)x$. On it the equation becomes

$$x^3 - \big((d - 1)^2 + (1 - c)(d - 1) + cd\big)x^2 + cd(d - 1)x = 0.$$

Its roots are 0, c and a third root x_3, counted with multiplicity, and $cx_3 = cd(d - 1)$. As $c \neq 0$, $x_3 = d(d - 1)$. By Proposition 45.2 the third k-point on the line is $-(P + 3P) = -4P = (d(d - 1), d(d - 1)^2)$, and

$$4P = \big(d(d - 1),\ -d(d - 1)^2 - (1 - c)d(d - 1) + cd\big) = \big(d^2 - d,\ d^2(c - d + 1)\big).$$

(7) With $m \geqslant 5$ and d as in (6), show that $m = 7$ if and only if P, $2P$ and $4P$ are collinear, and that this holds if and only if $c = d^2 - d$. Deduce a one-parameter family of elliptic curves with a k-point of order 7.

SOLUTION. **Here $c = d^2 - d$ and $b = d^3 - d^2$, with $d \in k$.** The k-points P, $2P$ and $4P$ are distinct, since P, $2P$ and $3P$ are not O. The line through P and $2P$ meets E again at $-3P$. If $4P$ lies on it, then $4P = -3P$, so $7P = O$ and $m = 7$, as 7 is prime. Conversely, if $m = 7$, then $-3P = 4P$ lies on the line. By (2) this line is $y = cx$, and by (6) the k-point $4P$ lies on it if and only if $d^2(c - d + 1) = cd(d - 1)$. As $d \neq 0$, this is $dc - d^2 + d = cd - c$, that is, $c = d^2 - d$. Then $b = cd = d^3 - d^2$.

Remark. This is the family of [Sil09, Remark VIII.7.8]:

$$y^2 + (1 + d - d^2)xy + (d^2 - d^3)y = x^3 + (d^2 - d^3)x^2, \qquad \Delta = d^7(d-1)^7(d^3 - 8d^2 + 5d + 1).$$

(8) With d as in (6), suppose that $m \geqslant 6$, so that $d \neq 1$, and put $f := c/(d-1)$. Show that $m = 9$ if and only if $2P$, $3P$ and $4P$ are collinear, and that this holds if and only if

$$c^2 - (d-1)c - (d-1)^3 = 0, \quad \text{that is,} \quad d = f^2 - f + 1.$$

Deduce a one-parameter family of elliptic curves with a k-point of order 9.

Solution. **Here $d = f^2 - f + 1$, $c = f^2(f-1)$ and $b = f^2(f-1)(f^2 - f + 1)$, with $f \in k$.** The k-points $2P$, $3P$ and $4P$ are distinct, since P and $2P$ are not O. The line through $2P$ and $3P$ meets E again at $-5P$. If $4P$ lies on it, then $4P = -5P$, so $9P = O$ and $m = 9$, as $m \geqslant 4$. Conversely, if $m = 9$, then $-5P = 4P$ lies on the line. Here $2P = (cd, c^2 d)$ and $3P = (c, c(d-1))$, so the line has slope $\lambda = (cd - d + 1)/(d-1)$, and $4P$ lies on it if and only if $(d-1)\big(y(4P) - y(3P)\big) = (d-1)\lambda\big(x(4P) - x(3P)\big)$, that is,

$$(d-1)\big(d^2(c - d + 1) - c(d-1)\big) = (cd - d + 1)(d^2 - d - c).$$

Expanding, the difference of the two sides is $d\big(c^2 - (d-1)c - (d-1)^3\big)$, and $d \neq 0$. Substituting $c = f(d-1)$ gives $(d-1)^2\big(f^2 - f - (d-1)\big) = 0$, that is, $d = f^2 - f + 1$. Then $c = f(d-1) = f^2(f-1)$ and $b = cd$.

Remark. The discriminant of this family vanishes exactly where $f(f-1)(f^2-f+1)(f^3-6f^2+3f+1) = 0$, and for the other $f \in k$ the k-point $(0, 0)$ has order exactly 9 (checked in Magma).

(9) With d and f as in (8), suppose that $m \geqslant 9$, so that $5P, \ldots, 8P$ are not O. The following code takes $b = cd$, works over the field $\mathbb{Q}(c, d)$ of rational functions, and factors the numerators of $x(4P) - x(8P)$ and $x(5P) - x(6P)$ after the substitution $c = f(d-1)$.

```
K<c, d> := FunctionField(Rationals(), 2);
E := EllipticCurve([1 - c, -c*d, -c*d, 0, 0]);    // b = c*d
P := E![0, 0];
4*P;                      // (d^2 - d : c*d^2 - d^3 + d^2 : 1)
R<f, d> := PolynomialRing(Rationals(), 2);
// the irreducible factors of the numerator of x(iP) - x(jP), with c = f(d - 1)
Factors := func<i, j | [g[1] : g in Factorization(
  Evaluate(Numerator((i*P)[1] - (j*P)[1]), [f*(d - 1), d]))]>;
Factors(4, 8);
// [ d, d - 1, f - 1, f^2 - f*d^2 - f*d + 3*d^2 - 3*d + 1 ]
Factors(5, 6);
// [ d - 1, f, f^3*d - 3*f^2*d + 4*f*d - f - d^2 ]
```

Solution. For $i \neq j$ with $0 < |i - j| < m$, the k-points iP and jP have the same x-coordinate if and only if $iP = -jP$, since $iP = jP$ would give $(i-j)P = O$. The numerators vanish exactly where the x-coordinates agree. Indeed the denominators of $x(5P)$, $x(6P)$ and $x(8P)$ are $(d-1)^2$, $(c - d + 1)^2$ and $\big(cd - (2d-1)(d-1)\big)^2$ (checked in Magma), which vanish only where $5P$, $6P$ or $8P$ is O. For $8P$, this holds as $8P = O$ means that $4P = -4P$, that is, $2y(4P) + (1-c)x(4P) - cd = d\big(cd - (2d-1)(d-1)\big) = 0$. And $d \neq 0$ (as $b \neq 0$), $d \neq 1$ (as $m \neq 5$), $f \neq 0$ (as $c \neq 0$) and $f \neq 1$ (as $f = 1$ means $c = d - 1$, that is, $b = c + c^2$ and $m = 6$).

(a) Show that $m = 12$ if and only if

$$f^2 - (d^2 + d)f + 3d^2 - 3d + 1 = 0.$$

Suppose that $\operatorname{char} k \neq 2$. Find a one-parameter family of elliptic curves with a k-point of order 12.

Solution. **Take $d = (v^2 - 2v + 2)/v$ and $f = (3v^2 - 6v + 4)/v^2$, with $v \in k^\times$.** Since $m \geqslant 9$, the x-coordinates of $4P$ and $8P$ agree if and only if $12P = O$, that is, $m = 12$. By the solution above and the first line of output, this holds if and only if $f^2 - (d^2 + d)f + 3d^2 - 3d + 1 = 0$. Its discriminant as a polynomial in f is

$$(d^2 + d)^2 - 4(3d^2 - 3d + 1) = d^4 + 2d^3 - 11d^2 + 12d - 4 = (d - 1)^2(d^2 + 4d - 4).$$

So, for $d \neq 1$, there is a solution $f \in k$ if and only if $d^2 + 4d - 4 = w^2$ for some $w \in k$, that is, $(d + 2 - w)(d + 2 + w) = 8$. Putting $d + 2 - w = 2v$ and $d + 2 + w = 4/v$, with $v \in k^\times$, gives

$$d = v + \frac{2}{v} - 2 = \frac{v^2 - 2v + 2}{v}, \qquad w = \frac{2}{v} - v, \qquad f = \frac{d^2 + d + (d - 1)w}{2} = \frac{3v^2 - 6v + 4}{v^2},$$

the other root f corresponding to $2/v$ in place of v. Then $c = f(d - 1)$ and $b = cd$.

Remark. The discriminant of this family vanishes exactly where

$$(v - 1)(v - 2)(3v^2 - 6v + 2)(3v^2 - 6v + 4)(v^2 - 2v + 2) = 0,$$

and for the other $v \in k^\times$ the k-point $(0, 0)$ has order exactly 12 (checked in Magma).

(b) Show that $m = 11$ if and only if

$$d^2 - (f^3 - 3f^2 + 4f)d + f = 0.$$

Suppose that $\operatorname{char} k \notin \{2, 11\}$. Show that this curve is birational to the curve

$$w^2 = f(f^3 - 4f^2 + 8f - 4),$$

and that the latter has genus one.

Solution. **Put $w := \big(2d - (f^3 - 3f^2 + 4f)\big)/(f - 1)$.** Since $m \geqslant 9$, the x-coordinates of $5P$ and $6P$ agree if and only if $11P = O$, that is, $m = 11$, as 11 is prime. By the solution above and the second line of output, this holds if and only if $d^2 - (f^3 - 3f^2 + 4f)d + f = 0$. Completing the square,

$$\big(2d - (f^3 - 3f^2 + 4f)\big)^2 = (f^3 - 3f^2 + 4f)^2 - 4f = f(f - 1)^2(f^3 - 4f^2 + 8f - 4).$$

As $f \neq 1$, dividing by $(f - 1)^2$ gives $w^2 = f(f^3 - 4f^2 + 8f - 4)$, and $d = \big((f - 1)w + f^3 - 3f^2 + 4f\big)/2$ recovers d, so the two curves are birational. The quartic $f(f^3 - 4f^2 + 8f - 4)$ has distinct roots, since 0 is not a root of the cubic, whose discriminant $-176 = -2^4 \cdot 11$ is not 0 in k. So the smooth projective model C of $w^2 = f(f^3 - 4f^2 + 8f - 4)$ has a map $f\colon C \to \mathbb{P}^1$ of degree 2, ramified exactly over the four roots, and not over ∞, as the degree 4 is even. As $\operatorname{char} k \neq 2$, Hurwitz's formula [Sil09, Theorem II.5.9] is an equality, $2g - 2 = 2(-2) + 4 = 0$, and C has genus one.

Remark. By (1) and (3), the pairs (E, P) of an elliptic curve over k and a k-point P of order $m \geqslant 4$, up to change of variables, correspond bijectively to the $(b, c) \in k^2$ with $\Delta \neq 0$ that satisfy the condition for m. For $m = 7, 9, 11, 12$ these are the k-points of an affine curve $Y_1(m)$, in the coordinates d, f or v above, and the Kubert–Tate curve over $Y_1(m)$, with its section $(0, 0)$, is a universal elliptic curve with a section of order m: every pair (E, P) over k is its fiber over exactly one k-point of $Y_1(m)$. The smooth projective model of $Y_1(m)$ is the modular curve $X_1(m)$, and the points added are the cusps [Sil09, Theorem C.13.1(b) and Remark VIII.7.8]. Example C.13.4 there is the family of (7), with $t = d$. For the orders $4 \leqslant m \leqslant 10$ and $m = 12$ of Theorem 45.1, the elliptic curves with a k-point of order m lie in a one-parameter family [Sil09, Remark VIII.7.8], and $X_1(m)$ is rational. The order 11 is the first one left out, and $X_1(11)$ has genus one. Over

$\mathbb{Q}$, Magma identifies C with the elliptic curve $y^2 - y = x^3 - x^2$ (Cremona's 11a3), of rank 0 with five rational points. The curve $X_1(11)$ has five rational cusps [Sil09, Example C.13.3]. So every rational point of $X_1(11)$ is a cusp, and no elliptic curve over $\mathbb{Q}$ has a rational point of order 11, as Theorem 45.1 says. In this language, Theorem 45.1 says that every rational point of $X_1(m)$ is a cusp for $m = 11$ and $m \geqslant 13$ [Sil09, Remark C.13.5].

46. No good reduction everywhere

This section is [Sil09, Exercise 8.15], and follows its hint. A Weierstrass equation over $\mathbb{Z}$ with $\Delta = \pm 1$ would have good reduction at every prime, and congruences modulo 8 rule it out.

In the following definitions and results, E is an elliptic curve over $\mathbb{Q}$ and p is a prime number, with valuation v_p. Write $\mathbb{Z}_{(p)}$ for the ring of rational numbers whose denominators are prime to p.

Definition 46.1 (Minimal equations [Sil09, §VII.1])**.** A Weierstrass equation for E is minimal at p if its coefficients lie in $\mathbb{Z}_{(p)}$ and $v_p(\Delta)$ is as small as possible among such equations for E.

Proposition 46.2 (Good reduction [Sil09, Proposition VII.5.1(a)])**.** *The curve E has good reduction at p if and only if a Weierstrass equation for E minimal at p has $v_p(\Delta) = 0$.*

Corollary 46.3 (Global minimal equations [Sil09, Corollary VIII.8.3])**.** *The curve E has a Weierstrass equation with coefficients in $\mathbb{Z}$ that is minimal at every prime p.*

Proposition 46.4 (Weierstrass formulas [Sil09, §III.1])**.** *Put*

$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 2a_4 + a_1a_3, \qquad b_6 = a_3^2 + 4a_6,$$

$$b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \qquad c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6.$$

Then $\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6$, and $1728\Delta = c_4^3 - c_6^2$.

Problem 54 (No good reduction everywhere)**.**
Let E be an elliptic curve over $\mathbb{Q}$. Suppose, for a contradiction, that E has good reduction at every prime p.

(1) Show that E has a Weierstrass equation with coefficients in $\mathbb{Z}$ and $\Delta = \pm 1$. Fix such an equation, and put $\varepsilon := \Delta$.

Solution. **Take the equation of Corollary 46.3.** By Corollary 46.3, E has a Weierstrass equation with coefficients in $\mathbb{Z}$ that is minimal at every prime p. By Proposition 46.2, $v_p(\Delta) = 0$ for every p. As $\Delta \in \mathbb{Z}$, it is ± 1.

(2) Show that a_1 is odd.

Solution. **If a_1 were even, then $\Delta \equiv 5 \pmod 8$.** Suppose that a_1 is even. Then $b_2 = a_1^2 + 4a_2 \equiv 0 \pmod 4$ and $b_4 = 2a_4 + a_1a_3$ is even. So $b_2^2b_8 \equiv 0 \pmod{16}$, $8b_4^3 \equiv 0 \pmod 8$ and $9b_2b_4b_6 \equiv 0 \pmod 8$, and Proposition 46.4 gives $\Delta \equiv -27b_6^2 \pmod 8$. As Δ is odd, so is b_6, and then $b_6^2 \equiv 1 \pmod 8$ and $\Delta \equiv -27 \equiv 5 \pmod 8$. But $\pm 1 \equiv 1$ or $7 \pmod 8$, a contradiction.

(3) Show that $c_4 \equiv 1 \pmod 8$.

Solution. **The integer b_2 is odd, so $b_2^2 \equiv 1 \pmod 8$.** By (2), $b_2 = a_1^2 + 4a_2$ is odd, so $b_2^2 \equiv 1 \pmod 8$. And $24b_4 \equiv 0 \pmod 8$. So $c_4 = b_2^2 - 24b_4 \equiv 1 \pmod 8$.

(4) Put $u := c_4 - 12\varepsilon$. Show that

$$c_6^2 = u\,(u^2 + 36\varepsilon u + 432),$$

and deduce that u is a positive square or three times a positive square.

Solution. **The two factors are positive, and their greatest common divisor divides 27.** As $\varepsilon^2 = 1$, we have $c_4^3 = (u + 12\varepsilon)^3 = u^3 + 36\varepsilon u^2 + 432u + 1728\varepsilon$. By Proposition 46.4,

$$c_6^2 = c_4^3 - 1728\varepsilon = u\,(u^2 + 36\varepsilon u + 432).$$

Here $u^2 + 36\varepsilon u + 432 = (u + 18\varepsilon)^2 + 108 > 0$, and u is odd, by (3), so $u \neq 0$. As $c_6^2 \geqslant 0$, $u > 0$. A common divisor of u and $u^2 + 36\varepsilon u + 432$ divides $432 = 2^4 \cdot 3^3$, and is odd, so it divides 27. So every prime $\ell \neq 3$ divides at most one of the two factors, and $v_\ell(u) = v_\ell(c_6^2)$ is even. Hence $u = 3^e s^2$ for some $e \geqslant 0$ and some positive integer s prime to 3, and u is a square if e is even and three times a square if e is odd.

(5) Show that $u \equiv 5 \pmod 8$, and conclude that E has bad reduction at some prime p.

Solution. **Odd squares are 1 modulo 8, and three times an odd square is 3.** By (3), and as $12\varepsilon \equiv 4 \pmod 8$ for $\varepsilon = \pm 1$,

$$u = c_4 - 12\varepsilon \equiv 1 - 4 \equiv 5 \pmod 8.$$

But u is odd, so by (4) it is an odd square, which is 1 modulo 8, or three times an odd square, which is 3 modulo 8. This contradiction shows that E has bad reduction at some prime p.

Remark. By Shafarevich's theorem, for every finite set S of primes there are only finitely many elliptic curves over $\mathbb{Q}$, up to isomorphism, with good reduction at every prime outside S [Sil09, Theorem IX.6.1]. This problem is the case $S = \varnothing$, with none. For $S = \{2\}$ there are 24, and for $S = \{2, 3\}$ there are 784 [Sil09, Example IX.6.1.1].

Part 15. The canonical height and the regulator

In this worksheet we see how the canonical height $\hat{h}$ on an elliptic curve E allows us to think about the Mordell–Weil group $E(\mathbb{Q})$ as a lattice Λ in an euclidean space V. As a nice application of this, we have Néron's count of the rational points of bounded height. We use `Magma` to double check our computations.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$, given by a Weierstrass equation.
- The height $h_x(P) := \log H(x(P))$ of a rational point $P \neq O$, where $H(u/w) := \max(|u|, |w|)$ for u/w in lowest terms, and $h_x(O) := 0$. `Magma` calls it `NaiveHeight`.
- The canonical height $\hat{h}(P) := \frac{1}{2} \lim_{n\to\infty} 4^{-n} h_x(2^n P)$ of a rational point P, and the height pairing $\langle P, Q\rangle := \hat{h}(P + Q) - \hat{h}(P) - \hat{h}(Q)$, so that $\langle P, P\rangle = 2\hat{h}(P)$ [Sil09, §VIII.9]. The limit exists by Problem 55(2). `Magma` computes them as `CanonicalHeight(P)`, which is $2\hat{h}(P)$, not $\hat{h}(P)$, and `HeightPairing(P, Q)`.

47. Tate's construction of the canonical height

This section is Tate's construction of the canonical height [Sil09, Proposition VIII.9.1], first on an abelian group, as in [HS00, Theorem B.4.1], then on the rational points of an elliptic curve. Problem 55 asks for the proofs, which need only the following ingredients.

Definition 47.1 (Quadratic heights)**.** Let A be an abelian group. A height on A is a function $h\colon A \to \mathbb{R}$. Two heights satisfy $h = h' + O(1)$ if $|h - h'|$ is bounded on A. A height h is quasi-quadratic if

$$h(P + Q) + h(P - Q) = 2h(P) + 2h(Q) + O(1),$$

that is, if the difference of the two sides is bounded on $A \times A$. It is quadratic if the two sides are equal for all $P, Q \in A$ (the parallelogram law).

Definition 47.2 (Heights on an elliptic curve [Sil09, §VIII.6])**.** Let K be a number field, h the absolute logarithmic height on $\mathbb{P}^1(K)$ [Sil09, §VIII.5], E an elliptic curve over K, and $f \in K(E)$ a nonconstant function. The height relative to f is the height $h_f(P) := h(f(P))$ on $E(K)$. The function f is even if $f(-P) = f(P)$ for all P, as x is. Over $\mathbb{Q}$, h_x is the height of the Standing notation.

Theorem 47.3 (Quasi-parallelogram law [Sil09, Theorem VIII.6.2])**.** *In Definition 47.2, if f is even, then h_f is a quasi-quadratic height on $E(K)$.*

Lemma 47.4 (Comparison of heights [Sil09, Lemma VIII.6.3])**.** *In Definition 47.2, let $f, g \in K(E)$ be nonconstant even functions. Then $(\deg g)\, h_f = (\deg f)\, h_g + O(1)$ on $E(K)$.*

Problem 55 (Tate's construction)**.**

(1) Let A be an abelian group, and h a quasi-quadratic height on A. Prove that for every $P \in A$ the limit

$$\hat{h}(P) := \lim_{n\to\infty} 4^{-n} h(2^n P)$$

exists, and that $\hat{h}$ is the unique quadratic height on A with $\hat{h} = h + O(1)$.

SOLUTION. **Telescoping gives a Cauchy sequence. A bounded quadratic height is 0.** Let $c \geqslant 0$ bound the difference of the two sides of the quasi-parallelogram law. With $Q = P$ it gives $|h(2P) + h(0) - 4h(P)| \leqslant c$, so $|h(2P) - 4h(P)| \leqslant C$ for all P, with $C := c + |h(0)|$. Apply this to $2^n P$, divide by 4^{n+1}, and sum over $M \leqslant n < N$:

$$\begin{aligned} \left|4^{-N} h(2^N P) - 4^{-M} h(2^M P)\right| &\leqslant \sum_{n=M}^{N-1} 4^{-(n+1)} \left|h(2^{n+1}P) - 4h(2^n P)\right| \\ &\leqslant C \sum_{n \geqslant M} 4^{-(n+1)} = \tfrac{1}{3} C\, 4^{-M}. \end{aligned}$$

So the sequence is Cauchy, and $\hat{h}(P)$ exists. With $M = 0$ and $N \to \infty$, $|\hat{h}(P) - h(P)| \leqslant C/3$. The quasi-parallelogram law for $2^n P$ and $2^n Q$, divided by 4^n, shows that the two sides of the parallelogram law for $4^{-n} h(2^n \cdot)$ differ by at most $c\, 4^{-n}$. In the limit they are equal, so $\hat{h}$ is quadratic.

For uniqueness, let q and q' be quadratic heights with $q = q' + O(1)$. Their difference $d := q - q'$ is quadratic and bounded. The parallelogram law for d with $P = Q = 0$ gives $d(0) = 0$, and with $Q = P$ it gives $d(2P) = 4d(P)$. So $d(P) = 4^{-n} d(2^n P)$ for every n, which tends to 0 since d is bounded. Hence $q = q'$.

(2) Let K, E and f be as in Definition 47.2, with f even. Deduce that for every $P \in E(K)$ the limit

$$\frac{1}{\deg f} \lim_{N\to\infty} 4^{-N} h_f(2^N P)$$

exists, that it is a quadratic height on $E(K)$, and that it does not depend on f.

SOLUTION. **Apply (1) to $h_f/\deg f$. Uniqueness compares two choices of f.** By Theorem 47.3, $h_f/\deg f$ is a quasi-quadratic height on $E(K)$. By (1), the limit exists and is the unique quadratic height $\hat{h}_f$ with $\hat{h}_f = h_f/\deg f + O(1)$. Let $g \in K(E)$ be another nonconstant even function. By Lemma 47.4, $h_f/\deg f = h_g/\deg g + O(1)$. So $\hat{h}_f$ and $\hat{h}_g$ are quadratic

heights equivalent to $h_g/\deg g$, and $\hat{h}_f = \hat{h}_g$ by the uniqueness in (1). Over $\mathbb{Q}$, with $f = x$ of degree 2, this is the canonical height $\hat{h}$ of the Standing notation.

Remark. Silverman proves the existence and the independence of f directly, by the same telescoping sum [Sil09, Proposition VIII.9.1], and the parallelogram law afterwards [Sil09, Theorem VIII.9.3]. Hindry and Silverman do the abstract construction for any map ϕ with $h \circ \phi = \alpha h + O(1)$ and $\alpha > 1$ [HS00, Theorem B.4.1].

48. The lattice of rational points

This section draws [Sil09, Proposition VIII.9.6] for the curve $y^2 = x^3 + 17$ of [Sil09, Example III.2.4], after [HS00, §B.5]. The rational points P_1 and P_3 of Problem 56 carry Silverman's names. He notes that they generate the Mordell–Weil group, and that this is true but not so easy to prove. Problem 56 proves it with `Magma`'s `CPSHeightBounds`, which returns real numbers l and u with $l \leqslant h_x(P) - 2\hat{h}(P) \leqslant u$ for every rational point P [BCP97]. First, two facts about lattices in general.

Definition 48.1 (Lattices and covolume [HS00, §B.5])**.** A lattice in a real vector space V of dimension r is a subgroup Λ generated by a basis $\lambda_1, \dots, \lambda_r$ of V. If V is Euclidean, the covolume $\operatorname{covol}(\Lambda)$ is the volume of $\{t_1\lambda_1 + \cdots + t_r\lambda_r : 0 \leqslant t_i < 1\}$.

Proposition 48.2 (The covolume [HS00, Exercise B.14(b)])**.** *In Definition 48.1,* $\operatorname{covol}(\Lambda)^2 = \det(\lambda_i \cdot \lambda_j)_{i,j}$.

From now on, V and Λ denote the following.

Proposition 48.3 (The height lattice [Sil09, Proposition VIII.9.6])**.** *Let* $V := E(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{R}$*, let* $v_P := P \otimes 1$ *for a rational point* P*, and let* Λ *be the image of* $E(\mathbb{Q})$ *in* V*. The pairing extends to an inner product on* V*, with* $|v_P|^2 = 2\hat{h}(P)$*. Then* Λ *is a lattice in* V*, and the kernel of* $E(\mathbb{Q}) \to \Lambda$ *is* $E(\mathbb{Q})_{\mathrm{tors}}$.

Definition 48.4 (The elliptic regulator [Sil09, §VIII.9])**.** Let r be the rank of E, and $Q_1, \dots, Q_r$ rational points whose images form a basis of Λ. The elliptic regulator of E is $R_E := \det\big(\langle Q_i, Q_j\rangle\big)_{1\leqslant i,j\leqslant r}$. By Proposition 48.2, $\sqrt{R_E} = \operatorname{covol}(\Lambda)$ is the covolume of Λ in V.

Problem 56 (Two rational points generate $E(\mathbb{Q})$)**.**
Let $E\colon y^2 = x^3 + 17$ over $\mathbb{Q}$, with LMFDB label `10404.f2`, and let $P_1 = (-2, 3)$ and $P_3 = (2, 5)$ be two of its rational points. Let $g_{ij} := \langle P_i, P_j\rangle$ for $i, j \in \{1, 3\}$, and let $G := (g_{ij})$ be the Gram matrix of P_1 and P_3. `Magma` computes the rank and the torsion of E, the matrix G and its determinant, the bounds l and u, and the x-coordinates of the rational points $P \neq O$ with $H(x(P)) \leqslant 5$:

```
R := RealField(6);
E := EllipticCurve([0, 0, 0, 0, 17]);
Rank(E); #TorsionSubgroup(E);                    // 2 true
                                                 // 1
P1 := E![-2, 3];  P3 := E![2, 5];
G := HeightPairingMatrix([P1, P3]);  ChangeRing(G, R);
// [0.454617 0.363961]
// [0.363961 1.06248]
R!Determinant(G), R!Regulator(E);                // 0.350554 0.350554
l, u := CPSHeightBounds(E);  R!l, R!u;           // -1.63999 1.18106
```

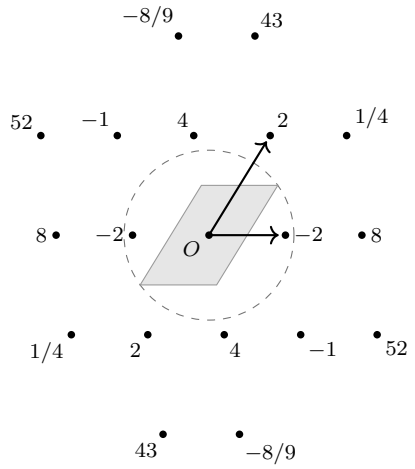


FIGURE 1. The rational points of the curve E of Problem 56 near the origin of the plane V. Each dot is the vector v_P of a rational point $P = aP_1 + bP_3$ with $2\hat{h}(P) \leqslant 3.4$, labeled by its x-coordinate. Since $|v_P|^2 = 2\hat{h}(P)$, the labels grow more complicated outward: integers near O, and the first denominators, $1/4$ and $-8/9$, farther out. The rational points P and $-P$ have the same x-coordinate and sit opposite each other. The arrows are v_{P_1} and v_{P_3}. The shaded parallelogram $\{sv_{P_1} + tv_{P_3} : |s|, |t| \leqslant \frac{1}{2}\}$ has area $\sqrt{\det G} = 0.592076$, and every vector of V is a vector of $\mathbb{Z}v_{P_1} + \mathbb{Z}v_{P_3}$ plus a vector in it. The dashed circle $2\hat{h} = 0.561255$ passes through its two farthest corners $\pm\frac{1}{2}(v_{P_1} + v_{P_3})$, so it contains the parallelogram.

```
X := {a/d : a in [-5..5], d in [1..5] | GCD(a, d) eq 1 and IsPoint(E, a/d)};
X;                                                  // { -2, -1, 1/4, 2, 4 }
X subset {(m*P1 + n*P3)[1] : m, n in [-2..2] | [m, n] ne [0, 0]};   // true
```

The items below show that P_1 and P_3 generate $E(\mathbb{Q})$.

(1) Show that v_{P_1} and v_{P_3} form a basis of V.

SOLUTION. **$\dim V = 2$, and $\det G \neq 0$.** The rank of E is 2, by the `Magma` computation, so V has dimension 2. A linear relation $\alpha v_{P_1} + \beta v_{P_3} = 0$ with $(\alpha, \beta) \neq (0, 0)$ would give $(\alpha, \beta)\, G = 0$, by pairing it with v_{P_1} and v_{P_3}. But $\det G = 0.350554 \neq 0$. So v_{P_1} and v_{P_3} are independent, hence a basis.

(2) Let P be a rational point. Show that there are $m, n \in \mathbb{Z}$ such that $Q := P - mP_1 - nP_3$ has v_Q in the shaded parallelogram of Fig. 1. Show that $Q \in \mathbb{Z}P_1 + \mathbb{Z}P_3$ if and only if $P \in \mathbb{Z}P_1 + \mathbb{Z}P_3$.

SOLUTION. **Take m, n nearest to the coordinates of v_P.** By (1), $v_P = sv_{P_1} + tv_{P_3}$ with $s, t \in \mathbb{R}$. Let m, n be integers nearest to s, t, so that $|s - m|, |t - n| \leqslant \frac{1}{2}$. Since $P \mapsto v_P$ is a homomorphism, $v_Q = (s - m)v_{P_1} + (t - n)v_{P_3}$, which lies in the shaded parallelogram. And $P - Q = mP_1 + nP_3$ lies in $\mathbb{Z}P_1 + \mathbb{Z}P_3$, so P and Q lie in it or not together.

(3) Show that every rational point Q with v_Q in the shaded parallelogram has

$$2\hat{h}(Q) \leqslant \tfrac{1}{4} \max(g_{11} + 2g_{13} + g_{33},\ g_{11} - 2g_{13} + g_{33}),$$

and deduce that $H(x(Q)) \leqslant 5$.

SOLUTION. **$2\hat{h}(Q) \leqslant 0.561255$, so $H(x(Q)) \leqslant e^{1.74231} = 5.71$.** The corners of the parallelogram are $\pm\frac{1}{2}(v_{P_1} + v_{P_3})$ and $\pm\frac{1}{2}(v_{P_1} - v_{P_3})$. By bilinearity, $|v_{P_1} \pm v_{P_3}|^2 = g_{11} \pm 2g_{13} + g_{33}$, which is 2.24502 and 0.789176. The convex function $|v|^2$ is largest on the parallelogram at a corner, so

$$2\hat{h}(Q) = |v_Q|^2 \leqslant \tfrac{1}{4}\max(2.24502, 0.789176) = 0.561255.$$

Then $h_x(Q) \leqslant 2\hat{h}(Q) + u \leqslant 0.561255 + 1.18106 = 1.74231$, so $H(x(Q)) \leqslant e^{1.74231} = 5.71$. Since $H(x(Q))$ is an integer, $H(x(Q)) \leqslant 5$.

(4) Conclude that P_1 and P_3 generate $E(\mathbb{Q})$.

SOLUTION. **Otherwise some $Q \notin \mathbb{Z}P_1 + \mathbb{Z}P_3$ has $H(x(Q)) \leqslant 5$, and there is none.** Suppose that a rational point P is not in $\mathbb{Z}P_1 + \mathbb{Z}P_3$. By (2) there is a rational point $Q \notin \mathbb{Z}P_1 + \mathbb{Z}P_3$ with v_Q in the shaded parallelogram. In particular $Q \neq O$, and $H(x(Q)) \leqslant 5$ by (3). The `Magma` search shows that $x(Q) = x(R)$ for some $R \in \mathbb{Z}P_1 + \mathbb{Z}P_3$. Then $Q = \pm R$ lies in $\mathbb{Z}P_1 + \mathbb{Z}P_3$, a contradiction.

Remark. This is the saturation step of a computation of $E(\mathbb{Q})$. `Magma`'s `Regulator(E)` runs it on its own, so its agreement with $\det G$ is a consistency check, not a proof. In general, if the images of rational points $Q_1, \dots, Q_r$ generate a subgroup of index N in Λ, then $\det(\langle Q_i, Q_j\rangle) = N^2 R_E$.

49. Néron's counting theorem

This section is Néron's counting theorem [HS00, Theorem B.6.3] for the curve of Problem 56, with its constant, through the count of lattice points in ellipsoids [HS00, Lemma B.6.4].

Lemma 49.1 (Lattice points in ellipsoids [HS00, Lemma B.6.4])**.** *Let r be the rank of E, suppose that $r \geqslant 1$, and let q be a positive definite quadratic form on V. Then, as $T \to \infty$,*

$$\#\{\lambda \in \Lambda : q(\lambda) \leqslant T\} = \frac{\operatorname{vol}\{v \in V : q(v) \leqslant 1\}}{\operatorname{covol}(\Lambda)}\, T^{r/2} + O\big(T^{(r-1)/2}\big).$$

Theorem 49.2 (Néron's counting theorem [HS00, Theorem B.6.3])**.** *Suppose that E has rank $r \geqslant 1$. There is a constant $c > 0$ such that, as $T \to \infty$,*

$$\#\{P \in E(\mathbb{Q}) : h_x(P) \leqslant T\} = c\,T^{r/2} + O\big(T^{(r-1)/2}\big).$$

Problem 57 (Counting rational points of bounded height)**.**
Let E, P_1, P_3, G, l and u be as in Problem 56. Continuing its listing, `Magma` counts the rational points with $h_x \leqslant T$:

```
Gi := G^-1;
count := function(T)
  N := [Floor(Sqrt((T - l)*Gi[i, i])) : i in [1, 2]];
  return #[0 : a in [-N[1]..N[1]], b in [-N[2]..N[2]] | NaiveHeight(a*P1 + b*P3) le T];
end function;
[<T, count(T)> : T in [10, 20, 50, 100]];
// [ <10, 53>, <20, 103>, <50, 263>, <100, 529> ]
```

The function `count` runs over a box that contains every $(a, b) \in \mathbb{Z}^2$ with $2\hat{h}(aP_1 + bP_3) \leqslant T - l$. The items below prove Theorem 49.2 for E, with $c = \pi/\sqrt{R_E}$.

(1) Show that $P \mapsto v_P$ is a bijection $E(\mathbb{Q}) \to \Lambda$, that $R_E = \det G$, and that $\operatorname{covol}(\Lambda) = \sqrt{R_E}$.

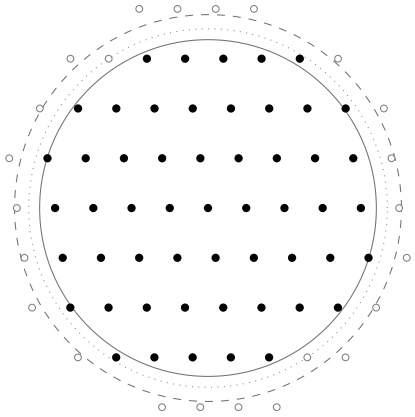


FIGURE 2. The rational points of the curve E of Problem 56 counted by Néron's theorem for $T = 10$. Each dot is the vector v_P of a rational point P with $|v_P|^2 = 2\hat{h}(P) \leqslant 14$. It is filled if $h_x(P) \leqslant 10$, and open otherwise. There are 53 filled dots. If h_x were equal to $2\hat{h}$, the filled dots would be exactly those inside the dotted circle $2\hat{h} = 10$. Instead, $h_x - 2\hat{h}$ is only known to lie between $l = -1.63999$ and $u = 1.18106$. So every rational point inside the solid circle $2\hat{h} = 10 - u$ has $h_x \leqslant 10$: all 49 dots there are filled. No rational point outside the dashed circle $2\hat{h} = 10 - l$ has $h_x \leqslant 10$: all dots there are open. In the ring between the two circles, the answer depends on $h_x - 2\hat{h}$, which has no pattern: 4 of its 10 dots are filled. The ring has area $\pi(u - l)$ for every T, while the disc inside it has area about πT. So, for large T, the count is about the area of the disc divided by the area $\sqrt{R_E}$ of a fundamental parallelogram.

SOLUTION. **$E(\mathbb{Q})_{\text{tors}} = 0$, and (P_1, P_3) is a basis by Problem 56.** The torsion of E is trivial, by the `Magma` computation of Problem 56, so $P \mapsto v_P$ is injective by Proposition 48.3, and it is onto Λ by definition. By Problem 56, P_1 and P_3 generate $E(\mathbb{Q})$, so their images form a basis of Λ, and $R_E = \det G = 0.350554$ by Definition 48.4. Then $\operatorname{covol}(\Lambda) = \sqrt{R_E}$ by Proposition 48.2.

(2) Show that

$$\{P : 2\hat{h}(P) \leqslant T - u\} \subseteq \{P : h_x(P) \leqslant T\} \subseteq \{P : 2\hat{h}(P) \leqslant T - l\},$$

and check it against Fig. 2.

SOLUTION. **$2\hat{h} + l \leqslant h_x \leqslant 2\hat{h} + u$.** If $2\hat{h}(P) \leqslant T - u$, then $h_x(P) \leqslant 2\hat{h}(P) + u \leqslant T$. If $h_x(P) \leqslant T$, then $2\hat{h}(P) \leqslant h_x(P) - l \leqslant T - l$. In Fig. 2, with $T = 10$, all 49 dots inside the solid circle are filled, and every dot outside the dashed circle is open.

(3) Deduce that the number of rational points P with $h_x(P) \leqslant T$ is $cT + O(\sqrt{T})$ with $c = \pi/\sqrt{R_E}$, and compare cT with the output.

SOLUTION. **$c = \pi/\sqrt{0.350554} = 5.30606$, and $cT = 53.1, 106.1, 265.3, 530.6$.** By (1), the outer sets of (2) correspond to the lattice points $\lambda \in \Lambda$ with $|\lambda|^2 \leqslant T - u$ and $|\lambda|^2 \leqslant T - l$. Apply Lemma 49.1 with $q(v) = |v|^2$, whose unit disc has area π, and $\operatorname{covol}(\Lambda) = \sqrt{R_E}$. Both outer sets have $\pi(T + O(1))/\sqrt{R_E} + O(\sqrt{T}) = cT + O(\sqrt{T})$ elements, and so does the middle one. The counts 53, 103, 263, 529 differ from cT by less than 4.

(4) Suppose that you did not know the rank of E. How would the counts reveal it?

SOLUTION. **Doubling T doubles the count, so $r/2 = 1$ and $r = 2$.** By Theorem 49.2, the count $N(T)$ grows like $T^{r/2}$. Here $N(20)/N(10)$, $N(100)/N(50)$ are close to 2, so $r/2 = 1$. In general $r = \lim 2 \log N(T)/\log T$: the rank can be read off from a count of rational points.

Remark. Néron stated the theorem with the multiplicative height $H_x = e^{h_x}$, where it reads $c(\log T)^{r/2}$ [HS00, Theorem B.6.3]. Rational points are sparse: on this curve, exactly 529 of them have an x-coordinate whose numerator and denominator are at most $e^{100} \approx 10^{43}$.

Part 16. Szpiro's conjecture and the ABC conjecture

This worksheet is a short version of [Sil09, §VIII.11]. The minimal discriminant Δ_E of an elliptic curve E over $\mathbb{Q}$ and its conductor N_E are divisible by the same primes, the primes of bad reduction. In N_E they appear to small powers, while in Δ_E a prime may appear to a large power. Szpiro's conjecture bounds $|\Delta_E|$ by a constant times $N_E^{6+\varepsilon}$. It would already imply Fermat's last theorem for all sufficiently large exponents [Sil09, Proposition VIII.11.2]. It is closely related to the ABC conjecture, which bounds the height of a rational point $[A : B : C]$ of the line $X+Y = Z$ by a power of the radical of ABC. Problem 58 computes, with the help of `Magma`, the minimal discriminant and the reduction types of the curve $y^2 = x(x + A)(x - B)$ attached to $A + B = C$. Problems 59 and 60 use it to show that each conjecture implies a form of the other.

Standing notation.

- An elliptic curve E over $\mathbb{Q}$. A Weierstrass equation for E has invariants c_4, c_6 and discriminant Δ [Sil09, §III.1], which lie in $\mathbb{Z}$ when its coefficients do.
- A prime p, with valuation v_p on $\mathbb{Q}$, and the ring $\mathbb{Z}_{(p)}$ of rational numbers whose denominators are prime to p.

50. The minimal discriminant of a Frey curve

Problem 58 proves [Sil09, Lemma VIII.11.3]. Silverman bounds the scaling factor u of a change of variables with two identities that write $288A^2$ and $288B^2$ in terms of c_4 and c_6. Here `Magma` finds identities in terms of c_4 and Δ instead, and they give both parts of the lemma at once.

Definition 50.1 (Minimal equations [Sil09, §§VII.1, VII.5])**.** A Weierstrass equation for E is minimal at p if its coefficients lie in $\mathbb{Z}_{(p)}$ and $v_p(\Delta)$ is as small as possible among such equations. It is global minimal if its coefficients lie in $\mathbb{Z}$ and it is minimal at every prime p. The curve E has good, multiplicative or additive reduction at p if the reduction modulo p of an equation minimal at p is nonsingular, has a node, or has a cusp.

Proposition 50.2 (Reduction types [Sil09, Remark VII.1.1 and Proposition VII.5.1])**.** *Consider a Weierstrass equation for E with coefficients in $\mathbb{Z}_{(p)}$.*

(1) If $v_p(\Delta) < 12$ or $v_p(c_4) < 4$, then the equation is minimal at p.
(2) Suppose that the equation is minimal at p. Then E has good reduction at p if $v_p(\Delta) = 0$, multiplicative reduction if $v_p(\Delta) > 0$ and $v_p(c_4) = 0$, and additive reduction if $v_p(\Delta) > 0$ and $v_p(c_4) > 0$.

Definition 50.3 (Minimal discriminant [Sil09, §VIII.11])**.** The curve E has a global minimal Weierstrass equation [Sil09, Corollary VIII.8.3]. Its discriminant is the minimal discriminant Δ_E of E. It does not depend on the equation, since two global minimal equations are related by a change of variables with $u = \pm 1$ [Sil09, Proposition VII.1.3(b)].

Proposition 50.4 (Changes of variables [Sil09, Proposition III.3.1(b) and Table 3.1])**.** *Any two Weierstrass equations for an elliptic curve over $\mathbb{Q}$ are related by a change of variables $x = u^2x' + r$, $y = u^3y' + su^2x' + t$ with $u \in \mathbb{Q}^\times$ and $r, s, t \in \mathbb{Q}$. If the first has discriminant Δ and invariant c_4, then the second has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$.*

In this section, A, B, C are nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$, and E is the elliptic curve

$$E\colon y^2 = x(x + A)(x - B). \tag{50.1}$$

The invariant c_4 and the discriminant Δ are those of this equation.

Problem 58 (The minimal discriminant of a Frey curve)**.**
The following code works in the polynomial ring $\mathbb{Z}[A, B]$. It checks that $c_4 = 16F$ and $\Delta = 16G^2$ for the polynomials F and G below, and writes A^3 and B^3 as combinations of F and G with coefficients in $\mathbb{Z}[A, B]$.

```
P<A, B> := PolynomialRing(Integers(), 2);
// y^2 = x(x + A)(x - B), over the field of fractions of P
E := EllipticCurve([FieldOfFractions(P) | 0, A - B, 0, -A*B, 0]);
c4 := P!cInvariants(E)[1];
Delta := P!Discriminant(E);
F := A^2 + A*B + B^2;
G := A*B*(A + B);
c4 eq 16*F, Delta eq 16*G^2;      // true true
I := IdealWithFixedBasis([F, G]);
Coordinates(I, A^3);              // [ A, -1 ]
Coordinates(I, B^3);              // [ B, -1 ]
```

(1) Show that A, B, C are pairwise coprime, and that $\gcd(c_4, \Delta) = 16$.

Solution. **Here $c_4 = 16F$ and $\Delta = 16(ABC)^2$, and F is prime to ABC.** A prime dividing two of A, B, C divides the third, as $C = A + B$. So, since $\gcd(A, B, C) = 1$, they are pairwise coprime. By the output, $c_4 = 16F$ and $\Delta = 16G^2$, where $G = AB(A + B) = ABC$, and

$$A^3 = AF - G, \qquad B^3 = BF - G$$

in $\mathbb{Z}[A, B]$, hence for our integers A and B. A prime p dividing F and G divides A^3 and B^3, so it divides A and B. As A and B are coprime, there is no such prime. So F is prime to G, hence to G^2, and

$$\gcd(c_4, \Delta) = \gcd(16F, 16G^2) = 16\gcd(F, G^2) = 16.$$

(2) Show that the minimal discriminant of E satisfies $|\Delta_E| = 2^4(ABC)^2$ or $|\Delta_E| = 2^{-8}(ABC)^2$. In particular, $|\Delta_E| \geqslant 2^{-8}(ABC)^2$.

Solution. **The scaling factor is $u = \pm 1$ or $u = \pm 2$.** By Proposition 50.4, a global minimal equation for E comes from (50.1) by a change of variables with some $u \in \mathbb{Q}^\times$, and it has discriminant $\Delta_E = u^{-12}\Delta$ and invariant $u^{-4}c_4$. As (50.1) has coefficients in $\mathbb{Z}$ and the global minimal equation is minimal at every prime p, $v_p(\Delta_E) \leqslant v_p(\Delta)$, that is, $v_p(u) \geqslant 0$. So $u \in \mathbb{Z}$. The global minimal equation has coefficients in $\mathbb{Z}$, hence so do $u^{-4}c_4$ and $u^{-12}\Delta$. So u^4 divides c_4 and Δ. By (1), u^4 divides 16, so $u = \pm 1$ or $u = \pm 2$. Hence $|\Delta_E| = |\Delta|/u^{12}$ is $16(ABC)^2 = 2^4(ABC)^2$ or $2^{-12} \cdot 16(ABC)^2 = 2^{-8}(ABC)^2$.

(3) Show that E has multiplicative reduction at every odd prime p dividing ABC.

Solution. **At an odd prime p dividing ABC, p divides Δ but not c_4.** Let p be an odd prime dividing ABC. Then p divides $\Delta = 16(ABC)^2$. By (1), $\gcd(c_4, \Delta) = 16$ is a power of 2, so p does not divide c_4. So $v_p(c_4) = 0 < 4$, and (50.1) is minimal at p by Proposition 50.2(1). As $v_p(\Delta) > 0$ and $v_p(c_4) = 0$, the curve E has multiplicative reduction at p by Proposition 50.2(2).

51. From Szpiro to ABC

Problem 59 is the proof of [Sil09, Proposition VIII.11.5(a)]. It applies Conjecture 51.4 to the curve (50.1) of Problem 58, whose minimal discriminant is large and whose conductor is small. Szpiro made his conjecture in 1983, and Masser and Oesterlé proposed the ABC conjecture in 1985. Silverman's displayed definition of the conductor takes $f_p(E) = 2$ at every prime of additive reduction, which he notes is not quite correct at 2 and 3. At these two primes the correct exponent has a technical definition, sketched in Remark 51.2. The two conductors differ by a factor dividing $2^6 \cdot 3^3$, which the constants in the conjectures absorb.

Definition 51.1 (Conductor [Sil09, §VIII.11])**.** The conductor of E is

$$N_E \coloneqq \prod_p p^{f_p(E)},$$

where the exponent $f_p(E)$ is 0 if E has good reduction at p, 1 if it has multiplicative reduction at p, and 2 if it has additive reduction at p and $p \geqslant 5$. If E has additive reduction at $p = 2$ or $p = 3$, then $f_p(E)$ is defined in [Sil94, §IV.10], and $2 \leqslant f_2(E) \leqslant 8$ and $2 \leqslant f_3(E) \leqslant 5$.

Remark 51.2 (The exponents f_2 and f_3 [Sil09, §C.16])**.** Let p be a prime at which E has additive reduction. The inertia group at p acts on the Tate modules of E, and an integer $\delta_p(E) \geqslant 0$ measures the wild ramification in this action. It vanishes for $p \geqslant 5$, and $f_p(E) = 2 + \delta_p(E)$.

The exponent can also be computed directly. For any prime p, let m_p be the number of irreducible components, ignoring multiplicities, of the special fiber of the minimal complete Néron model of E at p. Then the Ogg–Saito formula [Sil09, Proposition C.16.2] gives

$$f_p(E) = v_p(\Delta_E) + 1 - m_p.$$

Definition 51.3 (Height and radical)**.** Every rational point P of the line $X + Y = Z$ in $\mathbb{P}^2$ has coordinates $P = [A : B : C]$ with $A, B, C \in \mathbb{Z}$ and $\gcd(A, B, C) = 1$, unique up to sign. Then $A + B = C$. The height of P is

$$H(P) \coloneqq \max\bigl(|A|, |B|, |C|\bigr).$$

The radical of a nonzero integer n is

$$R(n) \coloneqq \prod_{p \mid n} p.$$

Conjecture 51.4 (Szpiro [Sil09, Conjecture VIII.11.1])**.** *For every $\varepsilon > 0$ there is a constant κ_ε such that every elliptic curve E over $\mathbb{Q}$ satisfies*

$$|\Delta_E| \leqslant \kappa_\varepsilon N_E^{6+\varepsilon}.$$

Conjecture 51.5 (ABC [Sil09, Conjecture VIII.11.4])**.** *For every $\varepsilon > 0$ there is a constant λ_ε with the following property. Let $P = [A : B : C]$ be a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$, with $A, B, C \in \mathbb{Z}$ coprime and $ABC \neq 0$. Then*

$$H(P) \leqslant \lambda_\varepsilon R(ABC)^{1+\varepsilon}.$$

Problem 59 (Szpiro implies ABC with exponent 3/2)**.**
Assume Conjecture 51.4. Let $P = [A : B : C]$ be a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$, with $A, B, C \in \mathbb{Z}$ coprime and $ABC \neq 0$. You will show that for every $\varepsilon > 0$ there is a constant λ_ε, independent of P, with

$$H(P) \leqslant \lambda_\varepsilon R(ABC)^{3/2+\varepsilon}.$$

(1) Show that the absolute values of A, B, C, suitably ordered, are the coordinates of a rational point P' of the same line with $H(P') = H(P)$, and with the same radical of the product of the coordinates. So you may assume that $0 < A \leqslant B < C$. Show that then $ABC \geqslant H(P)^2/2$.

Solution. **The largest absolute value is the sum of the other two.** The three nonzero integers A, B, $-C$ sum to 0. So two of them, say x and y, have the same sign, and the third, z, has the opposite sign and $|z| = |x| + |y|$. Order x and y so that $|x| \leqslant |y|$. Then $P' := [|x| : |y| : |z|]$ is a rational point of $X + Y = Z$ with coprime coordinates, $H(P') = |z| = H(P)$, and $R(|xyz|) = R(ABC)$. Replacing P by P', we may assume that $0 < A \leqslant B < C$. Then $H(P) = C$ and $2B \geqslant A + B = C$, so

$$ABC \geqslant 1 \cdot \frac{C}{2} \cdot C = \frac{H(P)^2}{2}.$$

(2) Let E be the curve (50.1). Show that $N_E \leqslant 2^7 R(ABC)$.

Solution. **Odd primes have $f_p(E) \leqslant 1$, $f_2(E) \leqslant 8$, and $2 \mid R(ABC)$.** Let p be an odd prime. If $p \mid ABC$, then $f_p(E) = 1$ by Problem 58(3). If $p \nmid ABC$, then $v_p(\Delta) = 0$ for $\Delta = 16(ABC)^2$. So (50.1) is minimal at p and E has good reduction at p (Proposition 50.2), and $f_p(E) = 0$. Finally $f_2(E) \leqslant 8$ by Definition 51.1. If A and B are both odd, then $C = A + B$ is even, so 2 divides ABC and $R(ABC)$. So

$$N_E \leqslant 2^8 \prod_{p \mid ABC,\ p \neq 2} p = 2^8 \cdot \frac{R(ABC)}{2} = 2^7 R(ABC).$$

(3) Let $\varepsilon > 0$. Use Conjecture 51.4 to show that $H(P)^4 \leqslant \mu_\varepsilon R(ABC)^{6+\varepsilon}$ for a constant μ_ε that depends only on ε, and conclude.

Solution. **Here $\mu_\varepsilon = 2^{52+7\varepsilon}\kappa_\varepsilon$.** By (1), Problem 58(2), Conjecture 51.4 and (2),

$$\frac{H(P)^4}{4} \leqslant (ABC)^2 \leqslant 2^8 |\Delta_E| \leqslant 2^8 \kappa_\varepsilon N_E^{6+\varepsilon} \leqslant 2^8 \kappa_\varepsilon \cdot 2^{7(6+\varepsilon)} R(ABC)^{6+\varepsilon}.$$

So $H(P)^4 \leqslant 2^{52+7\varepsilon}\kappa_\varepsilon R(ABC)^{6+\varepsilon}$. Taking fourth roots and replacing ε by 4ε gives the statement of the problem, with $\lambda_\varepsilon = \left(2^{52+28\varepsilon}\kappa_{4\varepsilon}\right)^{1/4}$.

Remark. The same argument, applied to the curve $y^2 = x^3 - 2(A - B)x^2 + C^2 x$, which is 2-isogenous to E and has discriminant $-2^8 ABC^4$, lowers the exponent 3/2 to 6/5 [Sil09, Exercise 8.20].

52. From ABC to Szpiro

Problem 60 follows the proof of [Sil09, Proposition VIII.11.5(b)], which assumes that $\gcd(c_4, c_6) = 1$. The general case is [Sil09, Exercise 8.21]. The proof applies Conjecture 51.5 to the rational point $[c_4^3 : -c_6^2 : 1728\Delta]$ of the line $X + Y = Z$. The book has Δ in place of 1728Δ, a misprint corrected in its errata.

Proposition 52.1 (Weierstrass formulas [Sil09, §III.1])**.** *The invariants and the discriminant of a Weierstrass equation satisfy* $1728\Delta = c_4^3 - c_6^2$.

Problem 60 (ABC implies Szpiro)**.**
Assume Conjecture 51.5. Let E be given by a global minimal Weierstrass equation, with invariants c_4, c_6 and discriminant $\Delta = \Delta_E$, and suppose that $\gcd(c_4, c_6) = 1$. You will show that for every $\delta > 0$ there is a constant κ_δ, independent of E, with

$$|\Delta_E| \leqslant \kappa_\delta N_E^{6+\delta}.$$

(1) Show that $c_4 c_6 \neq 0$, and that $P := [c_4^3 : -c_6^2 : 1728\Delta]$ is a rational point of the line $X + Y = Z$ in $\mathbb{P}^2$ with coprime integer coordinates. Deduce that

$$H(P) = \max\bigl(|c_4|^3, |c_6|^2, 1728|\Delta|\bigr).$$

Solution. **If $c_4 = 0$, then $c_6 = \pm 1$ and $1728\Delta = -1$.** By Proposition 52.1, $c_4^3 + (-c_6^2) = 1728\Delta$, so P lies on the line. If $c_4 = 0$ or $c_6 = 0$, then the other one has absolute value $\gcd(c_4, c_6) = 1$, and $1728\Delta = \pm 1$, which is impossible in $\mathbb{Z}$. So $c_4 c_6 \neq 0$, and also $\Delta \neq 0$. A prime dividing the three coordinates divides the coprime integers c_4 and c_6. So the coordinates are coprime, and Definition 51.3 gives $H(P)$.

(2) Show that $R(c_4^3 c_6^2 \cdot 1728\Delta) \leqslant 6|c_4 c_6| N_E$.

Solution. **Every prime dividing Δ_E is a prime of bad reduction.** The primes dividing $c_4^3 c_6^2 \cdot 1728\Delta$ divide c_4, c_6, 6 or Δ, so

$$R(c_4^3 c_6^2 \cdot 1728\Delta) \leqslant R(c_4) R(c_6) \cdot 6 \cdot R(\Delta) \leqslant 6|c_4 c_6| R(\Delta).$$

As the equation is minimal, every prime p dividing $\Delta = \Delta_E$ is a prime of bad reduction by Proposition 50.2(2). So $f_p(E) \geqslant 1$, and $R(\Delta) \leqslant N_E$.

(3) Let $0 < \varepsilon < 1/5$. Use Conjecture 51.5 and the inequalities $|c_4| \leqslant H(P)^{1/3}$ and $|c_6| \leqslant H(P)^{1/2}$ to show that

$$H(P)^{(1-5\varepsilon)/6} \leqslant 6^{1+\varepsilon} \lambda_\varepsilon N_E^{1+\varepsilon}.$$

Deduce the statement of the problem.

Solution. **Take $\varepsilon = \delta/(36 + 5\delta)$.** By Conjecture 51.5, (1) and (2),

$$H(P) \leqslant \lambda_\varepsilon R(c_4^3 c_6^2 \cdot 1728\Delta)^{1+\varepsilon} \leqslant \lambda_\varepsilon \bigl(6|c_4 c_6| N_E\bigr)^{1+\varepsilon} \leqslant \lambda_\varepsilon \bigl(6 H(P)^{5/6} N_E\bigr)^{1+\varepsilon}.$$

Dividing by $H(P)^{5(1+\varepsilon)/6}$ gives the inequality of the statement. As $1 - 5\varepsilon > 0$, raising it to the power $6/(1-5\varepsilon)$ gives

$$|\Delta_E| < H(P) \leqslant \bigl(6^{1+\varepsilon} \lambda_\varepsilon\bigr)^{6/(1-5\varepsilon)} N_E^{6(1+\varepsilon)/(1-5\varepsilon)}.$$

For $\varepsilon = \delta/(36 + 5\delta)$, which lies between 0 and $1/5$, the exponent is exactly $6 + \delta$, and the constant $\kappa_\delta = \bigl(6^{1+\varepsilon} \lambda_\varepsilon\bigr)^{6/(1-5\varepsilon)}$ depends only on δ.

Remark. Silverman instead writes the inequality of Conjecture 51.5 once for each coordinate of P:

$$|c_4|^{2-\varepsilon} \leqslant \mu_\varepsilon |c_6 N_E|^{1+\varepsilon}, \qquad |c_6|^{1-\varepsilon} \leqslant \mu_\varepsilon |c_4 N_E|^{1+\varepsilon}, \qquad |\Delta| \leqslant \mu_\varepsilon |c_4 c_6 N_E|^{1+\varepsilon},$$

with $\mu_\varepsilon = 6^{1+\varepsilon}\lambda_\varepsilon$. He raises them to the powers $2+2\varepsilon$, $3+3\varepsilon$ and $1-5\varepsilon$, and multiplies them. The powers of $|c_4|$ and $|c_6|$ cancel, and what is left is $|\Delta|^{1-5\varepsilon} \leqslant \mu_\varepsilon^6 N_E^{6+6\varepsilon}$, the bound above. Without $\gcd(c_4, c_6) = 1$, one divides the coordinates of P by $\gcd(c_4^3, c_6^2)$, which minimality bounds [Sil09, Exercise 8.21].

Part 17. Thue's equation

This worksheet works through [Sil09, Exercises 9.5 and 9.6(a)–(b)]. In 1909 Thue proved that an equation $F(x, y) = b$, where F is a binary form of degree at least 3 with integer coefficients and

nonzero discriminant and b is a nonzero integer, has only finitely many integer solutions. His idea was to turn the equation into a problem of Diophantine approximation. A solution with y large makes x/y very close to a root of $F(T,1)$, and algebraic numbers cannot be approximated too well by rational numbers. Problem 61 makes the first step quantitative: a polynomial is small only near its roots. Problem 62 proves Thue's theorem with Roth's theorem, and extends it to the S-integers of a number field for forms with an irreducible factor of degree at least 3. One part of each exercise is corrected or restricted, as explained at the start of each section.

Standing notation.

- A number field K, its set of places $\mathcal{P}_K$, and the subset $\mathcal{P}_K^\infty$ of archimedean places.
- For $v \in \mathcal{P}_K$, the completion K_v, the absolute value $|\ |_v$ on K_v extending the usual absolute value on $\mathbb{Q}_p$ or $\mathbb{R}$, and the local degree $n_v \coloneqq [K_v : \mathbb{Q}_v]$, where $\mathbb{Q}_v$ is the completion of $\mathbb{Q}$ at the restriction of v.
- An algebraic closure $\overline{K}$ of K. For $v \in \mathcal{P}_K$, an algebraic closure $\overline{K_v}$ of K_v, to which $|\ |_v$ extends uniquely, as K_v is complete, and a K-embedding $\iota_v \colon \overline{K} \to \overline{K_v}$. Then $|z|_v \coloneqq |\iota_v(z)|_v$ extends $|\ |_v$ to $\overline{K}$. The ι_v are arbitrary, and every statement below holds for every choice of them. For $K = \mathbb{Q}$, the field $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, ι_∞ is the inclusion $\overline{\mathbb{Q}} \subseteq \mathbb{C}$, and $|\ |$ is the usual absolute value.
- The height $H_K(P) \coloneqq \prod_{v\in\mathcal{P}_K} \max_i |x_i|_v^{n_v}$ of a point $P = [x_0 : \cdots : x_N]$ of $\mathbb{P}^N(K)$, which does not depend on the coordinates and satisfies $H_K(P) \geqslant 1$ [Sil09, §VIII.5]. For $x \in K$, $H_K(x) \coloneqq H_K([x:1])$. For coprime integers p and q with $q \neq 0$, $H_{\mathbb{Q}}(p/q) = \max(|p|,|q|)$.
- The product formula $\prod_{v\in\mathcal{P}_K} |a|_v^{n_v} = 1$ for $a \in K^\times$ [Sil09, Product Formula VIII.5.3].

53. Polynomials are small only near their roots

Problem 61 is [Sil09, Exercise 9.5]. If t is far from every root of a polynomial f, then $|f(t)|$ is not small. Quantitatively, $|f(t)|$ is at least a constant times the distance from t to the nearest root, and the constant depends on how close the roots are to each other, which the discriminant controls. Silverman's last part asks for a constant that depends only on the degree of f and on the projective height of its vector of coefficients. No such constant exists: dividing f by an integer N leaves this height unchanged, but at an archimedean place it divides $|f(t)|_v$ by N. So Problem 61(5) asks for a constant that depends on the affine height of the coefficients instead.

Definition 53.1 (Discriminant)**.** Let L be a field and $f = a_0T^n + a_1T^{n-1} + \cdots + a_n \in L[T]$ with $n \geqslant 1$ and $a_0 \neq 0$. Over an algebraic closure of L, $f = a_0\prod_{i=1}^n (T - \xi_i)$. The discriminant of f is

$$D(f) \coloneqq a_0^{2n-2} \prod_{i<j} (\xi_i - \xi_j)^2.$$

It is a polynomial with integer coefficients in $a_0, \ldots, a_n$, by the theorem on symmetric polynomials, since each ξ_i occurs in $\prod_{i<j}(\xi_i - \xi_j)^2$ to degree at most $2n-2$. So $D(f) \in L$, and $D(f) \in \mathbb{Z}$ when $f \in \mathbb{Z}[T]$. It is nonzero if and only if the roots of f are distinct.

Definition 53.2 (Mahler measure [HS00, §B.7])**.** The Mahler measure of $f = a_0\prod_{i=1}^n (T-\xi_i) \in \mathbb{C}[T]$, with $a_0 \neq 0$, is

$$M(f) \coloneqq |a_0| \prod_{i=1}^n \max(1, |\xi_i|).$$

Lemma 53.3 (Mahler measure and coefficients [HS00, Lemma B.7.3.1 and the formula after it])**.** *Let $f = a_0T^n + \cdots + a_n \in \mathbb{C}[T]$ with $a_0 \neq 0$. Then $M(f) \leqslant (n+1)^{1/2} \max_k |a_k|$.*

Proposition 53.4 (Gauss's lemma [HS00, proof of Proposition B.7.3]). *Let v be a non-archimedean place of K. For a polynomial g with coefficients in $\overline{K}$, let $|g|_v$ be the maximum of the absolute values of its coefficients. Then $|gh|_v = |g|_v|h|_v$. In particular, if $f = a_0\prod_{i=1}^n(T-\xi_i)$, then*

$$|f|_v = |a_0|_v\prod_{i=1}^n \max(1,|\xi_i|_v).$$

Proposition 53.5 (Hadamard's inequality). *For $z=(z_1,\dots,z_m)\in\mathbb{C}^m$, let $\|z\| \coloneqq \left(\sum_k |z_k|^2\right)^{1/2}$. Let V be an $m\times m$ complex matrix with rows $r_1,\dots,r_m$. Then $|\det V| \leqslant \prod_{i=1}^m \|r_i\|$.*

Problem 61 (Polynomials are small only near their roots).
Let $f = a_0T^n + a_1T^{n-1} + \cdots + a_n \in K[T]$ be a polynomial of degree $n\geqslant 2$, with $a_0a_n\neq 0$ and with distinct roots $\xi_1,\dots,\xi_n\in\overline{K}$. So $f = a_0\prod_{i=1}^n(T-\xi_i)$, and $D(f)\neq 0$. (Silverman allows $n=1$, where $f(t)=a_0(t-\xi_1)$ and every statement below is immediate.)

(1) Let $v\in\mathcal{P}_K$, and let $\varepsilon_v \coloneqq 1/2$ if v is archimedean and $\varepsilon_v\coloneqq 1$ if v is non-archimedean. Let $t\in\overline{K}$, and let ξ_i be a root of f nearest to t, that is, $|t-\xi_i|_v = \min_j|t-\xi_j|_v$. Show that

$$|f(t)|_v \geqslant \varepsilon_v^{n-1}|f'(\xi_i)|_v|t-\xi_i|_v.$$

Solution. **For $j\neq i$, $\varepsilon_v|\xi_i-\xi_j|_v \leqslant |t-\xi_j|_v$.** For $j\neq i$, the choice of ξ_i gives

$$|\xi_i-\xi_j|_v \leqslant |\xi_i-t|_v + |t-\xi_j|_v \leqslant 2|t-\xi_j|_v,$$

and if v is non-archimedean, then even $|\xi_i-\xi_j|_v \leqslant \max(|\xi_i-t|_v,|t-\xi_j|_v) = |t-\xi_j|_v$. So $\varepsilon_v|\xi_i-\xi_j|_v\leqslant|t-\xi_j|_v$ in both cases. By the product rule, $f'(\xi_i) = a_0\prod_{j\neq i}(\xi_i-\xi_j)$. So

$$|f(t)|_v = |a_0|_v|t-\xi_i|_v\prod_{j\neq i}|t-\xi_j|_v \geqslant \varepsilon_v^{n-1}|a_0|_v|t-\xi_i|_v\prod_{j\neq i}|\xi_i-\xi_j|_v = \varepsilon_v^{n-1}|f'(\xi_i)|_v|t-\xi_i|_v.$$

(2) In this item and the next, $K=\mathbb{Q}$, $f\in\mathbb{Z}[T]$, and $|\ |$ is the usual absolute value on $\mathbb{C}$. Show that every root ξ_i of f satisfies

$$|f'(\xi_i)| \geqslant (n-1)^{-(n-1)/2}M(f)^{-(n-2)}.$$

Solution. **Combine $|D(f)|\geqslant 1$ with Hadamard's inequality.** Number the roots so that $i=1$, and let V be the Vandermonde matrix of $\xi_2,\dots,\xi_n$, the $(n-1)\times(n-1)$ matrix

$$V \coloneqq \begin{pmatrix} 1 & \xi_2 & \xi_2^2 & \cdots & \xi_2^{n-2} \\ 1 & \xi_3 & \xi_3^2 & \cdots & \xi_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \xi_n & \xi_n^2 & \cdots & \xi_n^{n-2} \end{pmatrix}.$$

For $n=2$, $V=(1)$. As $f\in\mathbb{Z}[T]$ has distinct roots, $D(f)$ is a nonzero integer. So

$$1 \leqslant |D(f)|^{1/2}$$

Split the pairs $i<j$ in the definition of $D(f)$ according to whether $i=1$:

$$= |a_0|\prod_{j=2}^n|\xi_1-\xi_j|\cdot|a_0|^{n-2}\prod_{2\leqslant i<j\leqslant n}|\xi_i-\xi_j|$$

As $f'(\xi_1) = a_0\prod_{j=2}^n(\xi_1-\xi_j)$ and $|\det V| = \prod_{2\leqslant i<j\leqslant n}|\xi_i-\xi_j|$,

$$= |f'(\xi_1)||a_0|^{n-2}|\det V|$$

By Proposition 53.5, $|\det V|$ is at most the product of the norms of the rows of V:

$$\leqslant |f'(\xi_1)||a_0|^{n-2}\prod_{i=2}^{n}\Big(\sum_{k=0}^{n-2}|\xi_i|^{2k}\Big)^{1/2}$$

Each of the $n-1$ terms of the sum is at most $\max(1,|\xi_i|)^{2(n-2)}$:

$$\leqslant (n-1)^{(n-1)/2}|f'(\xi_1)|\Big(|a_0|\prod_{i=2}^{n}\max(1,|\xi_i|)\Big)^{n-2}$$

Finally, $\max(1,|\xi_1|)\geqslant 1$ and $n-2\geqslant 0$:

$$\leqslant (n-1)^{(n-1)/2}|f'(\xi_1)|M(f)^{n-2}.$$

Dividing by $(n-1)^{(n-1)/2}M(f)^{n-2}$ gives the bound.

(3) Let $A\coloneqq\max(|a_0|,\dots,|a_n|)$. Show that every rational number t satisfies

$$|f(t)|\geqslant\frac{\min\big(|t-\xi_1|,\dots,|t-\xi_n|\big)}{2^{n-1}(n+1)^{n-3/2}A^{n-2}}.$$

Solution. **Combine (1) and (2) with $M(f)\leqslant(n+1)^{1/2}A$.** Let ξ_i be a root nearest to t. By (1), with $\varepsilon_\infty=1/2$, (2) and Lemma 53.3,

$$|f(t)|\geqslant\frac{|f'(\xi_i)||t-\xi_i|}{2^{n-1}}\geqslant\frac{|t-\xi_i|}{2^{n-1}(n-1)^{(n-1)/2}(n+1)^{(n-2)/2}A^{n-2}}\geqslant\frac{|t-\xi_i|}{2^{n-1}(n+1)^{n-3/2}A^{n-2}}.$$

Remark. Silverman states the weaker bound $|f(t)|\geqslant(2n^2A)^{-n}\min_i|t-\xi_i|$. It follows, since $A\geqslant 1$ as a_0 is a nonzero integer, and $n+1\leqslant n^2$ as $n\geqslant 2$, so

$$2^{n-1}(n+1)^{n-3/2}A^{n-2}\leqslant 2^n(n^2)^nA^n=(2n^2A)^n.$$

The proof never uses $t\in\mathbb{Q}$, and the argument of (1) works verbatim for every $t\in\mathbb{C}$. Rationality enters only in Problem 62, through Roth's theorem.

(4) Let $S\subseteq\mathcal{P}_K$ be a finite set of places. Show that there is a constant $C_f>0$, depending only on f, such that every $t\in K$ satisfies

$$\prod_{v\in S}\min\big(1,|f(t)|_v^{n_v}\big)\geqslant C_f\prod_{v\in S}\min_{1\leqslant i\leqslant n}\min\big(1,|t-\xi_i|_v^{n_v}\big).$$

Solution. **One can take $C_f=\prod_{v\in S}\min(1,c_v)^{n_v}$ with $c_v=\varepsilon_v^{n-1}\min_i|f'(\xi_i)|_v$.** For $v\in S$, let $\delta_v\coloneqq\min_i|t-\xi_i|_v$. As the roots are distinct, $f'(\xi_i)\neq 0$ for every i, so $c_v>0$. By (1), $|f(t)|_v\geqslant c_v\delta_v$. For real numbers $x,y\geqslant 0$, $\min(1,xy)\geqslant\min(1,x)\min(1,y)$. So

$$\min\big(1,|f(t)|_v\big)\geqslant\min(1,c_v)\min(1,\delta_v).$$

Raising to the power n_v and multiplying over $v\in S$ gives the inequality, since $\min(1,\delta_v)^{n_v}=\min_i\min(1,|t-\xi_i|_v^{n_v})$.

(5) Let $H\coloneqq H_K([1:a_0:\cdots:a_n])$. Find a constant C_f for (4) that depends only on n, $[K:\mathbb{Q}]$ and H.

Solution. **One can take $C_f=(2n+2)^{-2n[K:\mathbb{Q}]}H^{-(2n-3)}$.** For $v\in\mathcal{P}_K$, let $|f|_v\coloneqq\max_k|a_k|_v$, let $M_v(f)\coloneqq|a_0|_v\prod_i\max(1,|\xi_i|_v)$, and let $e_v\coloneqq 1$ if v is archimedean and $e_v\coloneqq 0$ otherwise. If v is archimedean, $|\ |_v$ on $\overline{K}$ comes from an embedding of $\overline{K}$ in $\mathbb{C}$, so $M_v(f)\leqslant(n+1)^{1/2}|f|_v$ by Lemma 53.3. If v is non-archimedean, $M_v(f)=|f|_v$ by Proposition 53.4.

The derivative. Number the roots so that $i = 1$. The two equalities in the solution of (2) hold for every absolute value:

$$|D(f)|_v^{1/2} = |f'(\xi_1)|_v |a_0|_v^{n-2} \prod_{2 \leqslant i < j \leqslant n} |\xi_i - \xi_j|_v.$$

At an archimedean v, the rest of that chain bounds $|a_0|_v^{n-2} \prod_{2 \leqslant i < j \leqslant n} |\xi_i - \xi_j|_v$ by $(n-1)^{(n-1)/2} M_v(f)^{n-2}$. At a non-archimedean v, $|\xi_i - \xi_j|_v \leqslant \max(1, |\xi_i|_v) \max(1, |\xi_j|_v)$, and each $i \geqslant 2$ occurs in $n-2$ pairs, so it is bounded by $M_v(f)^{n-2}$. So in both cases

$$|D(f)|_v^{1/2} \leqslant (n-1)^{e_v(n-1)/2} |f'(\xi_1)|_v M_v(f)^{n-2} \leqslant \big((n-1)^{(n-1)/2} (n+1)^{(n-2)/2}\big)^{e_v} |f'(\xi_1)|_v |f|_v^{n-2}.$$

Let $\delta_v \coloneqq \min_i |t - \xi_i|_v$ and $\gamma_v \coloneqq \big(2^{n-1}(n+1)^{n-3/2}\big)^{e_v}$. With (1), applied to a root ξ_i nearest to t, this gives

$$|f(t)|_v \geqslant \frac{\delta_v |D(f)|_v^{1/2}}{\gamma_v |f|_v^{n-2}}.$$

As $\gamma_v \geqslant 1$,

$$\min\big(1, |f(t)|_v\big) \geqslant \gamma_v^{-1} \min(1, \delta_v) \min\big(1, |D(f)|_v\big)^{1/2} \max\big(1, |f|_v\big)^{-(n-2)}.$$

The product over S. Raise this to the power n_v and multiply over $v \in S$. Three factors need a bound.

- As $\sum_{v \in \mathcal{P}_K^\infty} n_v = [K : \mathbb{Q}]$, $\prod_{v \in S} \gamma_v^{n_v} \leqslant \big(2^{n-1}(n+1)^{n-3/2}\big)^{[K:\mathbb{Q}]}$.
- The number $\max(1, |f|_v)^{n_v}$ is the factor of H at v, and every factor of H is at least 1. So $\prod_{v \in S} \max(1, |f|_v)^{n_v} \leqslant H$.
- Let $D \coloneqq D(f) \in K^\times$. Then $\min(1, |D|_v) = \max(1, |D^{-1}|_v)^{-1}$, so as for H,

$$\prod_{v \in S} \min\big(1, |D|_v\big)^{n_v} \geqslant H_K(D^{-1})^{-1} = H_K(D)^{-1},$$

  because $[D^{-1} : 1] = [1 : D]$. Hadamard's inequality for the full Vandermonde matrix of $\xi_1, \dots, \xi_n$ at the archimedean places, and the ultrametric inequality at the others, give $|D|_v \leqslant n^{ne_v} M_v(f)^{2n-2} \leqslant \big(n^n (n+1)^{n-1}\big)^{e_v} |f|_v^{2n-2}$. Hence $H_K(D) \leqslant \big(n^n (n+1)^{n-1}\big)^{[K:\mathbb{Q}]} H^{2n-2}$.

Altogether, $\prod_{v \in S} \min(1, |f(t)|_v^{n_v})$ is at least $\prod_{v \in S} \min(1, \delta_v^{n_v})$ times

$$\begin{aligned} &\big(2^{n-1}(n+1)^{n-3/2}\big)^{-[K:\mathbb{Q}]} \big(n^n (n+1)^{n-1}\big)^{-[K:\mathbb{Q}]/2} H^{-(n-1)} H^{-(n-2)} \\ &\qquad = \big(2^{n-1} n^{n/2} (n+1)^{(3n-4)/2}\big)^{-[K:\mathbb{Q}]} H^{-(2n-3)}. \end{aligned}$$

Finally $2^{n-1} n^{n/2} (n+1)^{(3n-4)/2} \leqslant 2^{2n} (n+1)^{2n} = (2n+2)^{2n}$.

Remark. Silverman asks for a constant depending only on n and the projective height $H_K([a_0 : \cdots : a_n])$. There is none. Over $K = \mathbb{Q}$, with $S = \{\infty\}$, let $f = (T^2 - 2)/N$ for an integer $N \geqslant 2$ and $t = 0$. The left side is $\min(1, 2/N) = 2/N$, and the right side is $C_f \min(1, \sqrt{2}) = C_f$, so $C_f \leqslant 2/N$. But $H_{\mathbb{Q}}([1/N : 0 : -2/N]) = H_{\mathbb{Q}}([1 : 0 : -2]) = 2$ for every N, while the affine height $H_{\mathbb{Q}}([1 : 1/N : 0 : -2/N]) = N$ grows.

54. Thue's equation

Problem 62 is [Sil09, Exercise 9.6(a)–(b)], with Roth's theorem in the form of [HS00, Theorem D.2.1]. If y is large, then at each place of S the ratio x/y of a solution is close to a root of $F(T, 1)$, and this root may change from place to place. Theorem 54.3 has a single algebraic number ξ, but it holds for every choice of the embeddings ι_v, so ξ can stand for a different conjugate at each

place. So Problem 62(3) assumes that F has an irreducible factor of degree at least 3, whose roots are conjugate. Silverman states that part for every F of degree at least 3 with nonzero discriminant.

Definition 54.1 (Binary forms)**.** A binary form of degree d over K is a nonzero homogeneous polynomial $F \in K[X,Y]$ of degree d. Over $\overline{K}$ it is a product of linear forms,

$$F = \prod_{i=1}^{d} L_i, \qquad L_i = \beta_i X - \alpha_i Y.$$

Let $\Delta_{ij} \coloneqq \alpha_i\beta_j - \alpha_j\beta_i$. The discriminant of F is $\prod_{i<j} \Delta_{ij}^2$. It does not depend on the factorization, and if $F(1,0) \neq 0$, it is the discriminant $D(f)$ of $f(T) \coloneqq F(T,1)$. It is nonzero if and only if no two of the L_i are proportional.

Definition 54.2 (S-integers [Sil09, proof of Proposition VIII.1.6])**.** Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$. The ring of S-integers of K is

$$\mathcal{O}_S \coloneqq \{a \in K : |a|_v \leqslant 1 \text{ for all } v \notin S\}.$$

Theorem 54.3 (Roth [HS00, Theorem D.2.1])**.** *Let $S \subseteq \mathcal{P}_K$ be a finite set, $\xi \in \overline{K}$ and $\varepsilon > 0$. Then only finitely many $t \in K$ satisfy*

$$\prod_{v \in S} \min\bigl(1, |t-\xi|_v^{n_v}\bigr) \leqslant H_K(t)^{-(2+\varepsilon)}.$$

Theorem 54.4 (Northcott [Sil09, Theorem VIII.5.11])**.** *For every $B > 0$, only finitely many $t \in K$ satisfy $H_K(t) \leqslant B$.*

Problem 62 (Thue's equation)**.**
Let $F \in K[X,Y]$ be a binary form of degree $d \geqslant 3$ with nonzero discriminant, and let $b \in K^\times$.
(1) Let $K = \mathbb{Q}$, and suppose that $F \in \mathbb{Z}[X,Y]$ and $b \in \mathbb{Z}$. Show that $F(x,y) = b$ has only finitely many solutions $(x,y) \in \mathbb{Z}^2$.

Solution. **For large y, x/y is too close to a root of $F(T,1)$, as $d \geqslant 3$.** The proof has four steps.

Step 1. Changes of variables make $F(1,0)F(0,1) \neq 0$.
Step 2. For a solution with $|y|$ large, $t \coloneqq x/y$ is within $C' H_{\mathbb{Q}}(t)^{-d}$ of a root of $F(T,1)$, for a constant C'.
Step 3. Roth's theorem leaves only finitely many such t.
Step 4. Each t comes from finitely many solutions, and so do the small values of y.

Step 1: reduction. For $m \in \mathbb{Z}$, the maps $(x,y) \mapsto (x, y+mx)$ and $(x,y) \mapsto (x+my, y)$ are bijections of $\mathbb{Z}^2$, and composing F with them gives binary forms of degree d with integer coefficients and nonzero discriminant, since an invertible linear substitution keeps non-proportional linear forms non-proportional. As $F(X,Y) = X^d F(1, Y/X)$, the polynomial $F(1,T)$ is nonzero, so $F(1,m) \neq 0$ for some $m \in \mathbb{Z}$. Then $G(X,Y) \coloneqq F(X, Y+mX)$ has $G(1,0) \neq 0$. Likewise $G(k,1) \neq 0$ for some $k \in \mathbb{Z}$, and $G(X+kY, Y)$ takes the values $G(1,0)$ and $G(k,1)$ at $(1,0)$ and $(0,1)$. So we may assume that $a_0 \coloneqq F(1,0)$ and $a_d \coloneqq F(0,1)$ are nonzero. Then $f(T) \coloneqq F(T,1) = a_0 T^d + \cdots + a_d$ has degree d, $a_0 a_d \neq 0$, and $D(f) \neq 0$ by Definition 54.1. Let $\xi_1, \ldots, \xi_d \in \overline{\mathbb{Q}}$ be its roots.
Step 2: close to a root. Let $A \coloneqq \max_k |a_k|$, $\gamma \coloneqq 2^{d-1}(d+1)^{d-3/2} A^{d-2}$, $C \coloneqq \gamma |b|$ and $R \coloneqq \max_i |\xi_i|$. Let (x,y) be a solution with $|y|^d \geqslant C$, and $t \coloneqq x/y$. Then $f(t) = y^{-d} F(x,y) = b/y^d$, so by Problem 61(3) some root ξ_i satisfies

$$|t - \xi_i| \leqslant \gamma |f(t)| = C|y|^{-d} \leqslant 1.$$

So $|x| \leqslant (R+1)|y|$, and $H_{\mathbb{Q}}(t) \leqslant \max(|x|,|y|) \leqslant (R+1)|y|$, since x and y need not be coprime. Let $C' \coloneqq C(R+1)^d$. Then

$$|t - \xi_i| \leqslant C' H_{\mathbb{Q}}(t)^{-d}.$$

Step 3: finitely many t. As $d \geqslant 3$ and $H_{\mathbb{Q}}(t) \geqslant 1$, Step 2 gives

$$|t - \xi_i| \leqslant C' H_{\mathbb{Q}}(t)^{-1/2} \cdot H_{\mathbb{Q}}(t)^{-5/2}.$$

If $H_{\mathbb{Q}}(t) \geqslant C'^2$, then $\min(1, |t-\xi_i|) \leqslant H_{\mathbb{Q}}(t)^{-5/2}$, and Theorem 54.3, with $K = \mathbb{Q}$, $S = \{\infty\}$ and $\varepsilon = 1/2$, applied to each of the d roots, leaves only finitely many such t. Only finitely many $t \in \mathbb{Q}$ have $H_{\mathbb{Q}}(t) < C'^2$, as their numerators and denominators are bounded. So the solutions with $|y|^d \geqslant C$ give only finitely many values of t.

Step 4: back to the solutions. For each value of t from Step 3, $f(t) = b/y^d \neq 0$, so $|y|^d = |b/f(t)|$ leaves at most two values of y, and $x = ty$. The solutions with $|y|^d < C$ have finitely many values of y. For each, the polynomial $F(X,y) - b$ has leading coefficient $a_0 \neq 0$, so only finitely many x satisfy $F(x,y) = b$.

Remark. The hypothesis $d \geqslant 3$ is needed: the equation $x^2 - 2y^2 = 1$ has the infinitely many solutions given by $x + y\sqrt{2} = \pm(3+2\sqrt{2})^k$, $k \in \mathbb{Z}$. For $d = 2$ the argument gives $|t - \xi_i| \leqslant C' H_{\mathbb{Q}}(t)^{-2}$, which Dirichlet's theorem allows infinitely often [Sil09, Proposition IX.1.2].

(2) `Magma` solves Thue equations. The code below lists all the integer solutions of $x^3 - 2y^3 = b$ for $1 \leqslant b \leqslant 6$. Explain why the argument of (1) could not produce such a list.

```
R<x> := PolynomialRing(Integers());
t := Thue(x^3 - 2);                    // the form X^3 - 2*Y^3
[ <b, Solutions(t, b)> : b in [1..6] ];
// [ <1, [[-1, -1], [1, 0]]>, <2, [[0, -1]]>, <3, [[-5, -4], [1, -1]]>,
//   <4, []>, <5, []>, <6, [[2, 1]]> ]
```

SOLUTION. **Roth's theorem gives no bound for the height of the exceptions.** The argument of (1) bounds $|y|$ by an explicit constant except for the solutions with x/y in the finite set of Theorem 54.3. That theorem says the set is finite, but its proof gives no bound for the heights of its elements [Sil09, Remark IX.1.6]. So a search can find solutions, but never knows when it has found them all.

Remark. `Magma` uses instead Baker's lower bounds for linear forms in logarithms, which give an explicit, if huge, bound for the solutions, and then reduces it until a search is feasible [Sil09, §IX.5].

(3) Let $S \subseteq \mathcal{P}_K$ be a finite set containing $\mathcal{P}_K^\infty$, and suppose that F has an irreducible factor $G \in K[X,Y]$ of degree $g \geqslant 3$. Show that $F(x,y) = b$ has only finitely many solutions $(x,y) \in \mathcal{O}_S \times \mathcal{O}_S$.

SOLUTION. **Conjugating the nearest roots to one root reduces to Roth's theorem.** *Reductions.* The changes of variables of (1) are also bijections of $\mathcal{O}_S \times \mathcal{O}_S$. As there, $F(1,m) \neq 0$ for some $m \in \mathbb{Z}$, so after replacing F by $F(X, Y+mX)$ and G by $G(X, Y+mX)$ we may assume that $a_0 \coloneqq F(1,0)$ is nonzero. Then $G(1,0) \neq 0$, as G divides F. Write $F = GG'$ with $G' \in K[X,Y]$. A nonzero element of K has $|\ |_v > 1$ at only finitely many places, and enlarging S only enlarges $\mathcal{O}_S$, so we may assume that the coefficients of G' lie in $\mathcal{O}_S$. Since $G(1,0) \neq 0$, the polynomial $G(T,1)$ has degree g, and it is irreducible over K, as a factorization of it would homogenize to one of G. Its roots $\xi_1, \dots, \xi_g \in \overline{K}$ are distinct, since F has nonzero discriminant. They are nonzero, since $G(0,1) \neq 0$: otherwise X would divide G, which is irreducible of degree

at least 3. If $y = 0$, then $a_0 x^d = b$ leaves at most d values of x. So let (x, y) be a solution with $y \neq 0$, and let $t := x/y$ and $M_v := \max(|x|_v, |y|_v)$ for $v \in \mathcal{P}_K$.

The places of S. The element $c := G'(x, y)$ lies in $\mathcal{O}_S$, and $c \neq 0$ because $G(x, y)c = b$. By the product formula, $\prod_{v\in S} |c|_v^{n_v} = \prod_{v\notin S} |c|_v^{-n_v} \geqslant 1$, so

$$\prod_{v\in S} |G(x, y)|_v^{n_v} \leqslant B := \prod_{v\in S} |b|_v^{n_v}.$$

A local bound. Fix $v \in S$, and let $a := G(1, 0)$, $R_v := \max(1, 2\max_i |\xi_i|_v)$, $m_v := \min_i |\partial_T G(\xi_i, 1)|_v$, which is positive as the roots are distinct, and $\delta_v := \min_i \min(1, |t - \xi_i|_v)$. Let $\kappa_v := \min(2^{-g}|a|_v, \varepsilon_v^{g-1} R_v^{-g} m_v)$, with ε_v as in Problem 61(1). It is positive and does not depend on the solution. We show that $|G(x, y)|_v \geqslant \kappa_v M_v^g \delta_v$. Here $G(x, y) = a\prod_i (x - \xi_i y)$. If $|x|_v > R_v|y|_v$, then $M_v = |x|_v$ and $|x - \xi_i y|_v \geqslant |x|_v - |\xi_i|_v|y|_v \geqslant |x|_v/2$ for every i, so $|G(x, y)|_v \geqslant 2^{-g}|a|_v M_v^g$, which is enough as $\delta_v \leqslant 1$. Otherwise $M_v \leqslant R_v|y|_v$, and Problem 61(1), applied to $G(T, 1)$, gives

$$|G(x, y)|_v = |y|_v^g |G(t, 1)|_v \geqslant R_v^{-g} M_v^g \cdot \varepsilon_v^{g-1} m_v \delta_v.$$

The height. For $v \notin S$, $M_v \leqslant 1$, so $H_K(t) = H_K([x : y]) \leqslant \prod_{v\in S} M_v^{n_v}$. Raising the local bounds to the powers n_v and multiplying, with $\kappa := \prod_{v\in S} \kappa_v^{n_v}$,

$$\prod_{v\in S} \min_i \min\bigl(1, |t - \xi_i|_v^{n_v}\bigr) = \prod_{v\in S} \delta_v^{n_v} \leqslant \frac{B}{\kappa} \prod_{v\in S} M_v^{-gn_v} \leqslant \frac{B}{\kappa} H_K(t)^{-g}.$$

One root. For each $v \in S$, choose $i(v)$ with $|t - \xi_{i(v)}|_v = \min_i |t - \xi_i|_v$. There are only finitely many maps $i\colon S \to \{1, \dots, g\}$, so it suffices to fix one and show that only finitely many t have it. As $G(T, 1)$ is irreducible over K, for each $v \in S$ there is $\sigma_v \in \mathrm{Aut}(\overline{K}/K)$ with $\sigma_v(\xi_1) = \xi_{i(v)}$. Replace each ι_v by the K-embedding $\iota_v \circ \sigma_v$, and write $|\ |'_v$ for the resulting extension of $|\ |_v$ to $\overline{K}$, so that $|z|'_v = |\sigma_v(z)|_v$. As $\sigma_v(t) = t$, $|t - \xi_1|'_v = |t - \xi_{i(v)}|_v$. With $C := B/\kappa$, and since $g \geqslant 3$ and $H_K(t) \geqslant 1$,

$$\prod_{v\in S} \min\bigl(1, (|t - \xi_1|'_v)^{n_v}\bigr) \leqslant C H_K(t)^{-g} \leqslant C H_K(t)^{-1/2} \cdot H_K(t)^{-5/2}.$$

If $H_K(t) \geqslant C^2$, the right side is at most $H_K(t)^{-5/2}$, and Theorem 54.3, with the embeddings $\iota_v \circ \sigma_v$, $\xi = \xi_1$ and $\varepsilon = 1/2$, leaves only finitely many t. By Theorem 54.4, only finitely many t have $H_K(t) < C^2$.

Back to the solutions. For each value of t, $f(t) := F(t, 1) = b/y^d \neq 0$, so $y^d = b/f(t)$ leaves at most d values of y, and $x = ty$.

Remark. The irreducible factor G is used only to move the nearest roots at the places of S to a single root. If S is a single place, for instance if K is $\mathbb{Q}$ or an imaginary quadratic field and $S = \mathcal{P}_K^\infty$, there is nothing to move: the same argument, with $G = F$ and the nearest root fixed by the pigeonhole principle, works for every F of degree at least 3 with nonzero discriminant, as in (1). With several places, the general case needs Roth's theorem with a different algebraic number at each place, which Theorem 54.3 does not give. For $F = XY(X + Y)$, for instance, x/y may be close to 0 at some places of S and to -1 at others, and these roots are not conjugate. Silverman's statement still holds for such F, but its proof needs a further idea.

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